TOPIC -2 : WHOLE NUMBERS

Subtopic: place values and values.

Session 1.

Competences

- Place values and values of whole numbers
- Operations on place values and values of whole numbers
- Writing whole numbers in words and figures
  - Expanding whole numbers

Methods

- Guided discovery
- Group discussion
- Think, pair and share.
- explanation

Lesson activities

- **adding**
- **multiplying**
- **dividing**
  - **subtracting**

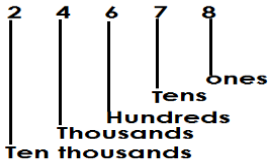
Relevance

The concept of place values and values helps a learner to identify the positions of objects.

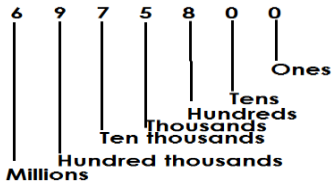
Content/facts

NUMERATION SYSTEMS AND PLACE VALUES
PLACE VALUES AND VALUES OF WHOLE NUMBERS
PLACE VALUE

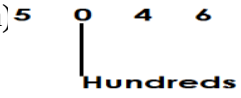
Place value is the position of a given digit in a given numeral.
Place values of whole digits end in letter "s"
1.Find the place value of each digit in the following
a)24678

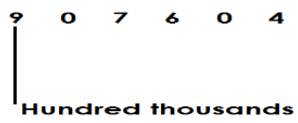


b). 6975800

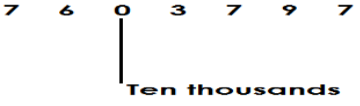


2.Finding the value of the underlined digits in the above numerals

(a) 

(b) 

(c)



Activity

1. Find the place value of each digit in the following

(a) 3467 (b) 907076 (c) 30464 (d) 9076036

2. Find the place value of the underlined digits.

(a) 904 (b) 6706 (c) 30467

(d) 907,632

FINDING VALUES OF WHOLE NUMBERS IN GIVEN NUMERAL

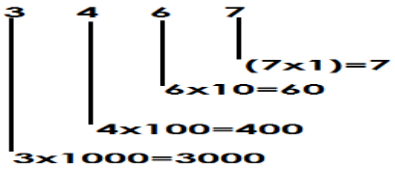
A value is how manyness of a given digit in a given numeral.

Value is got by multiplying a digit by is place value.

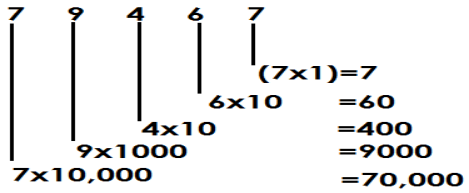
Value = digit x place value.

1. Find the value of each digit in the following.

a)



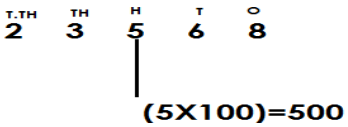
b)



3. Find the value of digit 5 in the number

23568

Solution



ACTIVITY

1. Find the value of each digit in the following numbers

(a) 423 (b) 6834

(c) 20168 (d) 24019

2. Find the value of underlined digits in the following numbers.

(a) 4<u>6</u>3 (b) 2<u>0</u>18 (c) 1324<u>9</u>

3. Find the value of digit 4in 3456.

FINDING SUM, DIFFERENCE, PRODUCT AND QUOTIENT OF DIGITS IN A GIVEN NUMBER.

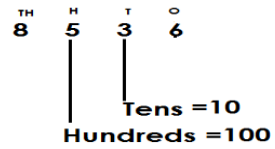
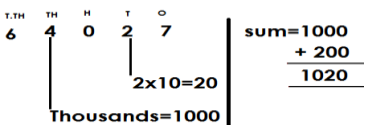
Sum- the answer got after adding the given numbers.

Difference- the answer obtained after subtracting

Product- the result of multiplication

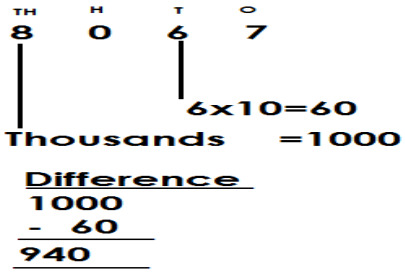
Quotient- the result of division.

Examples

1.Given the number 8536. Find the sum of the place value of 3 and place value of 5 **Sum =** 100 + 10 = 110 	2.What is the sum of the place value of 4 and the value of 2 in 64027? 

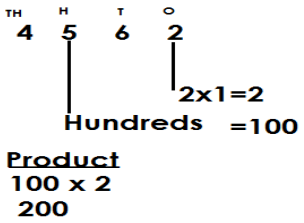
3.Find the difference between the place value of 8 and the value of 6 in 8067.

Solution



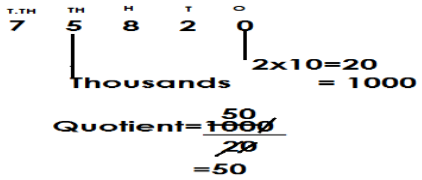
4.Find the product of the place value of 5 and the value of 2 in the number 4563.

Solution



5.Find the quotient of the place value of 5 and the value of 2 in 75820.

Solution



ACTIVITY

1.Given 4635. Find the sum of a place value of 4 and place of 3.
2.Find the sum of the value of 6 and value of 4 in 93461.
3.Find the difference between the place value of 8 and the value of 5 in 8057.
4.What is the difference between the value of 3 and the place value of 4 in 7324?
5.Find the product of the place value of 2 and the value of 3 in 42673?
6.Given the number 97265, find the quotient of the value of 7 and 5.
7.Given the number 8536, find the product of a place value of 3 and place value of

EXPANDING WHOLE NUMBERS

Whole numbers are expanded in three ways;

- Using place values
- Using value
- Using powers/indices/exponents

Examples

1.Expand 208 using place values

Solution

H T O
2 0 8

= (2x100) + (0x10) + (8x1)

2.Expand 439 using values

Solution

H T O
4 3 9

= (4x100) + (3x10) + (9x1)
= 400 + 30 + 9

3.Expand 650 using exponents

Solution

6	5	0
10^2	10^1	10^0

= $(6x10^2) + (5x10^1) + (0x10^0)$

Activity

1. Expand the following numbers using place values
   (a) 432 (b) 1068 (c) 4674
2. Expand the following numbers using values
   (a) 68 (b) 468 (c) 2074
3. Expand the following numbers using exponents
   (a) 146 (b) 4023 (c) 1742

scenario

Session 2

competences

- Rounding off whole numbers
- Converting Hindu Arabic to Roman Numerals and vice versa

Methods

- Guided discovery
- Group discussion
- Think, pair and share.
- explanation

Lesson activities

- **adding**
- **multiplying**
- **dividing**
- **subtracting**

Relevance

The concept of place values and values helps a learner to identify the positions of objects.

Content/facts

ROUNDING OFF OF WHOLE NUMBERS

Note:

Rounding off is a way of approximating numbers to the required place values

ROUNDING OFF WHOLE NUMBERS

- In rounding off, base on the digit on the right of the figure in the required place value to either roundup or round down.
- Round down in case the digit on the right of the figure in the required place value is either,0,1,2,3, or 4 by adding 0 to the digit in the required place value.
- Roundup in case the digit on the right of the figure in the required place value is either 5, 6, 7, 8, or 9 by adding the value of the digit in the required place value.
- All digits coming on the right of the digit in the required place value will turn into zero (0) when rounding off of whole numbers.

Examples

1.Round off 24643 to the nearest hundreds.

2 4 6 4 3=

Tth	TH	H	T	O
2	4	6	4	3
		RPV		

24600
+ 000
24600

2.Round off 4867 to the nearest thousands

Solution

4867=

TH	H	T	O
4	8	6	7
RPV			

4000
+1000
5000

3.Round off 75 the nearest hundreds

Solution

75=

TH	H	T	O
	0	7	5
	RPV		

000
+100
100

Activity

1. Round off to the nearest tens

(a) 24 (b) 337 (c) 4689 (d) 214

2. Round off to the nearest hundreds

(a) 263 (b) 586 (c) 1721 (d) 9296

3. Round off to the nearest thousands

(a) 8634 (b) 1945 (c) 57389 (d) 8634

ROUNDING OFF DECIMAL NUMBERS

Note: -

While rounding off decimals, all digits coming after the figure in the required place value are crossed out. I.e. the last digit must be in the mentioned place value.

Rounding off to the nearest whole numbers means to round off to the place value of ones.

Rounding off to one decimal place means to the place value of tenths.

Rounding off to two decimal places means to the place value of hundredths.

Rounding off to three decimal places means to the place value of thousandths.

Examples

1.Round off 39.45 to the nearest tenths

39.45=

T	O	.	Tths	Hths
3	9	.	4	5
			RPV	

39.4
+ 0.1
39.5

2.Round off 89.996 to the nearest hundredths

89.996 =

T	O	.	Tths	Hths	THths
8	**9**	**.**	**9**	**9**	**6**
				RPV	

89.99
+ 0.01
90.00

3.Correct 97.656 to 1 decimal place

97.656 =

T	O	.	Tths	Hths	THths
9	**7**	**.**	**6**	**5**	**6**
		RPV			

97.6
+ 0.1
97.7

Activity

1. Round off the following to one decimal place
2. (a) 54.67 (b) 503.906 (c) 576.96 (d) 709.63
3. Round off the following to the nearest tenths
4. (a) 907.63 (b) 123.99 (c) 54.076
5. Round off the following to the nearest hundredths
6. (a) 40.986 (b) 49.586 (c) 50.764

7. Round off 69.465 to 2 decimal places
8. Round off 909.3976 to the nearest thousandths.

TYPES OF NUMERALS

1.Hindu Arabic numerals

2.Roman numerals

Hindu Arabic numerals

These are numerals that are used in most parts of the world.

Major Hindu Arabic numerals

0,1,2,3,4,5,6,7,8,9

ROMAN NUMERALS

1.Basic Roman numerals

Hindu Arabic	1	5	10	50	100	500	1000
Roman numerals	I	V	X	L	C	D	M

NOTE:

- All roman numerals should be written in capital letters.
- All types of Roman numerals are derived from basic Roman numerals.

2.Repeated roman numerals

They are got by repeating some basic Roman numerals like I, X, C and M

Repeated roman numerals act on digits 2 and 3 in place values of ones, tens, hundreds and thousands.

(a)In the place value of ones	(b)In the place value of tens	(c)In the place value of hundreds	(d)In the place value of thousands
2 - II	20 - XX	200 - CC	2000-MM
3 - III	30 - XXX	300 - CCC	3000-MMM

3.Addition Roman numerals

Addition Roman numerals use digits 6, 7 and 8 in the place values of ones, tens, hundreds.

(a)In the place value of ones	(b)In the place value of tens	(c)In the place value of hundreds
6 = 5+1	60 = 50 +10	600 = 500 +100
= VI	= LX	= DC
7 = 5 + 2	70 = 50 + 20	700 = 500 +100
= VII	= LXX	= DCC
8 = 5+3	80 = 50 + 30	800 = 500 + 300
= VIII	= LXXX	= DCCC

4.SUBTRACTION ROMAN NUMERALS

These uses digits 4 and 9 in the place values of ones, tens and hundreds.

(a)In the place value of ones	(b)In the place value of tens	(c)In the place values of hundreds
4 = 5 -1	40 = 50 - 10	400 = 500 - 100
= IV	= XL	= CD
9 = 10 - 1	90 = 100 - 10	900 = 1000 -100
= IX	= XC	= CM

WRITING NUMBERS IN ROMAN NUMERALS

1.Write 149 in roman numerals

100	40	9
C	XL	IX

149 = CXLIX

2.Write the following in roman numerals;

(a) 38

30	8
XXX	VIII

38 = XXXVIII

b) 89

80	9
LXXX	IX

89 = LXXXIV

c) 499

400	90	9
CD	XC	IX

499 = CDXCIX

d) 649

600	40	9
DC	XL	IX

649 = DCXL IX

e) 782

700	80	2
DCC	LXX	II

782 = DCCLXXXII

Activity

Write the following in Roman numerals

(a) 49 (b) 36
(c) 29 (d) 69
(e) 326 (f) 99
(g) 436 (h) 223
(i) 648 (j) 562
(k) 888 (l) 1999

Changing from Roman numerals to Hindu Arabic numerals

1.Write LXXXIX in Hindu Arabic numerals

LXXXIX= LXXX IX
80 9
= 89

2.Write the following in Hindu Arabic numerals

(a)XVIII

X	VIII
10	8

XVIII =18

b) DCVII

DC	VII
600	7

DCVII = 607

(c)MCMXCIX=

M	CM	XC	IX
1000	900	90	9

1999

Activity

(a) XXXVI
(b) XCIV
(c) LVI
(d) CXXXVIII
(e) CDXCVIII
(f) CMXLIX
(g) MMCXI
(h) DCCCLXXXVIII
(j) MMMDCCCLXXXVIII

WORD PROBLEMS INVOLVING HINDU ARABIC AND ROMAN NUMERALS

Examples

1. Musa was born in 1996. Express his year of birth in roman numerals

10000	900	90	6
M	CM	XC	VI

1996 = MCMXCVI

2. Our teacher is 79 years old. Write his age in Roman numerals

70	9
LXX	IX

79 = LXXIX

3. Mugisha has CCXLIX animals on his farm. Write his number of animals in Hindu Arabic numerals

CC	XL	IX
200	40	9

CCXLIX = 249 animals

4. Mukasa's great grandfather died in MDCCCLXXXVIII. Write his year of death in Hindu Arabic numerals

M	DCCC	LXXX	VIII
1000	800	80	8

MDCCCLXXXVIII = 1888

Activity

1. 2. It is 206km from Town A to town B. Write that distance in roman numerals.
2. Alex was born in 2007. Write his year of birth in roman numerals.
3. There are 108 English books and 73 science books in the library. Write the total number of books in the library.
4. A bag contains XXXIX oranges. Write the number of oranges in Hindu Arabic number.
5. A trader sold CMLXIII bags of cotton. Write the number of bags of cotton in Hindu Arabic numeral.
6. A shop keeper sold XXIV kg of sugar on Monday and LIII kg on Tuesday. Find the total number of kg sold in Hindu Arabic numerals.

scenario

Session 3

competences

- Place values and values of bases.
- Converting non-decimal base to the decimal base (base ten)
- Converting from base ten to non-decimal bases.
- Converting non-decimal bases to non-decimal bases.

Methods

- Guided discovery
- Group discussion
- Think, pair and share.
- explanation

Lesson activities

- **adding**
- **multiplying**
- **dividing**
- **subtracting**

Relevance

The concept of place values and values helps a learner to identify the positions of objects.

Content/facts

BASES

A base is a counting system where a particular natural/counting number is taken to be the limit of its counting system.

Base names

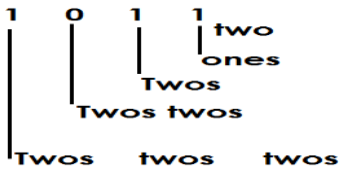
Base	Name	Digits used
2	binary	0,1
3	Trinary/ternary	0,1,2
4	quaternary	0,1,2,3
5	quinary	0,1,2,3,4
6	senary	0,1,2,3,4,5
7	Septenary/ heptanol	0,1,2,3,4,5,6
8	octal	0,1,2,3,4,5,6,7
9	nonary	0,1,2,3,4,5,6,7,8
10	Decimal/Standard/Normal/Denary	0,1,2,3,4,5,6,7,8,9

PLACE VALUES OF BASE NUMBERS

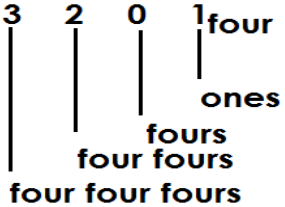
N.B: The starting place value of base numbers is ones.

1.Write the place value of each digit

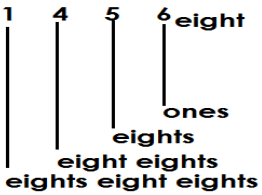
(a)



(b)

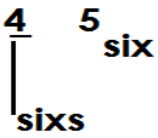


(c)

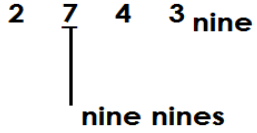


2.Find the place value of the underlined digits.

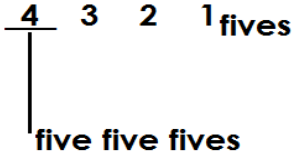
(a)



(b)



(c)



Activity

1. Find the place value of each digit

(a) 101_{two} (b) 231_{four} (c) 4031_{fives} (d) 567_{eight} (e) 678_{nine}

2. Find the place value of the underlined digits.

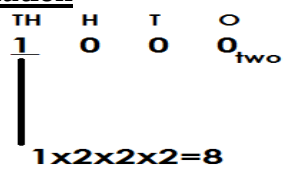
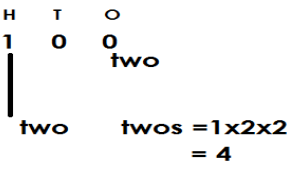
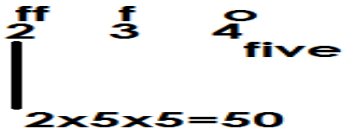
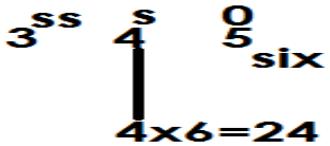
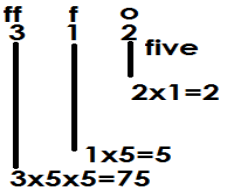
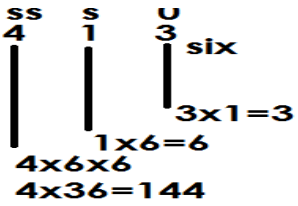
(a) 203_{four} (b) 4031_{five} (c) 367_{eight} (d) 405_{six}

Values of digits in base numbers

A value is a product of a digit and its place value

Value of a digit = digit x place value

Examples

1.Find the value of 1 in 1000 two **Solution**  	2.Find the value of 1 in 100 two  	3.Find the value of 2 in 234 five **Solution** 
4.Find the value of 4 in 345 six  	5.Find the value of each digit (a)  	(b) 

Activity

1. Find the value of each digit

(a) 101_{two} (b) 203_{three}

(c) 402_{five} (d) 121_{three}

2. Find the value of the underlined

(a) $\underline{1}02t_{hree}$(b)$1\underline{0}1_{two}$

(c) $7\underline{2}03_{nine}$ $\underline{1}101_{two}$

WRITING BASE NUMBERS IN WORDS

Note: Base number digits are read and written independently (one by one) while the base is written last.

Write the following in words

1. 1010two

One zero one zero base two

2. 10001

One zero zero zero base two

3. 4302five

Four three zero two base five

4. 6254seven

Six two five four base seven

Activity

Write the following base numbers in words

(a) 1000_{two} (b)11111_{two} (c) 2022_{three} (d)7654_{eight} (e) 5454_{six}

EXPANDING BASE NUMBERS

Base numbers are expanded in three ways

(i) Place values

(ii) Values

(iii) powers/exponents/indices

Expanding base numbers using place values

Expand the following using place values (a) 1 1 1 1 two TH H T O 1 1 1 1 two (1x2x2x2) + (1x2x2) + (1x2) + (1x1)	(b) H T O 1 0 1 two (1x2x2) + (0x2) + (1x1)	(c) fff ff f o 1 2 0 3 four (1x4x4x4) + (2x4x4) + (0x4) + (3x1)
(d) 567_{eight} ee e o 5 6 7 eight (5x8x8) + (6x8) + (7x1)		

Activity

Expand the following using place values

(a) 234_{five} (b) 101_{two} (c) 4201_{five}

(d) 465_{six} (e) 17_{ten} (f) 3102_{four}

Expanding base numbers using values

Expand the following using values (a) 111_{two} 1 0 0 1 two TH H T O 1 0 0 1 two (1x2x2x2) + (0x2x2) + (0x2) + (1x1) 8+0+0+1	(b) H T O 1 1 1 two (1x2x2) + (1x2) + (1x1) 4+2+1	c)124_{five} ff f o 1 2 4 five (1x5x5) + (2x5) + (4x1) 25+ 10+ 4
(d) 2 1 2 three H T O 2 1 2 three =(2x3x3) + (1x3) + (2x1) 18+3+2		

Activity

1. Expand the following base numbers using values

(a) 1010 two (b) 1011two
(c) 10001two (d) 203 four
(e) 103 four (f) 120_{three}
(g) 134_{five} (h) 2231_{five}
(i) 1003_{five} (j) 435_{six}

EXPANDING BASE NUMBERS USING POWERS/INDICES/EXPONENTS

1.Expand 1100_{two} using powers.

1	1	0	0
2^3	2^2	2^1	2^0

$(1x2^3) + (1x2^2) + (0x2^1) + (0x2^0)$

2.**Expand the following using powers**

(a) 214 five

2	1	4
5^5	5^1	5^0

$= (2x5^2) + (1x5^1) + (4x5^0)$

(b) 102 three

1	0	2
3^3	3^1	3^0

$= (1x3^2) + (0x3^1) + (2x3^0)$

(c) 175 ten (d) 46 eight

1	7	5
10^2	10^1	10^0

$= (1x10^2)$

4	6
8^1	8^0

$+ (7x10^1) + (5x10^0)$

$= (4x8^1) + (6x8^0)$

Activity

1.Expand the following base numbers using powers

(a) 1111_{two} (b) 1001_{two} (c) 1000_{two}

(d) 3012_{four} (e) 212_{three} (f) 5312_{seven}

(g) 43_{six} (h) 6735_{nine} (i) 257_{ten}

(j) 1010_{ten}

CONVERSION OF BASE NUMBERS

Note:

Base numbers are converted from;

(a) Base ten/decimal base to other bases/non-decimal bases

(b) Other bases/non-decimal bases to base ten/decimal bases.

(c) Other bases/non-decimal bases t other bases/non-decimal bases.

1.Change 29_{ten} to binary base

B	N	R
2	29	1
2	14	0
2	7	1
2	3	1
2	1	1
	0	

$29_{ten} = 11101_{two}$

2.Change 75ten to quinary base

B	N	R
5	75	0
5	15	0
5	3	3
	0	

$75_{ten} = 300_{five}$

3.Babirye scored 42 in a test. Express her marks in octal base

B	N	R
8	42	2
8	5	5
	0	

$42_{ten} = 52_{eight}$

Activity

1. Change the following base numbers to the required base

(a) 23ten to binary base

(b) 49ten to binary base

(c) 29ten to base three

(d) 45ten to quinary base

(e) 9ten to ternary base.

(f) 31ten to base four

2. Alex is 19yeaers old. Write his age in binary base.

3. A girl scored 63 marks in a test. Express her mark in nonary base.

4. The teacher is 28years. Write his age in base four.

5. Makanga was born in 2000. Write his year of birth to quinary base system.

Changing non-decimal base numbers to decimal base

Note

Here we expand a non-decimal base number using powers of the given non-decimal base number

1. Change 1001_{two} to decimal base

1	0	0	1
2^3	2^2	2^1	2^0

$= (1x2^3) + (0x2^2) + (0x2^1) + (1x2^0)$

$= (1x2x2x2) + (0x2x2) + (0x2) + (1x1)$

$= 8+0+0+1$

$= 9_{ten}$

2. Change 132_{five} to base ten

1	3	25
5^2	5^1	5^0

$= (1x5^2) + (3x5^1) + (2x5^0)$
$= (1x5x5) + (3x5) + (2x1)$
$= 25+15+2$
$= 42_{ten}$

3.Express 1122_{three} as a decimal base number

1	1	2	2
3^3	3^2	3^1	3^3

$= (1x3^3) + (1x3^2) + (2x3^1) + (2x3^0)$
$= (1x3x3x3) + (1x3x3) + (2x3) + (2x1)$
$= 27+9+6+2$
$= 44_{ten}$

Activity

1. Change the following non-decimal base numbers to decimal base.

(a) 1111_{two} (b) 1010_{two} (c) 1011_{two} (d) 1100_{two} (e) 1000_{two}
(f) 202_{three} (g) 123_{four} (h) 210_{six} (i) 143_{five} (j) 1012_{five}

CHANGING NON-DECIMAL BASE NUMBERS TO NON-DECIMAL BASE NUMBERS

Note:

First change a non-decimal base number to base ten by expanding then after to the required base by dividing

Examples

1. Change 110two to base three

1	1	0
2^2	2^1	2^0

$= (1X2^2) + (1X2^1) + (0X2^0)$
$= (1X2X2) + (1X2) + (0X1)$
$= 4+2+0$
$= 6_{ten}$

B	N	R
3	6	0
3	2	2
	0	

$110_{two} = 20_{three}$

---

2. Change 1111_{two} to quinary base.

$(1x2^3) + (1x2^2) + (1x2^1) + (1x2^0)$
$(1x2x2x2) + 1x2x2) + (1x2) + (1x1)$
$8+4+2+1$
$15ten$

B	N	R
5	15	0
5	3	3
	0	

$1111_{two}=30_{five}$

3.Change 102_{four} to binary base

1	0	2
4^2	4^1	4^0

$(1x4^2) + (0x4^1) + (0x4^0)$

$(1x4x4) + (0x4) + (0x1)$

$16+0+1$

17_{ten}

B	N	R
2	17	1
2	8	0
2	4	0
2	2	0
2	1	1
	0	

$102_{four}=10001_{two}$

Activity

1. Change the following to the required bases

(a) 1001two to base four

(b) 1000two to base three

(c) 1010two to quinary base

(d) 122three to binary base

(e) 102four to base three

(f) 300four to base five

Scenarios

Session 4

competences

- Operations on bases.
- Solving for unknowns in bases.

Methods

- Guided discovery
- Group discussion
- Think, pair and share.
- explanation

Lesson activities

- **adding**
- **multiplying**
- **dividing**
- **subtracting**

Relevance

The concept of place values and values helps a learner to identify the positions of objects.

Content/facts

Operations on bases.

ADDIT ION OF BASE NUMBERS

1.Add: 110two+11two	2.Add: 1111two+1111two	3.Workout: 310five+212five
$\begin{array}{r} {\scriptstyle 1\ 1}\\ 1\ 1\ 0_{two}\\ +\ \ 1\ 1_{two}\\ \hline 1\ 0\ 0\ 1_{two} \end{array}$ $\begin{array}{l} \frac{2}{2}=r0\\ \\ \frac{2}{2}=1r0 \end{array}$	$\begin{array}{r} 1\ 1\ 1\ 1_{two}\\ +\ 1\ 1\ 1\ 1_{two}\\ \hline 1\ 1\ 1\ 1\ 0_{two} \end{array}$ $\begin{array}{l} \frac{2}{2}=1r0\\ \\ \frac{3}{2}=1r1 \end{array}$	$\begin{array}{r} 3\ 1\ 0_{five}\\ +\ 2\ 1\ 2_{five}\\ \hline 1\ 0\ 2\ 2_{five} \end{array}$ $\frac{5}{5}=1r0$

Activity

1. **Add the following**

(a) 1000two + 101two

(b) 1100two+ 111two

(c) 1111two +1011two

(d) 222three +122three

(e) 413five+321five

(f) 1332four+233four

SUBTRACTION OF BASES

1. Subtract: 1111two-1000two

```
  1 1 1 1two
- 1 0 0 0two
-----------
  0 1 1 1two
```

2.Workout: 1000two-111two

```
    1 1
0   2 2 2
  1 0 0 0two
-   1 1 1two
-----------
  0 0 0 1 two
```

3.Workout: 300four-211four

```
  3 0 0 four
- 2 1 1 four
-----------
  0 2 3 four
```

Activity

1.Workout the following: -

(a) 1011two-110two **(b) 1010two-101two**

(c) 202three-21three **(d) 321four-132three**

(e) 403five-344three **(f) 342five-123five**

(g) 1000two-111two

MULTIPLICATION OF BASES

1.Multiply: 10twox11two

```
    1  0 two
  x 1  1 two
-----------
    1  0
+ 1  0
-----------
  1  1  0 two
```

2.Workout: 111two x 1 1two

```
      1 1 1 two
    x 1 1 two
-------------
      1¹ 1 1
+   1  1 1
-------------
  1 0 1 0 1 two
```

2=1r0
2

3=1r1
2

3.Workout: 202three x 12three

```
      2 0 2three
    x 1 2three
-------------
  1 1 1¹ 1 1
+ 2 0 2
-------------
  1 0 2 0 1three
```

2x2=4
4÷3=1r1
4÷3=1r1
3÷3=1r0

4.Multiply: 142five x 43five

solu

```
      2 1
    1 4 2 five
    x 4 3 five
-------------
    1 0 3 1
  1 2 2 3
-------------
  1 3 3 1 1 five
```

3x2=6
6÷5=1r1
3x4=12
12÷5=2r3
8÷5=1r3
17÷5=3r2

5.Workout: 312four x 32 four

```
      1 1
      3 1 2 four
    x   3 2 four
-------------
    1 2¹ 3 0
+ 2 2 0 2
-------------
  2 3 3 1 0four
```

4÷4=1r0
6÷4=1r2
6÷4=1r2
4÷4=1r0
10÷4=2r2
5÷4=1r1

Activity

1. Workout the following

(a) 1101two x 11 two **(b) 202three x12 three** **(c) 142five x 32 five**

(d) 43 six x 23 six **(e) 214 six x 23 six**

DIVISION OF BASES

Note:

- **Change the non-decimal bases to decimal bases**
- **Divide numbers in base ten**
- **Change the quotient in the given base.**

1.Workout the following

(a) 103nine ÷ 15 nine

1	0	3
9^2	9^1	9^0

÷

1	5
9^1	9^0

$(1x9^2) + (0x9^1) + (3x9^0) \div (1x9^1) + (5x9^0)$

(1x9x9) + (0x9) + (3x1) ÷ (1x9) + (5x1)

81+0+3 ÷ 9+5

84 ÷ 14

6ten

Therefore 103nine ÷ 15nine= 6 nine

(b) Workout: 220eight ÷ 14 eight

2	2	0
8^2	8^1	8^0

÷

1	4
8^1	8^0

$(2x8^2) + (2x8^1) + (0x8^0) \div (1x8^1) + (4x8^0)$

(2x8x8) + (2x8) + (0x1) ÷ (1x8) + (4x1)

2x64+16+0 ÷ 8+4

128+16+0 ÷ 12

144 ÷ 12

12ten

220 eight ÷ 14 eight=14 eight

B	NO	rem
8	12	4
8	1	1
	0	

2.Workout: 144five ÷ 12 five

1	4	4
5^2	5^1	5^0

÷

1	2
5^1	5^0

=

(1x52) + (4x51) + (4x5) ÷ (1x51) + (2x50)

= (1x5x5) + (4x5) + (4x1) ÷ (1x5) + (2x1)

= 25+20+4 ÷ 5+2

= 49 ÷ 7

= 7ten

B	NO	rem
5	7	2
5	1	1
	0	

Therefore 144 five ÷ 12 five=12 five

Workout: 124six ÷ 21 six

2	1	
6^2	6^1	

÷

1	2	4
6^2	6^1	6^0

=

(1x62) + (2x61) + (4x60) ÷ (2x61) + (1x60)

=

(1x6x6) + (2x6) + (4x1) ÷ (2x6) +1x1)

= 36+12+4 ÷ 12+1

= 52 ÷ 13

= 4ten

B	NO	rem
6	4	4
	0	

= 4six

Activity

1. **Workout: 204six ÷ 11five**
2. **Workout: 330four ÷ 30five**
3. **Workout: 44five ÷ 111 five**
4. **Divide 231six ÷ 21six**

FINDING UNKNOWN BASE

Examples

1. Find the unknown base in 20k=12ten

$(2xk^1) + (0xk^0) = 12$

$(2xk) + (0x1) = 12$

2	0	$=12$
k^1	k^0	

$2k+0 = 12$

$2k = 12$

$\frac{2k}{2} = \frac{12}{2}$

$k = 6$

The unknown base is senary.

---

2. Find the unknown base if 32y=1011two

$(1x2^3) + (0x2^2) + (1x2^1) + (1x2^0)$

3	2	$=$	1	0	1	1
y^1	y^0		2^3	2^2	2^1	2^0

$3y = 9$

$\frac{3y}{3} = \frac{9}{3}$

$y = 3$

The unknown base is Trinary.

---

3. Find the unknown base if 103m=28

1	0	3	$= 28$
m^2	m^1	m^0	

$(1xm^2) + (0xm^1) + (3xm^0) = 28$

$(1xm^2) + (0xm) + (3x1) = 28$

$m^2 + 0 + 3 = 28$

$m^2 + 3 = 28$

$m^2 + 3 - 3 = 28 - 3$

$m^2 = 25$

$\sqrt{m^2} = \sqrt{25}$

$m = 5$

The unknown base is Quinary.

4.If 203k =1222three, find the unknown base

2	0	3	$=$
K^2	K^1	K^0	

1	2	2	2
3^3	3^2	3^1	3^0

$(2xk^2) + (0xk^1) + (3xk^0) = (1x3^3) + (2x3^2) + (2x3^1) + (2x3^0)$

$(2k^2) + (0xk) + (3x1) = (1x3x3x3) + (2x3x3) + (2x3) + (2x1)$

$2k^2 + 0 + 3 = 27 + 18 + 6 + 2$

$2k^2 + 3 = 53$

$2k^2 + 3 - 3 = 53 - 3$

$2k^2 = 50$

$\frac{2k^2}{2} = \frac{50}{2}$

$k^2 = 25$

$\sqrt{k^2} = \sqrt{25}$

$k^2 = 5$

The unknown base is Quinary.

Activity

1. Find the unknown bases in the following given that;

(a) 23m=11 (b) 34y=25

(c) 46n=18ten (d) 20k=20three

(e) 50p=30five (f) 102k=1000three

(g) 103x=213four (h) 202k=110100two

(i) 203p=2210three

---

scenarios

TOPIC 2 - INTEGERS

SUBTOPIC: OPERATIONS ON INTEGERS.

Session 1

competences

- ✦ Adds and subtracts integers with and without a number line
- ✦ Multiplies integers with and without a number line

Methods

- Guided discovery
- Group discussion
- Think, pair and share.
- explanation

Lesson activities

- **adding**

- **multiplying**
- **dividing**
- **subtracting**

Content/facts

INTEGERS

Integers are positive or negative numbers including zero e.g. -3, -2, -1, 0, +1, +2…

Note;

Positives are written with a positive/plus sign or without e.g. + 2 or 2, +1 or 1, etc.

Negatives are written with a minus sign e.g. -1, -2, -8, etc.

Zero is neither a positive nor a negative and it is neutral.

ADDITION AND SUBTRACTION OF INTEGERS WITHOUT A NUMBER LINE.

Note:

- x - =+, + x += +, -x +=-, + x - = -

When adding integers with the same sign maintain the sign.

Examples

Work out the following a) +4 + +4 **Solution** +4 + (+4) 4+4 +8	b)-8 + -5 **Solution** -8+(-5) -8-5 -13	c) -k + -7k **Solution** -K + (-7k) -k -7k -8k	b) -4y+ (-5y + (-3y) -4y-5y-3y -12y c) -4+ (-10) -4 -10 -14

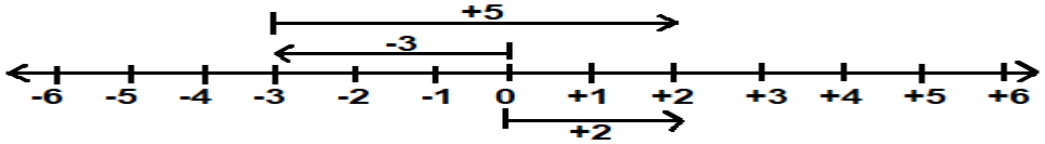
Activity.

Workout the following

1. +6 + +6
2. +10 + -4
3. -10 + -2
4. -14m + -16m + -5m
5. -10 +10
6. +7 +3
8. -10 +4
9. -2 +3

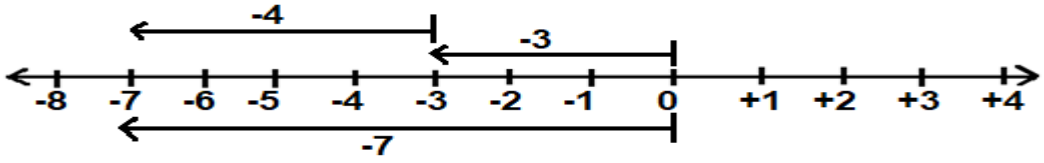
ADDITION OF INTEGERS USING A NUMBER LINE.

1. Add: -3 + +5 using a number line.



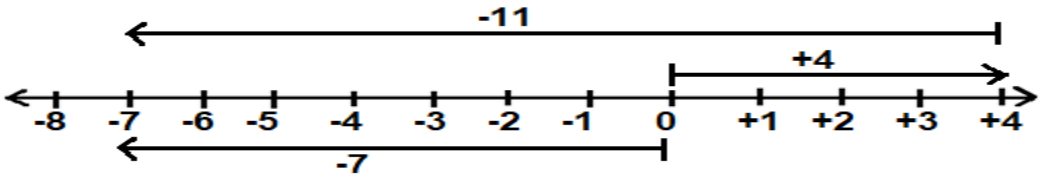
$^{-}3 + {}^{+}5 = {}^{+}2$

2. Add: -3 + -4 using a number line.



$^{-}3 + {}^{-}4 = {}^{-}7$

3. Workout +4 + -11 using number line



$^{+}4 + {}^{-}11 = {}^{-}7$

Activity.

Workout the following using a number line

1. -4 + +7
2. 7 + -4
3. -5 + -4
4. -9 +6
5. 6+ -10
6. 7 + -2

SUBTRACTION OF INTEGERS

Examples

Work out the following (a) +7 - +8 +7 – (+8) +7-8 -1	(b) -8 - -10 -8 – (-10) -8 +10 +2	(c) -5y - +10y -5y – (+10) y -5y – 10 y -15y	(d) 9-13 9-13 -4

Activity.

Workout the following

1. 5 -2
2. 2-5
3. 4 - +10
4. 5- 10
5. -1 -2 -3
6. 11- -13
7. 14n - -16n
8. -7 - -13
9. -4 -11- +2

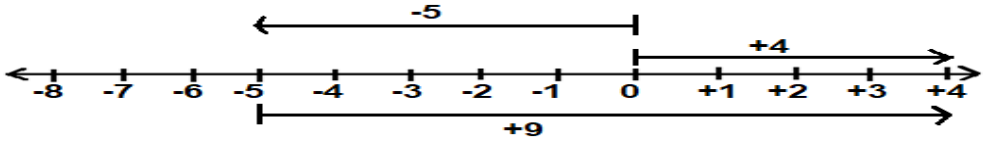
SUBTRACTION OF INTEGER USING A NUMBER LINE.

Note:

- When subtracting, the two arrows begin from a common point.
- An answer arrow is drawn from where the last integer arrow ended to where the first integer arrow ended.

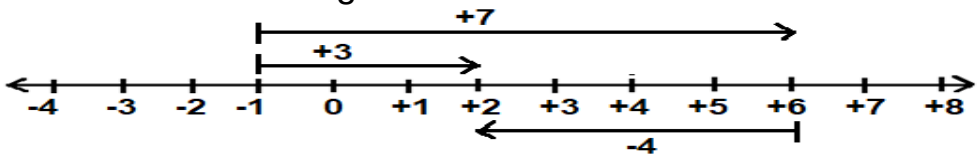
Examples

1. Workout 4 - ⁻5 using a number line.



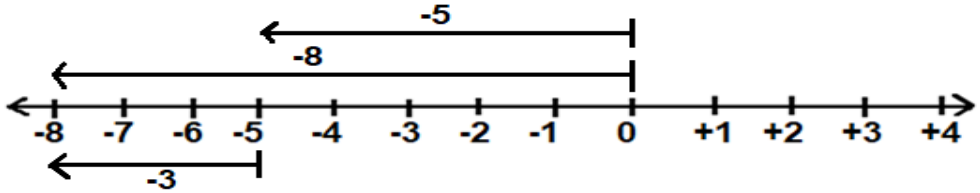
4 - ⁻ 5 = ⁺9

2. Subtract +3 - +7 using a number line.



+3 - +7 = ⁻4

3. Workout ⁻8 - ⁻5 using a number line.



⁻8 - ⁻5 = ⁻3

Activity.

Subtract the following using a number line

(a) -3 - -7

(b) 4 - -5

(c) -5 - +5

(d) -4 - -5

(e) -6 - -5

(f) 5 - -5

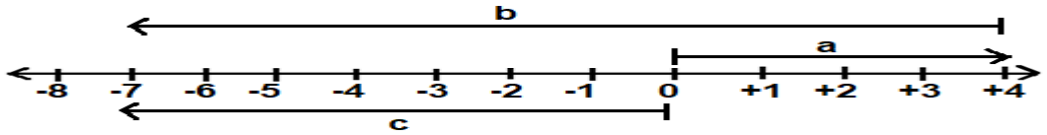
(g) -6 -5

(h) -4 - -6

MATHEMATICAL STATEMENTS

Addition mathematical statements.

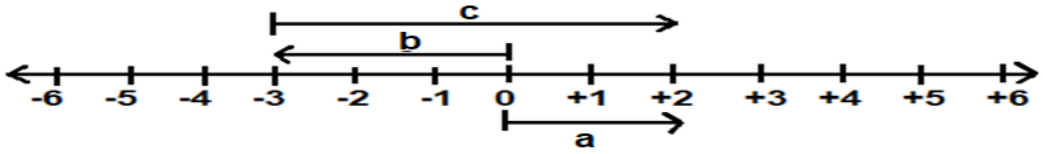
1. Write down the mathematical statements shown on the number line below.



a + b = c

$^{+}4 + {}^{-}11 = {}^{-}7$

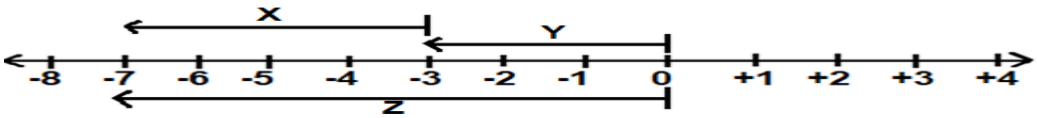
2. Write the mathematical sentence represented on the number lie below.



b + c = a

$^{-}3 + {}^{+}5 = {}^{+}2$

3. Write the mathematical statement represented by the above number line.



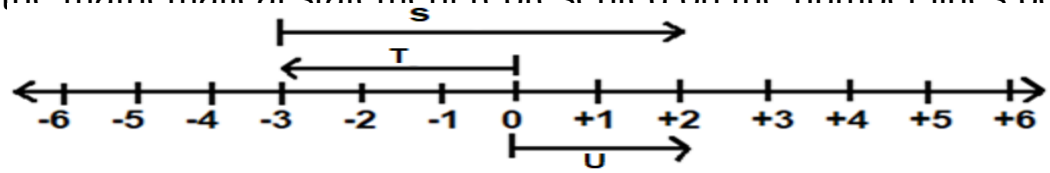
Y + X = Z

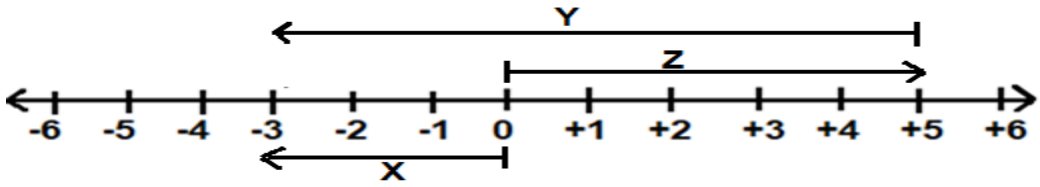
$^{-}3 + {}^{-}4 = {}^{-}7$

Activity.

Write the mathematical statement represented on the number lines below.

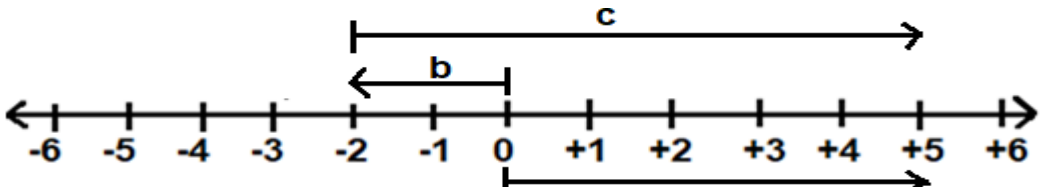
a)



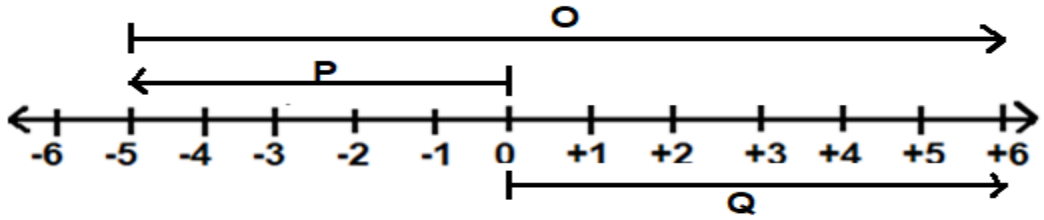


b)

c)

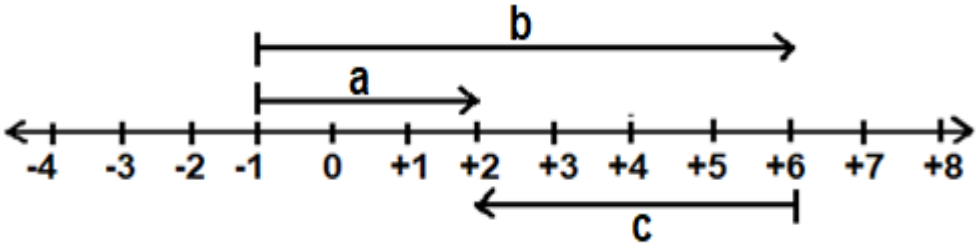


d)



Subtraction mathematical statements

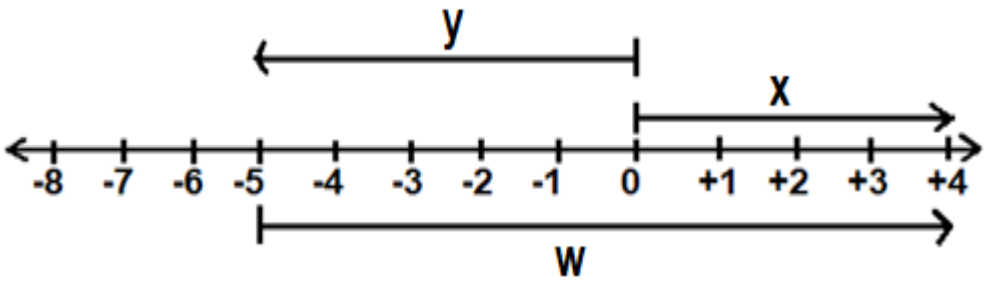
1. Write down the mathematical statements shown on the number line below.



a – b = c

$^{-}3 - {}^{+}7 = {}^{-}4$

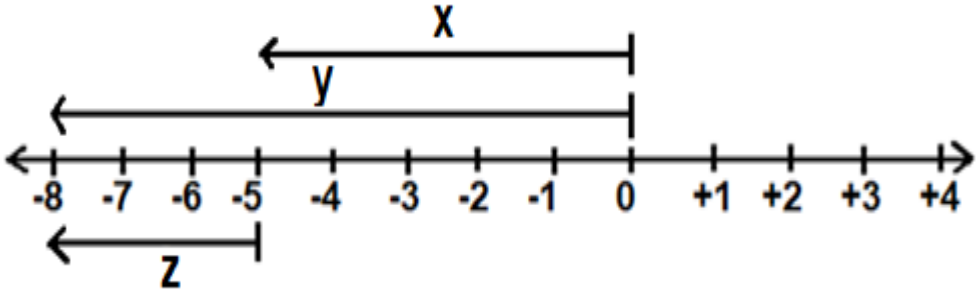
2. Write down the mathematical statements shown on the number line below.



x – y – w

$^{+}4 - {}^{-}5 = {}^{+}9$

3. Write down the mathematical statements shown on the number line below.



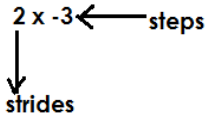
y – x = z

-8 - - 5 = -3

MULTIPLICATION OF INTEGERS ON A NUMBER LINE

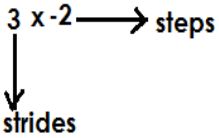
Note:

- Given 2 x -3, 2 represents number of strides/ laps/groups and -3 represents number of steps. That is to say;



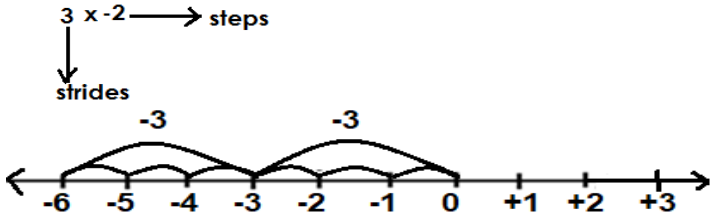
- Given -2 x3, first rewrite starting with a positive i.e. 3 x -2

0



Examples

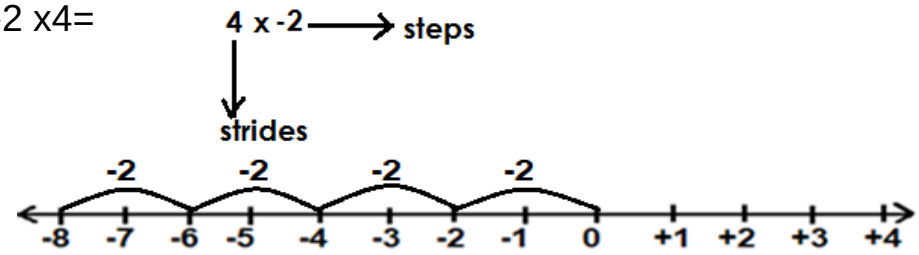
Workout 2x-3 using a number line



2x -3 = -6

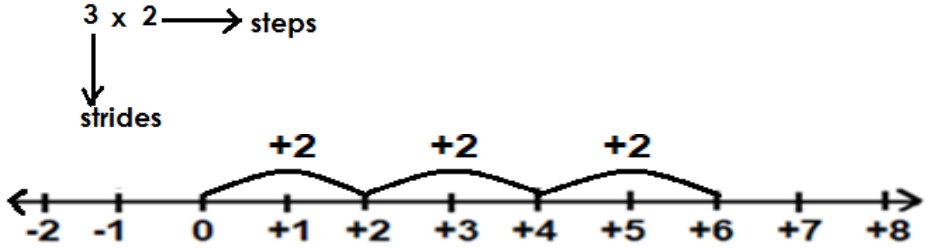
Workout: -2 x4 using a number line.

-2 x4=



-2 x 4= -8

Using a number line, workout 3x2



3x2=+6

Activity

1. Workout the following using a number line.

(a) 2 x 3 (b) 4 x 3 (c) 3 x $^{-}$3 (d) 2 x $^{-}$3 (e) 5 x $^{-}$2

(f) $^{-}$3 x 4 (g) $^{-}$5 x 2 (h) $^{-}$2 x 5 (i) $^{-}$4 x 2

Scenarios

Session 2

competences

- ✦ Solves word statements involving temperature.
- ✦ Solves word statements involving years
- ✦ Solving word statements involving marks

.

Methods

- Guided discovery
- Group discussion
- Think, pair and share.
- explanation

Lesson activities

- **adding**
- **multiplying**
- **dividing**
- **subtracting**

Content/facts

Solves word statements involving temperature and years

1.A hot metallic rod at 40^0c was cooled in a refrigerator after 2 minutes, the temperature was 15^0c. How many degrees did the rod lose?

$+40^0$c -15^0c

25^0c

2.A fruit seller had 100 oranges, he sold off 60 oranges and later bought 150 more oranges. How many oranges does he have now?

100-60 +150

(100 +150)- 60

250 -60

190 oranges

3. The temperature of a place was 20^0F. If it dropped by 230c, what is the temperature of the place now?

Solution

$^{+}20^0$F – ($^{+}23^0$F)

$^{+}20^0$F - 23^0F

$^{-}3^0$F

4. A man was born in 30BC and he died in 20Ad. How old was he?

Solution

AD-BD

20 – (-30)

20 +30

50years

Solving word statements involving marks

Examples

1. On a farm there are hens and cows altogether. There are 12 heads and 34 legs. How many cows are on the farm?

Soln.

Let the No. of cows be M

	Cows	Hens	Sum
Heads	M	12 – m	12
Legs	4 x m 4m	2 (12 –m)	34

4m + 2 (12 – m) = 34
4m + 24 – 2m = 34
4m – 2m + 24 = 34
2m + 24 = 34
2m _ 24 – 24 = 3y – 24
$\frac{2m}{2} = \frac{10}{2}$

M = 5

a) How many hens are on the farm?

No. of hens= 12 – m
= 12 – 5
= 7

2.A teacher awards 3 marks for every correct answer and subtracts 2 marks for every wrong answer. Out of the 20 questions, Nabusoko wrote 15 correct answers. How many marks did she get?

Soln.

Let the marks be K.

No. of marks = c – w
Wrong = 20 – 15
= 5
Marks = (3 x 15) – (2 x 5)
= 45 – 10
= 35 marks

b) Ndole got marks. How many correct answers did he write?

Soln.

Let the correct answers be K.

	Correct answer	Wrong answers	Sum
Question	K	20 – k	20
Marks	3 x k	2 (20 – k)	40

3k – 2(20 – k) = 40
3k – 40 + 2k = 40
3k – 2k – 40 = 40
5k – 40 = 40
5k – 40 + 40 = 40 + 40

$\frac{5k}{5} = \frac{80}{5}$

K = 16

scenarios

Session 3

competences

- ✦ Adds and subtracts integers with and without a dial.
- ✦ Multiplies integers with and without a dial.
- ✦ Divides integers without a dial.

Methods

- Guided discovery
- Group discussion
- Think, pair and share.
- explanation

Lesson activities

- **adding**
- **multiplying**
- **dividing**
- **subtracting**

Content/facts

FINITE SYSTEM

Definition:

This is a way of expressing numbers as remainders.

ADDITION IN FINITE SYSTEM

Note:

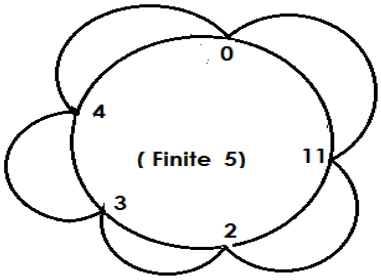
- Identify elements in a given modular system.
- In addition, using a dial we take a clockwise direction.

Examples:

1. Work out: 2 + 4 = (finite 5)

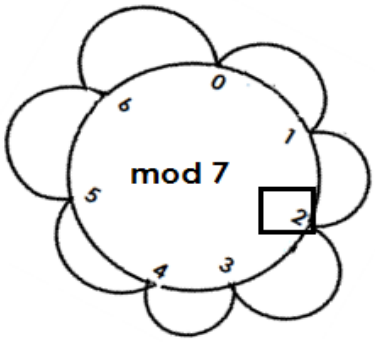
Soln.

Finite 5 = (0 , 1 , 2 , 3 , 4)



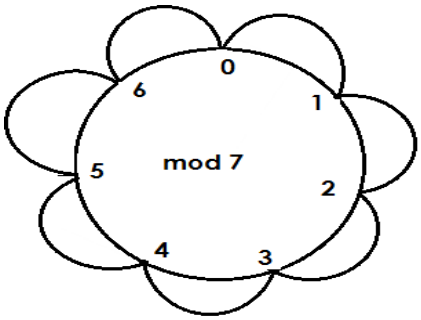
$\therefore$ 2 + 4 = 1 (finite 5)

2. Simplify 3 + 6 = _____ (mod 7)
   Mod 7 = {0, 1, 2, 3, 4, 5, 6}



$\therefore$ 3 + 6 = 2 (mod 7)

3. work out: 3 + 4 + 3 = _____ (mod 7)



$\therefore$ 3 + 4 + 3 = 3 (mod 7)

Activity:

1. 2 + 3 = _____ (finite 5)
2. 1 + 4 = _____ (finite 5)
3. 3 + 5 = _____ (finite 6)
4. 1 + 2 + 3 = _____ (mod 7)
5. 2 + 6 + 3 = _____ (mod 8)
6. 1 + 7 = _____ (mod 8)
7. 2 + 3 + 4 = _____ (mod 9)

8. 2 + 3 + 1 = _____ (finite 7)

Simplify without using a dial

Note:

Addition in finite system without using a dial, consider remainders.

Examples

1. Add: 2 + 3 = _____ (finite5)

2 + 3 = _____ (finite 5)

5 = _____ (finite 5)

5 ÷ 5 = _____ (finite 5)

1 rem 0

∴ 2 + 3 =0 (finite 5)

2. Add: 3 + 4 + 3 = _____ (finite 7)

Soln.

3 + 4 + 3 = _____ (finite 7)

10 = _____ (finite 7)

10 ÷ 7 = _____ (finite 7)

1 rem 5 = _____ (finite 7)

∴ 3 + 4 + 3 = 3 (finite 7)

3. Workout: $2^3 + 4^2$= _____ (finite 5)

2 x 2 x 2 + 4 x 4 = _____ (finite 5)

8 + 16 = _____ (finite 5)

24 = _____ (finite 5)

24 ÷ 5 = _____ (finite 5)

4 rem 4 = _____ (finite 5)

∴23 + 42 = 4 (finite 5)

Activity

Work out:

1. 3 + 4 = _____ (finite 5)

2. 1 + 2 + 3 = _____ (finite 6)

3. $2^3 + 4^3$ = _____ (finite 7)

4. 33 + 6 = _____ (mod 7)

5. 2 + 3 + 4 = _____ (finite 5)

6. 6 + 20 = _____ (mod 8)

7. 2 + 6 + 3 = _____ (mod 7)

8. 0 + 3 + 2 + 4 = _____ (mod9)

SUBTRACTION IN FINITE SYSTEM

A. USING A DAIL

Note:

- In subtraction using a dial, take anti-clockwise direction.

steps to be moved

a - b = (finite k)

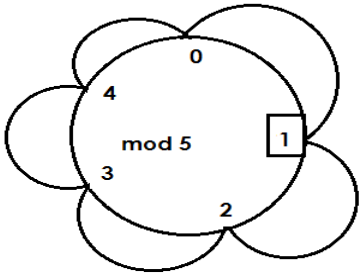
starting point

Examples:

1. Work out: 2 – 6 = _____ (mod 5)

Soln.

Mod 5 = {0,1, 2, 3, 4}

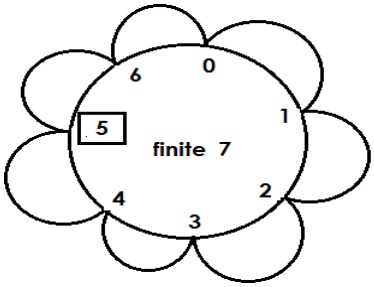


∴2 – 6 = 1 (mod 5)

2. 3 – 5 = _____ (finite 7)

Soln.

Finite 7 = (0 , 1 , 2 , 3 , 4 , 5 , 6)

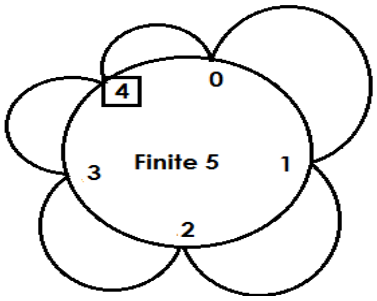


∴ 3 – 5 =5 (finite 7)

3. Simplify: 2 – 3 = _____ (finite 5). Use a dial.

Soln.

Finite 5 = {0, 1, 2, 3, 4}

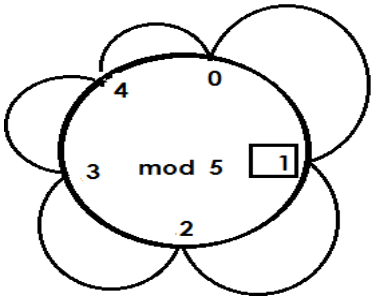


2 – 3 = 4 (finite 5)

4. work out; 1 – 2 – 3 = _____ (mod 5)

Soln.

Finite 5 = {0, 1, 2, 3, 4)



∴1 – 2 - 3 = 1 (mod 5)

<u>Activity:</u>

Work out the following using a dial.

1. 2 – 3 = _____ (finite 5)
2. 3 – 8 = _____ (mod 7)
3. 2 – 5 = _____ (finite 6)
4. 1 – 4 = _____ (mod 5)
5. 3 – 7 = _____ (mod 6)
6. 1 – 3 - 4 = _____ (finite 7)
7. 2 – 3 – 4 = _____ (finite 8)
8. 0 - 3 = _____ (mod 5)

B. SUBTRACTION IN FINITE SYSTEM WITHOUT USING A DIAL

<u>Note:</u>

1. If the number you are subtracting is bigger than the original but when it is a member of a given finite system, increase the original by the given modular system.
2. If the number you are subtracting is not a member of the given finite system convert the number to the given finite system by dividing and consider the remainder.

Examples:

1. work out 4 – 6 = _____ (finite 5)

   Soln.

   4 – (6 – 5) = _____ (finite 5)

   4 – (1 rem 1) = _____ (finite 5)

   4 – 1 = 3 (finite 5)

   ∴ **4 – 6 = 3 (finite 5)**

2. simplify: 1 – 24 = _____ (finite 5)

   Soln.

   1 – (24 ÷ 5) = _____ (finite 5)

   1 – 4 rem (4) = _____ (finite 5)

   1 – 4=_____ (finite 5)

   (1 + 5) – 4 = _____ (finite 5)

   6 – 4= 2 (finite 5)

   ∴**1 – 24 = 2 (finite 5)**

3. work out: 1 – 2 -3 = _____ (mod 6)

Soln.

1(- 2 – 3) = _____ (mode 6)

1 – 5 = _____ (mod 6)

(1 + 6) – 5 = _____ (mod 6)

7 – 5 = 2 (mod 6)

∴ **1 – 2 – 3 = 2 (mod 6)**

Activity:

Simplify:

1. 2 – 5 = _____ (finite 7)
2. 1 – 3 = _____ (mod 5)
3. 1 – 8 = _____ (mod 5)
4. 4 – 5 = _____ (finite 7)
5. 2 – 16 = _____ (mod 5)

6. 4 – 20 = _____ (finite 6)
7. 3 – 19 = _____ (finite 5)
8. 1 – 3 – 4 = _____ (mod 7)
9. 3 – 4 – 5 = _____ (finite 9)
10. 0 - 2 = _____ (finite 5)

MULTIPLICATION IN FINITE SYSTEM

Using a dial

Note:

a x b = ____ (mod k)

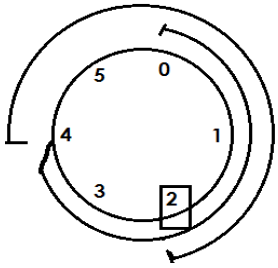
laps steps

- In multiplication we take a clockwise direction starting from zero.

Examples

1. Work out: 2 x 4 = ____ (finite 6)

(Finite 6) = {0, 1, 2, 3, 4, 5)



∴2 x 4 = 2 (finite 6)

2. Work out: 3 x 2 = _____ (finite5)

Soln.

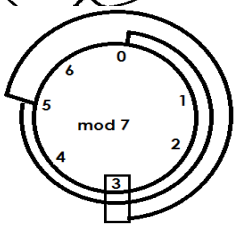
Finite 5 = {0, 1, 2, 3, 4}

∴3 x 2 = 1 (finite 5)

3. 2 x 5 = _____ (mod 7)

Soln.

Mod 7 = {0, 1, 2, 3, 4, 5, 6}



∴2 x 5 = 3 (mod 7)

Activity:

Work out the following using a dial

1. 2 x 3 = _____ (finite 5)
2. 3 x 4 = _____ (finite 6)
3. 3 x 5 = _____ (finite 8)
4. 4 x 3 = _____ (finite 5)
5. 6 x 2 = _____ (mod 7)
6. 1 x 4 = _____ (mod 6)
7. 2 x 5 = _____ (mod 8)
8. 2 x 4 = _____ (mod 7)

MULTIPLYING IN FINITE SYSTEM (WITHOUT USING A DIAL)

Note:

Just like addition, when multiplying numbers, consider remainders.

Examples:

1. Work out: 2 x 8 = _____ (finite 7)

   Soln.

   2 x 8 = _____ (finite 7)

   16 = _____ (finite 7)

   16 ÷ 7= _____ (finite 7)

   2 rem. (2) = _____ (finite 7)

   ∴2 x 8 = 2 (finite 7)
2. Workout: 3 x 2 = _____ (finite 5)

   Soln.

   3 x 2 = _____ (finite 5)

   6 = _____ (finite 5)

   6 ÷ 5 = _____ (finite 5)

   1 rem (1) = _____ (finite 5)

   ∴ 3 x 2 = 1 (finite 5)
3. Work out: 3 x 2 x 4 = _____ (finite 7)

   3 x 2 x 4 = _____ (finite 7)

   24 = _____ (finite 7)

   24 ÷ 7 = _____ (finite 7)

   3 rem. (3) = _____ (finite 7)

   ∴ 3 x 2 x 4 = 3 (finite 7)

Activity:

Work out the following without using and dial

1. 2 x 5 = _____ (mod 6)
2. 4 x 5 = _____ (mod 7)
3. 4 X 3 = _____ (mod 7)

4. 3 x 5 = _____ (mod 8)
5. 1 x 2 x 3 = _____ (mod 5)
6. 2 x 1 x 4 = _____ (mod 5)
7. 2^2 X 3^2 = _____ (finite 8)
8. 4^2 = _____ (mod 7)
9. 4 X 2 = _____ (finite 6)
10. 3 x 2 x 4 = _____ (mod 8)

DIVISION IN FINITE SYSTEM

- In division of finite system, do not consider remainder.
- Take the quoted to be the final answer.
- If the dividend is not a multiple of the divisor, increase it by a given modular system until a multiple of the dividend is got.

Examples

1. Work out: 2 ÷ 3 = 4 (finite 5)

   Soln.

   (2 + 5) ÷ 3 = _____ (finite 5)

   1 = ÷ 3 = _____ (finite 5)

   (7 + 5) ÷ 3 = _____ (finite 5)

   12 ÷ 3= 4 (finite 5)

2. Work out: 4 ÷ 3 = 6___ (mod 7)

Soln.

(4 + 7) ÷ 3 = _____ (mod 7)

11 ÷ 3 = _____ (mod 7)

(11 + 7) ÷ 3 = _____ (mod 7)

18 ÷ 3 = 6 (mod 7)

∴4 ÷3 = 6 (mod 7)

3. Workout: 5 ÷7 = _____ (finite 12)

   Soln.

   (5 + 12) ÷7 = _____ (finite 12)

   17 ÷7 = ____ (finite 12)

   (17 + 12) ÷ 7 = _____ (finite 12)

   29 + 7 = _____ (finite 12)

   (29 + 12) ÷ 7 = _____ (finite 12)

41 ÷ 7 = _____ (finite 12)

(41 + 12) ÷ 7 = _____ (finite 12)

63 ÷7 = _____ (finite 12)

63 ÷7 = 9 (finite 19)

5 ÷ 7 = 9 (finite 12)

Activity:

Work out:

1. 3 ÷4 = _____ (mod 7)
2. 3 ÷ 5 = _____ (mod 7)
3. 3 ÷ 2 = _____ (finite 5)
4. 1 ÷ 3 = 3 = _____ (mod 5)
5. 4 ÷ 5 = _____ (mod 6)
6. 1 ÷ 4 = _____ (finite 7)
7. 1 ÷ 6 = _____ (mod 7)
8. 6 ÷9 = _____ (mod 12)
9. 5 ÷3 = _____ (finite 7)
10. 5 ÷ 6 = _____ (finite 7)

scenarios

Session 4

competences

- ✦ Applies finite system on days of the week
- ✦ Applies finite system on months of the year
- ✦ Applies finite system on time of the day

Methods

- Guided discovery
- Group discussion
- Think, pair and share.
- explanation

Lesson activities

- **adding**
- **multiplying**
- **dividing**
- **subtracting**

Content/facts

APPLICATION OF FINITE SYSTEM

Note:

- Days of the week are in finite 7. Therefore;

Mon	Tue	Wed	Thurs.	Fri	Sat	Sun
1	2	3	4	5	6	0

Months of the year are in mod 12.

J	F	M	A	M	J	J	A	S	O	N	D

1	2	3	4	5	6	7	8	9	10	11	12

- Time of the day is in finite 12.
- Questions stated in respect to future, we add.
- Questions stated in respect to past we shall subtract.

Example I

If today is Wednesday, what day of the week will it be after 64 days?

Soln.

Day + 64 = ____ (mod 7)
Wed + 64 = _____ (mod 7)
3 + 64 = _____ (mod 7)
67 = _____ (mod 7)
67 ÷ 7 = _____ (mod 7)
9 rem (4) mod 7.
3 + 6 = 4 (mod 7)
∴ **It will be Thursday.**

Example II

If today is Monday, what day of the week was it 20 days ago.

Soln.

Day -20 = _____ (mod 7)
Mon – 20 = _____ mod 7)
1 – 20 ÷ 7 = ____ (mod 7)
1 – (2 rem 6) = _____ (mod 7)
1 – 6 = _____ (mod 7)
(1 + 7) -6 = _____ (mod 7)
8 - 6 = 2 (mod 7)

∴ **It was Tuesday**

Example III

If this is march, what month of the year will it be after 40 months.

Soln.

Month + 40 = _____ (mod 12)
March + 40 = _____ (mod 12)
3 + 40 = _____ (mod 12)

43 = _____ (mod 12)
43 ÷ 12 = _____ (mod 12
3 rem (9) mod 12)
3 + 40 = 9 (mod 12)

∴**It will be September.**

Example IV

If this is may, which month of the year was it 46 months ago?

Soln.

Month – 46 = _____ (finite 12)
May – 46 = _____ (finite 12)
5 – 46 = _____ (finite 12)
5 – 46 ÷ 12 = _____ (finite 12)
5 – (3 rem 10) finite 12
5 – 46 = 10 (finite 12)

∴ **It was October.**

Activity:

1. If today is Saturday, what day of the week will it be after 72 days?
2. Otieno will celebrate his birthday 56 days to come if today is Wednesday, what day of the week will he celebrate?
3. If today is Thursday, what day of the week will it be after 96 days?
4. If today is Monday, what day of the week will be after 20 days?
5. If today is Thursday, which day of the week was it 36 days ago?
6. Matama was born 142 days ago, if today is Thursday, which day of the week was it?
7. If today is Saturday, which day of the week was it 76 days ago?
8. This is April, which month of the year was it 18 months ago?
9. Wandera celebrated his birth day 10 months ago. If this is October, which month of the year was?

10. Okello went to London 46 months ago, if this is January, which month of the year did he go?
11. If this is May, which month of the year will it be after 19 months?
12. This is September, which month of the year will it be after 96 months?
13. Isaac is to go to London in 86 months of this is June, which month of the year will he go?
14. Okello will celebrate his birth day 52 months to come, if we are in November, which month of the year will be celebrated

APPLICATION OF FINITE SYSTEM GIVEN DAYS AND DATES OF THE MONTH

Note:

- Months with 30 days are September, April, June and November all the rest have 31 except February, which has 28 and 29 days in each leap year.
- A leap year is the year having 366 days.

Examples

If today is Thursday, 28th March, what day of the week will it be on 16th July of the same year?

Soln.

March 31 – 28 ____ 03
April _____ 30
May _____ 31
June _____ 30
July _____ + 16
Total = 110 days
Day + 110 = _____ (mod 7)
Thurs + 110 = _____ (mod 7)
4 + 110 = _____ (mod 7)
114 = _____ (mod 7)
114 ÷ 7 = _____ (mod 7)
16 rem (2)(mod 7)
∴ **it will be Tuesday.**

If today is Tuesday 17th April, what day of the week was it 13th January of the same year?

Soln.

April _____ 17
March _____ 31
February _____ 28
January 31 – 13 + 18
Total = 94 days
Day – 94 = _____ (mod 7)
Tue – 94 = _____ (mod 7)

2 – 94 = _____ (mod 7)
2 – (94 ÷ 7) = _____ (mod 7)
2 – 13 rem (3) (mod 7)
2 – 3 =_____ (mod 7)
(2 + 7) – 3 = _____ (mod 7)
9 – 3 = 6 (mod 7)
2 – 94 = 6 (mod 7)
∴ It was Saturday.

Activity:

1. If today is 20th May, Friday, what day of the week will it be 10th November of the same year?
2. If today is Monday 10th March, what day of the week will it be on 21st May of the same year?
3. If today is Thursday 2nd May, what day of the week will it be on 18th August of the same year?
4. If today is Sunday 31st March, what day of the week will it be 20th July of the same year?
5. If today is Tuesday 9th September, which day of the week was it on 11th June, 11th the same year?
6. If today is Tuesday 18th September, what day of the week was it 10th June of the same year?
7. If today is Monday 6th July, what day of the week was it 9th march of the same year?
8. If today is 8th Thursday July which day of the week was it 10th April of the same year?

APPLICATION OF FINITE SYSTEM (ABOUT TIME)

- In order to consider part of the day, i.e., pm or am, we have to base on the quotient.
- If the quotient is even, it leaves am as am and pm as pm.
- If the quotient is odd it changes am to pm and pm to am.
- Application of finite system in time we use finite 12.

Examples

- If it's 9:00a.m, what time of the day will it be after 32 hours?

Soln.
9 + 32 = _____ (mod 12)
41 = _____ (mod 12)
41 ÷ 12 = _____ (mod 12)

Odd (3) rem 5 (mod 12)

∴ **it will be 5:00pm.**

It is 9:40pm, what time of the day will it be after 41 hours?

Soln.

9 + 41 = _____ (finite 12)

50 = _____ (finite 12)

50 ÷ 12= _____ (finite 12)

Even (4) rem 4 (finite 12)

∴**It will be 2:40pm.**

Activity:

1. It is now 7:00pm, what time of the day will it be after 32 hours?
2. It is 5:30am, what time of the day will it be after 38 hours?
3. I have a flight in 52 hours, if it's now 3:20pm, what time of the day will it be?
4. It is 11:00am and we are to start the function in 70 hours, at what time of the day shall we start?
5. It is now 1:40pm what time of the day will it be after 23 hours?
6. If it's 4:35am, what time of the day will it be after 63 hours?

scenarios

Session 5

- ✦ Finds equivalences in finite system
- ✦ Applies equivalences in more than one finite
- ✦ Finds unknowns in finite system

Methods

- Guided discovery
- Group discussion
- Think, pair and share.
- explanation

Lesson activities

- **adding**

- **multiplying**
- **dividing**
  - **subtracting**

Content/facts

APPLICATION OF FINITE SYSTEM INVOLVING MORE GROUPS

Examples

1.Find the smallest number of pens that when shared by 5 pupils 2 pens remain and when shared by 8 pupils 5 pens remain?

Soln.

2 (mod 5) 7, 12, 17, 22, 27,32

3 mode 11, 19, 27, 35.

∴ **The number is 27**

2.Find the smallest number that when divided by 5, 3 remains, when divided by 6, 4 remain and when divided by 4, 0 remains.

Soln.

3 (mod 5) 8,13 , 18 , 23 , 28, 33

4 (mod 6) ,10,16 , 22 , 28 , 34.

∴ **The number is 28.**

Activity:

1. Find the least number of mangoes that when shared by 6, 7 remain and when shared by 5, 2 remain.
2. A man had some oranges, when he shared them among 3 groups, 2 remained, when he shared in 4 groups 3 remained and when he shared them in 5 groups 3 remained. Find the smallest number of oranges that he had.
3. Find the smallest number that when divided by 8, 2 remain and when shared by 9, 8 remain.

4. Find the smallest number that when shared by 7, 3 remain, when shared by 4, 2 remain and when shared by 8, 6 remain.
5. Find the least number of sweets such that when divided by 3, 2 remain, when divided by 4, 3 remain and when divided by 5, 3 remain.
6. A teacher bought some pens and put them in groups of 95 and 7 pens were left, when he put them in groups of 8s only 4 pens were left but when he put them in group of 3, only, 1 pen was left. Find the smallest number of pens that the teacher bought.

FINDING UNKNOWNS IN FINITES SYSTEM

Examples

1. Solve: m + 6 = 2 (finite 7)

   Soln.

   M + 6 – 6 = 2 – 6 (finite 7)
   M = (2 + 7) – 6 (finite 7)
   M = (9 – 6 (finite 7)
   M = 3 (finite 7)

2. Solve: K – 6 = 2 (mod 4)

   Soln.

   K – 6 + 6 = 2 + 6 (mod 4)
   K = 8 (mod 4)
   K =8 ÷ 4 (mod 4)
   K = 2 rem (0) (mod 4)
   K = 0 (mod 4)

3**.** Solve: 2p = 5 (mod 7)

soln.

2p = 5 + 5 (mod 7)

2p = 12 (mod 7)

$\frac{\mathbf{2p}}{\mathbf{2}} = \frac{\mathbf{12}}{\mathbf{2}}$ (mod 7)

P = 6 (mod 7)

4. Solve: 3y + 1 = 6 (finite 7)

Soln.

3y + 1 – 1 = 6 – 1 (finite 7)

3y = 5 (finite 7)

3y = 12 (finite 7)

$\frac{3y}{3} = \frac{12}{3}$ (finite 7)

y = 4 (finite 7)

5. solve: 6 (m – 3) = 4 (mod 8)

6m – 18 = 4(mod 8)

6m – 18 + 18 = 4 + 18 (mod 8)

6m = 22 (mod 8)

6m = 22 + 8 (mod 8)

$\frac{6m}{6} = \frac{30}{6}$(mod 8)

$\frac{\mathbf{6m}}{\mathbf{6}} = \frac{\mathbf{30}}{\mathbf{6}}$ (mod 8)

M = 5 (mod 8)

6. Solve: $\frac{\mathbf{2p-4}}{\mathbf{2}}$ = 5 (finite 12)

Soln.

$\frac{2p-4}{2}$x 2 = 5 x 2 (finite 12)

2p – 4 = 10 (finite 12)

2p – 4 + 4 = 10 + 4 (finite 12)

$\frac{2p}{2} = \frac{14}{2}$ (finite 12)

P = 7 (finite 12)

<u>Activity:</u>

Solve:

1. k + 8 = 3 (mod 6)
2. y + 4 = 1 (finite 5)
3. m + 6 = 1 (mod 7)
4. 3 + y = 2 (mod 7)

5. p – 8 = 4 (mod 8)
6. n – 3 = 7 (mod 8)
7. 3k = 4 (mod 7)
8. 5m = 3 (finite 6)
9. 3y – 4 = 5 (mod 6)
10. 4(y – 2) = 3 (finite 5)
11. $\frac{3y-4}{2}$ = 3 (finite 8)
12. $\frac{4k-2}{4}$ = 1 (finite 7)

scenarios

TOPIC 3 -ALGEBRA

Subtopic: collecting like terms and unlike terms

Session 1

2025

- ✦ Collects like terms
- ✦ Simplifies algebraic expressions
- ✦ Substitution
- ✦ Removes brackets and simplifies algebraic expressions.
- ✦ Adds and subtracts algebraic terms.
- ✦ Solves equations involving whole numbers

Methods

- Guided discovery
- Group discussion
- Think, pair and share.
- explanation

Lesson activities

- **adding**
- **multiplying**
- **dividing**
- **subtracting**

Content/facts

ALGEBRA

Definition

Algebra is a branch of mathematics dealing with the use of symbols or unknown to represent quantity or value of something.

Collecting like terms

Note:

- **Like terms are terms that are exactly the same.**
- **When collecting like terms each term must be transferred with its sign before and any term without a sign is a positive number.**
- **Terms are separated by a comma, a plus, minus signs or equal signs.**
- **Multiplication, division and a bracket make single terms.**

Examples

1. Simplify: 4K – 3 + K + 8

Soln.

= 4k – 3 + k + 8

= 4k + k – 3 + 8

= 5k + 5

2. Simplify: 9 x y – 3 x y

Soln.

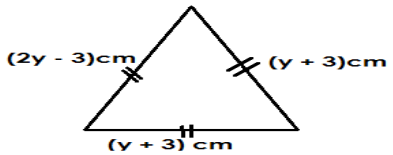
= 9x y – xy – 3xy

= 9 x y = 3 x y – xy

= 6 x y – xy

= 5xy

3. Find the perimeter of the following:



Soln.

Perimeter = S + S + S

= (y + 3) cm + (y + 3) cm + (2y-3) cm

= y + 3cm + y + 3cm + 2y – 3cm

= y + y + 2y + 3cm + 3cm – 3cm

= 4y + 6cm – 3cm

= (4y + 3) cm

Activity:

1. Simplify:

a) a + a + a

b) $Xy^2 + xy^2$

c) 9 x y – xy – 3xy

d) 4p – 6p – 3p

e) 4 x m + 8 – xm

f) $Pq^2 + 4 + 3Pq – 8$

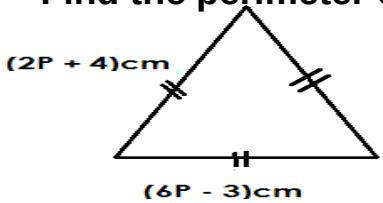
g) 7y + 8 – 2y – 14

h) P + 4 + 3P – 13

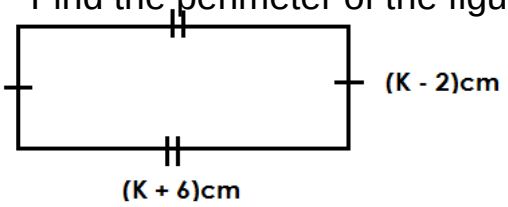
i) $Xy^2 + 3\ xy^2 – 2xy^2$

j) 4 x m + 8 – xm– 10

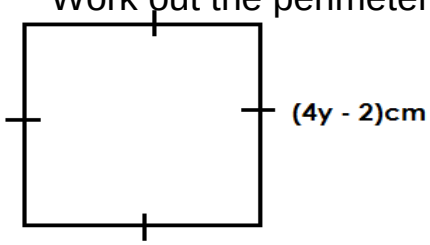
2. Find the perimeter of the figure below.



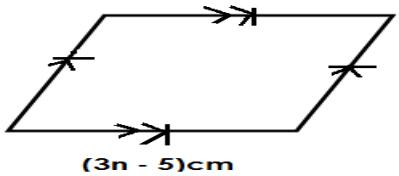
3. Find the perimeter of the figure below.



4. Work out the perimeter of the figure below.



5. Find the perimeter of the figure below.



FACTORIZING COMPLETELY

Examples

1. Factorize completely: XP + MX

Soln.

X	XP + MX
	P + M

x(p + m)

2. Factorize completely: PQ2y – MPQ

Soln.

P	PQ^2y - MPQ
Q	PQY - MQ
	QY - M

P X Q(PY - M)

PQ(PY - M)

ACTIVITY

Factorize the following completely

1. ma + my
2. 4ap + 18 x p
3. $3y^2p$ – 12ym

4. 18wy – 12 x y
5. 12yp + 15px
6. $am^2p + pmQ$
7. 14XM + 16ym
8. 20nx + 12pn
9. 4ay – 2ay
10. 18axy - 9yx

OPENING BRACKETS

Note:

To open brackets, multiply the figure outside by all items in brackets.

Opening of brackets with positive sign outside

Examples

1. Simplify: 4 (3k – 3)

Soln.

4 x 3k – 4 x3
12k – 12

2. Simplify: 2y (y – 2)

Soln.

2y x y – 2y x 2
$2y^2 – 4y$

Example III

Simplify: 2 (P – 4) + 3 (P + 2)

Soln.

2 x p - 2 x 4 + 3xp + 2 x 3
2p – 8 + 3p + 6
2p + 3p – 8 + 6
5p – 2

Activity:

Simplify the following:

1) 3(k – 3)
2) 4(m + 6)

3) 2p (4- p)
4) 2(a – 4) + 2(a + 3)
5) 3(k + 2) + (4 (k2)
6) 4(y + 2) + 2 (y + 3)
7) 2y (y + 6) + 2(y + 4)
8) 4(m – 3) + 3 (2m + 4)

OPENIONG BRACKETS WITH A NEGATIVE SIGN OUT SIDE

Note:

- A negative sign outside brackets changes all signs inside brackets.
- After the third step, there is a check point about integers.

Example 1

Simplify: - 3(w + 8)

Soln.

-3 x w – 3 x 8
-3w – 24

Example II

Simplify: 4(k – 3) – 2(k + 4)

Soln.

4 xk – 3 x 4 – 2 x k – 2 x 4
4k – 12 – 2k – 8
4k – 2k – 12 – 8
4k – 20

Example III

Simplify: 3(2y – 6) – 4(y – 12)

Soln.

3 x 2y – 6 x 3 – 4 x y + 12 x 4
6y – 18 – 4y + 48
6y – 4y – 18 + 48
2y + 30

Activity:

Simplify the following:

1. 3(y – 2) – 2(y – 3)
2. 4(x + 4) – 2(x – 3)
3. 3(y + 4) – 2(y – 6)
4. 5(k + 3) – 4(k + 2)

5. $3(y - 2) - (y + 4)$
6. $6(w - 2) - 3(w - 2)$
7. $2(3p - 4) - 2(y - 4)$
8. $5(y + 2) - 2(y - 4)$
9. $5(m + 2) - 3(m + 7)$

ADDITION OF COMPOUND TERMS

Example I

Add: $6P + 4$ to $3p + 2$

Soln.

$(3p + 2) + (6p + 4)$

$3p + 2 + 6p + 4$

$3p + 6p + 2 + 4$

$9p + 6$

Example II

Add: $3y - 6$ to $3y + 16$

Soln.

$3y + 16 + 3y - 6$

$3y + 3y + 16 - 6$

$6y + 10$

Activity:

Add the following:

1. $8k + 2$ to $2k - 8$
2. $4p + 6$ to $2p - 12$
3. $3y + 4$ to $2y + 7$
4. $3m - 6$ to $2m - 4$
5. $2w - 4$ to $w - 6$
6. $4y + 3$ to $2y - 11$
7. $3y - 10$ to $2y + 4$
8. $16p + 10$ to $2y + 4$
9. $12y - 11$ to $3y + 17$
10. $6w - 12$ to $2w + 20$

SUBTRACTION OF COMPOUND TERMS

Note:
While subtracting compound terms, we first introduce brackets to the given terms and then we follow instructions.

Example I

Subtract 2p + 4 from 3p + 6
Soln.
(3p + 6) – (2p + 4)
3p + 6 – 2p – 4
3p – 2p + 6 – 4
P + 2

Example II
Subtract: 2y – 10 from 5y – 6
Soln.
(5y – 6) – (2y – 10)
5y – 6 – 2y + 10
5y – 2y – 6 + 10
3y + 4

Activity:
Subtract the following: -

1. y + 4 from 2y – 6
2. 3m + 2 from 5m + 6
3. 2(m + 4) from 3(m + 6)
4. y – 8 from y + 10
5. 6y – 10 from 10y – 13
6. 3p + 4 from 4p + 8
7. 4n + 4 from 6n – 4
8. x + 4 from 2x + 9

ADDITION AND SUBTRACTION OF FRACTIONAL ALGEBRAIC TERMS

Example I

Note:

Find the LCD which is 20.

Simplify: $\frac{K}{4} + \frac{K}{5}$

soln.

$\frac{K}{4} + \frac{K}{5}$

$= \frac{\frac{(Kx\ 20)}{4} + \frac{(Kx\ 20)}{5}}{20}$

$= \frac{5k + 4k}{20}$

$= \frac{9K}{20}$

Example II

Simplify: $\frac{3p}{5} + \frac{P}{5}$

Soln. LCD =5

$\frac{3p}{5} + \frac{P}{5} = \frac{3P+P}{5}$

$= \frac{4P}{5}$

Example III

Simplify: $\frac{2y}{4} - \frac{2y}{3}$

Soln. LCD = 12

$\frac{3y}{4} - \frac{2y}{3} = \frac{(\frac{3y\ x\ 12}{4}) - (\frac{2y\ x\ 12}{3})}{12}$

$= \frac{9y-8y}{12}$

$= \frac{y}{12}$

Example IV

Simplify: $\frac{2y}{5} - \frac{y}{6}$

Soln = LCD = 30

$\frac{2y}{5} - \frac{y}{6} = (\frac{24}{8}\ x\ 30) - (\frac{4}{6}\ x\ 30)$

$$= \frac{12y-5y}{30}$$

$$= \frac{7y}{30}$$

Example V

Simplify:

$$\frac{k+2}{3} + \frac{K-3}{4}$$

Soln. LCD = 12

$$\frac{(K+2)}{3} + \frac{(K-3)}{4} = \frac{4(k+2) + 3(k-3)}{12}$$

$$= \frac{4 \times K + 4 \times 2 + 3 \times K - 3 \times 3}{12}$$

$$= \frac{4k + 8 + 3k - 9}{12}$$

$$= \frac{4K + 3K + 8 - 9}{12}$$

$$= \frac{7k - 1}{12}$$

Example VI

Simplify:

$$\frac{(P-2)}{3} - \frac{(P-3)}{4} = \frac{4(P-2) - 3(P-3)}{12}$$

$$= \frac{4 \times P - 2 \times 4 - 3 \times p + 3 \times 3}{12}$$

$$= \frac{4P - 8 - 3P + 9}{12}$$

$$= \frac{P + 1}{12}$$

Activity:

Simplify the following.

1. $\frac{K}{3} + \frac{K}{5}$
2. $\frac{n+2}{3} + \frac{n+4}{4}$

3. $\frac{m+2}{2} + \frac{m-3}{4}$

4. $\frac{y-2}{4} + \frac{6+y}{3}$

5. $\frac{2P+2}{4} + \frac{P-4}{3}$

6. $\frac{p+3}{4} - \frac{P+2}{6}$

7. $\frac{y+6}{4} - \frac{y-4}{5}$

8. $\frac{2x-4}{2} - \frac{X+2}{3}$

9. $\frac{n+4}{4} - \frac{n-2}{7}$

10. $\frac{2w-6}{5} - \frac{w-3}{6}$

REMOVING BRACKETS INVOLVING FRACTIONS

Example I

Simplify: ¾ (12y – 8)

Soln.

¾ x 12y – ¾ - 8

3 x 3y – 3 x2

9y – 6

Example II

Simplify: $^3/_7$ (14w + 28)

Soln.

$\frac{3}{7}$ (14w + 28)

$=\frac{3\ x14w}{7} + \frac{3\ x\ 28}{7}$

= 3 x 2w + 3 4

= 6w + 12

Example III

Simplify: ¼ (8k – 12) + $^1/_3$ (9k + 15)

Soln.

= ¼ x 8k – ¼ x 12 + $^1/_3$ x 9k + $^1/_3$ x 15

= 2k – 3 + 3k + 5

=` 2k + 3k – 3 + 5

= 5k + 2

-3	+5
	+
	+
+2	

Example IV

Simplify: $^1/_5$(15y – 10) – ¼ (8y – 12)

Soln.

$^1/_5$ x 15y – $^1/_8$ x 10 – ¼ x 8y + ¼ x 12

3y – 2 – 2y + 3

3y – 2y – 2 + 3

Y + 1

Activity:

Simplify the following:

1. 2/3 (6m + 9)
2. ¼ (8k – 16)
3. $^3/_7$ (7h + 14) + $^2/_9$ (18h – 36)
4. $^2/_5$ (15y + 20) + ¼ (16y + 4)
5. $^3/_8$ (16p – 24) + $^1/_3$ (9P + 12)
6. ¾ (12m – 4) – $^1/_3$ (3m + 3)
7. ½ (2a – 4) – $^1/_3$ (6a – 12b)

8. $2/5 (K + 4) - {}^{1}/_{3} (k + 4)$

9. $¾ (12w - 16) - {}^{1}/_{5} (10w + 5)$

SUBSTITUTION

Substitution means to replace.

Example I

Given that a = 4, b = 3 and c = 12, find the value of:

i) a + b
a + b = 4 + 3
= 7
ii) a2 + b2
soln
a2 + b2 = (a x a) + (b x b)
= (4 x 4) + (3 x 3)
= 16 + 9
= 25
iii) ½ ab + 1/3ca
Soln.
½ ab + 1/3 ca = (1/2 x a x b) + $^{1}/_{3}$ x c x a
= (½ x 4 x 3) + $^{1}/_{3}$ x 12 x 4
= 3 x 2 + 3 x 4
= 6 + 16
= 22

Example II

Given that a = $^{-}$2, b = 4 and c = 3. Find;

i) ab+ c
soln.
ab + c = (a x b) + c
= ($^{-}$2 x 4) + 3
=$^{-}$8 + 3
=$^{-}$5
ii) $\frac{a^2 + b^2}{a}$
soln.
$a^2 + b^2$ = $\frac{a \times a + b \times b}{a}$
= ($^{-}$2 x $^{-}$2) + (4 x 4)
= 4 + 16
= 20

iii) ab – c

ab – c = (a x b) - c
= (-2 x 4) – 3
= -8 – 3
= -11

iv) bc – 2a

soln.

bc- 2a = b x c – 2 x a
= (4 x 3) – (2x – 2)
= 12 – (-4)
= 12 + 4
= 16

Example III

If k = 2x and x = 8. Find 2k + 8.

Soln.

2k + x = (2 × 2x) + x
= (2 × 2 × 8) +8
= 32 + 8
= 40

Activity:

1. Given that P = 12, Q = -3 and y = 4

Find $\frac{PQ}{Y}$

i) $2Q^2 + 4y$ ii) 2 (P + Q) iii) ½ $P + Q^2$

2. If a = 5,b =10,c =6, d =½ andc = 1/5
Find;

i) 4(a + b) ii) ad (b- c) iii) e (a + b) iv) -3(a + 3c)

3. Find the value of $\frac{c^2 + b^2}{4}$ if a =1,b = 2 and c = 4.

4. Given that p = a^2 and a = 4. Find the value of 2p + 4a

SUBSTITUTION INVOLVING FRACTIONS

Example I

Given that y =½ b = ¾ and m = $^1/_3$. Find the value of;

a) y + b – m

y + b – m = ½ + ¾ - $^1/_3$

$$= \frac{(½ \times 12) + (^3/_4 \times 12) - (^1/_3 \times 12)}{12}$$

$$= \frac{(6 + 9) - 4}{12}$$

$$= \frac{15 - 4}{12}$$

$$= \frac{11}{12}$$

Activity:

1. If a = ½, C = $^2/_3$ and d = ¼. Find the value of ac +d.
2. Given that x = $^2/_3$, y = $^1/_3$. Find the value of x + y
3. If a = $^3/_{10}$,b = $5/_8$ and c = $^4/_5$. Find the value of a + bc.
4. Given that x = 1/8,y = 2/3 and Z = 4/5. Find the value of $\frac{xy}{z}$.
5. If P = 1 $^1/_3$,q = 2 $^1/_3$. Find the value of PQ – $^1/_3$
6. Given that a = 1/8,b = ¼ and c = 2/5. Evaluate ac + bc.

SOLVING SIMPLE EQUATIONS

Example I

Solve: 3 + 4p = 15

Soln.

3 – 3 + 4p = 15 – 3

$\frac{4p}{4} = \frac{12}{4}$

P = 3

Example II

Solve: 15 – 3w = 6

Soln.

15 – 15 – 3w = 6 – 15

-3w = -9

$\frac{-3w}{-3} = \frac{-9}{-3}$

W = 3

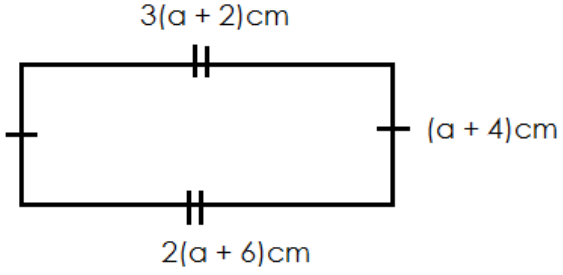
Activity:
Solve the following equations below.

1. 2 + 4y = 18
2. 4 + 2p = 10
3. 8 – 2x = 12
4. 7 – 3p = 11
5. 13 – 2m = 3
6. 8 + 5n = 28
7. 10 – 2y = 12
8. 4 + 2p = 12
9. 9 – 3y = 18
10. 18 + 3k = 27

SOLVING EQUATIONS INVOLVING UNKNOWNS ON BOTH SIDES

Examples:
1.solve: 3m – 6 = m + 12.
Soln.
3m – 6 + 6 = m + 12 + 6
3m = m + 18
3m – m = m – m + 18
$\frac{2m}{2} = \frac{18}{2}$
M = 9
2. solve: 4(m – 2) = 2(m + 6)
Soln.
4 x m – 4 x2 = 2 x m + 2 x 6
4m – 8 = 2m + 12
4m = 8 + 8 = 2m + 12 + 8
4m = 2m + 20
4m – 2m = 2m – 2m +20
$\frac{2m}{2} = \frac{20}{2}$
M = 10

3. Use the diagram below to answer questions that follow.



a) Find the value of a.

soln.

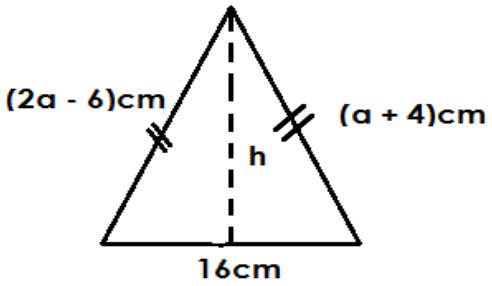
length = length

3(a + 2) = 2 (a + 6)
3a + 6 = 2a + 12
3a + 6 = 2a + 12 – 6
3a = 2a + 6
3a – 2a = 2a – 2a + 6
a = 6

b) Find its area.

Length	Width	Area
2(a + 6) cm 2 x a +2 x 6 2 x 6 + 2 x 6 12 + 12 14cm	(a + 4) cm (6 + 4) cm 10cm	L X W 14cm x 10cm 140cm^2

4. Below is an isosceles triangle. Use it to answer questions below.

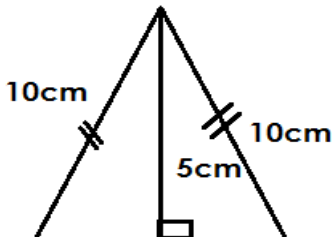


a) Find the value of a.

soln.

2a – 6 = a + 4
2a – 6 + 6 = a + 4 + 6
2a = a + 10
2a – a = a – a + 10
a = 10cm

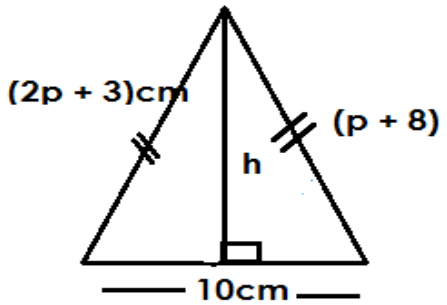
b) Work out the area of the triangle above.



Soln.

A = ½ x b x h
= ½ x 16cm x 5cm
= 8cm x 5cm
= 40cm^2

5. Below is a triangle. Use it answer questions below.



a) Find the value of P.

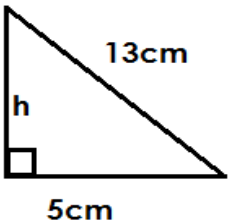
Soln.

2p + 3cm = p + 8cm
2p + 3cm – 3cm= p +8cm + 3cm
2p = P + 5cm
2p – p = p – p + 5cm
P = 5cm

b) Work out the area of the triangle.

Soln.

First find the value of h.



$a^2 + b^2 = c^2$
$h^2 + 5^2 = 13^2$
h2 + 5 x 5 = 13 x 13

$h^2 + 25 = 169$

$h^2 + 25 - 25 = 169 - 25$

$\sqrt{h^2} = \sqrt{144}$

$h = 12cm$

area = ½ x b x h

= ½ x 10cm x 12cm

= 5cm x 12cm

= 60cm²

Activity:

1. Solve the equations below:

a) $5y + 4 = 3y + 8$

b) $4(m - 3) = 2(m + 6)$

c) $3p - 4 = p + 6$

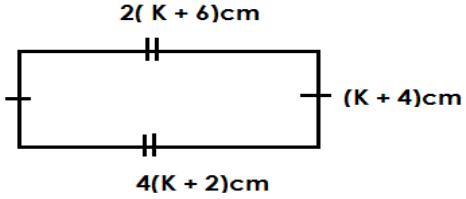
d) $3y - 2 = 2y + 4$

e) $3(m + 2) = 2(m + 8)$

f) $5(y - 2) = 3(y + 2)$

g) $6m + 4 = m + 19$

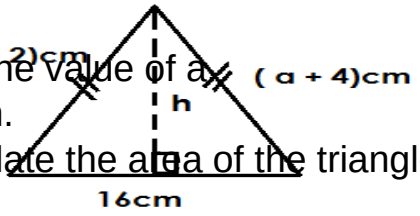
2. Use the diagram below to answer questions that follow.



a) Find the value of K.

b) Find the area of the figure above.

3. Below is triangle. Use it to answer questions that follow

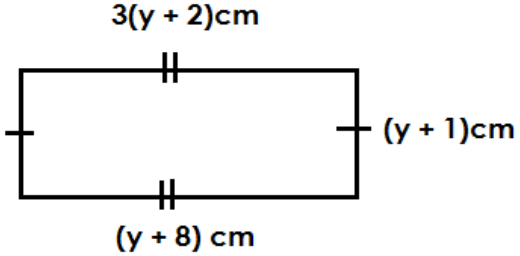


a) Find the value of a.

b) Find h.

c) Calculate the area of the triangle above.

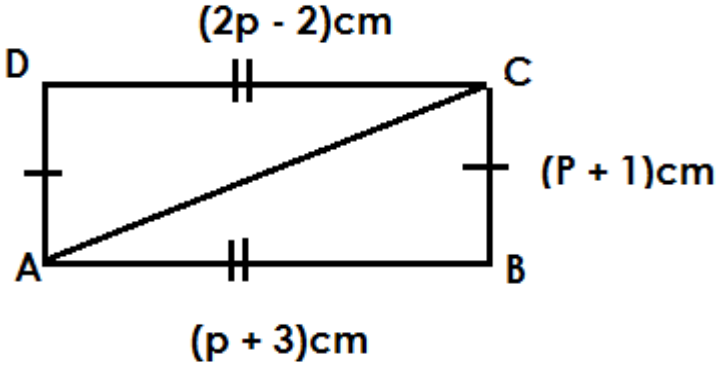
4. Below is a rectangle. Use it to answer questions that follow.



a) Find the value of y.

b) Find the perimeter of the figure.

5. Study the figure below and answer the questions that follow.



a) Find the value of P.

b) Find the length of AC.

scenarios

Session 2

- ✦ Solves equations with unknowns on both sides.
- ✦ Solves equations on fractional terms
- ✦ Solving fraction algebraic equations

Methods

- Guided discovery
- Group discussion
- Think, pair and share.
- explanation

Lesson activities

- **adding**
- **multiplying**
- **dividing**
- **subtracting**

Con tent/facts

SOLVING EQUATIONS INVOLVING DOUBLE BRACKETS

1. Solve: $3(p - 4) + 2(p + 3) = 14$

Soln.

$3 \times p - 3 \times 4 + 2 \times p + 2 \times 3 = 14$

$3p - 12 + 2p + 6 = 14$

$3p + 2p - 12 + 6 = 14$

$5p - 6 = 14$

$5p - 6 + 6 = 14 + 6$

$\frac{5p}{5} = \frac{20}{5}$

$P = 4$

2. Solve $3(y - 2) - 2(y + 4) = 3$

Soln.

$3 \times y - 3 \times 2 - 2xy - 2 \times 4 = 3$

$3y - 6 - 2y - 8 = 3$

$3y - 2y - 6 - 8 = 3$

$y - 14 = 3$

$y - 14 + 14 = 3 + 14$

$y = 17$

Activity:

Solve the following equations

1. $4(3y - 2) - 5(y - 3) = 28$
2. $3(2p + 1) + 4(p + 3) = 35$
3. $2(m + 1) + 3(m + 1) = 10$
4. $3(2w + 1) - 2(w + 4) = 36$
5. $2(2p - 1) - 2(p - 3) = 4$
6. $2(3 \times -1) - 4(x - 1) = 4$
7. $5(x + 2) - 3(x - 3) = 27$
8. $6(y + 1) + 3(y + 2) = 39$
9. $4(p + 2) + 2(p - 3) = 16$
10. $4(P + 2) + 2(p + 3) = 12$

SOLVING EQUATIONS INVOLVING FRACTIONS

Examples

Solve: $\frac{K}{3}$= 4

$\frac{K}{3}$x 3 = 4 x 3

K = 12

Soln.

4 x $\frac{1}{4}$P – 6 x 4 = 7 x 4

P – 24 = 28

P – 24 + 24=28 + 24

P =52

3. Solve: 3 - $\frac{2m}{3}$ = 7

Soln.

3 x $\frac{3}{1}$– $\frac{2m}{3}$x 3= 7 x 3

9 – 2m = 21

9 – 9 – 2m = 21 – 9

$\frac{-2m}{-2}$ = $\frac{12}{-2}$

m = -6

Activity:

Solve the following equations.

1. $\frac{3p}{2}$= 12
2. 2 ½ K = 10
3. 1/4P = 2
4. $\frac{2}{3}$P = 8
5. $\frac{3}{4}$y +2 = 8
6. $\frac{m}{7}$+ 11 = 9
7. $\frac{1}{9}$y– 13 = 14

8. $\frac{2K}{3}$+ 8 = 12

9. $\frac{3m}{4}$+2m = 11

10. 4 – $\frac{3p}{4}$= 12

SOLVING EQUATIONS INVOLVING DECIMALS

Examples

1. Solve: 0.9p + 0.5 = 5.7

Soln.

$\frac{9}{10}$P + $\frac{5}{10}$ = $\frac{57}{10}$ LCD=10

10 x $\frac{9}{10}$P + $\frac{5}{10}$x 10 = $\frac{57}{10}$ X 10

9p + 5 = 57

9p + 5 – 5 = 57 – 5

$\frac{9P}{9}$= $\frac{52}{9}$

P = $5\frac{7}{9}$

ACTIVITY:

Solve the following equations

1. 0.6y – 0.3
2. 0.4x – 0.8 = 2.4
3. 0.4p + 0.5 = 2.1
4. 0.2m – 1.4 = 2.2
5. 0.3k + 0.4 = 1.9
6. 0.2n – 0.2 = 1.8
7. 0.4y – 0.8 = 3.4
8. 0.2x – 0.1 = 0.4

SOLVING EQUATIONS INVOLVING COMPOUND FRACTIONS.

Examples:

1. Solve:

$\frac{y-5}{2}$ + P = $\frac{y}{8}$= 13

Soln.

$8 \times \frac{y-5}{2} + \frac{y}{8} = 13 \times 8$

$4(y - 5) + y = 104$

$4y - 20 + y = 104$

$4y + y - 20 = 104$

$5y - 20 = 104$

$5y - 20 + 20 = 104 + 20$

$\frac{5y}{5} = \frac{124}{5}$

$Y = 24\frac{4}{5}$

2. Solve: $\frac{m-2}{3} = \frac{m+4}{5}$

soln.

$15 \times (\frac{m-2)}{3}) = (\frac{m+4}{5}) \times 15$ **LCD =15**

$5(m - 2) = 3(m + 4)$

$5m - 10 = 3m + 12$

$5m - 10 + 10 = 3m + 12 + 10$

$5m = 3m + 22$

$\frac{2m}{2} = \frac{22}{2}$

$m = 11$

<u>Activity:</u>

Solve the following equations.

1. $\frac{P+6}{8} + \frac{P}{4} = 3$

2. $\frac{X+2}{3} + \frac{X}{3} = 4$

3. $\frac{m+5}{5} + \frac{m}{5} = 5$

4. $\frac{m+5}{4} + \frac{m}{5} = 2$

5. $\frac{P+3}{4} + \frac{p}{2} = 6$

6. $\frac{k+5}{3} + \frac{k}{2} = 10$

7. $\frac{w-2}{3} - \frac{w}{4} = 2$

8. $\frac{2k+1}{2} = \frac{K-3}{3}$

9. $\frac{X+3}{3} = \frac{5X+1}{7}$

10. $\frac{2}{P+6} = \frac{2}{p-2}$

SOLVING EQUATIONS INVOLVING SQUARES

Examples

1.solve: $\frac{p^2}{4} = 25$

Soln.

$4 \times \frac{P^2}{4} = 25 \times 4$

$P^2 = 100$

$\sqrt{P^2} = \sqrt{100}$

P = 10

2. Solve: $\frac{2k^2}{6} = 12$

Soln.

$6 \times \frac{2k^2}{6} = 12 \times 6$

$\frac{2k}{2} = \frac{72}{2}$

$P^2 = 36$

$\sqrt{k^2} = \sqrt{36}$

k = 6

3. Solve: $\frac{2p^2}{3}$ -2 = 22

Soln.

3 x $\frac{2p^2}{3}$ - $\frac{2}{1}$ x 3 = 22 x 3 LCD = 3

$2p^2 - 6 = 66$

$2p^2 - 6 + 6 = 66 + 6$

$\frac{2p^2}{2} = \frac{72}{2}$

$\sqrt{P^2} = \sqrt{36}$

P = 6

Activity:

Solve the following equations

1. $\frac{1}{4}P^2 = 4$

2. $\frac{1}{8}m^2 = 2$

3. $\frac{1}{7}k^2 = 28$

4. $\frac{1}{4}m^2 = 400$

5. $\frac{3k^2}{5} = 15$

6. $\frac{2m}{7}^2 = 14$

7. $\frac{3m^2}{4} = 12$

8. $\frac{3k^2}{5} - 1 = 14$

9. $\frac{2m^2}{7} + 2 = 16$

10. $\frac{2p^2}{6} - 1 = 11$

Scenarios

Session 3

- Forms and solves equations about age involving quotient, sum, difference and product
- Forms and solves algebraic equations about age involving comparison.
- Finding time to come and time past involving algebra on time
- Forms and solves equations involving algebra on costs.

Methods

- Guided discovery
- Group discussion
- Think, pair and share.
- explanation

Lesson activities

- **forming algebraic equations**
- **adding**
- **multiplying**
- **dividing**
- **subtracting**

Content/facts

APPLICATION OF ALGEBRA (ABOUT AGE)

Note:

1. Older than (+)
2. Younger than (-)
3. In 10 years', time (+ 10)
4. 5 years ago (-5)
5. Twice (x2)
6. Four times (x4)
7. Difference (-)

Examples

1. Matama is 6 years older than Kabelga. If their total age is 38 years, how old is each?

Soln.

Let Kabalega's age be m.

Kabalega	Matama	Sum
M	M + 6	38 years

$M + m + 6 = 38$

$2m + 6 = 38$

$2m + 6 - 6 = 38 - 6$

$\frac{2m}{2} = \frac{32}{2}$

$m = 16$

Kabalega	Matama
= m = 16 years	= m + 6 = (16 + 6) years = 22 years

2. Peter is 10 years younger than Musa. Their total age is 30 years. How old is each of them?

Soln.

Let Musa's age be K.

Musa	Peter	Total age
K	(K – 10)	30 years

K + K – 10 = 30 years
2k – 10 = 30
2k – 10 + 10 = 30 + 10
$\frac{2k}{2} = \frac{40}{2}$
K = 20 years

Musa	Peter
=K =20yrs	= (k – 10) yrs. = (20 – 10) yrs. = 10yrs

3. Wandera is 10 yrs. older than Kakama. In 6 yrs.time their total age will be 46 yrs. How old is Wandera?

Let Kakamas' age be M.

Time	Kakama	Wandera	Total age
Now	M	M+ 10	----------
In 6yrs	M + 6	M + 10 + 6 M + 16	46yrs

M + 6 + m + 16 = 46yrs
2m + 22 = 46yrs
2m + 22 – 22 = 46 – 22
$\frac{2m}{2} = \frac{24}{2}$ yrs
M = 12 yrs.
Wandera = (m + 10) yrs.
= (12 + 10) years

= 22yrs

b) How old will Kakama be then?

Soln.

Kakama = (m + 6) yrs.
= (12 + 6) yrs.
= 18 yrs.

4. Matama is 5 years older than Opio. 7 years ago, their total age was 19yrs. how old is each of them?

Let Opio's age be m.

Time	Opio	Matama	Sum
Now	M	M + 5	-
7yrs ago	M – 7	(m + 5 – 7) (m – 2)	19 yrs.

m – 7 + m – 2 = 19
m + m – 7 – 2 = 19
2m – 9 = 19
2m – 9 + 9 = 19 + 9

$\frac{2m}{2} = \frac{28}{2}$

m = 14

Opio.	Matama.
K 14yrs	(K + 5) yrs. (14 + 5) yrs. 19 yrs.

Activity:

1. Mandevu is 10 yrs. older than Kasule. In 5 yrs. time, their total age will be 40yrs. How old is each of them?
2. Mbaliga is 7 years older than Martha. In 9 years' time their total age will be 45years. How old is Mbaliga?
3. Okello is 6 years older than Opio. If their total age is 20yrs, how old is each of them?
4. Kasira is 15yrs older than Mbede. In 5yrs time their total age will be 55yrs. How old is Kasira?
5. Owino is 6 yrs. younger than Matu. If their total age is 18 years, how old is Owino?
6. Koire is ten years younger than Kontwe. If their total age is 38 years, how old will Koire be in 6 years' time?
7. Kasana is 4 years younger than Magabali. In 10 yrs time their total age will be 48yrs. How old is Kasna?
8. Wandera is 6 years younger than Wandhuru. 10 years ago, their total age was 14 years. How old were each of them by then?
9. James is 4 times as old as Martha. If their total age is 40 yrs. How old is each of them?

10. Nalule is 6 times as old as her daughter. The difference in their ages is 15 years in 10 years' time to come. How old is Nalule?

APPLICATION OF ALGEBRA WITHOUT TOTAL OR DIFFERENCE

James is 11 years older than Kyamutwe. In 8 years',timeJames will be twice as old as Kyamutwe. How old is James.

Soln.

Let Kyamutwe's age be n.

Time	Kyamutwe	James
No	n	n + 11
In 8 yrs	(n + 8)	n + 11 + 8 n + 19

2 (n + 8) = n + 19
2 x n + 2 x 8 = n + 19
2n + 16 = n + 19
2n + 16 – 16 = n +19 – 16
2n = n + 3
n = 3

James = (n + 11) yrs
= (3 + 11) yrs
= (3 + 11) yrs
= 14 years

Example II

Martha is 12 years older than Jane4 years ago Martha was four times as old as Jane. How old is Jane?

Let Jane be K.

Time	Jane	Martha
Now	K	K + 12
4 years ago,	K – 4	K + 12 – 4 K + 8

4 (k – 4) = K + 8
4 x K – 4 x 4 = k + 8
4k – 16 = k + 8
4k – 16 + 16 = k + 8 + 16
4k = k + 24
4k – k = k – k + 24
3k = 24
$\frac{3k}{3} = \frac{24}{3}$

K = 8

Example III

Okello is 11 years younger than Sarah in 7 years' time he will be a half as old as Sarah. How old is Okello?

Soln.

Let Sarah's age be y**.**

Time	Sarah	Okello
Now	Y	Y - 11
7yrs	y + 7	y - 4

½ (y + 7)= y – 4

$\frac{y+7}{2}$= y - 4

$\frac{y+7}{2}$x 2 = 2 (y – 4)

y + 7 = 2 (y – 4)

y + 7 = 2 (y – 4)

y + 7 = 2y – 8

y – 2y + 7 = 2y – 2y – 8

$^{-}$y + y = $^{-}$ 8

$^{-}$y + 7 – 7 = 8 – 7

$^{-}$y = 15

$\frac{-y}{-1}$ = $\frac{-15}{-1}$

y = 15

Okello = (y – 11) years

= 15 – 11 years

= 4 years

Activity:

1. Nakazi is 14 years older than Ssekazi in 7 years' time, Nakazi will be twice as old as Ssekazi. How old is Nakazi?
2. Owinny is 12 years older than winne, in 2 years' timeOwinny will be twice as old as Owinny. How old is each of them?
3. Matama is 20 years younger than Kalule, in 8 years' time, she will be twice as old as Kalule. How old is Matama.
4. Kakama is 12 years younger than Kawala 5 years ago Kawala was 5 times as old as Kakama. How old is each of them?
5. John is 17 years young than Mukasa in 7 years' time John will be twice as old as Mukasa. How old is Mukasa?
6. Wandera is 13 years younger than Fataki in 8 years' timeWandera will be a half Fataki age. Find Fataki age.

FINDING YEAR'S TIME

Example I

Okello is 4 years and Moses is 15 years in how many years' time will Moses be twice as old as Okello?

Soln.

Let the year time be K

Time	**Okello**	**Moses**
Now	4	15
K yrs	K + 4	K + 15

$2(K + 4) = K + 15$

$2 \times k + 2 \times 4 = k + 15$

$2k + 8 = k + 15$

$2k + 8 - 8 = k + 15 - 8$

$2k = k + 7$

$2k - k = k - k + 7$

$K = 7$

Example II

Marth is 6 years old and Owino is 20 years old in how many years' time will Marth be a half Owino's age?

Soln.

Let years' time be M.

Time	Martha	Owino
New	6	20
M yrs	M + 6	M + 20

$\frac{1}{2}(m + \frac{20}{2}) = m + 6$

$M + 20 = 2m + 12$

$M + 20 - 20 = 2m + 12 - 20$

$M = 2m - 8$

$M - 2m = 2m - 2m - 8$

$-m = -8$

$$\frac{-m}{-1} = \frac{8}{-1}$$

$M = 8$

Activity:

1. Okello is 5 years and Peter is 17 years in how many years' time will Peter be twice as old as Okello?
2. Nakato is 30 years and her son is 12 years. In how many years' time will Nakato be twice as old as the son?
3. Alice is 18 years and Becca is 5 years. In how many years' time will Alice be twice as old as Becca?
4. Betty is 7 years and Martha is 19 years old. After how many years will Betty be a half Martha's age?
5. Wako is 26 years old and Panko is 9 years. After how many years will Panko be a half of Wako's age?
6. Ronald is 10 years old and Moses is 4 years old, in how many years' time will Moses be a half Ronald's age?

APPLICATION OF ALGEBRA IN SHOPPING

Example I

At a supermarket, a book costs sh. 400 more than a pen and a set is thrice the price of a pen. If Mandevu bought all the three items at sh. 3,900. Find the cost of a set.

Soln.

Let the pen's cost be m

Pen	Book	Set	Sum
M	M + 400	3m	3,900

M + m + 400 + 3m = 3900
M + m + 3m + 400 = 3900
5m + 400 = 3900
5m + 400 – 400 = 3900 – 400
5m = 3500

$\frac{5m}{5} = \frac{3500}{5}$

M = 700

Set = 3m
= 3 x sh. 700
= sh. 210

Example II

At the shop of a set cost 5p and a book costs p +400. Okello bought 2 sets and 4 books if he paid sh. 7,200, how much did he spend on books?

Soln.

	Set	**Book**	**Sum**
No.	2	4	…….

Cost	5p x 2 10p	4 (p +400) 4p + 600	Sh. 7200

10p + 4p + 1600 = sh. 7200
14p + 1600 = sh. 7200
14P + 1600 – 1600 = sh. 7200 – sh. 1600
$\frac{14P}{14}$ = sh. $\frac{5600}{14}$

P = sh. 400

Books = 4p + 1600
= (4 x 400) + 600 = sh. 3200

Activity:

1. The cost of a book is sh. 400 less than that of a set and a pen costs twice the cost of a set. If the total cost is sh. 1200. Find the cost of a pen.
2. An apple costs sh.(92p – 200) and an orange costs sh. (p +300). If Mark bought 6 apples and 2 oranges at a cost of sh. 7800 altogether, Find the cost of an apple.

b) How much did he spend on oranges?

3. A pen costs sh. 700 less than the cost of a book and a set costs sh. 3700. Find the cost of each item.
4. In a market a shirt cost sh. y and a dress cost sh. (y – 500). If a man bought 7 shirts and 3 dresses at sh. 38500 altogether, how much did he spend on shirts?

4. In a market, a goal cost (3k + 2000) and a cock costs sh. 9k + 5000). If a trader bought 2 goats and3 cocks altogether at sh. 235000. Find the cost of a goat.

MORE ABOUT APPLICATION OF ALGEBRA

Examples

1. On a farm there are hens and cows altogether. There are 12 heads and 34 legs. How many cows are on the farm?

Soln.

Let the No. of cows be M

	Cows	Hens	Sum
Heads	M	12 – m	12
Legs	4 x m 4m	2 (12 –m)	34

$4m + 2(12 - m) = 34$
$4m + 24 - 2m = 34$
$4m - 2m + 24 = 34$
$2m + 24 = 34$
$2m _ 24 - 24 = 3y - 24$
$\frac{2m}{2} = \frac{10}{2}$

M = 5

b) How many hens are on the farm?

No. of hens = 12 – m
= 12 – 5
= 7

1. A teacher awards 3 marks for every correct answer and subtracts 2 marks for every wrong answer. Out of the 20 questions, Nabusoko wrote 15 correct answers. How many marks did she get?

Soln.
Let the marks be K.
No. of marks = c – w
Wrong = 20 – 15
= 5
Marks = (3 x 15) – (2 x 5)
= 45 – 10
= 35 marks

b) Ndole got marks. How many correct answers did he write?

Soln.
Let the correct answers be K.

	Correct answer	Wrong answers	Sum
Question	K	20 – k	20
Marks	3 x k	2 (20 – k)	40

$3k - 2(20 - k) = 40$
$3k - 40 + 2k = 40$
$3k - 2k - 40 = 40$
$5k - 40 = 40$
$5k - 40 + 40 = 40 + 40$
$\frac{5k}{5} = \frac{80}{5}$

K = 16

Activity:

1. A pen costs sh. 300 and every book costs sh. 400 Mandela bought five items and paid sh. 1700. How many book and pens did he buy?
2. On a farm, there are twenty animals. Some are birds and others are goats, altogether there are 56 legs. How many birds and goats are on the farm?

Scenarios

Session 4

- ✦ Solves simple inequalities and finds solution sets
- ✦ Solves inequalities involving fractions
- ✦ Solves and finds compound sets of compound inequalities.

Methods

- Guided discovery
- Group discussion
- Think, pair and share.
- explanation

Lesson activities

- **finding solution sets.**
- **adding**
- **multiplying**
- **dividing**
- **subtracting**

Content/facts

INEQUALITIES AND SOLUTION SETS

Symbols

$>$ - Greater than

$\geq$ - Greater than or equal to

$<$ - less than

≤ - less than or equal to.

Note:
A solution set is a set of all possible values and integers that satisfies, a given inequality.

Examples

1. Given that Y ≤ 6. Find the solution set for Y.
   Soln.

   Y = {5,4,3,2,1, 0, -1, …}

2. If 3> k < 8. Find the solution set for K.
   Soln.

   K ={4,5,6,7}

Activity:

1. If M < 8. Find the solution set form if M is a positive integer.
2. Given that Y > 1. Find the solution set for Y if Y is a prime number less than 10.
3. If P ≥ 7 give a solution set for Y if Y is a positive number less than 12.
4. If Y ≥7, give a solution set for Y if it is an odd number less than 12.
5. If 3 < k 6, find a solution set for K.
6. Given that 5 ≤ Q < 12. Find the solution set for Q.
7. 3 < x < 10. If X is an even number, find the solution set for X.
8. $^{-}2 \leq W \leq 5$. Write down the solution set for W.
9. 3 ≤ K < 7, find the solution set for K.
10. If $^{-}2 \leq Y \leq 4$;

a) Find the solution set for Y.
b) Find n (Y)

SOLVING INEQUALITIES

Note:

- When dividing both sides by a negative, the sign changes to its opposite. That is to say;
  $^{-}2x \leq {}^{-}8$
  $\frac{-2x}{-2} \geq \frac{-8}{-2}$
  X ≥ 4

Examples.

1. Solve the inequality.2X ≤ 8

Soln.

$$\frac{2X}{2} \leq \frac{8}{2}$$

X =≤ 4

2. Solve the inequality.

$4X + 3 \leq 5X + 2 - 3$

$4X \leq 5X - 1$

$4X - 5X \leq 5X - 5X - 1$

$-X \leq -1$

$$\frac{+X}{+X} \leq \frac{+1}{+1}$$

X = 1

3. Solve and give the solution set for the following:

   $P + 4 \leq 6$

   Soln.

   $P + 4 - 4 \leq 6 - 4$

   $P \leq 2$

   $P = \{\ldots, -1,0,1,2\}$

4. Solve and find the solution set.

   $3 - 3p \leq 9$

   Soln.

   $3 - 3 - 3p \leq 9 - 3$

   $-3p \leq 6$

   $$\frac{-3P}{-3} \geq \frac{6}{-3}$$

   $P \geq -2$

   $P = \{ -2, -1,0,1,2,3, \ldots\}$

Activity:

Solve and give the solution sets for the following inequalities: -

1. $P + 2 < 6.$
2. $4 - P > 7$
3. $Y - 4 > 3$
4. $m + 6 \geq 4$
5. $2P - 6 \leq 8$
6. $4y - 4 < 8$
7. $6 - 5k \geq 16.$
8. $3 - 6X > 21$
9. $3K + 6 \leq 5K + 8$
10. $4n - 2 \geq 6n - 8$

SOLVING INEQUALITIES INVOLVING FRACTIONS

Examples

1. Solve and find the solution set

$\frac{3m}{5} + 2 \leq 5$

Soln

$5 \times \frac{3m}{5} + \frac{2X5}{1} \leq \frac{5 \times 5}{1}$ LCD = 5

$3m + 10 \leq 25$

$3m + 10 - 10 \leq 25 - 10$

$\frac{3m}{3} \leq \frac{15}{3}$

$M \leq 5$

$M = \{5,4,3,\ldots\}$

Activity:

Solve and find the solution set

1. $\frac{2y}{5} + 6 \geq 2$
2. $\frac{3n}{4} - 8 \geq 10$
3. $\frac{3P}{5} - 3 \geq -6$
4. $\frac{3W}{4} - 2 \leq 4$
5. $\frac{5X}{6} - 4 < 6$
6. $\frac{2y}{5} - 2 \geq 4$
7. $\frac{X}{2} + 1 \leq 2.$

Compound inequalities

Examples

1. Solve and give the solution set.

$4 < 4P < 20.$

$\frac{4}{4} < \frac{4P}{4} < \frac{20}{4}$

$1 < P < 5$

$P = \{ 2, 3 , 4 \}$

2. Solve and give the solution set for $4 \leq 2X + 6 < 12.$

Soln.

$4 \leq 2X + 6 < 12.$

$4 - 6 \leq 2X + 6 - 6 < 12 - 6$

$-2 \leq 2X < 6$

$\frac{-2}{2} \leq \frac{2X}{2} < \frac{6}{2}$

$-1 \leq X < 3$

$X = \{ -1,0,1,2\}$

3. Solve and give the solution set.

$8 \leq 2x - 4 \leq 10$

$8 + 4 \leq 2X - 4 + 4 \leq 10 + 4$

$\frac{12}{2} \leq \frac{2X}{2} \leq \frac{14}{2}$

$\frac{12}{2} \leq \frac{2X}{2} \leq \frac{14}{2}$

$6 \leq x \leq 7$

$X = \{6,7\}$

4. Solve and give the solution set.

$12 < 2 (m + 2) \leq 20.$

Soln.

$12 < 2m + 4 \leq 20$

$12 - 4 < 2m + 4 - 4 \leq 20 - 4$

$8 < 2m \leq 16$

$\frac{8}{2} < \frac{2m}{2} \leq \frac{16}{2}$

$4 < m \leq 8$

M ={5,6,7,8}

Activity:

Solve and find the solution set.

1. $6 \leq 3m < 15$
2. $8 < 4y \leq 20$
3. $2 \leq 6 - 2P \leq 20$
4. $2 < 3P - 1 \leq 8$
5. $3 < X - 3 < 6$
6. $4 > P + 6 > 10$
7. $6 \leq 2(2X + 1) \leq 14$
8. $4 < \frac{2P}{3} + 2 < 8$
9. $16 \geq 2X + 4 \geq 10$
10. $10 \leq 2X - 6 \leq 4$

SOLVING INEQUALITIES INVOLVING UNKNOWNS ON BOTH SIDES

Examples

1. Solve and give the solution set for K if K is a composite number less than15;

$2K - 2 + 2 \geq K + 2$

Soln.

$2K - 2 + 2 \geq K + 2$
$2K \geq K + 4$
$2K - K \geq K - K + 4$
$K \geq 4$
$K = \{4,6,8,9,10,12,14\}$

Activity

Solve and give the solution set for the following inequalities.

1. $2(m - 3) \leq 2(2m + 4)$
2. $3X + 4 < 4X + 2$
3. $5Y \geq 6Y + 3$
4. $3(n + 2) \leq 2(n + 2)$

5. $4(2y + 2) > 3(3y + 2)$
6. $2(4P – 3) \geq 2(2P – 9)$
7. $2(2X – 2) \geq 2(X + 2)$
8. $3(P – 1) \leq 2(P + 2)$

Scenarios

TOPIC 4 - SET CONCEPTS;

Session 1

- ✦ describes finite and infinite sets
- ✦ finds number of subsets and proper subsets
- ✦ finds number of elements given number of subsets

Methods

- Guided discovery
- Group discussion
- Think, pair and share.
- explanation

Lesson activities

- **adding**

- **multiplying**
- **dividing**
  - **subtracting**

Con tent/facts

A finite set **is a set whose members can be listed down and has an end. e.g**

set T = {even numbers between 4 and 12}

T = {6, 8, 10}

An infinite set is a set whose members are endless. e.g.

a set of odd numbers, a set of all stars in the sky, a set of all integers.

Set K= {all prime numbers}

K= {2, 3, 5, 7, 11, 13…..}

Activity 1. State whether the following sets are finite or infinite:

G = {girls in your class}

B = {boys in your class}

W= {days of the week}

K = {square numbers}

P= {pupils who like football}

T = {multiples of three}

H = {all alphabetical letters}

F = {months of the yea}

FINDING NUMBER OF SUBSETS

Number of subsets = 2^n

Where: - **n** is number of elements is a given set

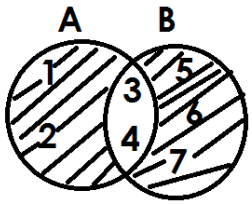
2 indicates that, number of subsets are in powers of 2.

Example1

Given that set P= { }, find number of subsets in set P. **Solution** **No. of subsets = 2^n** **= 2^0** **= 1 subset.**	**Example2** **If set T= {5}. Find number of subsets in set T** **Solution** **No. of subsets = 2^n** **= 2^1** **= 2 subsets**

Example 3

Below is set A and B



Find number of subsets in set (A n B) ^{I}

No. of subsets = 2^n

= 2^5

= 2x2x2x2x2

= 32 subsets

Examples 4

If set m= {1, 2, 3, 4, 5, 6, 7, 8} and set

P= {all factors of 12}. Find number of subsets in M n P

M= {1, 2, 3, 4, 5, 6, 7, 8}

P= {1, 2, 3, 4, 6, 12}

M n p= {1, 2, 3, 4, 6}

No. of subsets = 2^n

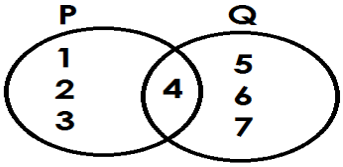
= 2^5

= 2x2x2x2x2

= 32

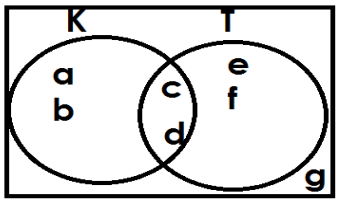
ACTIVITY

1. **Find number of subsets in set P with 4 elements.**
2. **Given that set y= {all vowel letters}. Find number of subsets in set y.**
3. **If set K= {a, b, c, d, e, f}, find number of subsets in set K.**
4. **Below is set P and Q**



Find number of subsets in P U Q

6. **If set A= {All even numbers less than 10} and set B= {all factors of 12}. Find number of subsets in A n B.**
7. **Use the Venn diagram below to answer question**



Find number of subsets in K^{l}

FINDING NUMBER OF ELEMENTS GIVEN NUMBER OF SUBSETS

Formula:

2^n = No. of subsets

Examples

1. Set K has 1 subset. How many elements are in set K?

Solution

2^n = no of subsets

$2^n = 1$ but $1 = 2^0$

$2^n = 2^0$

n = 0

No element

2. Set M has 64 subsets. How many elements are in set M?

Solution

2n = No. of subsets

2n = 64

2n =

2^n = 2x2x2x2x2x2

$2^n = 2^6$

n = 6

2	64
2	32
2	16
2	8
2	4
2	2
	1

Example 3

3. Set K has 128 subsets. How many elements are in set K?

Solution

2^n = No. of subsets

2^n = 128

2^n =

$2^n = 2^7$

n = 7

2	128
2	64
2	32
2	16
2	8
2	4
2	2
	1

Example 4

Set M has 256 subsets. How many elements are in set M?

2^n = No. of subsets

2^n = 356

2^n =

$2^n = 2^8$

n = 8

2	256
2	128
2	64
2	32
2	16
2	8
2	4
2	2
	1

ACTIVITY.

1. Set A has 8 subsets. Find the number of elements in set A.
2. There are 4 subsets in a set. How many elements are in that set.
3. Find the number of members in a set with the following number of subsets.

(a) 32 subsets

(b) 64 subsets

(c) 16 subsets

FINDING PROPER SUBSETS

(a) **By listing.**

1. If P= {a, b, c}, how many proper subsets has set P?

Solution

P= {a, b, c}

{ }, {a}, {b}, {c}, {a, b}, {a, c}, {b, c}

7 proper subsets.

2. Given that M= {p, q, r, s,}. List the number of proper subsets in set M.

Solution

M= {p, q, r, s}

{ }, {p}, {q}, {r}, {s}, {p,q},{p, r}, {p,s} {q,r},{q,s}, {r,s},{p,q,r},{p,q,s}, {p,r,s}, {q,r,s}

15 proper subsets

ACTIVITY

List the proper subsets for each of the following sets

1. B= {a,b}
2. C= {x, y, z}
3. D= {t}
4. p= {a, b, c, d}
5. q= { }
6. Z= {1, 2, 3}

(b) **By using the formula.**

Formula:

No. of proper subsets = $2^n - 1$

Where n is number of elements in a given set and 1 is the mother being subtracted from the formula.

Examples

1. If x= {1, 2, 3}. Find the number of proper subsets in set X.

Solution

Set X has 3 elements, how many proper subsets are in set X?

Proper subsets = 2^n-1

= 2^3-1

= {2x2x2}-1

= 8-1

= 7 proper subsets.

2. Set M = {1}, how many proper subsets are in set M?

Set M has 1 element

No. of proper subsets = 2^n-1

= 2^1-1

= 2-1

= 1 proper subset.

3. How many proper subsets are in a set with 5 elements?

Solution

No. of proper subsets = 2^n - 1

= 2^5 - 1

= {2x2x2x2x2} -1

= 32 - 1

= 31 proper subsets.

Activity

1. **If B= {1, 2}, how many proper subsets are in set B?**
2. **Given that R= {a, b, c, d, e, f}. Find the number of proper subsets in set R.**
3. **Find the number of proper subsets of a set which has;**

(a) 4 elements
(b) 3 elements
(c) 7 elements
(d) 9 elements

FINDING NUMBER OF ELEMENTS GIVEN NUMBER OF PROPER SUBSETS

Formula;

2^n - 1 = No. of proper subsets

Examples

1. Set M has 15 proper subsets. How many elements are in set M?

Solution

2^n – 1 = No of proper subsets

2^n - 1 = 15

2^n-1+1=15+1

2^n = 16

2^n =

2	16
2	8
2	4
2	2
	1

2^n = 2^4

n =4

2. Set K has 255 proper subsets. How many elements are in set k?

Solution

2^n-1=no of proper subsets

2^n - 1 = 255

2^n-1+1=255+1

2^n = 256

2^n =

2	256
2	128
2	64
2	32
2	16
2	8
2	4
2	2
	1

2^n = 2x2x2x2x2x2x2x2

2^n = 2^8

$n = 8$

3. Set R has 63 proper subsets. Find the number of elements in set R

$2^n - 1$ = No of proper subsets

$2^n - 1 = 63$

$2^n - 1 + 1 = 63 + 1$

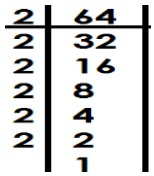
(0)

$2^n = 64$

$2^n =$

$2^n = 2^6$

$n = 6$



Activity

1. Set H has 3 proper subsets. How many elements are in set H?
2. There are 15 proper subsets in set D. How many elements are in set D?
3. Find the number of members in a set with;

(a) 31 proper subsets

(b) 7 proper subsets

(c) 511 proper subsets

(d) 1023 proper subsets

Scenarios

Session 2

competences

- ✦ represents information on a Venn diagram.
- ✦ Interpreting and solving word statements on Venn diagrams
- ✦ Probability on sets.

Methods

- Guided discovery
- Group discussion
- Think, pair and share.
- explanation

Lesson activities

- **adding**
- **multiplying**
- **dividing**
- **subtracting**

Content/facts

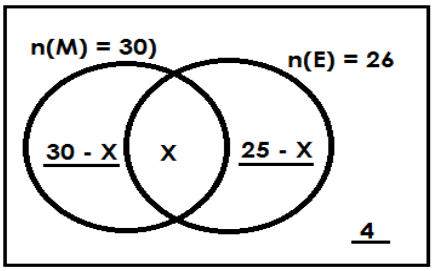
SOLVING PROBLEMS USING A VENN DIAGRAM

A). Given the interesting as the unknown

Examples I

In a class of 40 pupils, 30 lime math (M), 25 like English (E), X pupils like both subjects and 4 pupils like neither.

Complete the Venn diagram.



b) **Find the value of X.**

solution

30 – X + X + 25 – X + 4 = 40
30 + 25 + 4 – X = 40
59– X=40
59 – 59 – X = 40 - 59
-X = -19
-X= -19
-1 -1
X= 19
b) How many pupils like only one subject?
Solution
Only one = n(m) only + n (E) only.

= 30 – X + 25 – X
= 30 – 19 + 25 – 19
= 11 + 6
= 17 pupils

c) Find the probability of picking a pupil who does not like English from the class at random.

Solution

Prob = n(E) - pupils who do not like English
n(P) - universal

$$= \frac{25 - X + 4}{40}$$

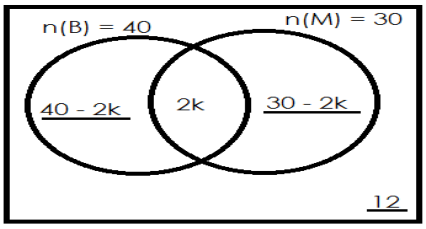
$$= \frac{(25 - 19) + 4}{40}$$

$$= \frac{6 + 6}{40}$$

$$= \frac{12}{40}$$

In a village of 60 farmers, 40 grow beans (B), 30 grow maize (M) and 2k farmers grow both crops while 12 do not grow any of the mentioned.

a) Complete the Venn diagram below.



b) Find the value of K.

40 + 30 + 12 – 2k = 60
82 – 2k = 60
82 – 82 - 2k = 60 – 82
- 2k = -22
-2 -2
K = 11

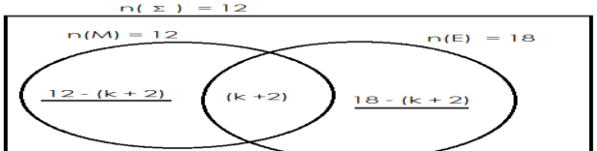
c) How many farmers grow maize but not beans?

Maize only = 30 – 2k
= 30 – (2 x k)
= 30 – (2 x 11)
= 30 – 22
= 8 farmers

Example 3

In a class of 28 pupils, 12 take math (M) and 18 take English (E). given that (K + 2) pupils take both math and English and 3 pupils take neither subject.

a) complete the Venn diagram below.



b) How many pupils like both subjects?

Solution.

$12 - (k+2) + (k+2) + 18 - (K+2) + 3 = 28$

$12 + 3 + 18 - (k+2) = 28$

$15 + 18 - k - 2 = 28$

$33 - 2 - k = 28$

$31 - k = 28$

$31 - 31 - k = 28 - 31$

$\frac{-k}{-1} = \frac{-3}{-1}$

$K = 3$

Both = $k + 2$

$= 3 + 2$

$= 5$ pupils

c) How many pupils do not like English?

Soln.

$n(E) = 12 - (k+2) + 3$

$= 12 - (3+2) + 3$

$= 12 - 5 + 3$

$= 7 + 3$

$= 10$

d. Find the probability of picking a person who likes math only.

$Prob = \frac{n(E)}{n(p)}$

$= \frac{12 - (k+2)}{28}$

$= \frac{12 - (3+2)}{28}$

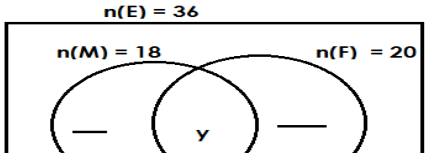
$= \frac{12 - 5}{28}$

$= \frac{7}{28}$

Activity:

1. In a class of 36 pupils, 18 like meat (M), 20 like fish (F) and y pupils like both meat and fish while 3 pupils don't like any of the mentioned.

a) complete the Venn diagram below.



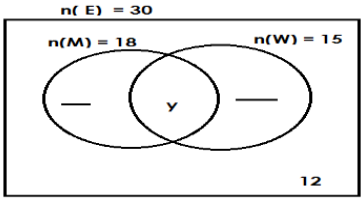
b) Find the value of y.

c) How many pupils like one type of food only?

d) Find the probability of picking a pupil who likes fish only?

2. At a birth day party attended by 30 guests, 18 took soda (S) and 15 took water (W). some guests took both and 12 guests did not take any drink.

a) Complete the Venn diagram below

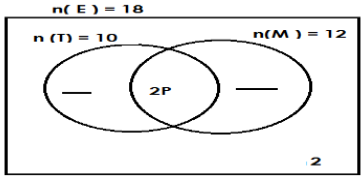


b) How many guests took both types of drink?

c) How many guests did not take water?

3. In a village of 18 farmers, 10 grow tobacco (T) and 12 grow millet (M). given that 2p farmers grow both and 2 farmers grow neither of the two crops.

a) Complete the Venn diagram below.

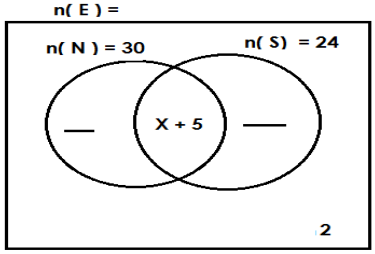


b) How many farmers grow both types of crops?

c) How many farmers grow one type of crops only?

4. During a trip to Jinja town, 30 students visited Nile breweries (N) and 24 students visited steel rolling mills (S), (X + 5) students visited both and 2 students did not visit any of the two factories.

a) Complete the Venn diagram below:



b) Given that 41 students altogether visited Jinja Town, find the value of X.

c) How many students visited Nile Breweries only?

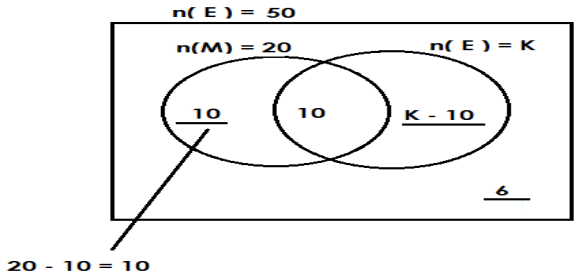
d) If a student is picked at random, what is the chance that he / she visited one type of factory only

B GIVEN ONE OF THE SETS AT THE UNKNOWN

Examples

1. In a class of 50 pupils, 20 like mathematics (M) and K pupils like English (E). given that 10 pupils like both subjects and 6 pupils like neither of the two.

a) Complete the Venn diagram below.



b) Find the value of K

soln.

K – 10 + 10 + 10 + 6 = 50

K + 10 + 6 = 50

K + 6 + 16 = 50 – 16

K = 34

c) How many pupils like one subject only.

Soln.

One subject only = 10 + (K – 10)

= 10 + (34 – 10)

= 10 + 24

= 34 pupils

e) Find the probability of picking a pupil who like English but not math.

Prob = n(E)

n(P)

= k- 10

50

= 34 – 10

50

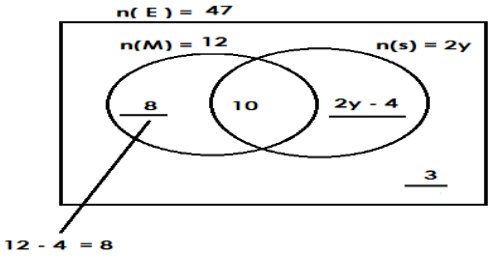
= 24

50

Example 2.

2. In a class of 47 pupils, 12 like music (M) and 2y like swimming (S). 4 pupils like both and 3 do not like any of the two activities.

a) Complete the Venn diagram below.



b) How many pupils like swimming only?

2y – 4 + 4 + 8 + 3 = 47
2y + 11 = 47
2y + 11 = 47 – 11
$\frac{2y}{2}$ = $\frac{38}{2}$
y = 18
swimming only = 2y – 4
= (2 x y) - 4
= (2 x 18) -4
= (36 – 4)
= 32 pupils

c) Find the probability of picking a pupil who like one subject only?

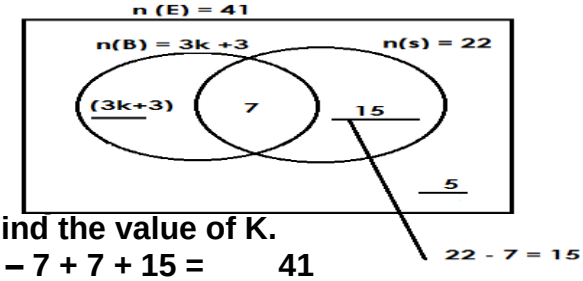
One subject only= 8 + (2y – 4)
= 8 + (2 x 18) – 4
= 8 + (36 – 4)
= 8 + 32
= 40

Prob = $\frac{n(E)}{n(P)}$
= $\frac{40}{47}$

Example 3.

In a family of 41 people, 3K + 3 drink beers (B) and 22 drink sodas (S). 7 people drink both drink and 5 people neither drink beer nor soda.

a) Complete the Venn diagram below.



b) Find the value of K.

(3k + 3) – 7 + 7 + 15 = 41
3k + 3 + 15 + 5 = 41
3k + 23 = 41
3k + 23 – 23 = 41 – 23
3k = 18
3 = 3
K = 6

c) How many people drink only one type?

Soln.

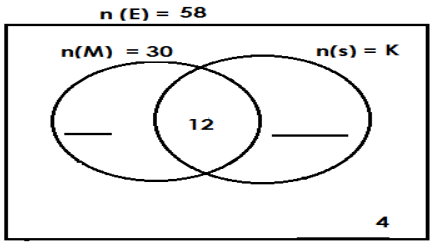
Only one =
= (2k – 3) – 7 + 15
= (2 x 6) + 3 – 7 + 15
=18 + 3 – 7 + 15

=21 – 7 + 15
=21 + 15) – 7
=36 – 7
=29 people

Activity

1. In a class of 58 pupils, 30 like math (M) and K like science (S). given that 12 pupils like both and 4 do not like any of the two subjects.

a) Complete the Venn diagram below.



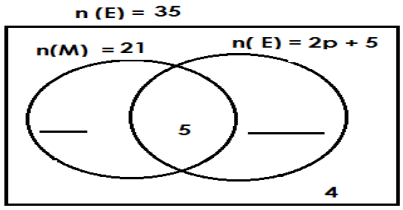
b) Find the value of K.

c) How many pupils like one type of subject only?

d) Find the probability of getting a pupil who like science only.

2. In a class of 35 pupils, 21 like math, 2pt 5 like both subjects while 4 pupils do not like any of the mentioned subject.

a) Complete the Venn diagram below.

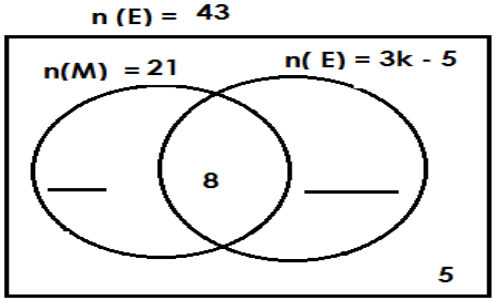


b) How many pupils like English?

c) Find the probability of picking a pupil who like English only.

3. In a village of 43 farmers, 21 grow maize, 3k – 5 grow beans, 8 grow both while 5 farmers do not grow any of the mentioned crops.

a) Complete the Venn diagram below.

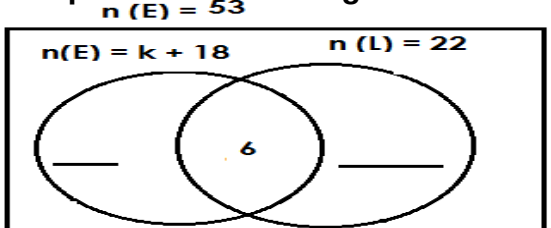


b) Find the value of K.

c) How many farmers who grow one type of crops only.

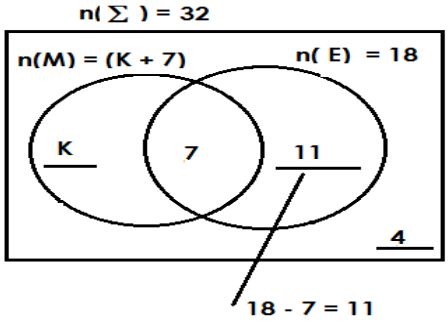
4. In a village of 53 people, k + 18 speak English, 22 speak Luganda and 6 people speak both languages while 4 people do not speak any of the two languages.

a) Complete the Venn diagram below.



b) How many people speak English only?
c) How many people speak Luganda but not English.
d) Find the probability of picking a person who speak one type of language?

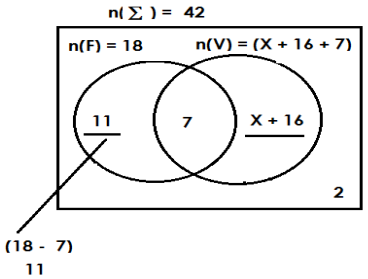
C). GIVEN THE DIFFERENCE OF SETS AS UNKNOWN.
Examples
1. In a class of 32 pupils, K pupils like math (M) only and 18 like English (E). given that 7 pupils like both while 4 pupils like neither.
a) Complete the Venn diagram below.



b) Find the value of K
Soln.
----
c) How many pupils like one subject only?
Soln.
One subject only

----
d) Find the probability of picking a pupil who likes math.
Probability

In a class of 42 pupils, X + 16 play volley ball only, 18 play football and 7 play both games while 2 pupils do not only of the two games.
a) Complete the Venn diagram below.



b) Find the value of X.

Soln.

$X + 16 + 7 + 11 + 2 = 42$

$X + 36 = 42$

$X + 36 - 36 = 42 - 36$

$X = 6$

c) Find the probability of picking a pupil who likes one subject only.

Soln.

One subject only $= X + 16 + 11$

$= 6 + 16 + 11$

$= 33$

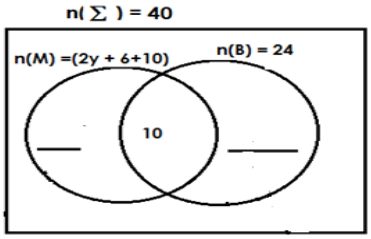
$\therefore$ Probability $= \frac{n(E)}{n(p)}$

$= \frac{33}{42}$

Activity

1. **In a village of 40 farmers 2y + 6 grow maize (M) only, 24 grow beans (B) and 10 farmers grow both crops. Given that 6 farmers grow neither.**

a) Complete the Venn diagram below

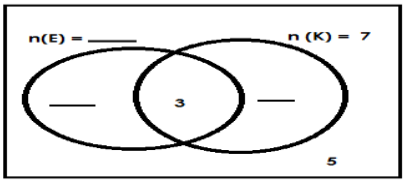


b) How many farmers grow maize?

c) How many farmers grow one type of crop only?

2. In a group of 18 taxi drivers X driver take Entebbe (E) route only any 7 drivers take Kampala (K) route. Given that 3 drivers take both routes and 5 drivers take none.

a) Complete the Venn diagram below:



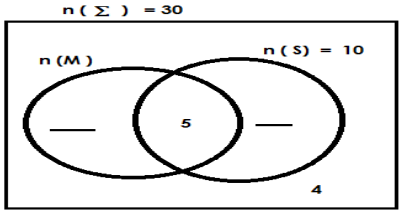
b) Find the value of X.

c) How many drivers take one route only?

d) Find the probability of picking a driver who takes Entebbe route only?

3. In a class of 30 pupils, (K + 2) like math only, 10 like SST (S) and 5 like both. Given that 4 like neither.

a) Complete the Venn diagram below.

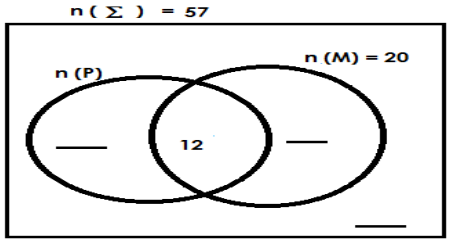


b) How many pupils like math?

c) How many pupils like one subject only?

4. At a party attended by 57 people, (2X + 2) took pepsi cola (P) only and 20 took Mirinda (M). 12 people took both and one person did not take any.

a) Complete the Venn diagram below:



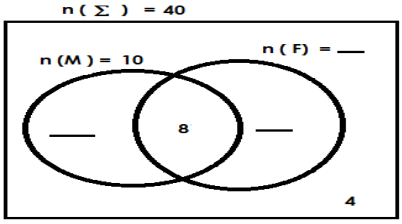
b) Find the value of X.

c) How many people took Pepsi only?

d) Find the probability of picking a person who took one type of drink only

5. In a group of 40 teachers, at St. Anthony Primary school, 10 eat meat (M) and (2y – 2) eat fish only. Given that 4 teachers eat neither, and 8 teachers eat both.

a) Complete the Venn diagram below: -



b) How many teachers eat fish.

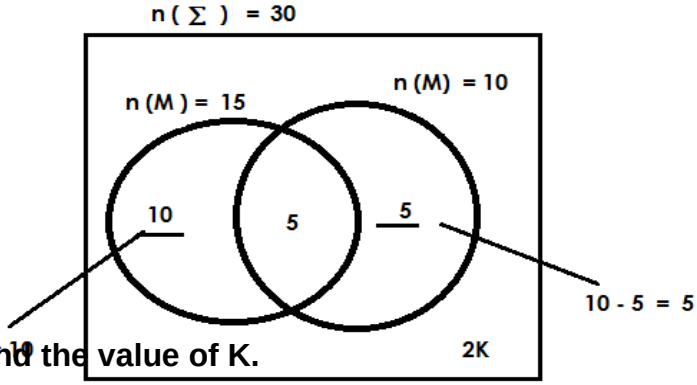
c) Find the probability of picking a teacher who eats only one type.

D). SOLVING PROBLEMS USING A VENN DIAGRAM GIVEN UNION COMPLEMENT AS THE UNKNOWN.

<u>Examples</u>

In a class of 30 pupils, 15 eat fish (F) 10 eat meat (M) 5 eat both dishes and 2K do not eat any of the mentioned.

a) <u>Complete the Venn diagram</u>



b) Find the value of K.

Solution.

2k + 10 + 5 + 5 = 30

2k + 20 = 30

2k + 20 – 20 = 30 – 20
2k = 10
$\frac{2k}{2} = \frac{10}{2}$
K = 5

b) Find the probability of picking a pupil who does not eat meat.

Solution:

Prob = $\frac{n(E)}{n(P)}$

= $\frac{n(M)'}{n(\Sigma)}$

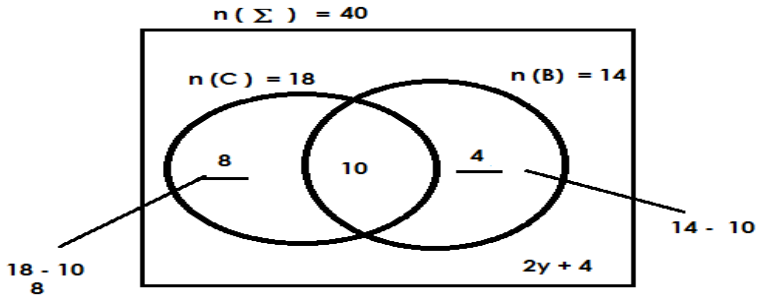
= $\frac{10 + 2k}{30}$

= $\frac{10 + (2 \times 5)}{30}$

= $\frac{10 + 10}{30}$

= $\frac{20}{30}$

In a village of 40 farmers, 18 grow cassava (C), 14 grow beans (B), 10 grow both while 2y + 4 farmers grow none of the mentioned. Complete the Venn diagram.



c) How many farmers do not grow any of the mentioned crop?

Solution.

2y + 4 + 4 + 10 + 8 = 40
2y + 26 = 40
2y + 26 – 26 = 40 – 26
2y = 14
$\frac{2y}{2} = \frac{14}{2}$
Y = 7

Farmers who don't grow any of the
= 2y + 4
= (2 x 7) + 4
= 14 +4
= 18

d) Find the probability of picking a farmer who grows only one crop.

Solution

Prob = $\frac{n(E)}{n(P)}$

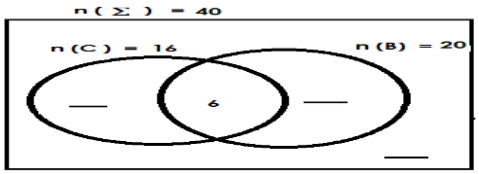
= $\frac{8 + 4}{40}$

= $\frac{12}{}$

Activity:

1. In a class of 40 pupils, 16 like English (E) 20 like math (M), 6 pupils like both subjects and K pupils do not like any of the mentioned subjects.

a) Complete the Venn diagram.

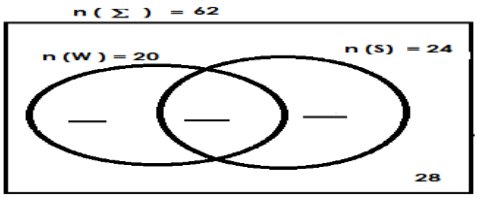


b) Find the value of K.

c) Find the probability of picking a pupil who do not like English from the class at a random.

2. In a bus of 62 passengers, 20 bought water (W), 24 bought soda (S0 and 8 bought both drinks while 2P passengers did not buy any of the mentioned drinks.

a) Complete the Venn diagram

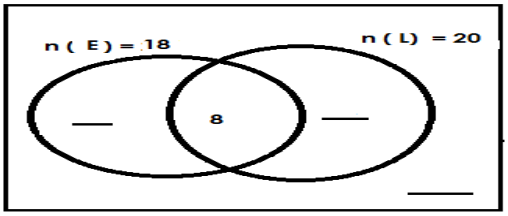


b) Find the value of P.

c) How many passengers bought only one drink?

3. In a class of 40 pupils, 18 speak English (E), 20 speak Luganda (L) 8 pupils speak both any of the mentioned.

a) Complete the Venn diagram.



b) Find the value of y.

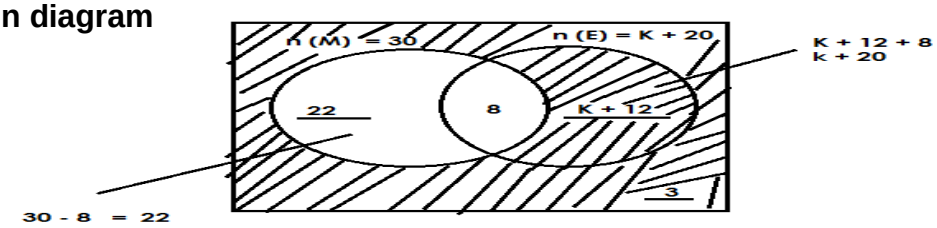
c) Find the probability of picking a pupil who does not speak English from the class at a random.

SOLVING PROBLEMS USING VENN DIAGRAMS BY COMPARING SETS

Examples

In a class, 30 pupils like math (M) K + 12 pupils like English only (E) and 8 pupils like both subjects while 3 pupils do not like any of the mentioned.

a) Complete the Venn diagram



b) If 36 pupils do not like math. Find the value of K

solution

K +12 + 3 = 36

K + 15 = 36

K + 15 – 15 = 36 – 15

K = 21

b) How many pupils like English?

Solution

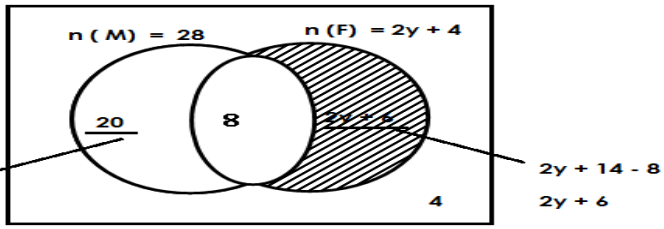
n(E) = K + 12 + 8

= 21 + 20

= 41

In a class 28 pupils like meat (M) 2y + 14 like. Fish and 8 pupils like both dishes while 4 pupils do not like any of the mentioned.

a) Complete the venn diagram.



1

c) Given the 42 pupils like only one dish, how many pupils like fish.

Solution

2y + 6 + 20 = 42

2y + 26 = 42

2y + 26 – 26

2y = 16

2y = +68

2 2

y = 8

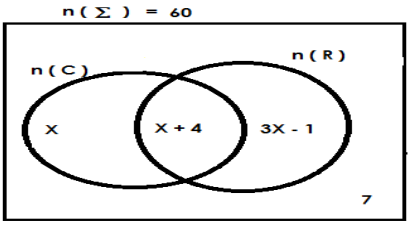
n (E) = 2y + 14

= (2 x 8) + 14

= 16 + 14

= 30

The Venn diagram below shows the number of farmers who grow cassava (C) and Rice (R) use it to answer questions.



a) If 24 farmers grow cassava, find the value of X.

b) How many farmers grow only one type?

only one = n (c) only + n (R) only.

= X + 3X - 1

= 10 + (3 x 10) - 1

= 10 + 30 – 1

= 10 + 29

= 39

c) If a farmer is picked at random, to lead the farmers. Find the probability of picking a farmer who does not grow cassava.

Soln.

$$\text{Prob} = \frac{n(E)}{n(P)}$$

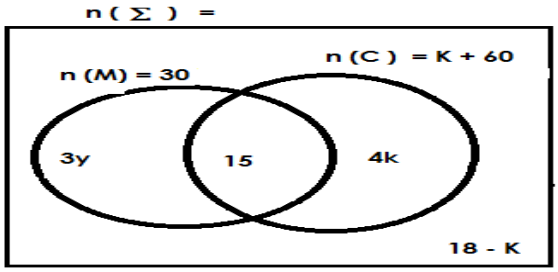
$$= \frac{3X - 1 + 7}{X + X + 4 + 3X - 1 + 7}$$

$$= \frac{(3 \times 10) - 1 + 7}{10+10+4+30 - 1 + 7}$$

$$= \frac{30 - 1 + 7}{6}$$

$$= \frac{36}{60}$$

In the Venn diagram below, 30 farmers grow maize (M), 15 grow both crops. Study it and answer the questions that follow.



a) Find the value of y.

soln.

$$3y + 15 = 30$$

$$3y + 15 - 15 = 30 - 15$$

$$3y = 15$$

$$\frac{3y}{3} = \frac{15}{3}$$

$$y = 5$$

b) Find the value of K.

soln.

$$4k + 15 = k + 60$$

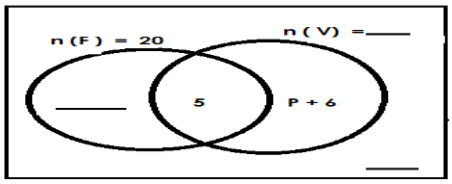
$$4k + 15 = 15 = k + 60 - 15$$

4k = k + 45
4k - k = k – k + 45
3k = 45
3k = 45
3 3
K = 15

Activity:

1. **In a class, 20 play football (F) P + 6 play volley ball (V) only and 5 pupils play both while 2 pupils play neither of the two.**

a) Complete the Venn diagram.

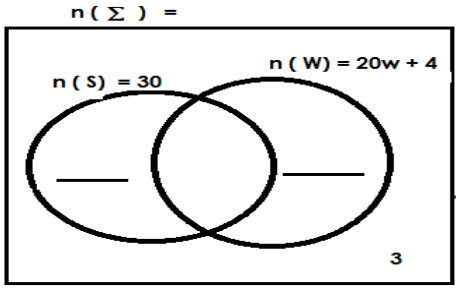


b) If 28 pupils play volley ball, find the value of P.

c) How many pupils play only one game?

2. In a group, 30 people take soda (S), 2W – 4 people take water (W) and 4 people take both drinks while 3 people do not take any of the mentioned.

a) Complete the Venn diagram.

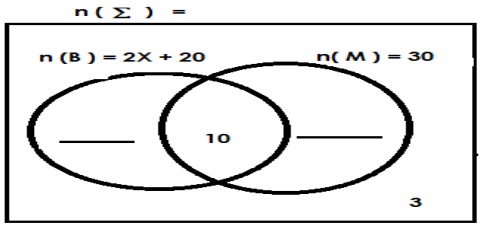


b) Given that 40 people take only one drink. Find the number of people who take water in the group.

c) Find the probability of picking a person who take soda only from the group at a random.

3. In a village, 2x + 10 farmers grow beans only (B), 30 farmers grow maize (M) , 10 farmers grow both crops and X+16 farmers do not grow any.

a) Complete the Venn diagram.



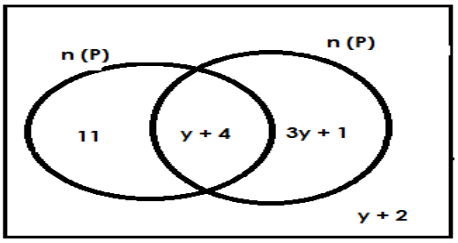
b) Given that the number of farmers who grow beans only is the same as those who do not grow any of the mentioned crops. Find the value of X,

c) How many farmers grow only one crop?

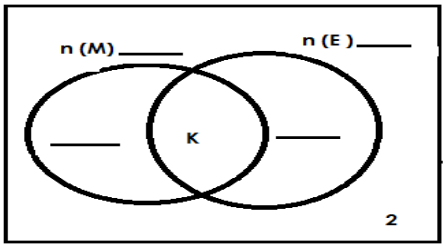
d) Find the number of farmers in the whole village.

4. In the Venn diagram, the number of people who like posho only is equal

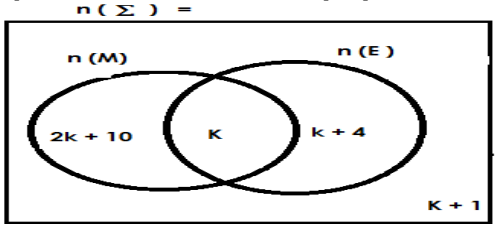
to the number of people who like rice only.



a) Find the value of y.
b) Find the probability of picking a pupil who does not like posho from the class at a random.
5. In a class, 13 pupils like English (E) only, K like both math (M) and English K + 15 like math but not English while 2 like neither of the two subjects.
a) Complete the Venn diagram



b) If 33 pupils like only one subject, find the value of K.
c) Find the probability of picking a pupil who like math from the class.
6. The Venn diagram below shows pupils who like math (M) and English (E). use it to answer questions if 16 more pupils like math than English.



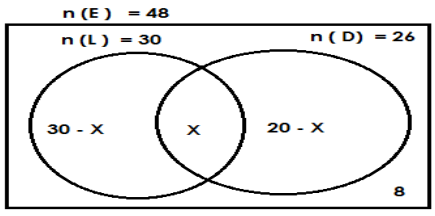
a) Find the value of K.
b) How many pupils who do not like English (E)?

SOLVING PROBLEMS USING A VENN DIAGRAM GIVEN ONE OF THE SETS AS THE UNIVERSAL

<u>Example I</u>

In a class of 48 pupils, all speak English 9E), 30 pupils speak Luganda (L) 20 pupils speak drinks (D) and X pupils speak all the three languages while 8 pupils speak English only.

a) complete the Venn diagram.



b) How many pupils speak all the three languages?

30 – X + X + 20 – X + 8 = 48

30 + 20 + 8 = X = 48

58 – X = 48

58 – 58 – X = 48 – 58

-x = -10

$\frac{-x}{-1} = \frac{-10}{-1}$

X = 10

c) How many pupils speak only two?

Solution

Only two = 30 – x + 20 – X

= 30 – 10 + 20 – 10

= 20 + 16

= 30 pupils

Find the probability of picking a pupil who speak only one language.

Solution

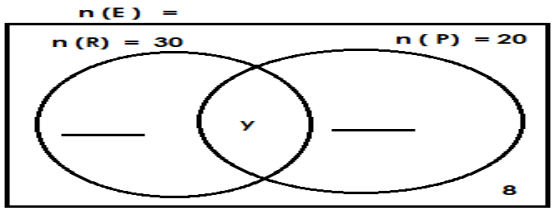
Prob = $\frac{n(E)}{n(P)}$

= $\frac{8}{48}$

Activity

1. In a village of 50 farmers, all grow maize (M), 30 grow beans (B) 26 grow peas (P) and y farmers grow all the three crops while 8 farmers grow maize only.

a) Complete the diagram.

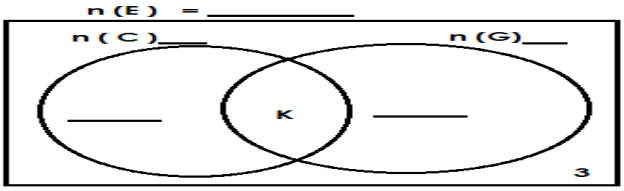


b) Find the value of X.

c) What is the probability of picking a farmer who do not grows only two crops?

2. In a bus of 38 passengers, all bought water (W), 20 bought chicken (c) 24 bought Ganga (G) and K passengers bought water only.

a) Complete the Venn diagram

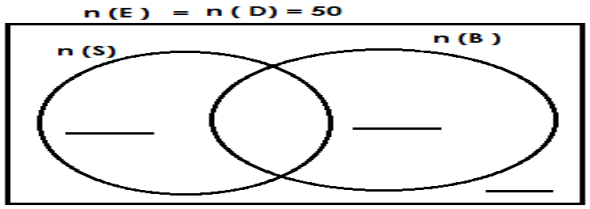


b) Find the value of K

c) Find the probability of picking a passenger who bought chicken and water only.

3. In a market of 50 traders, all sell dresses (D), 36 sell shirts (S), 28 sell bags (B) while M traders sell all the three items and six traders sell dresses only.

a) **Complete the Venn diagram.**



Scenarios

TOPIC 5 – LENGTH MASS AND CAPACITY;

Session 1

- ✦ Areas of all geometric figures
- ✦ Application of area examinable figures
  - ✦ Solving word statements on trapezium and isosceles triangle

Methods

- Guided discovery
- Group discussion
- Think, pair and share.
- explanation

Lesson activities

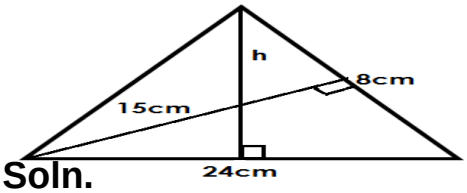
- **adding**
- **multiplying**
- **dividing**
- **subtracting**

Content/facts

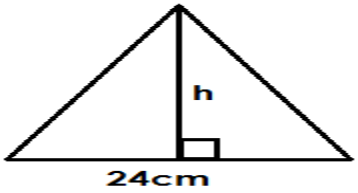
FINDING BASE OR HEIGHT OF A TRIANGLE BY COMPARING AREA

Example I

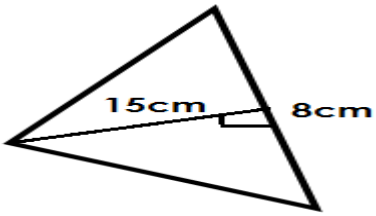
Below is a triangle use it to find A.



Soln.



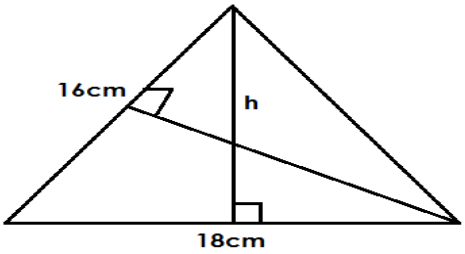
=

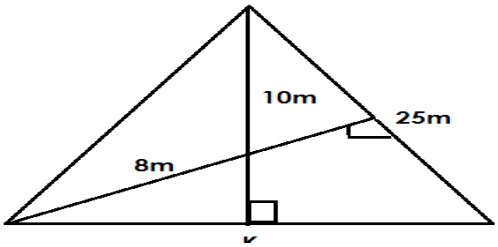


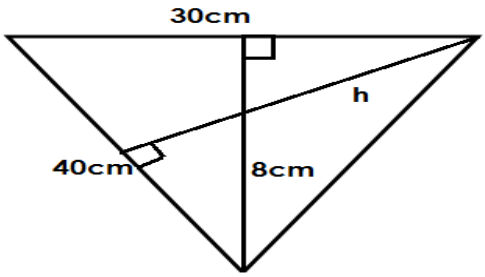
Area	**=**	**Area**
$\frac{1}{2}$ **x b x h**	**=**	$\frac{1}{2}$ **x b x h**
$\frac{1}{2}$ **x 24cm x h**	**=**	$\frac{1}{2}$ **X 8cm x 15cm,**
12cm x h	**=**	$\frac{1}{2}$ **x 8cm x 15cm**
$\frac{\text{12cm x h}}{\text{12cm}}$	**=**	$\frac{\text{4cm x 15}}{\text{12cm}}$
h	**=**	**5cm**

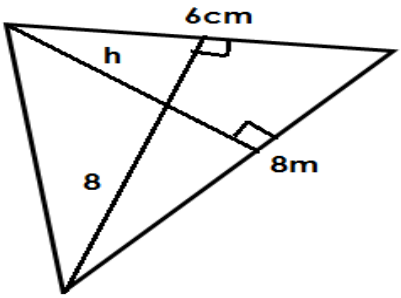
Activity:

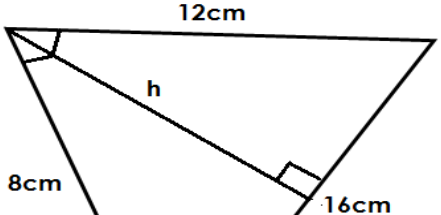
Find the value of unknowns in the following figures.

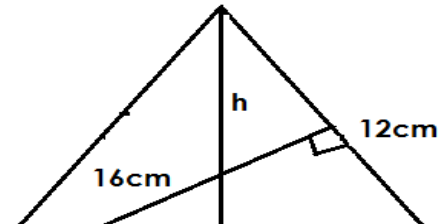








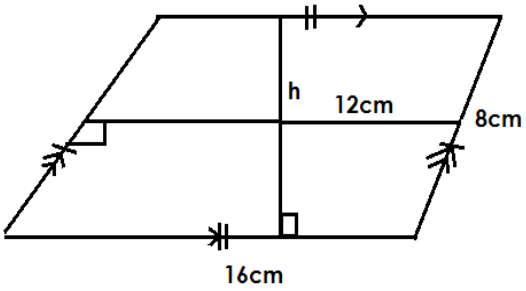




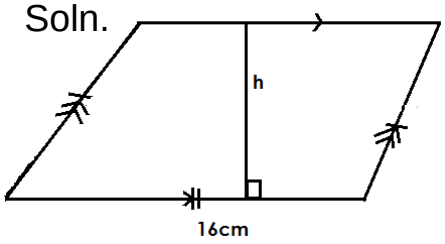
FINDING HEIGHT OR BASE OF A PARALLELOGRAM BY COMPARING AREA

Example I

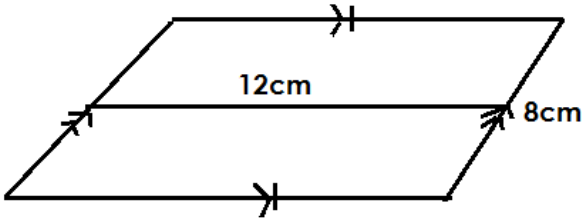
Below is a parallelogram. Use it to find the value of h.



Soln.



=



Area	=	Area
B x H	=	B x H
16cm x h	=	8cm x 12 cm

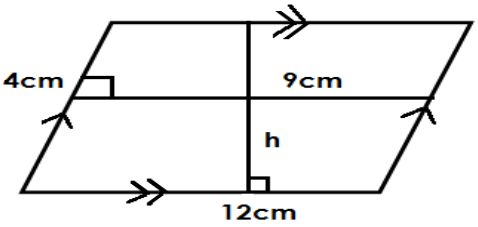
$\frac{16\text{cm x h}}{16\text{cm}} = \frac{8\text{cm x 12cm}}{16\text{cm}}$

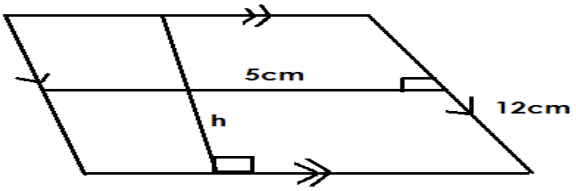
h = 6cm

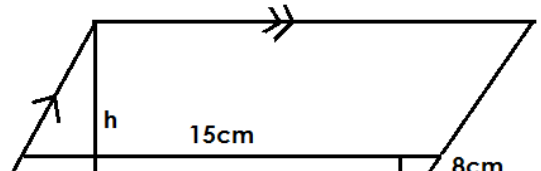
Find the value of unknowns in the following diagrams.

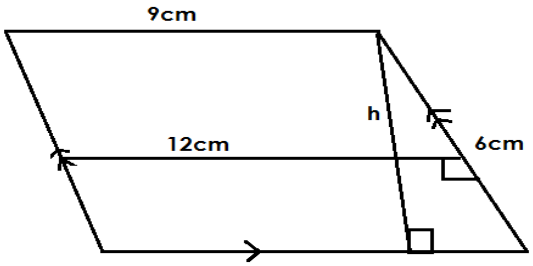
Activity:

1





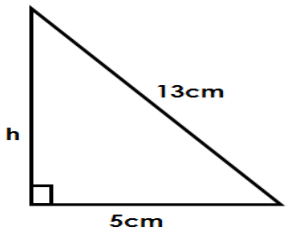




PHYTHAGORAS THEOREM

Example I

Below is a triangle. Use it to answer questions



Find the value of h.

Soln.

$a^2 + b^2 = c^2$

$5^2 + h^2 = 132$

$5 \times 5 + h^2 = 13 \times 13$

$25 \times h^2 = 169$

$25 - 25 + h^2 = 169 - 25$

$h^2 = 144$

$\sqrt{h^2} = \sqrt{144}$

$h = 12cm$

Calculate its area

Soln.

Area = $\frac{1}{2}$ x b x h

= $\frac{1}{2}$ x 5cm x 12cm

= $30cm^2$

Find its perimeter

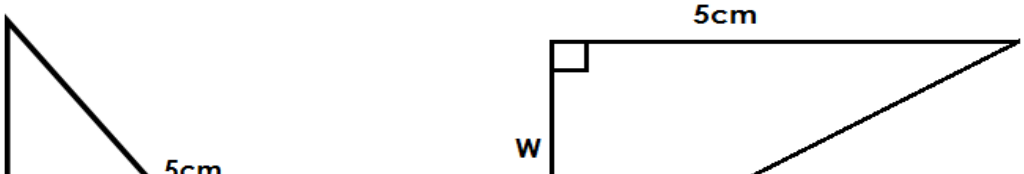
Soln.

P = s + s + s

= 5cm + 12cm + 13cm

= 30cm

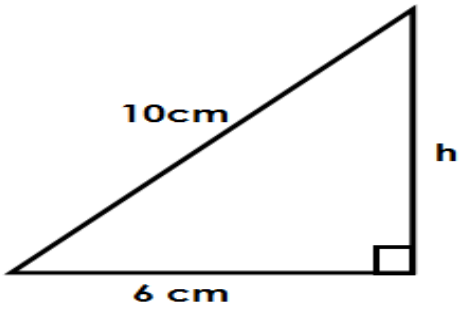
The diagram below, find the unknown side, area and perimeter.

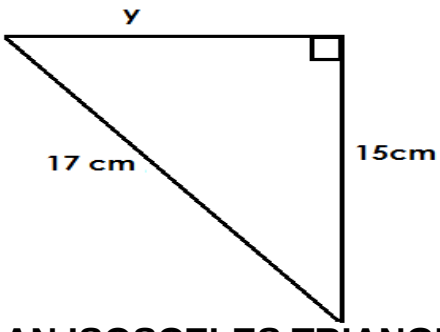


1.

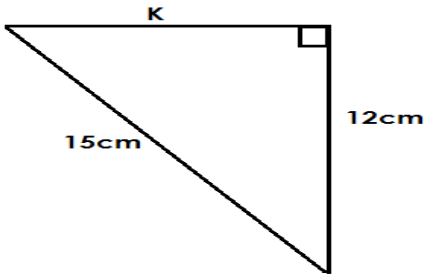
2.

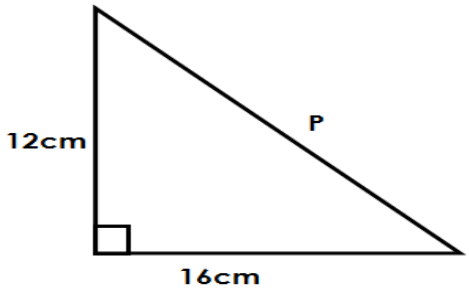
3.





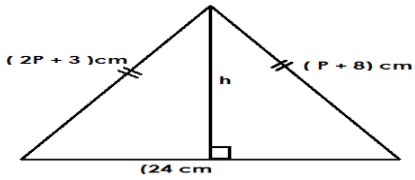
4.





AN ISOSCELES TRIANGLE AND PHYTHAGORAS THEOREM

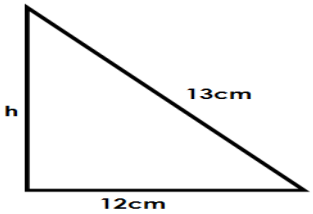
Below is a triangle, use it to answer questions.



Find the value of P.

Side = side
2P + 3 = P + 8
2p + 3 – 3 = P + 8 – 3
2P = P + 5
2P – P = P – P + 5
P = 5

Calculate the area of the figure.



side
(P + 8) cm
(5 + 8) cm
13cm

$a^2 + b^2 = c^2$

$12^2 + h^2 = 1c^2$

$(12 \times 12) + h^2 = 13 \times 13$

$144 - 144 + h^2 = 169 - 144$

$h^2 = 25$

$\sqrt{h^2} = 252$

h =5 cm

Area = $\frac{1}{2}$ x b x h

= $\frac{1}{2}$ x 24cm x 5cm

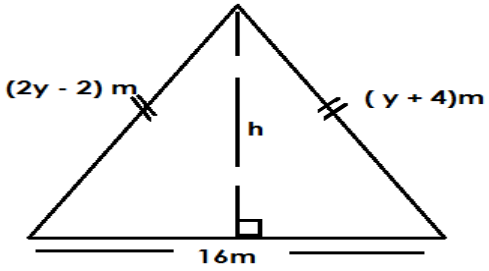
= 12cm x 5cm

= **$60cm^2$**

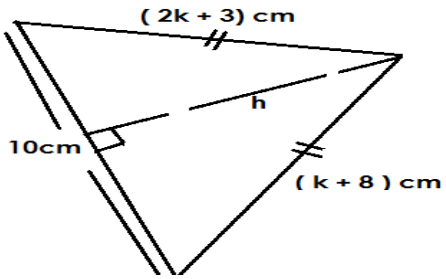
Activity

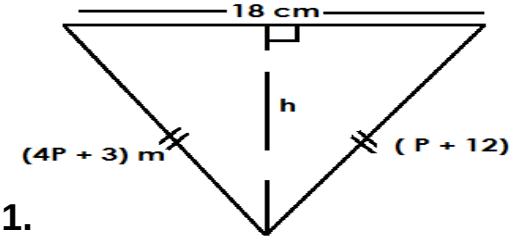
In the diagrams below, Find the value of unknown area and perimeter.

1.



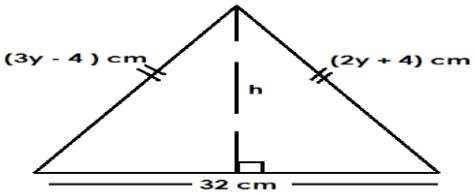
2

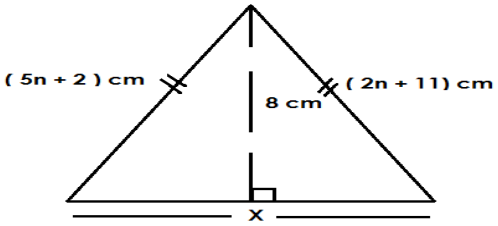


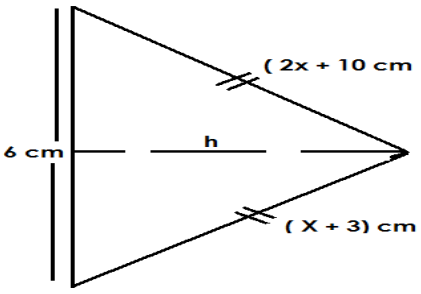


1.

4





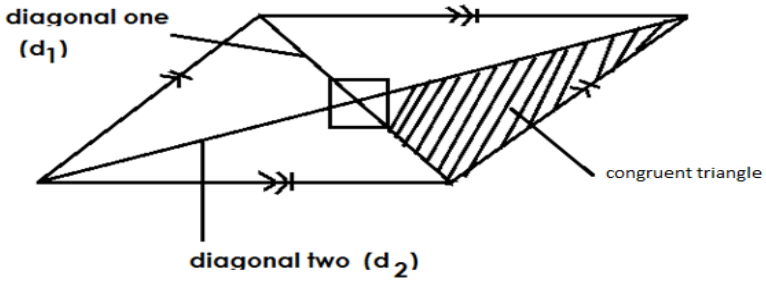


RHOMBUS

Note:

- **A rhombus is a slanted square i.e. all its sides are equal.**

- **A rhombus has its opposite side parallel to each other.**
- **A rhombus has two diagonals that bisect each other at an angle of 90°.**
- **A rhombus has two line of folding symmetry which are its diagonals.**
- **The opposite angles of a rhombus are equal.**



FINDING AREA AND PERIMETER OF A RHOMBUS

Area = $\frac{1}{2}$ x diagonal one x diagonal two

$= {}^{1}/_{2} \times d_1 \times d_2$

Perimeter = side +side +side +side +side

= s +s +s + s

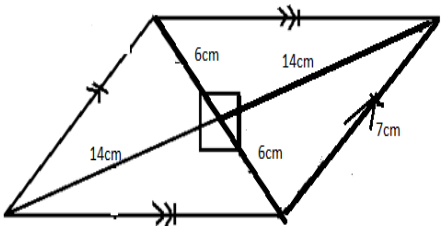
OR

Perimeter = 4 x S

Example 1

Below is a rhombus use it to answer questions.

a) Find its area.



$d_1 = 6cm + 6cm$

$= 12cm$

$d_2 = 14cm + 14cm$

$= 28cm$

Area = $\frac{1}{2}$ x 12cm x 28cm

= 6cm x 28cm

= $168cm^2$

b) Calculate its perimeter.

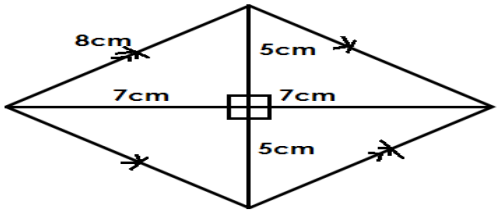
Perimeter = S + S + S + S

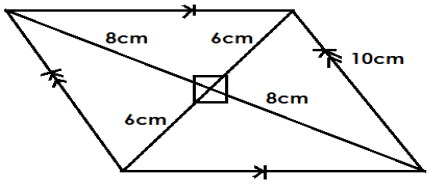
= 7cm + 7cm + 7cm + 7cm

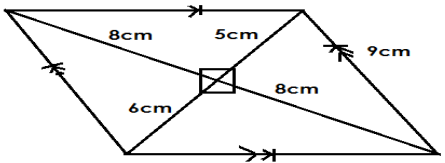
= 28cm

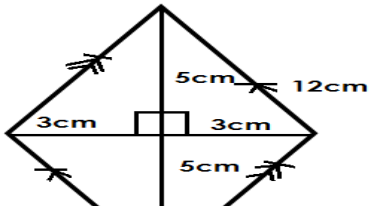
Activity:

Find the area and perimeter of the following.



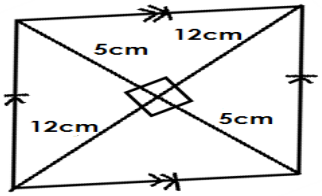






A RHOMBUS AND PYTHAGORAS THEOREM

The figure below is a rhombus



d_1 = 5cm + 5cm
= 10cm

d_2 = 12cm + 12cm
= 24cm

a) Find its area.

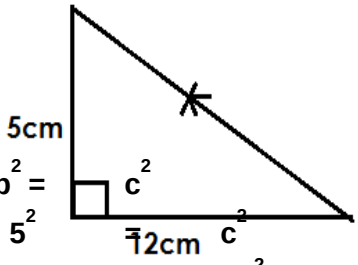
Soln.

A = $\frac{1}{2}$ x d_1 x d_2

= $\frac{1}{2}$ x 10cm x 24cm

= 120cm^2

b) Calculate its perimeter



$a^2 + b^2 = c^2$

$12^2 + 5^2 = c^2$

(12 x 12) + (5 x 5) = c^2

144 + 25 = c^2

169 = C^2

$\sqrt{169} = \sqrt{c^2}$

13 = c

C = 13

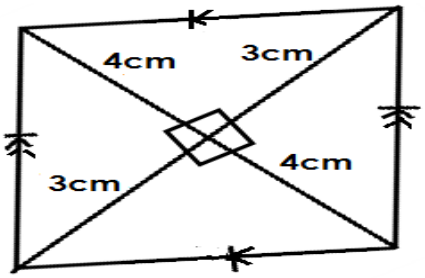
P = S + S + S + S

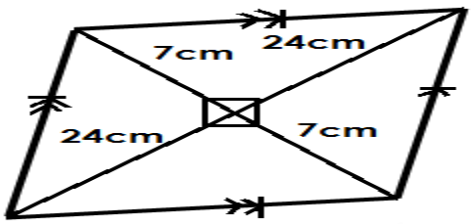
= 13cm + 13cm + 13cm + 13cm

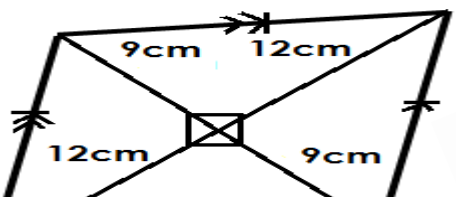
= 52cm

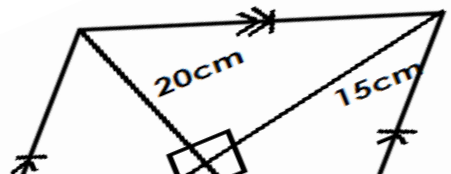
Activity

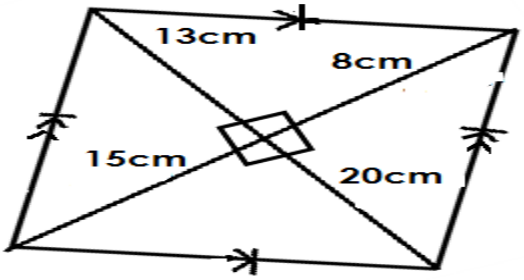
Find area and perimeter of the following figures.

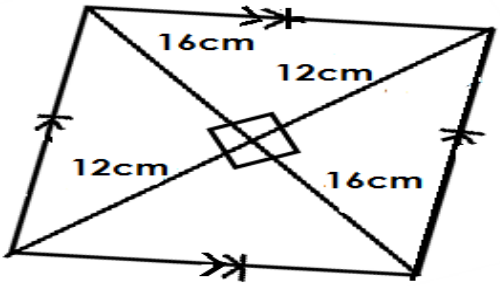






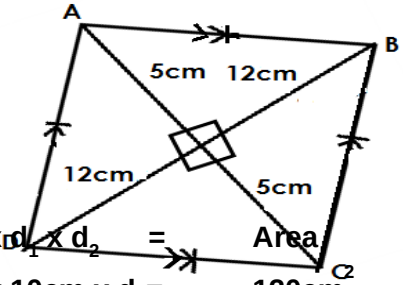






MORE ABOUT ARHOMBUS

Area of the figure below is 12cm^2. If AC = $\overline{10}$cm. find line BD



$\frac{1}{2}$ x d_1 x d_2 = Area

$\frac{1}{2}$ x 10cm x d_2 = 120cm

5cm x d_2 = 120cm^2

$\frac{5cm \ x \ d2}{5cm}$ = $\frac{120cm \ x \ cm}{5cm}$

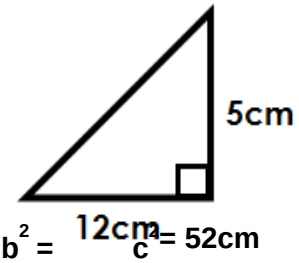
d_2 = 24cm

∴BD = 24cm

D = $\frac{10}{2}$

= 5cm

b) Calculate its perimeter.



$a^2 + b^2$ = c = 52cm

$12^2 + 5^2$ = c^2

(12 x 12) + (5 x 5) = c^2

144 + 25 = c^2

169 = C^2

169 = C^2

$\sqrt{169}$ = $\sqrt{c^2}$

13 = c

C = 13cm

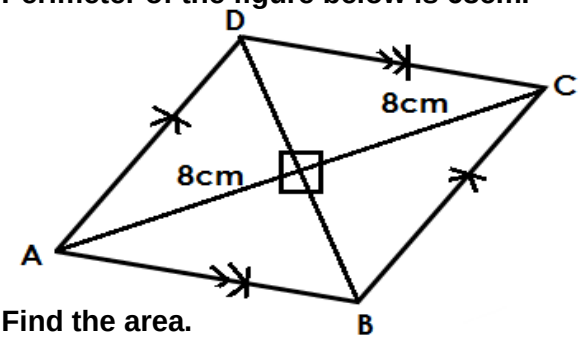
P = S + S + S + S

=13CM + 13CM +13CM + 13CM

= 52cm

Example II

Perimeter of the figure below is 68cm.

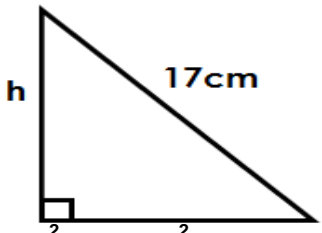


Find the area.

S + S + S + S = Perimeter

$\frac{45}{4}$ **=** $\frac{68}{4}$**cm**

S =17cm



$a^2 + b^2 = 8\ cm^2$ c

$8^2 + h^2 = c^2$

$(8x\ 8) + h^2 = 17 \times 17$

$64 + h^2 =$ 289

$64 – 64 + h^2$ = 289 – 64

h^2 = 225

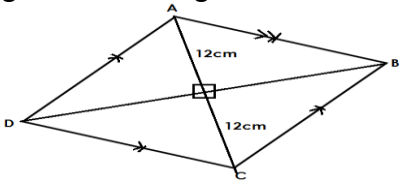
$\sqrt{h^2} = \sqrt{225}$

h = 15cm

Area = $\frac{1}{2}$ **d_1 x d_2**

= $60cm^2$

Diagonal of the figure below is 3:4 if AC = 24cm. Calculate its area.



Total ratio = 3 + 4

= 7

Let the total length of the two diagonals be K

$\frac{3}{7}$ x k = 24cm

$\frac{3k}{7}$ = 24cm

$\frac{3k}{7}$ x 7 = 24cm x 7

$\frac{3k}{3}$ = $\frac{24cm}{3}$ x 7

K = 56cm

BD = $\frac{4}{7}$ x 56cm

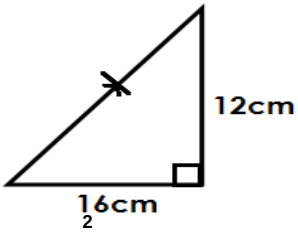
= 32cm

Area = $\frac{1}{2}$ x d_1 x d_2

= $\frac{1}{2}$ x 24cm x 32cm

= $384cm^2$

b) Find its perimeter



$a^2 + b^2 = c^2$

$16^2 + 12^2 = c^2$

$(16 \times 16) + (12 \times 12) = c^2$

$256 + 144 = c^2$

$400 = C^2$

$\sqrt{400} = \sqrt{c^2}$

20 = c

C = 20cm

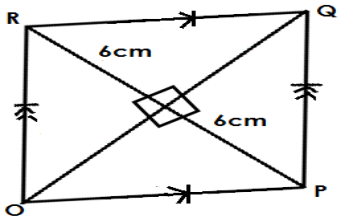
P= S + S + S + S

= 20 cm + 20cm + 20cm + 20cm

= 80cm

Activity:

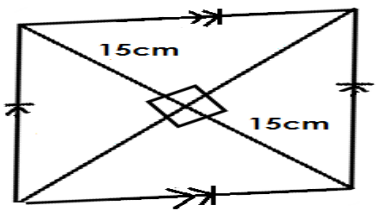
1. The area of the figure below is $96cm^2$



a) If RP = 12cm, find diagonal $\overrightarrow{OC}$

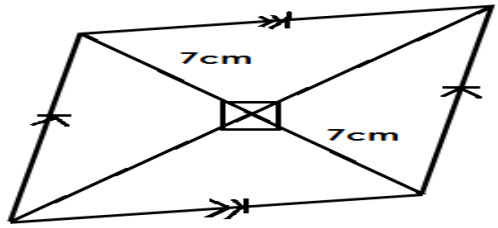
b) Calculate the perimeter of the figure above.

2. The area of the figure below is $240cm^2$.



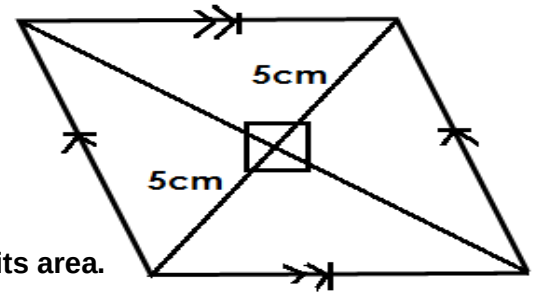
Find the perimeter of the figure above.

3. The perimeter of the figure below is 100cm.



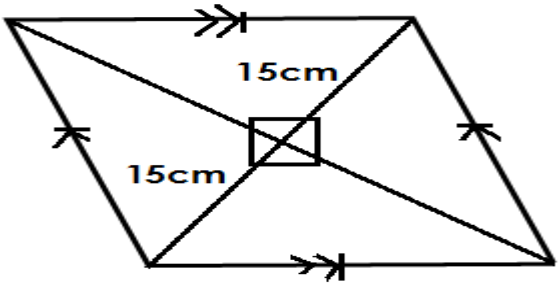
Find the area of the figure above.

2. The perimeter of the figure below is 52cm.



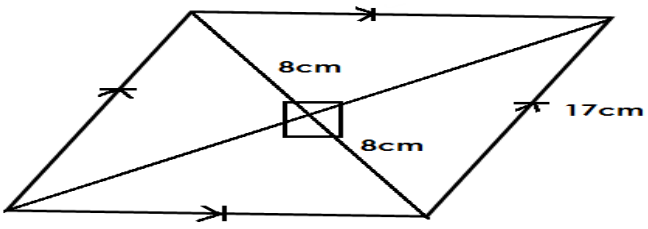
Find its area.

5. The diagonal of the figure below are in a ratio of 3:4. Calculate its area.



b) Find the perimeter of the above figure.

6. The perimeter of the figure below is 68cm.

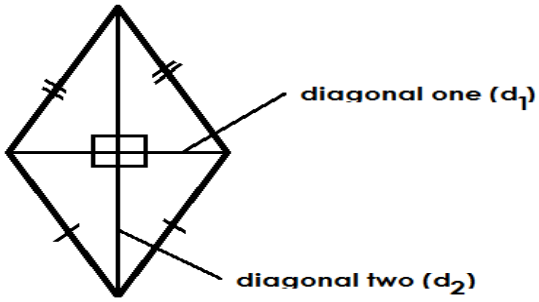


Calculate the area of the above figure

KITE

- **A kite is a 4-sided polygon.**
- **It is made up of 2 different isosceles triangles.**
- **It has two diagonals.**
- **The longer diagonal bisects the shorter diagonal at an angle of 90^0**
- **It has one line of folding symmetry (longer diagonal)**
- **Its adjacent sides are equal.**
- **Diagonals of a kite are not equal.**

Illustration



Area of a kite $= \frac{1}{2} \times d_1 \times d_2$

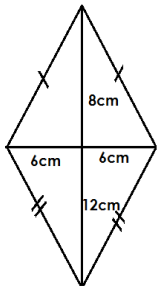
or

Area $= \frac{1}{2} \times bh \times 2$.

Perimeter of a kite = S + S + S + S

Examples

1. Study the figure below and answer the questions that follow.



a) Find the area of the figure above,

soln.

d_1 = 6cm + 6cm

= 12

d_2 = 12cm + 8cm

= 20cm

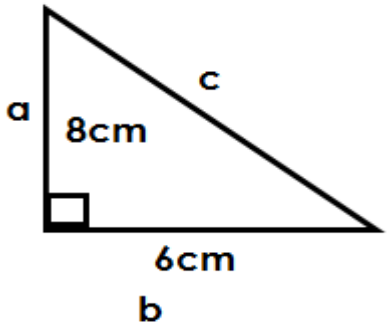
Area = $\frac{1}{2}$ x 12cm x 12cm

= 6cm x 20

= $120cm^2$

b).Find the perimeter of the figure.

Soln.



$a^2 + b^2 = c^2$

$= c^2$

$= c^2$

$= \sqrt{c^2}$

c

∴ Perimeter = S + S + S + S

= 10cm + 10cm + 20cm + 20cm

= 60cm

2. Find the area of a kite whose diagonals are 16cm and 14cm.

Soln

Area = $\frac{1}{2}$ x d_1 x d_2

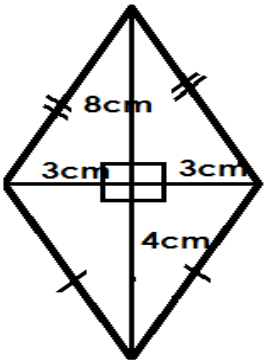
= $\frac{1}{2}$ x 16cm x 14cm

= 8cm x 14cm

= 112cm^2

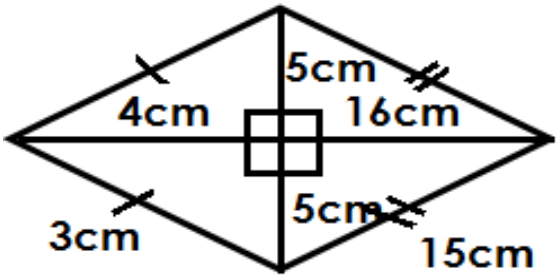
Activity:

1. Given the kite below:



Find the area of the figure above.

2. Study the figure below and answer the questions that follow.

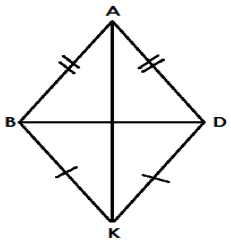


a) **Find the area of the figure above.**

b) What is the perimeter of the above figure.

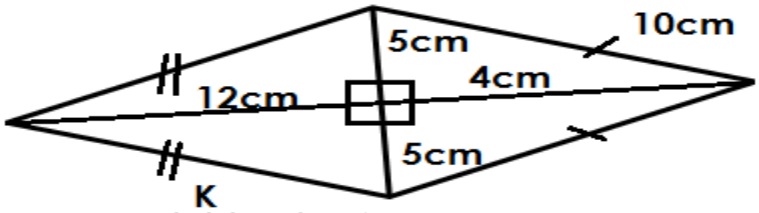
3.Find the area of a kite whose diagonals are 10cm and 6cm.

4.In the figure below, AK = 30cm and BD = 6cm



Find the area of the above figure.

5.Study the figure below and answer questions that follow.

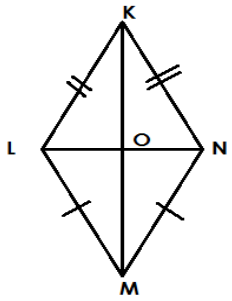


a) Find the value of K.

b) Find the area of the above figure

c) Find the perimeter of the above figure.

6.Find the area of a kite whose diagonals are 18cm and 22cm.
7.The diagonals of the kite below measure 14cm and 22cm, KO = 8cm



a) Find the length of OM and LO.

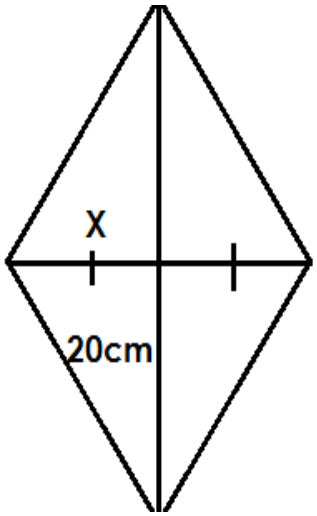
b) Find the area of LMN

c) Find the area of the kite

FINDING THE UNKNOWN SIDE AND AREA OF KITES

Examples

1. The area of the kite below is 160cm2. One of the diagonals is 20cm. find the length of the second diagonal.



$\frac{1}{2}$ **x d1 x d2 = Area**

$\frac{1}{2}$ **x 20 x d = 160cm**

$\frac{10}{10}$ **d^2 =** $\frac{160cm}{10}$

d^2 = 16cm

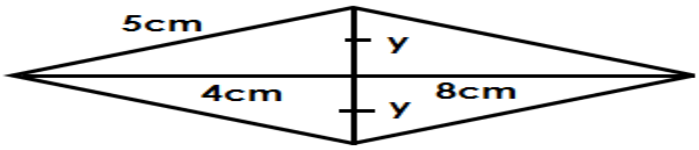
b) Find the value of X.

X = $\frac{d}{2}$

= $\frac{16}{2}$**cm**

= 8cm

2. The diagram below is a kite.



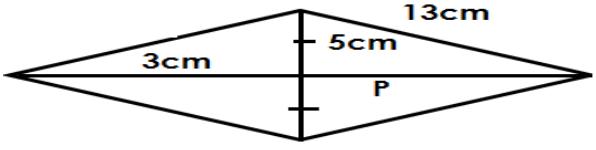
a) Find the value of y.

b) Calculate the area of the kite above.

Activity:

1. A kite has an area of 72cm2. One of its two diagonals is 8cm. find the length of the second diagonals.

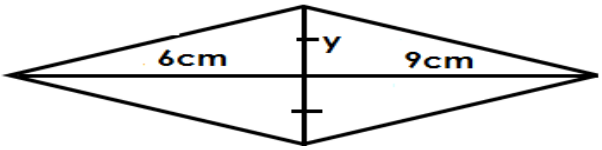
2. Study the kite below and answer questions below.



a) Find the value of P.
b) Find the length of the longer diagonal.
c) Calculate the area of the kite.

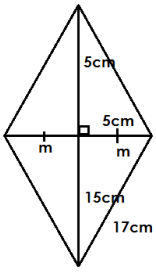
3. The area of a kite is 120cm^2. If one of its diagonals is 8cm, find the length of the second diagonal.

1. The figure below has an area of 45cm^2.



Find the length of the second diagonal.

2. Study the kite below and find the following: -



a) Find the value of m.
b) Find the length of the shorter diagonal.
c) Work out the area of the kite.

Scenarios

Session 2

- ✦ Area of shaded parts involving circles
- ✦ Word statements on perimeter and area
- ✦ Relationship between revolution, distance and circumference
- ✦ Relationship between cylinder and the rectangle (folding and unfolding
- ✦ Finding volume of concrete in a cylinder and Total surface area of solid figures
  - ✦ Finding volume of concrete in a cylinder

Methods

- Guided discovery
- Group discussion
- Think, pair and share.
- explanation

Lesson activities

- **adding**
- **multiplying**
- **dividing**
- **subtracting**

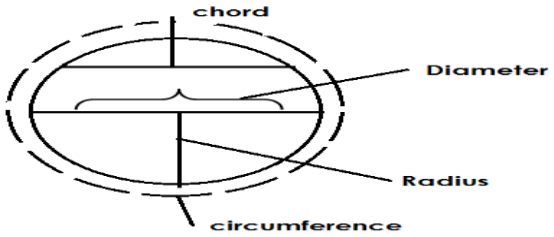
Content/facts

MEASURES

CIRCLES

A circle is a shape made up of a complete curve / arc.

Parts of a circle



CHORD

- **A chord is a line running from one end to another end of a circle.**

DIAMETER

- **This is a line of a circle running from one end to another end through the Centre.**
- **Diameter is the longest chord of the circle.**
- **Diameter is twice the radius i.e.**
- **D = r + r**

RADIUS

- **Radius is a line of a circle running from the centre to the circumference.**
- **Radius is a half diameter i.e.**

R $=\frac{D}{2}$ or D ÷ 2 if given fractions.

CIRCUMFERENCE

- **Circumference is the total distance round a circle or a circular object.**

Circumference = πD

Where;

π = Pi

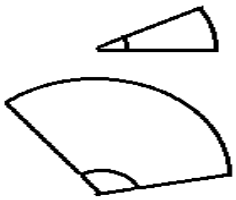
D = Diameter

Pi (π)

- **Pi is the quotient of circumference and diameter of a circle.**
- **Pi is taken as $\frac{22}{7}$ or 3 $\frac{1}{7}$ or 3.14**
- **We use pi as $\frac{22}{7}$ or 3 $\frac{1}{7}$ if diameter, radius or height of a circular object is a multiple of 7.**
- **We use pi as 3.14 if diameter or radius given is not a multiple of 7.**

FORMULAS TO BE USED

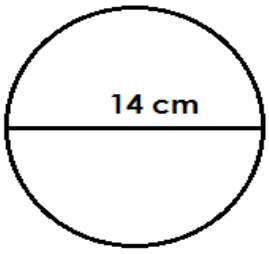
SHAPE	CIRCUMFERENCE	PERIMETER	AREA
CIRCLE	πD	_____	πr^2

SEMI CIRCLE 	**½πD**	**½ πD +D**	**½πr^2**
QUADRANT 	**¼ πD**	**¼π + D**	**¼ πr^2**
OTHER SECTORS 	$\frac{A}{360}\pi$**D**	$\frac{A}{360}\pi$**D +D**	$\frac{A}{360}\pi r^2$

FINDING CIRCUMFERENCE AND AREA OF A CIRCLE

Example I

Below is a circle.



a)Find its circumference

soln.

cir = πD

=$\frac{22}{7}$ X 14cm

= 22 x 2 cm

= 44 cm

r = $\frac{D}{2}$

= $\frac{14}{2}$ cm

= 7 cm

b). calculate its area.

Solution.

Area = πr^2

Area=$\frac{22}{7}$ x 7cm x 7 cm

=154 cm^2

Example II

a) Find circumference of a circle whose diameter is 40 m.

Soln.

Cir = πD

= 3.14 X 40m

= $\frac{314}{100}$X 40m

= $\frac{1256m}{10}$

= 125.6m

b) Calculate its area.

Soln.

r = $\frac{D}{2}$

= $\frac{40\ m}{2}$

= 20m

Area = πr^2

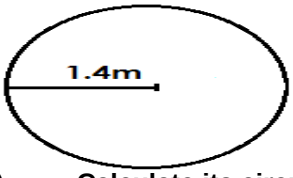
= 3.14 x 20m x 20m

= $\frac{314}{100}$ x 20m x 20m

= 1256m^2

Example III

Below is a circle, use it to answer questions.



D = r + r
= 1.4m
+ 1.4m
= 2.8m

a) Calculate its circumference.

Soln.

Cir = πD

= $\frac{22}{7}$ x 2.8m

= $\frac{22}{7}$ x $\frac{28m}{10}$

= $\frac{88m}{10}$

= 8.8m

b) Find its area.

$A = \pi r^2$

= $\frac{22}{7}$ x 1.4m x 1.4m

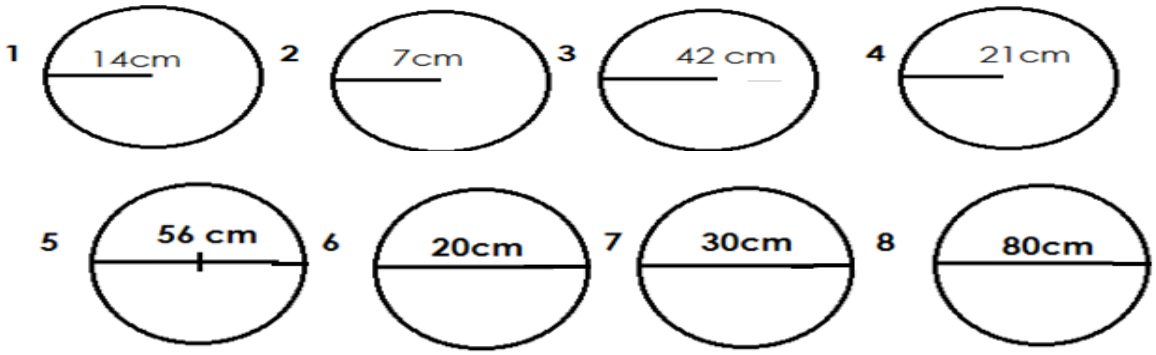
= $\frac{22}{7}$ x $\frac{14m}{10}$ x $\frac{14m}{10}$

= $\frac{6116m^2}{100}$

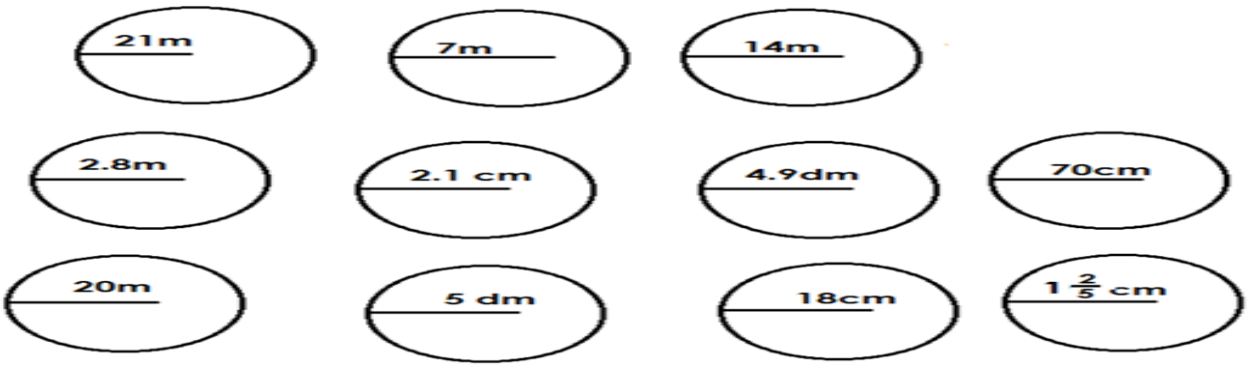
= 6.16m²

Activity:

1. Find circumference and area of the following.



2 Calculate circumference and area of the following circles.



Fll

1.

Soln.

πD = circumference

$\frac{22D}{7}$ = 176m

$\frac{22D}{7}$ x 7 = 176 m x 7

$\frac{22D}{22}$ = $\frac{116m \times 7}{22}$

D = 56M.

R = $\frac{D}{2}$

= $\frac{56m}{2}$

= 28m.

iii) Calculate its area.

Soln.

$A = \pi r^2$

$= \frac{22}{7}$ x 28m x 28m

$=2464m^2$

Example II

Circumference of a circle is 8.8cm, calculate its area

Soln.

$\pi D =$ circumference

$\frac{22D}{7} =$ 8.8cm

$\frac{22D}{7}$ x 7 = 8.8cm x 7

22D = 8.8cm x 7

22D = 61.6CM

22D x 10 = $\frac{616m}{10}$ x 10

22D x 10 = 616m

$\frac{22D \times 10}{22 \times 10} = \frac{616m}{22 \times 10}$

D = $\frac{28m}{10}$

= 2.8m

R = D ÷ 2

= 2.8 ÷ 2

= $\frac{28}{10} \div \frac{2}{1}$

= $\frac{28}{10}$ X $\frac{1}{2}$

= $\frac{14}{10}$

= 1.4m

$A = \pi r^2$

= $\frac{22}{7}$ x 1.4m x 1.4m

= $\frac{22}{7}$ x $\frac{14}{10}$m x $\frac{14}{10}$m

= $\frac{616m^2}{100}$

= $6.16m^2$

Find the circumference of a circle whose area is $154cm^2$.

π = Area

$\frac{22}{7}r^2 = 154cm^2$

$\frac{22}{7}r^2 = 154cm^2$ x 7

$22r^2 = 154cm^2$ x 7

$\frac{22r^2}{22}$ = $\frac{154}{22}$cm^2 x 7

r^2 = 49cm^2

$\sqrt{r^2}$ = $\sqrt{49}$cm^2

r = 7cm

D = r + r

= 7cm + 7cm

= 14cm

Circumference = πD

= $\frac{22}{7}$ x 14cm

= 44cm

Area of a circle is 38½ m^2. Find its diameter.

πr^2 = Area

$\frac{22}{7}r^2$ = 38 ½ m^2

$\frac{22}{7}r^2$ x 7 = 38 ½ m^2 x 7

$22r^2$ = $\frac{77}{2}m^2$ x 7

$22r^2$ = $\frac{539}{2}$m^2 x 2

22r^2 x 2 = 539m^2

$\frac{22r2}{22\ x\ 2}$ = $\frac{539}{22\ x\ 2}$m

r^2 = $\frac{539}{44}$

r^2 = $\frac{49}{4}$

$\sqrt{r^2}$ = $\sqrt{\frac{49}{4}}$

r = $\sqrt{\frac{49}{4}}$

r = $\frac{7}{2}$

r = 3 ½ m

D = 2r

= 2 x $\frac{7}{2}$ m

= 7m

Activity:

1. Circumference of a circle is 8.8cm. calculate its area.
2. Find diameter of a circle whose circumference is 2.2cm.
3. Find area of a circle whose circumference is 880cm.

4. Circumference of a circle is 66m. find its area.

5. Find the area of a circle whose circumference is 132cm.

6. A goat was tied on a stump such that the furthest distance round its grazing place is 44m.

a) Find length of the rope.

b) Calculate the area of the grazing ground.

7. Find the area of the garden whose distance round is 2.2m.

8. Find circumference of a circle whose area is 154cm^2.

9. The area of a circular field is 616m^2. Calculate its circumference.

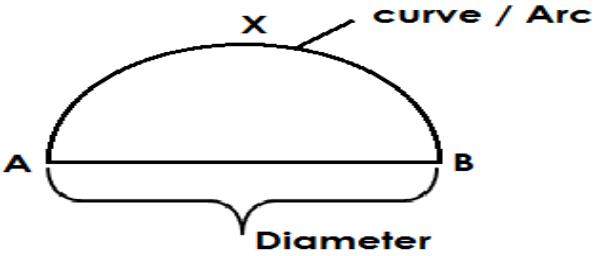
10. Find radius of a circle whose area is 2464cm^2.

11. Area of a circle is $9\frac{5}{8}$ cm^2. Find its diameter.

12. Area of a circle is 17,850m^2, find its radius take it as 3.14

SEMI CIRCLE

A semi-circle is a half of a circle.



Area of a semicircle= $\frac{1}{2}\pi r^2$

Circumference / length of the curve = $\frac{1}{2}\pi D$

Perimeter = $\frac{1}{2}\pi D + D$

FINDING AREA PERIMETER AND CIRCUMFERENCE OF A SEMI CIRCLE

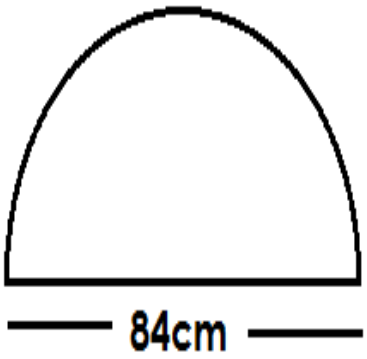
Examples

Below is a semi circle use it to answer questions.

r = $\frac{D}{2}$

= $\frac{84}{2}$ =

42cm



Find its area

Soln.

$A = \frac{1}{2}\pi r^2$

= $\frac{1}{2}$ x $\frac{22}{7}$ x 42cm x 42cm

= 2772cm^2

b) Find length of the curved line.

Soln.

Length = ½ πD

= $\frac{1}{2}$ X $\frac{22}{7}$ X 84cm

= 132 cm

c)Calculate its perimeter

soln

perimeter = ½ πD + D

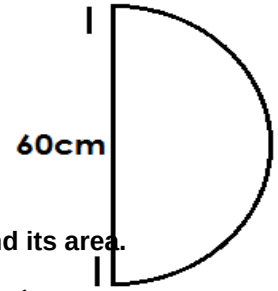
= ($\frac{1}{2}$ x $\frac{22}{7}$ x 84cm) + 84cm

= 131cm + 84cm

= 216cm

<u>Example II</u>

Use the figure below to answer questions.



R = ½

= $\frac{60cm}{2}$

= 30cm

Find its area.

$A = \frac{1}{2}\pi r^2$

= $\frac{1}{2}$ x 3.14 x 30cm x 30cm

= $\frac{1}{2}$ x $\frac{314}{100}$ x 30cm x 30cm

=1413 cm^2

b) Find length of the curved line.

Soln

Length=πD

= $\frac{1}{2}\pi D$

= ½ x 3.14 x 60cm

= ½ x $\frac{314}{100}$ x 60cm

= $\frac{942}{10}$ cm

= 94.2cm

c) calculate its perimeter

soln.

perimeter = ½ πD + D

= (½ x 3.14 x 60cm) + 60cm

= (½ x $\frac{314}{100}$ x 60cm) + 60 cm

= $\frac{942}{10}$ + 60 cm

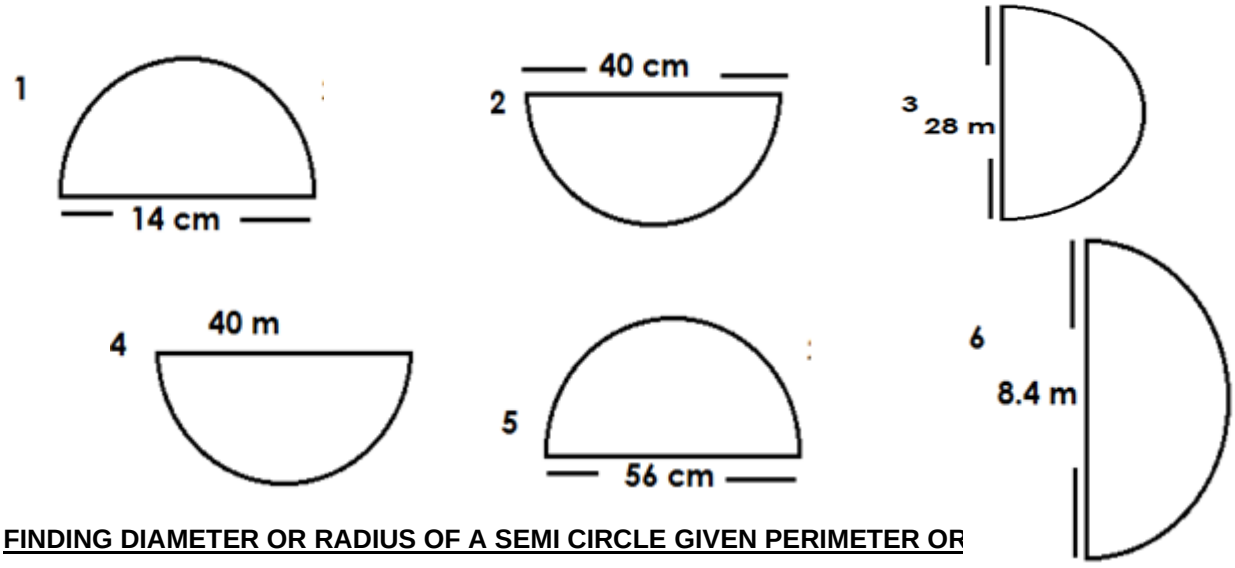
= 94.2 cm + 60 cm

= 154.2 cm

94.2
+ 60.0
154.2

Activity:

Find area, length of the curved line and perimeter of the following semi circles.



FINDING DIAMETER OR RADIUS OF A SEMI CIRCLE GIVEN PERIMETER OR

Example I

Perimeter of a semi-circle is 108m. Find its area.

Soln.

½ $\pi D + D$ =perimeter

½ x $\frac{22}{7}$ D + D = 108 m

$\frac{11}{7}$D + D = 108 m

$\frac{11D}{7}$ x 7 + D x 7 = 108 m x 7

11D + 7D = 108m x 7

18D = 108m x 7

$\frac{18D}{18}$ = $\frac{108m}{18}$ x 7

D = 42m

r = $\frac{D}{2}$

= $\frac{42m}{2}$

= 21m

$A = \pi r^2$

= $\frac{1}{2}$ x $\frac{22}{7}$ x 21m x 21m

= 693m^2

Example II

Area of a semi-circle is 628cm^2. Find its radius. Take π as 3.14.

Soln.

$\frac{1}{2}\pi r^2$ = Area

$\frac{1}{2}$ x 3.14 x r^2 = 62.8cm^2

$\frac{1}{2}$ x $\frac{314}{100}$r^2 = 628cm^2

$\frac{157r^2}{100}$= 628 cm^2

$\frac{157r^2}{100}$ x 100 = 628cm^2 x 100

157r^2 = 628cm^2 x100

$\frac{157r^2}{157}$ = $\frac{628}{157}$ cm^2 x 100

r^2 = 400cm^2

$\sqrt{r^2}$ = $\sqrt{400cm^2}$

r = 20cm

Activity:

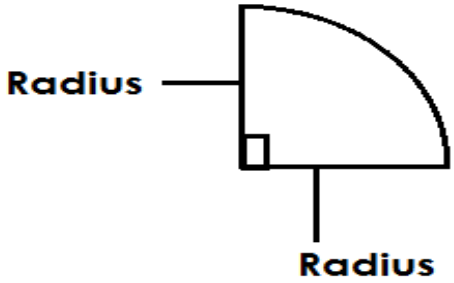
1. Perimeter of a semi-circle is 72cm, find its diameter.

b) Calculate its area.

2. Calculate the area of a semi-circle whose perimeter is 36cm.
3. Find area of a semi-circle whose perimeter is 252m.
4. Area of a semi-circle is 693cm^2. Find its radius.
5. Find perimeter of a semi-circle whose area is 1232 cm^2.
6. Area of a semi-circle is 2772m^2. Calculate its perimeter.
7. Find area of a semi-circle whose perimeter is 108m.
8. Given that perimeter of a semi-circle is 27.72. find its diameter.

A QUADRANT (AQUARTER)

- A quadrant is a quart part of a circle.
- A quadrant is made up of two radii and an arc.



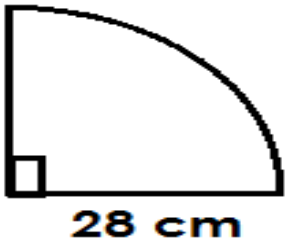
Area = ¼ πr^2

Length of the curve = ¼ πD

Perimeter = ¼ πD + D

Example I

Below is quadrant use it to answer questions.a) Find its area.



$A = \frac{1}{4}\ \pi r^2$

= $\frac{1}{4}$ X $\frac{22}{7}$ X 28 cm x 28cm

= 616cm^2

b). Find length of the curved line.
Soln.

D = r +r
= 28cm + 28cm
= 56 cm

Length = $\frac{1}{4}\pi D$

= $\frac{1}{4}$ **x** $\frac{22}{7}$ **x 56cm**

= 44 cm

c)Calculate its perimeter.

Soln.

P = $\frac{1}{4}\pi D$ **+ D**

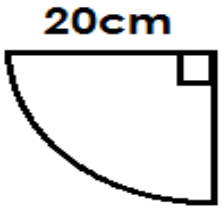
= ($\frac{1}{4}$ **x** $\frac{22}{7}$ **x 56cm) + 56 cm**

= 44cm + 56 cm

= 100cm

Example 2

Below is a quadrant, use it to answer questions.



D= r + r

=20cm + 20cm
= 40cm

a) Find its area.
Soln.

$A = \frac{1}{4}\pi r^2$

= $\frac{1}{4}$ **x 3.14 x 20cm x 20cm**

= $\frac{1}{4}$ **x** $\frac{314}{100}$ **x 20cm x 20cm**

= 314 cm^2

b) Find length of its curved line.

Length = $\frac{1}{4}\pi D$

= $\frac{1}{4}$ **X 3.14 X 40CM**

= $\frac{1}{4}$ **x** $\frac{314}{100}$ **x 40cm**

= 31.4 cm

Calculate its perimeter

Perimeter = $\frac{1}{4}\pi D$ **+ D**

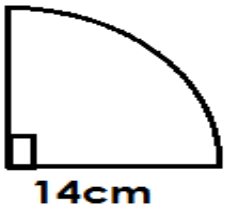
= ($\frac{1}{4}$ **x 3.14 x 40cm) + 40cm**

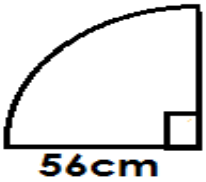
= ($\frac{1}{4}$ **x** $\frac{314}{100}$ **x 40cm) + 40cm**

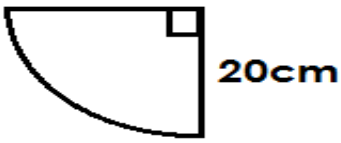
= 31.4 cm + 40cm

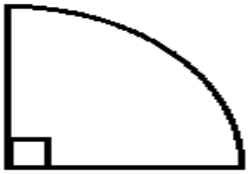
= 71.4 cm

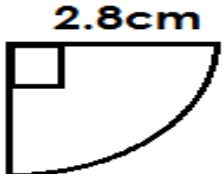
Activity:
Find areas, length of the curved line and perimeter of the following quadrants.

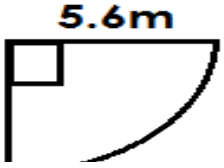




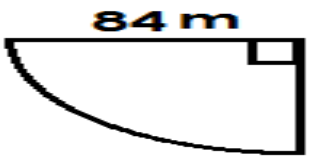


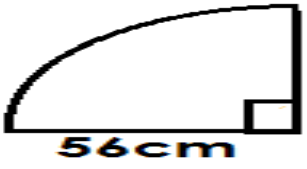












FINDING RADIUS GIVEN PERIMETER OR AREA OF A QUADRANT

Perimeter of a quadrant is 300cm. find its radius.

Soln.

$\frac{1}{4}\pi$D + D = perimeter

($\frac{1}{4}$ x $\frac{22}{7}$ x D) + D = 300 cm

$\frac{11D}{14}$ + D= 300 cm

$\frac{1}{4}$ x $\frac{11D}{14}$ + D x 14= 300 cm x 14

11D + 14D = 300 cm x 14

250 = 300 cm x 14

$\frac{25D}{25}$= $\frac{300\text{ cm x }14}{25}$

D = 168 cm

Radius = $\frac{D}{2}$

= $\frac{168\text{ cm}}{2}$

= 84 cm

b) Calculate its area.

Soln.

Area = $\frac{1}{4}\pi r^2$

=$\frac{1}{4}$ x $\frac{22}{7}$ x 84 cm x 84 cm

= 5544 cm^2

Example II

Area of a quadrant is 7850 cm2, find its radius (Take π = 3.14)

$\frac{1}{4}\pi r^2$ = Area. d = r + r

= 100 cm + 100cm

$\frac{1}{4}$ x 3.14 x r^2 =7850 cm^2 = 200 cm

$\frac{1}{4}$ x $\frac{314}{100}r^2$= 7850 cm^2

$\frac{157r^2}{200}$= 7850 cm^2

$\frac{1}{200}$ x $\frac{157}{200}r^2$ =7850 cm^2 x 200

157 r^2 = 7850 cm^2 x 200

$\frac{157\ r^2}{157\ r^2} = \frac{7850\ cm2\ x\ 200}{157}$

r^2 = 10,00cm2

$\sqrt{r^2}$ = $\sqrt{10000\ cm^2}$

r = 100 cm

Find its perimeter

Soln.

P = $\frac{1}{4}\pi$D + D

= ($\frac{1}{4}$ x 3.14 x 200 cm) + 200 cm

= ($\frac{1}{4}$ x $\frac{314}{100}$ x 200 cm) + 200 cm

= 157 cm + 200 cm

= 357 cm

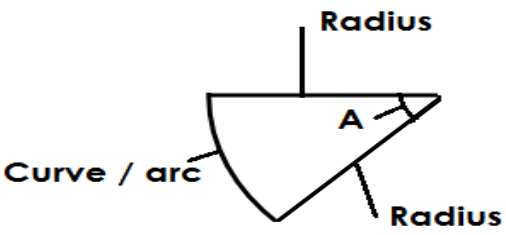
Activity:

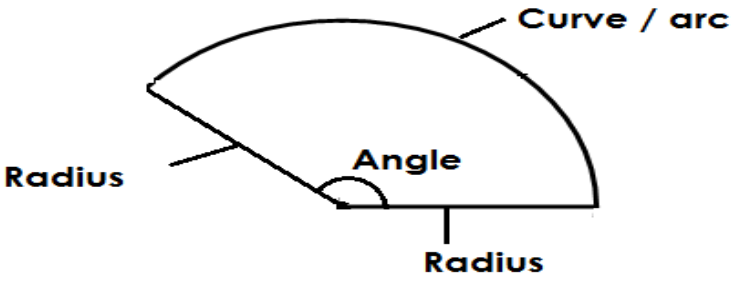
1. Perimeter of a quadrant is 44cm. find its radius.

b) Calculate its area.

2. Find area of a quadrant whose perimeter is 22m.
3. Perimeter of a quadrant is 50 cm, calculate its area.
4. Find radius of quadrant whose area is 2464 cm^2.
5. Area of a quadrant is 1256 cm^2. Find its perimeter. (Take π as 3.14).

OTHER SECTORS





Area = $\frac{A}{360}\pi r^2$

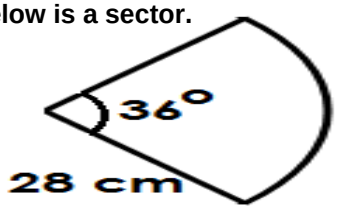
Length of the curve = $\frac{A}{360}\pi D$

Perimeter = $\frac{A}{360}\pi D + D$

Where A is the given angle in a sector.

FINDING AREA PERIMETER AND CIRCUMFERENCE OF OTHER SECTOR

1. Below is a sector.



D = r + r
= 28 cm + 28 cm
= 56 cm

a) Find its area.

Soln.

Area = $\frac{A}{360}\pi r^2$

= $\frac{360}{360}$ x $\frac{22}{7}$ x 28 cm x 28 cm

= $\frac{2464}{10}$

= 246.4 cm^2

b) Calculate its perimeter.

Soln.

P = $\frac{A}{360}\pi D$ + D

= ($\frac{36}{360}$ x $\frac{22}{7}$ x 56 cm) + 56cm

= $\frac{176\ cm}{10}$ + 56 cm

= 17.6 cm + 56 cm

= 73.6 cm

2. Below is a sector.



a) Find the area of the above sector. Use π = 3.14

soln.

A = $\frac{A}{360}\ \pi r^2$

= $\frac{36}{360}$ **x 3.14 x 40cm x 40cm**

= $\frac{5024}{10}$ **cm**

= **502.4 cm**^{2}

b) Find the perimeter of the above sector.

P = $\frac{A}{360}\pi D\ x\ D$

= ($\frac{36}{360}$ **x 3.14 x 80cm) + 80cm**

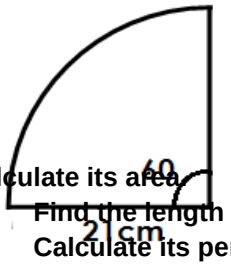
= $\frac{2512}{100}$ **+ 80 cm**

= **25.12 + 80cm**

= **105.12 cm**

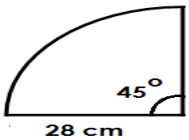
Activity:

1. Below is a sector. Use it to answer questions that follow. (Use π as $\frac{22}{7}$)



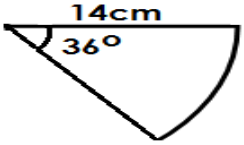
a)Calculate its area.
b) Find the length of its curved line.
c) Calculate its perimeter.

2. Find the area, length of the curved line and perimeter.



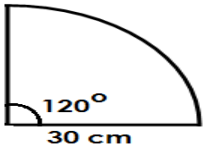
a) Find the area.
b) Find the length of the curved line (circumference)
c) Find the perimeter.

3. Given the sector below.



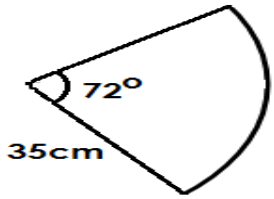
a) Find the area of the above sector.
b) What is the circumference of the sector above?
c) Find the perimeter of the figure above.

4. Study the figure below and answer questions that follow.



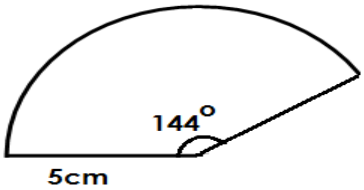
a) What is the area of the figure above?
b) Find the circumference of the figure above.
c) Find its perimeter

5. Study the sector and answer the questions that follow.



a) Find its area.
b) Find the circumference of the figure.
c) What is the perimeter of the above figure?

6. The figure below is a sector. Study it to answer the questions below,



a) Find the area of the figure above.
b) Find the circumference.
c) What is the perimeter of the figure above?

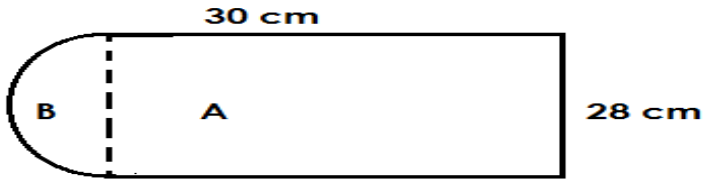
FINDING AREA AND PERIMETER OF COMBINED FIGURES

Note:

While finding perimeter of combined dotted line are not considered since they are taken to be imaginary lines.

Examples

1. Below is a figure. Use it to answer questions below.



a) Find its area.

Soln.

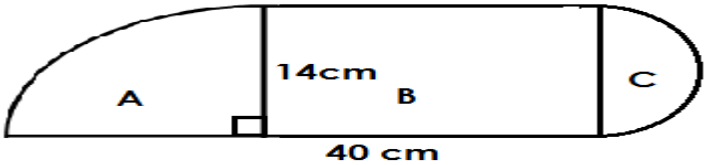
Area of A	Area of B	Total
L X W 30cm x 28cm 840cm^2	$\frac{1}{2}\pi r^2$ $\frac{1}{2}$ x $\frac{22}{7}$ x $\frac{28CM}{2}$ x $\frac{28CM}{2}$ 308 cm^2	840cm^2 + 308cm^2 1148cm^2

b) Find its perimeter

Lines	Curve	Perimeter
28cm + 30cm + 30cm	$\frac{1}{2}\pi d$	88cm + 44cm

88cm	$\frac{1}{2}$ x $\frac{22}{7}$ x 28cm 44 cm	132cm

2. Study the figure below and answer questions that follow.



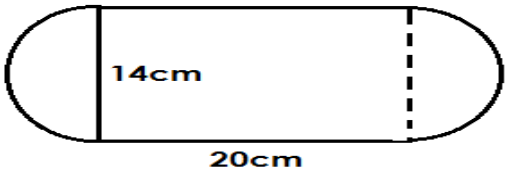
a) Find the area of the figure above.

Soln.

Area of A	Area of B	Area of C	Total
A = $\frac{1}{4}\pi r^2$ = $\frac{1}{4}$ x $\frac{22}{7}$ x 14cm x 14cm	L x W 40cm x 14cm 560 cm^2	A = $\frac{1}{2}\pi r^2$ = $\frac{1}{2}$ x $\frac{22}{7}$ x 7cm x7cm 77cm^2	154cm^2 + 569cm^2 77cm^2 791cm^2

Activity:

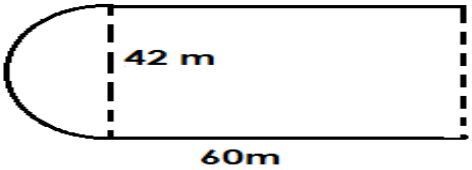
1. Find the area and perimeter of the figure below.



a) Find the area of the figure above.

b) Find the perimeter of the figure above.

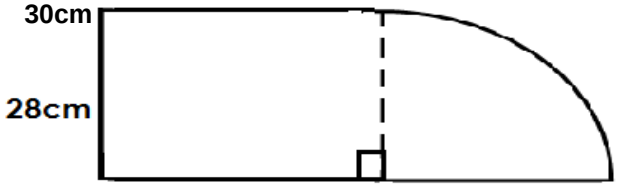
2.



a) Find the area of the above figure.

b) Find the perimeter of the figure.

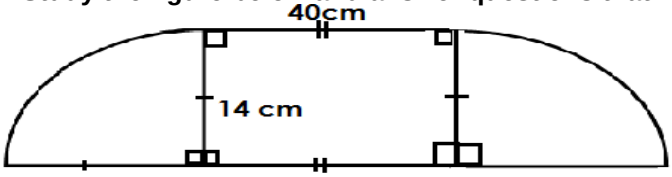
3. Use the figure below to answer questions below it.



a) Find the area of the figure.

b) Find the perimeter of the figure.

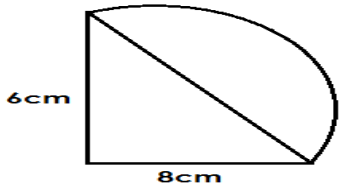
4. Study the figure below and answer questions that follow.



a) Find the area of the figure above.

b) Find its perimeter.

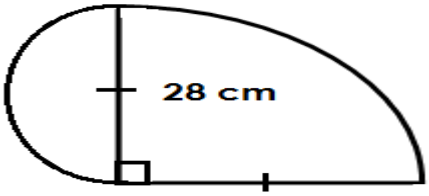
5. Below is a figure. Use it to answer questions below.



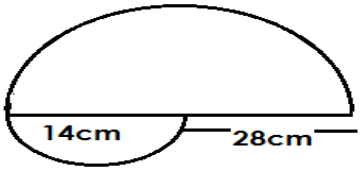
a) Find the area of the figure.

b) Find the perimeter of the figure.

6. Study the figure below and answer questions that follow.



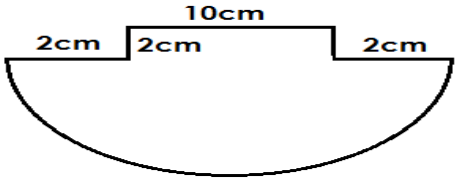
7. Study the figure below and answer questions that follow.



a) Find the area of the figure

b) Find the perimeter of the figure.

8) Below is a figure. Study it and answer questions below.



a) Find the area of the above figure.

b) Work out its perimeter.

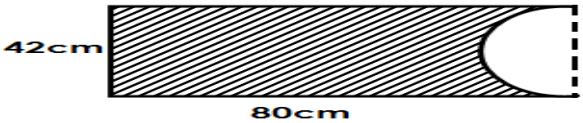
FINDING AREA AND PERIMETER OF SHADED PARTS

Note:

While finding perimeter of shaded parts, dotted lines are not considered.

Examples

1. Use the diagram below to answer questions that follow.



a) Find the area of the shaded part.

Soln.

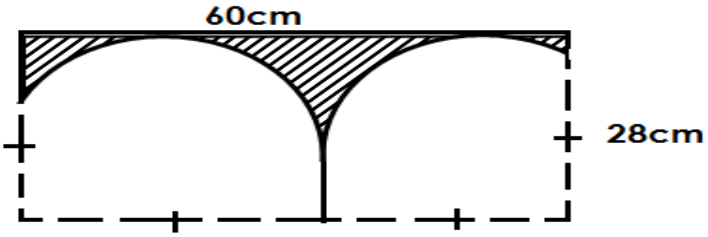
Area of outer	Area of inner	Shaded area
A = L x W = 80cm x 42cm =336 cm^2	A = $\frac{1}{2}\pi r^2$ = $\frac{1}{2}$ x $\frac{22}{7}$ x $\frac{42cm}{2}$ x $\frac{42cm}{2}$ =693cm^2	360cm^2 - 693cm^2 2667cm^2

b) Find its perimeter. (Perimeter of the shaded part)

soln.

Lines	Curve	Total
80cm + 42cm + 80cm 202cm	$\frac{1}{2}\pi r^2$ $\frac{1}{2}$ x $\frac{22}{7}$ x 42cm 66cm	202cm + 66cm 268cm

2. Study the figure below and answer questions that follow.



a) Find the area of the figure.

Soln.

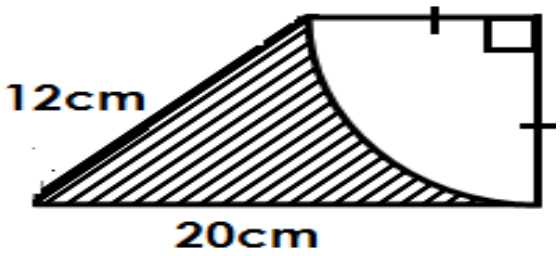
Area of outer	Area of inner	Shaded area
L x W 60cm x 28cm 168cm^2	$\frac{1}{4}\pi r^2$x 2 $\frac{1}{4}$ x $\frac{22}{7}$ x 28 x 28 x 2 1232cm^2	1680cm^2 - 1232cm^2 448cm^2

b) Find the perimeter of the shaded part.

Soln.

Lines	Curves	Perimeter
60cm	$\frac{1}{4}\pi$d x 2 $\frac{1}{4}$ X $\frac{22}{7}$ X 56cm x 2 88cm	60cm + 88cm 148cm

3. Study the figure and answer questions below.



a) Find the area of the figure above.

Soln.

Area of outer	Area of inner	Shaded area
A = $\frac{1}{2}$h (a + b) = $\frac{1}{2}$ x 14cm x (14cm + 20cm) = 238cm^2	$\frac{1}{4}\pi r^2$ $\frac{1}{4}$ x $\frac{22}{7}$ x 14cm x 14cm 154cm^2	238cm^2 - 154cm^2 84cm^2

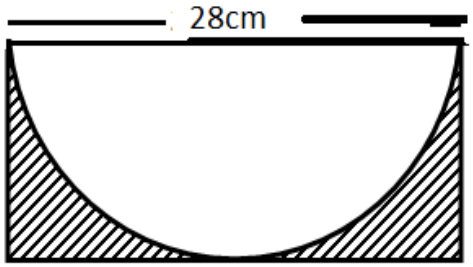
b) Find the perimeter of the shaded part.

Soln

Lines	Curve	Perimeter
12cm + 20cm 32cm	$\frac{1}{4}\pi d$ $\frac{1}{4}$ x $\frac{22}{7}$ x 28cm 22cm	32cm + 22cm 54cm

Activity:

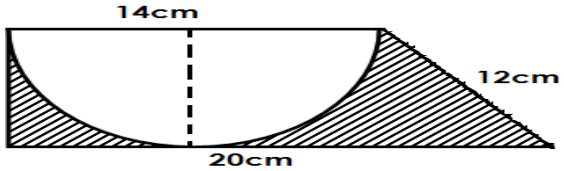
1. Find the area of the shaded parts of the figures below and perimeter of the shaded part ($\pi = \frac{22}{7}$)



a) Find the area of the shaded part.

b) Find the perimeter of the shaded part.

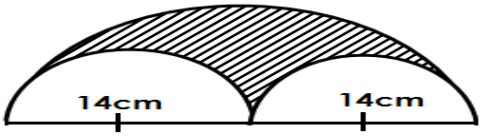
2. Below is a figure, use it to answer questions below. (Use $\pi = \frac{22}{7}$)



a) Find the area of the shaded part.

b) Find the perimeter of the shaded parts of the figure.

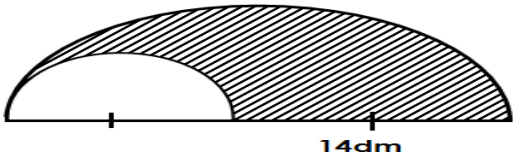
3. Study the figure below and answer questions that follow. (Use $\pi = \frac{22}{7}$)



a) Find the area of the shaded part.

b) Find the perimeter of the shaded parts of the figure above.

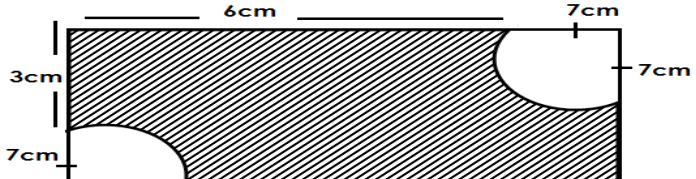
5. Below is a circle enclosed in a semi-circle (Use $\pi = \frac{22}{7}$)



a) Find the area of the shaded part.

b) Find the total distance around the shaded parts.

6. Use the diagram below to answer questions that follow.



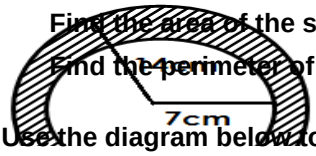
a) Find the area of the shaded part of the figure.

b) Find the total distance round the shaded parts of the figure.

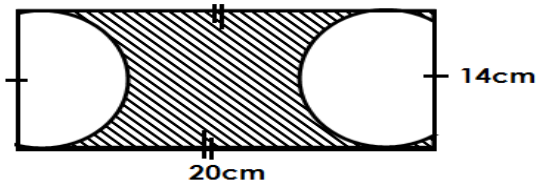
8. Given the figure below.

a) Find the area of the shaded part. (Use $\pi = \frac{22}{7}$)

b) Find the perimeter of the shaded part of the figure above.



9. Use the diagram below to answer questions below.



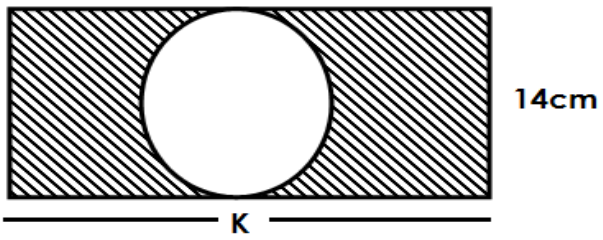
a) Find the area of the shaded part. (Use $\pi = \frac{22}{7}$)

b) Find the perimeter of the shaded parts. (Use $\pi = \frac{22}{7}$)

MORE ABOUT SHADED PARTS

Examples

1. Area of the shaded part in the figure below is $126cm^2$. Find the length of the rectangle.



Soln.

Outer - inner = shaded area.

L x W - πr^2 = $126cm^2$

K x 14cm - $\frac{22}{7}$ x 7cm x7cm = $126cm^2$

K x 14cm - $154cm^2$ = $126cm^2$

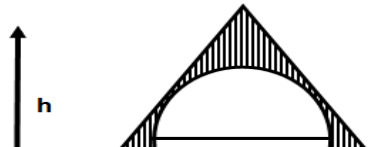
K x 14cm - 154cm + $154cm^2$ = $126cm^2$ + $154cm^2$

K x 14cm = $280cm^2$

$\frac{14k\ cm}{14cm}$ = $\frac{280cm\ x\ cm}{14cm}$

K = 20cm

2. Given the diagram below.



If the area of the shaded part in the above figure is 206cm^2. Find the height (h) of the triangle.

Soln.

Area of inner

$=\pi r^2$

$= \frac{22}{7}$ x 7cm x 7cm

=154 cm^2

Area of triangle = Area of shaded + Area of inner.

= 206cm^2

+ 154cm^2

360cm^2

$\frac{1}{2}$ x b x h = Area

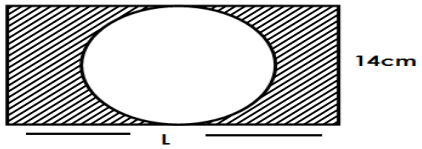
$\frac{1}{2}$ x 40cm x h = 360cm^2

$\frac{20\text{cm x h}}{20\text{cm}}$ = $\frac{360cm}{20cm}$ x cm

h = 18cm

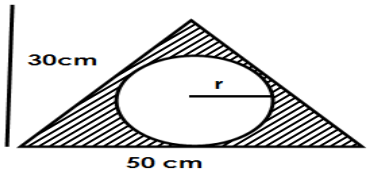
Activity:

1. Study the figure below and answer questions that follow.



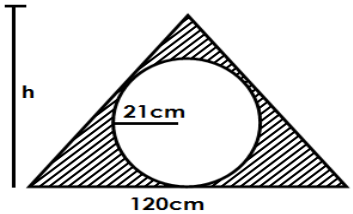
Given that the shaded area is 306cm^2. Find the value of L.

2. In the figure below, the shaded area is 134cm^2.

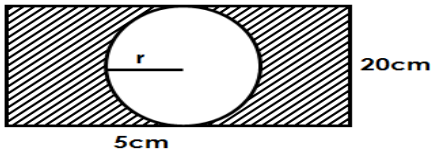


Find the value of r.

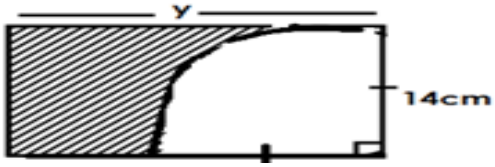
3. Find the value of h in the figure below given that the shaded area is 1014cm^2



4. In the figure below the shaded area is 846cm^2. Find the value of r.



5. In the figure below, the area of the shaded part 126cm^2.



a) Find the value of Y.

b) Calculate the perimeter of the shaded part.

REVOLUTIONS / TURNS

- **A revolution is a complete turn.**
- **A revolution is the turning made from a given point and back to that very point.**
- **A complete revolution is equal to 360°.**

FINDING NUMBER OF REVOLUTIONS

Number of revolutions = $\frac{\text{distance}}{\text{circumference}}$

Note:

Distance must be in the same units with diameter or radius of the given circular surface.

Example I

A bicycle wheel of diameter 14cm was used to cover a distance of 8.8km, in how many revolutions was the distance covered?

Soln.

No. of revolutions = $\frac{\text{distance}}{\text{circumference}}$

Cir = πD

= $\frac{22}{7}$ **x 14cm**

= 44cm

1km = 100,000cm

8.8km= (8.8 X 100,000) cm.

= $\left(\frac{88}{10}\right.$ **x 100,000) cm**

= 880,000cm

No. of revolutions = $\frac{880{,}000\text{ cm}}{44\text{cm}}$

= 20,000 revolutions

Example II

Find number of revolutions that a wheel barrow wheel of diameter 350cm can cover in a distance of 440m.

Soln.

No. of revolutions = $\frac{\text{distance}}{\text{circumference}}$

Soln.

Cir = πD

= $\frac{22}{7}$ x 35cm

= 22 x 5cm

= 110cm

1m = 100cm

440cm= 44,000cm

No. of revolutions = $\frac{440{,}000\text{cm}}{110\text{cm}}$

= 400 Revolutions

Activity:

1. Bernard rolled a car wheel of diameter 70cm from home to town through a distance of 6.6km. find the number of revolutions that he made.
2. A drum of diameter 42cm was rolled through a distance of 66cm. in how many revolutions was the drum rolled?
3. A car wheel of diameter 35cm covered a distance of 4.4km. in how many revolutions was the distance covered?
4. How many revolutions can a car wheel of radius 70cm make to cover a distance of 88km.
5. Odongo rolled a drum of radius 14cm through a distance of 13.2km. find the number of revolutions made.
6. A motorbike wheel of diameter 70cm covered a distance at a speed of 60km/hr. in 3 $\frac{2}{3}$. Find the number of revolutions made.
7. How many revolutions can a car wheel of diameter 210cm make to cover a certain distance at a speed of 48km/hr. in 5 $\frac{1}{2}$ hours.
8. Opoka rides a bicycle of diameter 63cm through a distance of 2.97km. find the number of revolutions that the distance was made.

FINDING RADIUS OR DIAMETER GIVEN NUMBER OF REVOLUTIONS

Note:

- Circumference = 1 revolution
- Circumference = $\frac{\text{Distance}}{\text{No.of revolutions}}$

It is better to change the given distance to centimeters (cm).

Example I

A distance of 4.4km was covered by a bicycle wheel in 4000 revolutions. What was the diameter of the bicycle wheel?

Soln.

1km = 100,000km

4.4km = (4.4 x 100,000) cm

= ($\frac{44}{10}$ x 100,000) cm

= 440,000cm.

Circumference = $\frac{\text{Distance}}{\text{No.of revolutions}}$

= $\frac{440{,}000\ cm}{4{,}000}$

=110cm

From:

πD = circumference

$\frac{22D}{7}$ = 110cm

$\frac{22D}{7}$ x 7 = 110cm x 7

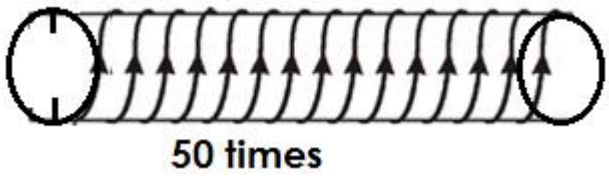
22D = 110cm x 7

$\frac{22D}{22}$ = 110cm x 7

D = 35cm

Example II

A thread of length 4400cm was wound round a cylindrical pipe shown below.



a) Find diameter of the wheel.

Circumference = $\frac{\text{Distance}}{\text{No.of revolutions}}$

= $\frac{4400cm}{50}$ 4400cm

= 88cm

πD = circumference

$\frac{22D}{7}$ = 88cm

$\frac{22D}{7}$ x 7 = 88cm x 7

22D = 88cm x 7

$\frac{22D}{22}$ = $\frac{88xm \times 7}{22}$

D = 4cm x 7

D = 28cm

b) Work out the area of its circular end.

r = $\frac{D}{2}$

= $\frac{28cm}{2}$

Soln. = 14cm

Area = πr^2

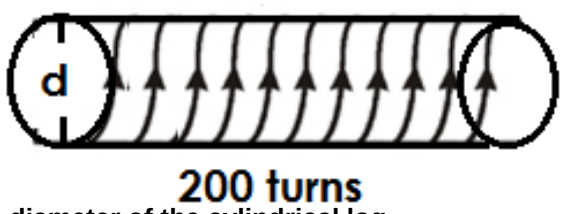
= $\frac{22}{7}$ x 14cm x 14cm

= 616cm^2

Activity:

1. **A lorry covered a distance of 8.8km in 20,000 revolutions. Find radius of the wheel**
2. **Opio's car covered a distance of 4.4km in 4000 revolutions, what was the diameter of the wheel.**
3. **A wheel barrow was pushed through a distance of 17.6m in 40 revolutions. Find the diameter of its wheel.**
4. **Odongo rolled a drum through a distance of 13.2km in 15000 revolutions. Find radius of the drum.**

5. A thread of length 88m was wound round the cylindrical log shown below.



Find the diameter of the cylindrical log.

5. **A drum was rolled from the kitchen to the office through a distance of 176m in 400 revolutions. Find the diameter of the drum.**

6. **Find the diameter of a bicycle wheel that was rolled through a distance of 440m in 2000 revolutions.**

FINDING DISTANCE GIVEN NUMBER OF REVOLUTIONS AND DIAMETER OR RADIUS

Note:

- **Distance = circumference x No. of revolutions.**
- **Length = circumference x No. of turns.**

Example I

A bicycle wheel of diameter 70cm covered a distance from Town A to town B in 1000 revolutions. Find the distance covered in km.
Soln.

Distance = circumference x No. of revolution
Cir = πD

= $\frac{22}{7}$ x 70cm

= 220cm

Distance = 220cm x 1000

= 220,000cm

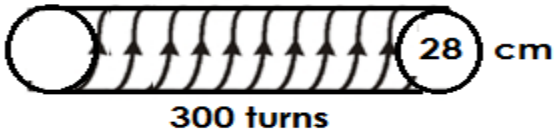
100,000cm = 1km
1cm = $\frac{1}{100,000}$km

220,000 = ($\frac{1}{100,000}$ x 220,000) Km

= $\frac{22}{10}$ km

= 2.2km

Example II
The diagram below shows a wire that was wound round a cylindrical log.



a) Find the length of the wire.

Soln.

Length = circumference x No. of turns

Cir = πD

= $\frac{22}{7}$ x 28cm

= 88cm

Length = 88cm x 800

= 26400cm

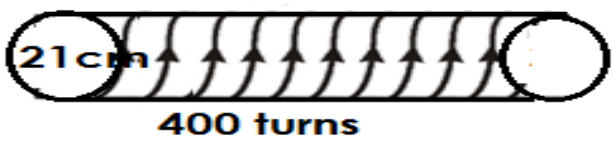
Activity:

1. Find the distance covered by a motorcycle whose diameter is 35cm in 200 revolutions. Give your answer in metres.

2. A drum of diameter 42cm was rolled 1000 times. Find the distance covered.

3. A car wheel of radius 35cm covered a certain distance in 400 revolutions. Find the distance covered.

4. A thread was wound round a cylindrical log below.



Find the length of the thread.

4. **A bicycle wheel has a diameter of 14cm. what distance can it cover in 500 revolutions.**
5. **A thread has been wound on a cylindrical tin 2000 times. Find the length of the thread if diameter of the tin was 40cm takeπ as 3.14.**
6. **A bicycle wheel of diameter 49cm covered a certain distance in 4000 revolutions. Find the distance covered in km.**

APPLICATION OF LENGTH

a) Given standing on a line.

No. of pupils = (Position x No. of ends) – excess

b) Given two positions .

No. of pupils = (sum of position) – 1

Examples

1. Peter is the 3rd pupil from either side of a line. How many pupils are in the line?

Soln.

No. of pupils = (Position x No. of ends) – Excess

= (3 X 2) – 1

= 6 – 1

= 5

2. Alex is the 4th pupil from one side and is the 6 from the other side of the line. How many pupils are on the line?

Soln.

No. of pupil = (sum of position) – 1

= (4 + 6) – 1

= 10 – 1

= 9

3. Pupils were told to stand in a line at interval of 4 metres apart. John was the 6th from either side of the line.

a) How many pupils were in a line?

Soln.

No. of pupils = (Position x No. of ends) – Excess

= (6 x 2) – 1

= 12 – 1

= 11

b) Calculate the length of the line

soln.

No. of space = No. of pupils – 1

= 11 – 1

= 10 spaces

Each space = 4m

10 spaces = 10 x 4m

= 40 metres

Activity

1. Kato is the 5th pupil from either side of the line. How many pupils are on the line?

2. John is the fourth pupil from either side on a line. How many pupils are there on the line?

3. Children were told to stand on a line and Isa was the 6th from one side and was the 8th from the other side. How many children were on the line?

4. Pupils were told to stand on a line at an interval of 2 metres apart. James was the 3rd from either side.

a) How many pupils were on the line?

b) Calculate the length of the line.

5. During elections of village chairperson, voters were told to stand on a line. The former chairperson was the 6th from one side and 9th from the other side.

a) How many voters were there altogether?

b) If the voters stood at an interval of 3 metres apart, find the length of the line in centimeters.

6. Kamoga is the 6th boy from either side on a line. How many boys are there on the line?

7. Soldiers stood on a line at intervals of 4 metres apart where by their commander was the 7th from one side and 5th from the other side of the line.

a) How many soldiers were on the line?

b) Calculate the length of the line.

PLANTING POLES / POSTS IN A STRAIGHT LINE

Note:

1. No of poles = $(\frac{\text{Distance}}{\text{interval}}) + 1$

(2) No. of spaces = No. of posts -- 1.

Example

3. 14 posts were planted along a straight road at an interval of 5 metres apart. What distance was covered?

Soln.

No. of spaces = No. of posts – 1

= 14 – 1

= 13

Each space = 5 metres

13 spaces = 13 x 5 metres

= 65 metres

4. Posts are planted at an interval of 5m apart along a straight road of length 65m. how many posts were used?

Soln.

No. of posts = $(\frac{\text{Distance}}{\text{interval}}) + 1$

= $(\frac{65\text{ m}}{5\text{m}}) + 1$

= 13 posts + 1 post
= 14 posts

5. 14 posts are planted along a straight road of distance 65m. find the *interval at which they are planted*.
Soln.

No. of spaces = No. of posts – 1
= 14 posts – 1
= 13

Interval = $\frac{\text{Distance}}{\text{No.of spaces}}$
= $\frac{65\text{m}}{13}$
= 5m

Activity:

1. 23 posts were planted along a straight road at an interval of 5 metres apart. Calculate the distance covered.
2. Electric poles were planted in a straight line at an interval of 20 metres apart to cover a distance of 5km.
a) Calculate the number of poles which were used.
b) If each pole was bought at sh. 5,000, how much money was used to purchase all the poles?
3. 16 posts are planted a long a straight road of distance 7.5km. find the interval at which the poles were planted in metres.
4. 88 electric poles were planted along a road at an interval of 50 metres apart. Find the distance covered.

5. Electric poles are planted 50 metres along a straight road. Kinto moved from 3 pole to the 10 poles. What distance did he cover?
6. Poles are planted at an interval of 3 metres a part a long a straight road of length 90 metres. How many poles were used?

PLANTING POSTS / POLES AROUND ARECTANGULAR SHAPE

Examples

3. A rectangular garden of 40 metres by 30 metres is fenced using posts placed at an interval of 5 metres apart.
a) How many posts were used?

Soln.

No. of posts = $\frac{\text{perimeter}}{\text{interval}}$
= 2 ($\frac{\text{L+W}}{\text{interval}}$)
= 2 ($\frac{40\text{m}+30\text{m}}{5\text{m}}$)
= 2 ($\frac{70\text{m}}{5\text{m}}$)
= 2 x 14
= 28

Activity:

1. A rectangular garden which measures 60 metres by 20 metres was fenced with poles placed at an interval of 5 metres apart.
a) Calculate the number of poles which were used.

b) If each pole was bought at sh. 1500, how much money was used to fence the garden.

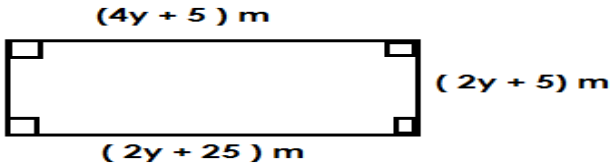
2.A rectangular plot of land measuring 100 metres by 50 metres is fenced using poles placed at the intervals of 10 metres apart.

a) Find the number of poles needed to fence the whole plot.

b) If each pole costs sh. 5000, how much is required to buy all the poles needed?

3.A rectangular plot was fenced with 20 poles placed at an interval of 5 metres apart. Find the distance round the plot.

4. The figure below is a garden. Study it carefully and answer the questions that follow.



The above garden is fenced using poles placed at an interval of 5 metres apart. Calculate the number of poles used to fence it.

5A square garden of perimeter 40 metres was fenced with 20 poles. Calculate the interval at which the poles were planted.

6.A rectangular piece of land has a perimeter of 36 metres. It was fenced using poles placed at the intervals of 6 metres apart. How many poles were used?

7.The total distance round a squared field is 100m. if it was fenced using poles placed at the intervals of 25 metres apart. How many poles were used?

8.How many poles are needed to fence a rectangular garden which measures 30 m by 20 m. if the poles are placed at an interval of 5 metres apart?

PLANTING POSTS / POLES AROUND A CURCULAR FIELD

Note:

1. No. of spaces = No. of poles

No. of poles = $\frac{\text{Distance}}{\text{interval}}$

2. With circular objects, we don't add the one (1) pole.

3. Circumference = No. of spaces x interval

Examples

1. Poles are planted round a circular flower garden of circumference 88m at an interval of 8 metres apart. How many poles are used?

Soln.

No. of poles = $\frac{\text{circumference}}{\text{interval}}$

= $\frac{88\text{m}}{8\text{m}}$

= 11

2. A circular garden of radius 7cm is fenced using posts placed at intervals of 2 metres apart. Find the number of poles required to fence the garden.

Soln.

Circumference = $2\pi r$

= 2 x $\frac{22}{7}$ x 7m

= 44m

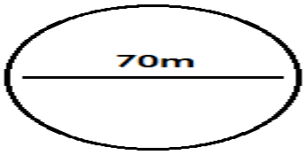
No. of poles = $\frac{\text{circumference}}{\text{interval}}$

= $\frac{44\text{m}}{2\text{m}}$

= 22 posts

Activity:

1. Poles are planted around a circular garden of circumference 154m at an interval of 7 metres apart. How many posts are used?
2. Short poles are to be planted around a circular pond of diameter 49m at an interval of 70cm.
   a) How many poles are needed?
   b) If each pole costs sh.1,100, how much money is needed for all the poles?
3. A circular garden is fenced using 22 posts placed at an interval of 2 metres apart. Find the circumference of the garden.
4. A circular garden of circumference 44m is fenced with poles placed at an interval of 4m apart. Find the number of poles used.
5. The shape below is Musisi's garden. Study it and answer questions that follow.



The above garden is fenced using poles placed at an interval of 20m apart. Calculate the number of poles used to fence it.

6. A circular flower garden of radius 28m is fenced using poles placed at an interval of 4 metres apart.
   a) Find the number of poles used.
   b) If each pole is bought at sh. 1200. Find the total cost.
7. A circular piece of land is fenced using 20 poles placed at an interval of 5 metres apart.
   a) Find the circumference of the piece of land.

PLANTING POLES AROUND EQUILATERALS

Note:

a) Number of spaces = Number of poles

b) Distance around the figure = No. of spaces x internal

c) Number of poles = $\frac{\text{perimeter}}{\text{interval}}$

Examples

1. An equilateral triangular plot is fenced with 8 poles, on each side placed at an interval of 2 metres apart. Find the perimeter of the triangle.

Soln.

Length of one side = No. of spaces x interval.

No. of spaces = 8 – 1

= 7 spaces

Length = 4 x 2m

= 8m

Perimeter = 8m x 2

= 16m

4. An equilateral triangular plot of perimeter 36 metres is fenced with poles placed at an interval of 3 metres apart. Find the number of poles which were used to fence the garden.

Soln.

No. of poles = $\frac{\text{Perimeter}}{\text{interval}}$

$= \frac{36\text{m}}{3\text{m}}$

= 12 poles

b) How many poles are placed on each side?

Soln.

Poles on each side $= \frac{12 \text{ poles}}{3}$

= 4 poles

Activity:

1. An eq at an interval of 4 metres apart. Find the perimeter of the garden.

2. The distance around an equilateral triangular house is 30 metres. It is constructed using poles placed at an interval of 2 metres apart.

a) Find the total number of poles a triangular garden is fenced with 5 poles on each side placed sed to construct the house.

b) How many poles are placed on each side?

3. The distance around an equilateral triangular plot is 36 metres. It is fenced using 7 poles on each side.

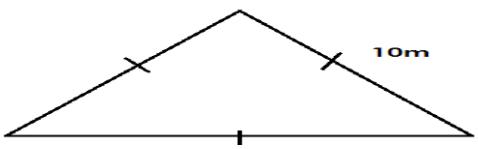
a) Find the interval at which the poles are fixed.

b) Find the length of each side of the plot.

4. Katunda designed an equilateral triangular compound which she fenced with 6 poles on each side placed at an interval of 5 metres apart.

5. An equilateral triangular plot is fenced with 10 poles on each side placed at an interval of 8m apart. Find the perimeter of the plot.

1. **The distance round an equilateral triangular piece of land is 64 metres. It is fenced using 9 poles on each side. Find the interval at which the poles are fixed.**
2. **Below is Mr. Kintu's equilateral triangular piece of land. Study it and answer questions that follow.**

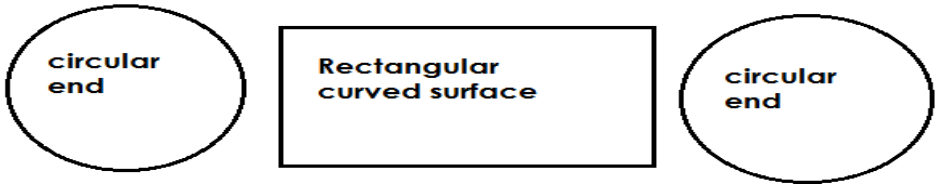


The above land is fenced using 6 poles on each side. Find the interval at which the poles are fixed.

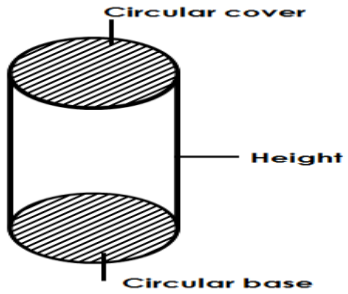
CIRCULAR BASED PRISM / CYLINDER

Note:

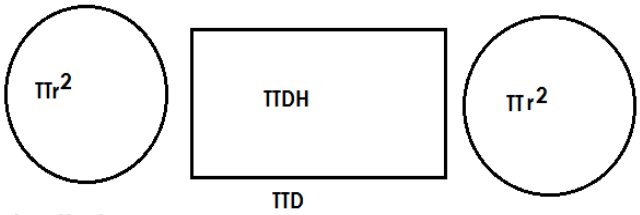
- **It has three surfaces i.e. 2 circular ends and one curved surface (Rectangular surface).**
- **Net of a cylinder**



Cylinder



FINDING THE TOTAL SURFACE AREA OF A CYLINDER



Closed cylinder

Soln.

$\pi r^2 + \pi DH + \pi r^2$

$\pi r^2 + \pi r^2 + \pi DH$

$2\pi r^2 + \pi DH$

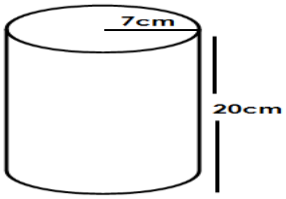
∴ T.S. A = $2\pi r^2 + \pi DH$

Open (one end opened cylinder)

T.S. A = $\pi r^2 + \pi DH$

Examples

1. Find the total surface area of the cylinder below.



d = 2r
= 2 x 7cm
= 14cm

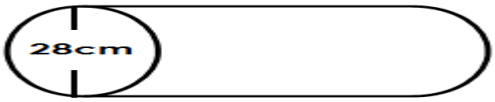
T.S. A = $2\pi r^2 + \pi DH$

=(2 x $\frac{22}{7}$ x 7cm x 7cm) + ($\frac{22}{7}$ x 14cm x 20cm)

=308cm² + 880cm²

=118cm²

2.Below is a cylinder. Use it to answer questions that follow.



a) Find the area of the curved surface.

A = + πDH

= $\frac{22}{7}$ x 28cm x 40cm

= 22 x 4cm x 40cm

= 3520cm.

b) Calculate its total surface area.

T.S. A =$2\pi r^2 + \pi DH$

=(2 x $\frac{22}{7}$ x 14cm x 14cm) + ($\frac{22}{7}$ x 28cm x 40cm)

= 1232cm²+3520cm²

= 4752cm²

3.Calculate the total surface area of an open cylinder whose radius is 14cm and height is 40cm.

Soln.

T.S. A =$2\pi r^2 + \pi DH$

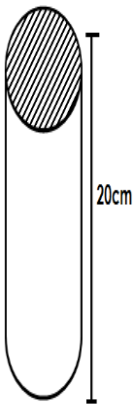
=($\frac{22}{7}$ x 14cm x 14cm) + ($\frac{22}{7}$ x 28cm x 40cm)

=616cm² + 3520cm²

=4136cm²

The area of the shaded part in the figure below is 154cm².

a)Calculate its T.S.A



soln. $A = \pi r^2$

$\frac{22\,r^2}{7}$ =154cm2 x 7

$\frac{22\,r^2}{22}$ =$\frac{154cm^2}{22}$ x 7

$\sqrt{r^2}$ = $\sqrt{49cm^2}$

r = 7cm

T.S. A= $2\pi r^2 + \pi DH$

=(2 x $\frac{22}{7}$ x 7cm x 7cm) + ($\frac{22}{7}$ x 14cm x 20cm)

=2 x 22cm x 7cm + 22 x 2cm x 20cm

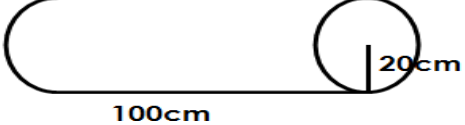
= 398cm² + 880cm²

= 1188cm²

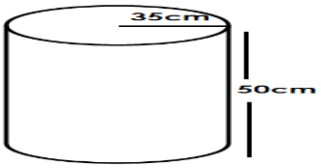
d = 2r
= 2 x 7cm
= 14cm

Activity:

1. **Find the total surface area of the figure below.**



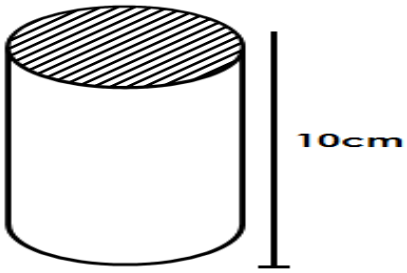
2. **Find the total surface area of the cylinder below.**



2. Find the total surface area of a cylinder with one open end whose diameter is 14cm and height is 20cm.

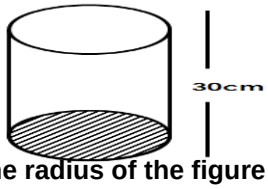
4. Calculate the total surface area of a cylinder whose radius is 28cm and height is 40cm.

5. The area of the shaded part in the figure below is 616cm^2.

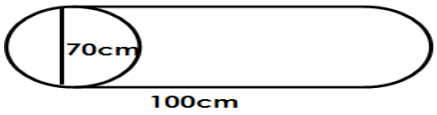


a) Find the total surface area of the figure.

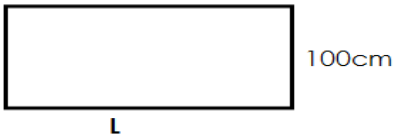
6. The circumference of the shaded part in the figure below is 88cm.



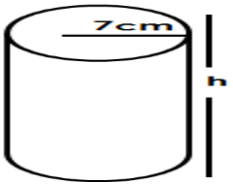
a) Find the radius of the figure above.
b) Find the area of the curved surface of the figure above.
c) Calculate the total surface area of the figure above.
7. The figure below is a cylinder drum.



a) If it was cut open to form a rectangular sheet of metal hat has used to form a rectangular door. Find the length of the door formed.
b) Calculate the perimeter of the rectangular door formed.
8. Below is a rectangular sheet of metal that was cut from a cylinder drum of radius 70cm.



a) Find its length.
b) What was the total surface area of the cylinder drum before cutting it open?
9. The area of the curved surface of the cylinder below is 880cm^2.



a) Calculate its height.

b) Find its curved surface

10. Calculate the total surface area of an open cylinder with a radius of 14cm and height 20cm.

VOLUME AND CAPACITY

Note:

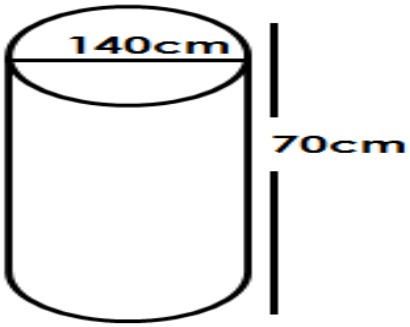
- **Volume is the amount of space occupies by an object.**
- **Capacity is the amount of contents that a given container can accommodate.**
- **To find capacity, you must have volume.**

Scale:

- **1 litre = 1000cm3**
- **Volume of a prism = Base area x height**
- **Volume of a cylinder = $\pi r^2 h$**
- **Volume of all prisms = Base area x height**

Examples:

1. Find the volume of the cylinder below.



Soln.

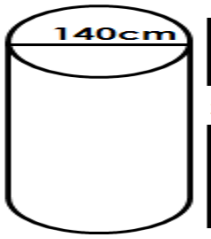
Volume = $\pi r^2 h$

= $\frac{22}{7}$ x $\frac{140cm}{2}$ x $\frac{140cm}{2}$ x 70

= 22 x 10cm x 70cm x 70cm

= 1078000cm^3.

2. Below is a cylinder tank



r = $\frac{d}{2}$

= $\frac{140cm}{2}$

= 70cm

a) Find its volume.

Soln.

1m = 100cm

2m = 2 x 100cm

= 200cm

Volume = $\pi r^2 h$

= $\frac{22}{7}$ x 70cm x 70cm x 200cm

= 22 x 10cm x 70cm x 200cm

= 308000cm^3

b) How many litres of water can a tank hold when it is ½ full.

Soln.

1000cm3 = 1litre

$1cm^3$ = $\frac{1}{1000}$L

3080000cm = ($\frac{1}{1000}$ x 3080000) litres

= 3080 litres

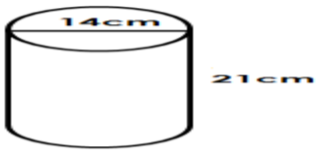
When ½ full = $\frac{1}{2}$ x 3080 litres

= 1540 litres

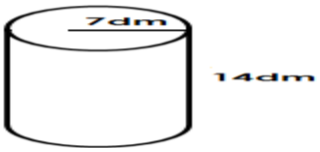
Activity:

1. Find the volume of the cylinder below.

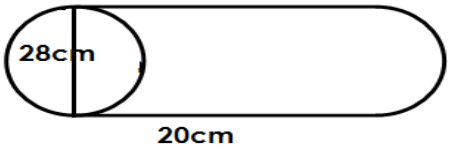
a)



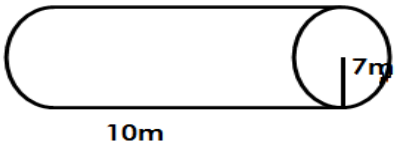
b)



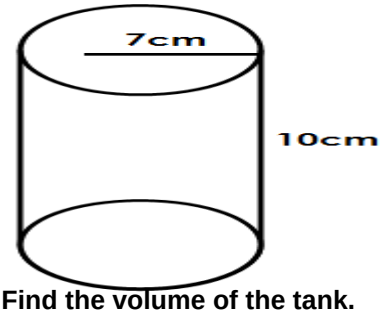
e)



d)



2. Below is a cylindrical tank.



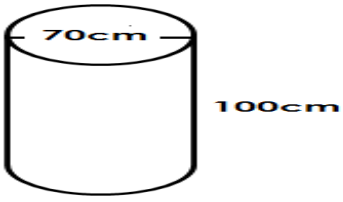
a) Find the volume of the tank.

b) How many litres of water can a tank above hold when $\frac{1}{2}$ full?

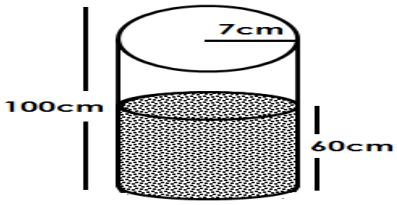
3a) Find the volume of a cylinder whose radius is 20cm and heigh is 50cm.

b). How many litres of water can the above cylinder hold?

c). Find the capacity of the tank below.



5. The tank below is holding some water.

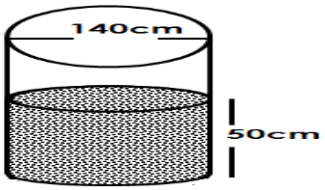


a) How many litres of water are in the tank?

b) How many litres of water are needed to fill the tank?

6. How many litres of water can a cylindrical tank of diameter 42cm and height of 100cm hold?

7. Below is a tank holding some water.



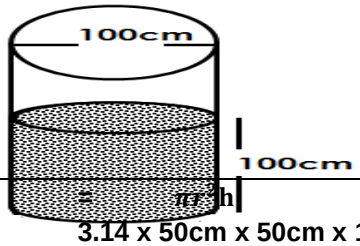
a) How many litres of water are in the tank?

b) If the tank above is $\frac{5}{8}$ full of water, how many litres of water can it hold when completely full?

MORE ON FINDING CAPACITY

Examples:

1. The tank below is $\frac{2}{3}$ full of water. How many litres of water can it hold when completely full. (Use π as 3.14).



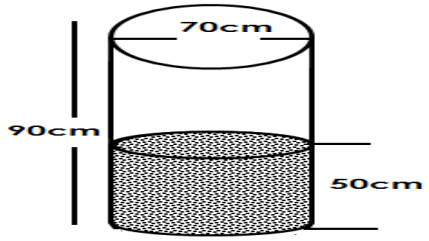
$r = \frac{D}{2}$

$= \frac{100cm}{2}$

$= 50cm$

$V = \pi r^2 h$	But $1000cm^3$ = 1 litre	Let the litres be m
= 3.14 x 50cm x 50cm x 100cm		$\frac{2}{3}$ of m = 785 litres
= $\frac{314}{100}$ x 50cm x 50cm x 100cm	$1cm^3$ = $(\frac{1}{1000}$)Litres	$\frac{2}{3}$ m = 785 L x 3
= 314 x 5cm x 5cm x 100cm		$\frac{2}{3}$ m = $\frac{785L \times 3}{2}$
= $785000cm^3$	785000cm3 = $(\frac{1}{1000}$ x 785000) L	m = 1177 $\frac{1}{2}$ litres
	= 785 litres	

2. The tank below is holding some water.



$R = \frac{D}{2}$

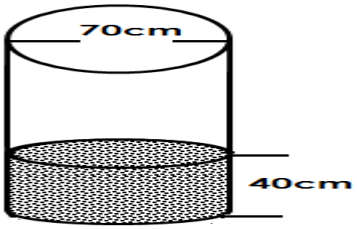
$= \frac{70cm}{2}$

$= 35cm$

a) Find the volume of water in the tank. **Soln.** **Volume = $\pi r^2 h$** **= $\frac{22}{7}$ x 35cm x 35cm x 50cm** **= 22 x 5cm x 35cm x 50cm** **= 192500cm^3** **1000cm^3 = 1 litre** **1cm^3 = ($\frac{1}{1000}$)litres** **= 192.5 litres**	**a) How many litres of water are needed to fill the tank?** **Soln.** **Height of empty space = 90cm – 50cm** **= 40cm** **Volume = $\pi r^2 h$ 5cm** **= $\frac{22}{7}$ x 35cm x 35cm x 40cm** **= 22 x 5cm x 35cm x 40cm** **= 154000cm^3** **1000cm^3 = 1 litre** **1cm^3 = ($\frac{1}{1000}$) litres** **= ($\frac{1}{1000}$ x 154000) litres** **= 154 litres**

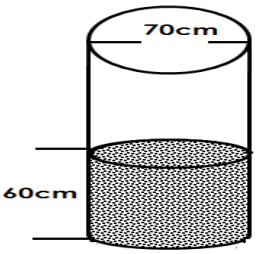
Activity:

1. The tank below is $\frac{2}{3}$ full of water.



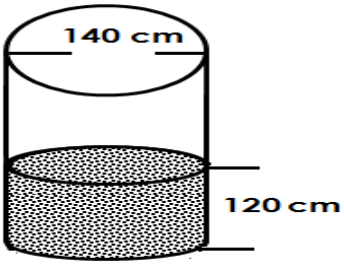
a) How many litres of water can the tank hold when capacity full of water?
b) How many litres of water are needed to fill the tank?

2. Below is a tank which is partly full of water.

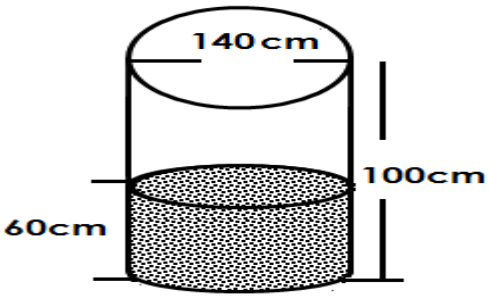


a) If it is $\frac{1}{3}$ full of water, how many litres can the tank hold when it is $\frac{1}{2}$ full?
b) How many litres of water are needed to fill the tank?

3. The tank below is $\frac{3}{4}$ full of water, find its full capacity.



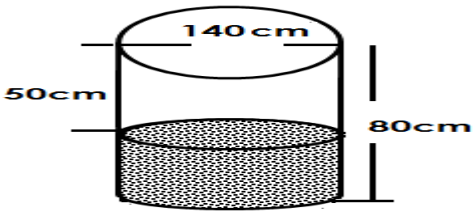
4. Use the tank below to answer questions.



a) How many litres are in the tank?

b) How many litres of water are needed to fill the tank?

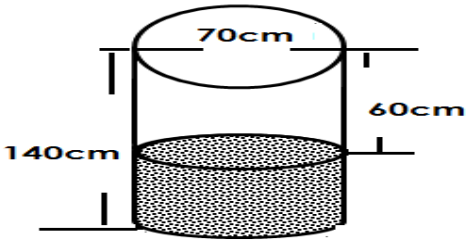
5. The tank below is holding some water.



a) How many litres of water are in the tank?

b) How many litres of water are needed to fill the tank.

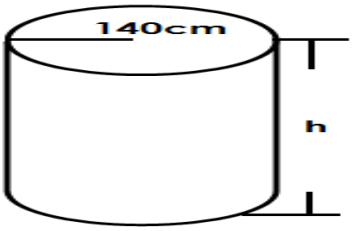
6) How many litres of water are in the tank below.



FINDING HEIGHT OR RADIUS OF A CYLINDER GIVEN CAPACITY OF A CYLINDER

Example I

A tank below is hold 154 litres.



$r = \frac{140cm}{2}$

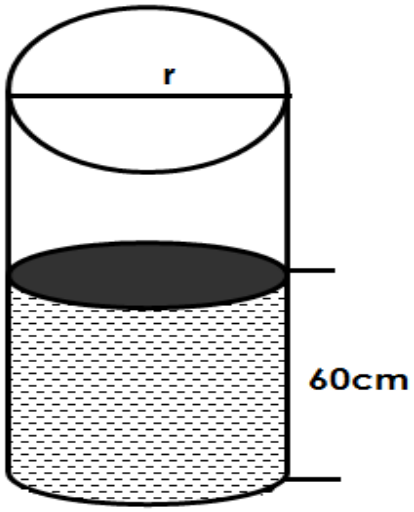
$= 70cm$

Find the height of the tank. Soln. 1 L = (1540 x 100) cm^3 1540 L = (1540 x 1000) cm^3 = 1540000cm^3	πrh^2 = vol $\frac{22}{7}$ x 70cm x 70cm x h = 1540,000cm^3. 22 x 10cm x 70 x h = 1540,000cm^3 $\frac{22}{22}$ x $\frac{10cm}{10cm}$ x $\frac{70cm}{70cm}$ x h = $\frac{1540{,}000cm \; x \; 70cm \; x \; 70cm}{22 \; x \; 70cm}$ h = 100cm

Example II

Find the radius of the tank below holding 75.36 litres (Take π as 3.14)

Soln.



$\pi r h^2$= vol

3.14r²h = 75.360cm³

$\frac{314}{100}$r² x 60cm = 75360cm³.

$\frac{1}{100}$ x $\frac{314}{100}$r² x 60cm = 75360cm³ x 100

314r² x 60cm = 75360cm³ x 100

$\frac{314}{314}$r² x 60cm = $\frac{75360cm2 \times 100}{314 \times 60cm}$

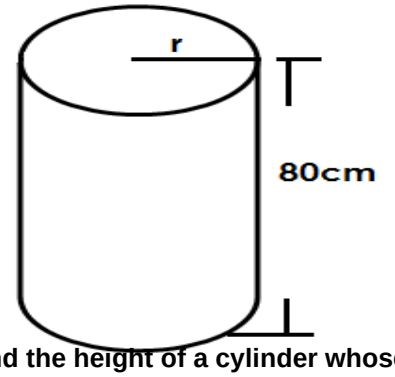
r² = 400cm²

$\sqrt{r^2}$ = $\sqrt{400cm^2}$

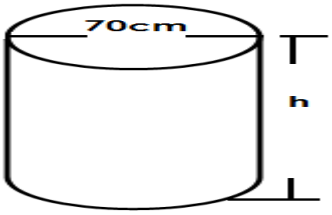
r = 20cm

Activity:

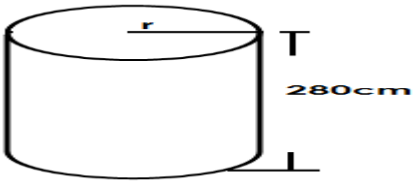
1. Find the radius of the tank below if its capacity is 308 litres.



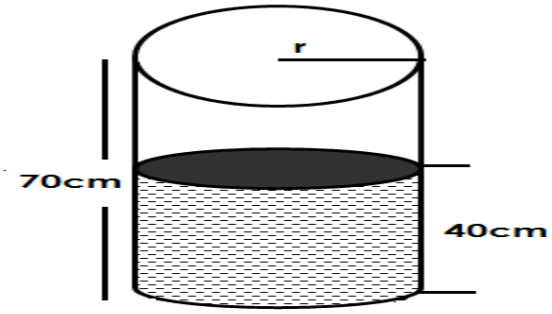
2. Find the height of a cylinder whose diameter is 28cm, if its volume is 6160cm³.
3. Below is a cylinder. Find its height if its diameter is 70cm and capacity is 385 litres.



4. Find the radius of the tank below. Take π = $\frac{22}{7}$



5. Study the diagram below and answer questions that follow.



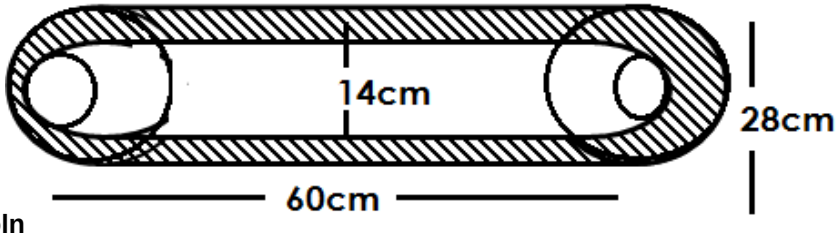
a) Find its radius

b) **How many litres of water can the tank hold when completely full of water.**

COMPARING VOLUME

Examples

1. **The figure below shows a cylindrical hole pipe. Find the volume of the concrete.**

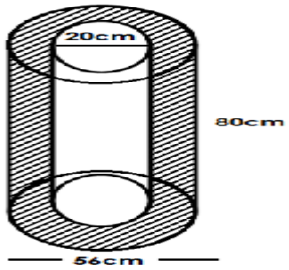


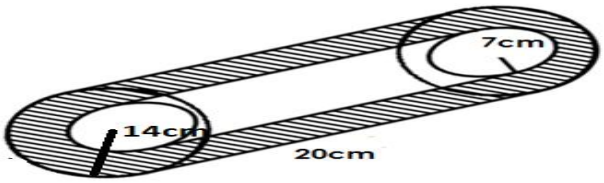
Soln

Volume of outer	Volume of inner	Volume of concrete
πr^2h $\frac{22}{7}$ x $\frac{28cm}{2}\frac{28cm}{2}$ x 60cm 36960cm^3	πr^2h $\frac{22}{7}$ x $\frac{14cm}{2}\frac{14cm}{2}$ x 60cm 9240cm^3	36960cm^3 - 9240cm^3 27720cm^3

Activity:

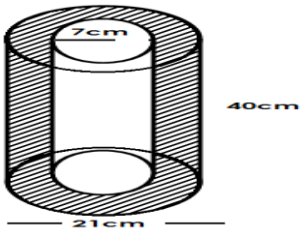
1. **Below is a hole pipe. Calculate the volume of the metal.**

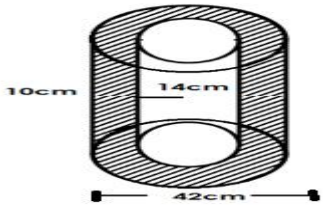




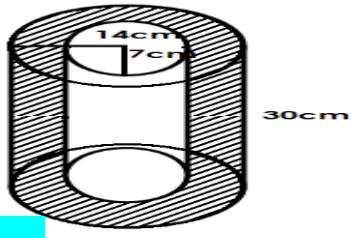
2.Find the volume of the concrete in the figure above.

3. **Find the volume of the concrete in the figures below.**





6. **The figure below shows a cylindrical hollow pipe of concrete. Calculate the volume of the concrete**.



Scenarios

Session 3

- ✦ Comparing of different solid figures.
- ✦ Packing

Methods

- Guided discovery
- Group discussion
- Think, pair and share.
- explanation

Lesson activities

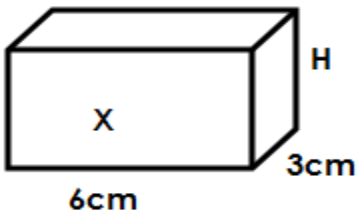
- **adding**
- **multiplying**
- **dividing**
- **subtracting**

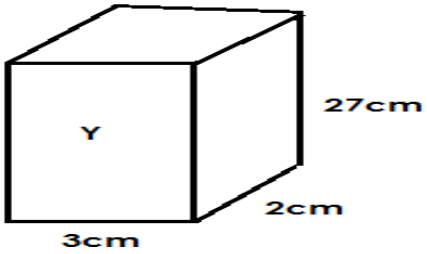
Content/facts

COMPARING VOLUME OF DIFFERENT CONTAINERS

Example I

Container X has the same volume with container Y. Find the volume of H.





Soln.

Volume = volume

L X W X h = L X W X h

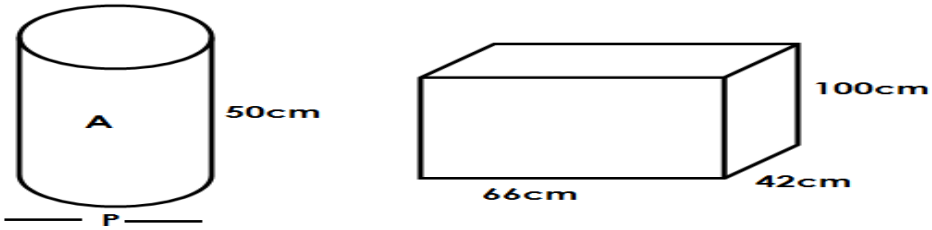
6cm x 3cm x h = 27cn x 2cm x 3cm

$\frac{6cm}{6cm} \times \frac{3cm}{3cm} \times h = \frac{27cm \times 2cm \times 3cm}{6cm \times 3cm}$

h = 9cm

Example II

Tank A has the same volume with tank B. find the volume of P.



Volume = volume

$\pi r^2 h$ = L X W X h

$\frac{22}{7} r^2 \times 50cm$ = 66cm x 42cm x 100cm

$7 \times \frac{22}{7} r^2 \times 50cm$ = 66cm x 42cm x 100cm x7

$22r2 \times 50cm$ = 66cm x 42cm x 100cm x 7

$\frac{22}{22} r^2 \times \frac{50cm}{50cm}$ = $\frac{66cm \times 42cm \times 100cm \times 7}{22 \times 50cm}$

r^2 = 3cm x 42cm x 2 x 7

r^2 = $1764cm^2$

$\sqrt{r^2}$ = $\sqrt{1764cm^2}$

R = 42cm

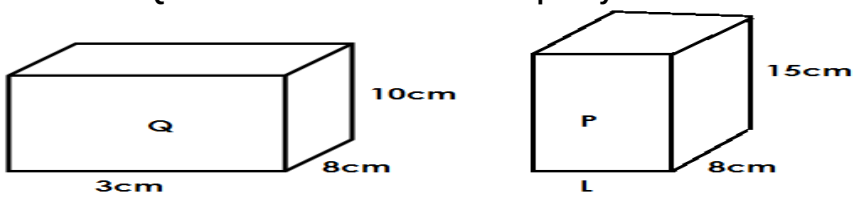
P = r + r

= 42cm + 42cm

= 84cm

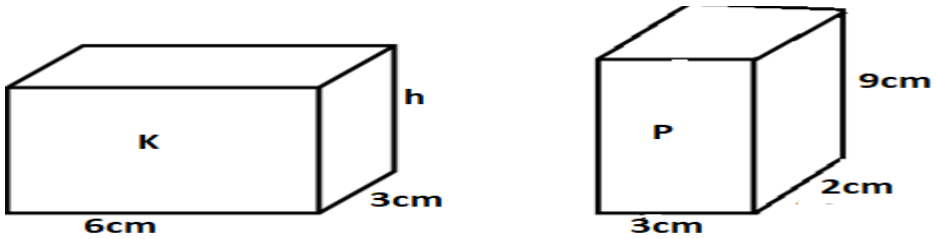
Activity:

1. **Tank Q and tank P have the same capacity.**

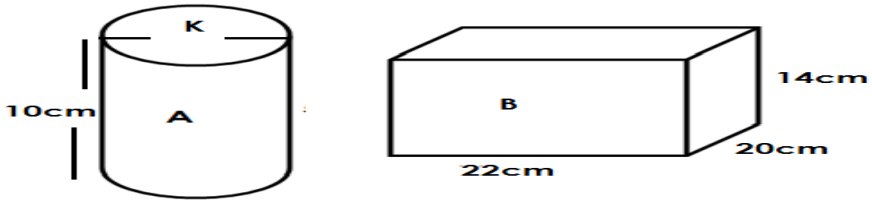


Find the value of L

2. **Tank K and tank P have the same volume. Find the height of tank K.**

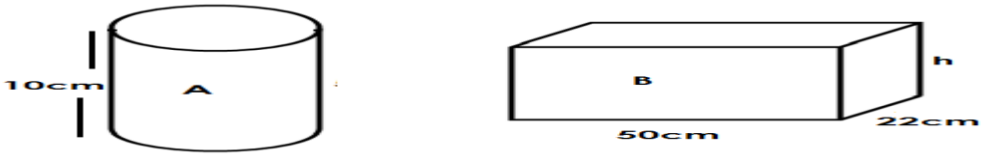


3. **Tank A and Tank B hold the same amount of water when completely full.**

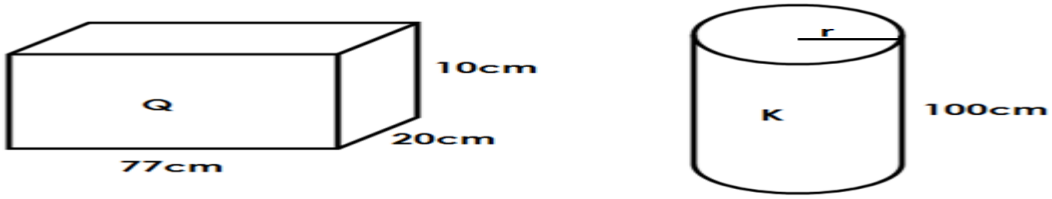


Find the value of P.

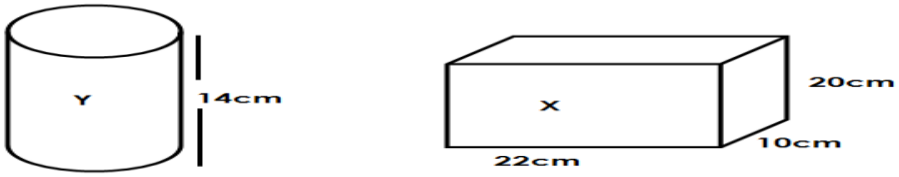
4. A trader poured cooking oil from tank A into tank B. if the two tanks hold the same capacity, what was the new height of cooking oil in tank B?



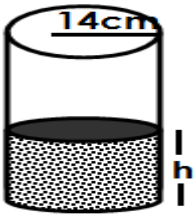
2. **Find radius of tank K if the two tanks have the same capacity.**



3. **Container X has the same volume with container Y. find the diameter of Y**



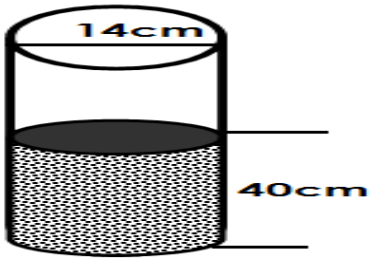
4. **The tank below contains 18480cm^3 of water.**



a) Find the value of h.

b) If Othieno used 2240cm3 of the water and poured the remaining in a rectangular tank of length 20cm and width 14cm, find the new height of water in the rectangular tank.

8. The tank below is holding some water.



If water in the tank is poured in a rectangular tank measuring 22cm long, 20 cm wide. Find the height of the water in the rectangular tank.

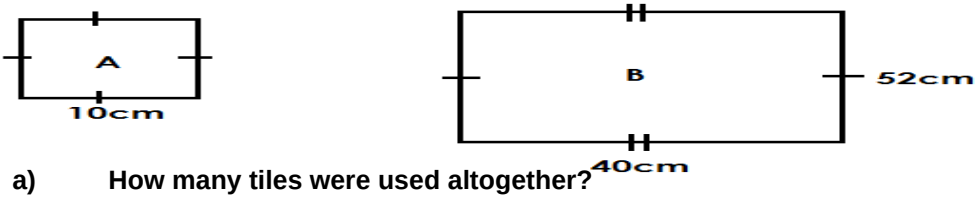
PACKING (AREA)

Note:

- **Remainder are not considered in packing; hence whole numbers are taken to be the final answers.**
- **Length is packed along length and width is packed along the width.**
- **Diameter is packed along the length and width.**

Example I

Square tiles A were packed on the flow marked B



a) How many tiles were used altogether?

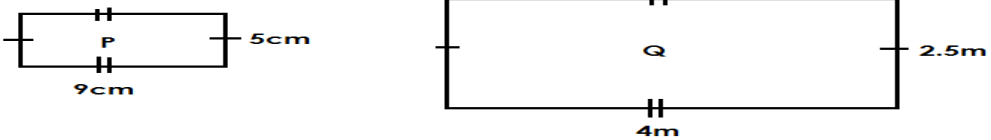
A long length	A long width	Total no. of files
$\frac{40cm}{10cm}$ 4 tiles	5 rem 2 $\frac{52cm}{10cm}$ 5 tiles	(4 x 5) tiles 20 tiles

Find the area left un covered with files.

Area of flow	Area of tiles	Area uncovered
L x W 40cm x 52cm $2080cm^2$	S x S x 20 10cm x 1mcm x 20 $2000cm^2$	$2080cm^2$ $-2000cm^2$ $80cm^2$

Example II

Rectangular tiles P were used to cover room Q. How many tiles are needed?



1m = 100cm 4m = 4 x 100cm = 400cm	1m = 100cm 2.5cm = $\frac{25}{10}$ x 100cm = 250cm

Along length	Along width	No. of tiles

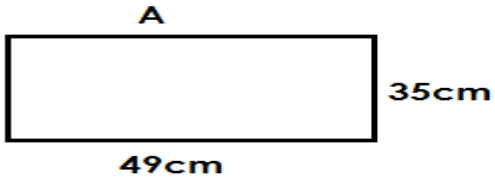
$\frac{400cm}{9}$ 44 tiles	$\frac{250cm}{5cm}$	(44 x 50) tiles 2200 tiles

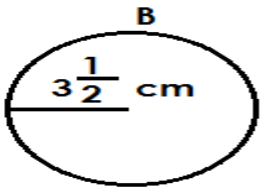
Find the area of the flow left un covered.

Area of room	Area of tiles	Area uncovered
L X W 400cm x 250cm 100,000cm^2	L x W x 2200 9cm x 5cm x 22cm 99000cm^2	100,000cm^2 - 99,000cm^2 1,000cm^2

Example III

Circular cards B were cut from rectangular manilla paper A.





D = r + r

= $\frac{7cm}{2} + \frac{7cm}{2}$

= $\frac{14cm}{2}$

= 7cm

a) How many cards were made?

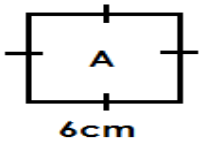
Along length	Along width	Total no. of cards
L x W 49cm x 35cm 1715cm^2	$\frac{35cm}{7cm}$ 5cm	(7 x 5) cards 35 cards

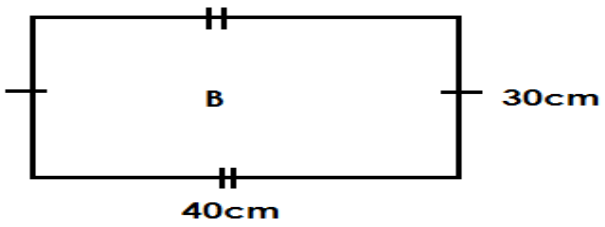
b) Find the area of the parts wasted.

Area of manila	Area of all cards	Area wasted
L X W 49cm x 35cm 1715cm2	πr^2x 35 $\frac{22}{7}$ x 3 $\frac{1}{2}$ cm x 3 $\frac{1}{2}$ cm x 35 $\frac{22}{7}$ x $\frac{7}{2}$ cm x $\frac{7}{2}$ cm x 35 $\frac{2693}{2}$cm^2	(1715 – $\frac{2695}{2}$) cm^2 $\frac{3430 - 2695}{2}$ $\frac{735}{2}$ = 367½ cm^2

ACTIVITY:

1. How many square tiles A are needed to covered a floor B.

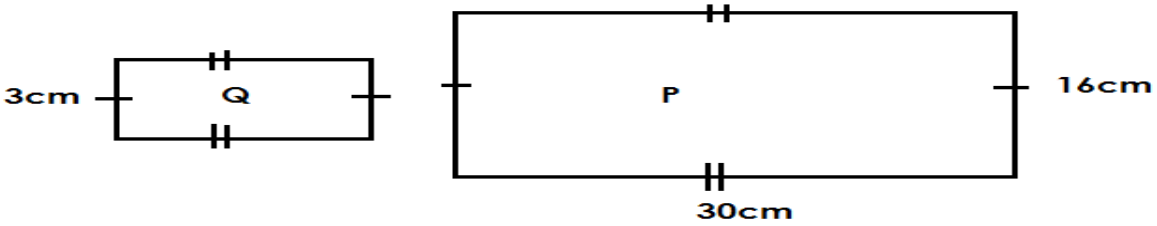




b) Find the area of the floor left uncovered.

2. How many squares tiles each 30cm are needed to cover a floor of 15cm long and 9m wide.

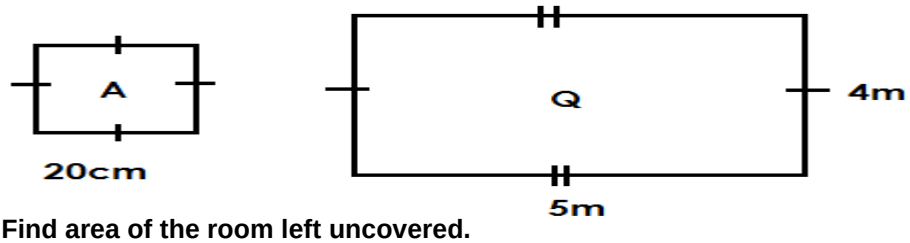
3. Small cards Q were made from rectangular manilla paper P.



a) Find number of cards made.

b) Find area of the manilla paper wasted.

4. Square tiles A were used to cover rectangular room Q.



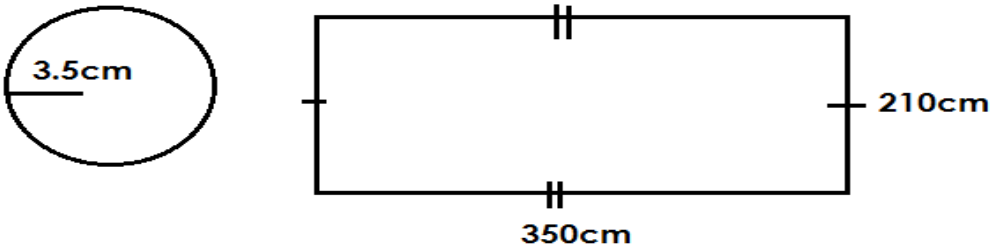
a) Find area of the room left uncovered.

5. The diagram below shows a rectangular piece of metal 150cm by 130cm. if circular table mats of diameter 28cm are to be cut, how many mats will be made?

b) How much material can remain after cutting out the mats?

6. Bwenjje made a rectangular designed mat measuring 70cm by 50cm, he later cut circular table mats of radius 14cm. how many circular table mats will be cut from rectangular mat?

b) Jacos designed a rectangular cloth measuring 350cm by 210cm. he designed it with circular parts of radius 3.5cm. what is the area of parts left undersigned?



PACKING (VOLUME)

Note:

- **Length is packed along length.**
- **Width is packed along width**
- **Height is packed along height.**

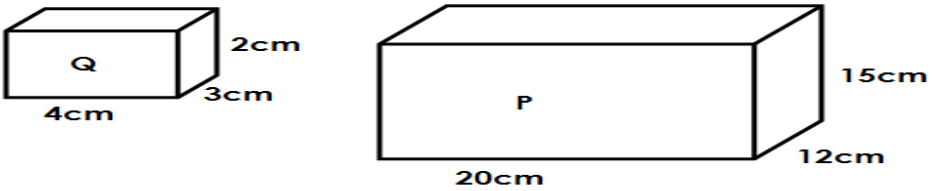
- **No. of tins = $\frac{L}{L} X \frac{W}{W} x \frac{h}{h}$**

- **Remainders are not considered.**
- **Number of layers is equal to the number of objects packed along the height.**

  = $\frac{H}{h}$

Example I

1. Small pack Q are to be packed in big packed P.



a) How many packs can be packed altogether?

Soln.

Along length	Along width	Along height	Total no. of packs.
$\frac{L}{L}$ $\frac{20cm}{4cm}$ 5 cuboids	$\frac{W}{W}$ $\frac{12cm}{5cm}$ 4 cuboids	$\frac{h}{h}$ $\frac{15cm}{2cm}$ 7 cuboids	(5 x 4 x 7) pack 140 packets

b) How many packs are packed on the first layer?

Length	Width	Height	Packs
5	4	1	5 x 4 x 1 20 packets

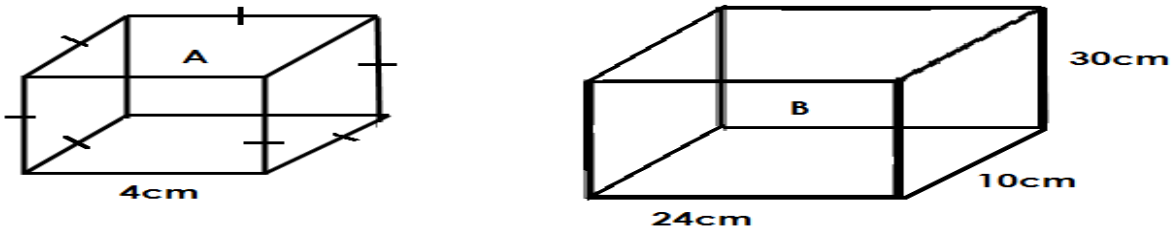
c) Find the volume of space left un occupied.

Soln.

Volume of box P	Volume of all packs	Volume un occupied
L X W X h 20cm x 12cm x 15cm $3600cm^3$	L X W X h x 140cm 4cm x 3cm x 2cm x 140cm $3360cm^3$	$3600cm^3$ $-3360cm^3$ $240cm^3$

Activity:

1.

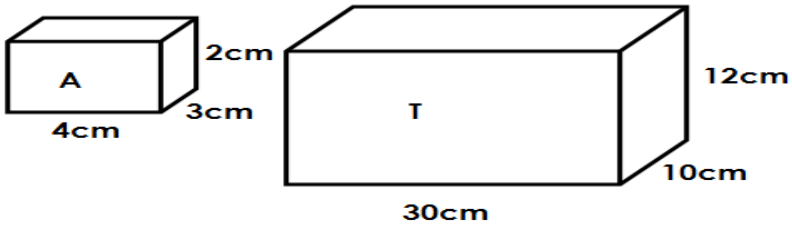


a) small cubes A are to be packed in big box B. how many cubes of A can fit in B?

b) Find the number of cubes that were packed up to the 4th layer.

c) Find the volume of space left un occupied.

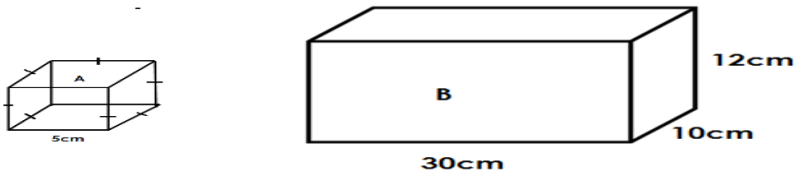
2. Small boxes of chalk A were packed in big box T.



a) How may boxes are packed altogether?

b) **Find the volume of space left un occupied.**

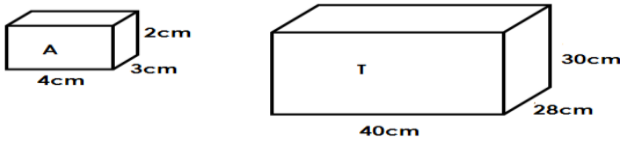
3. **Small boxes A are to be packed in big box B.**



a) **How many layers of small boxes A were packed in big box B?**

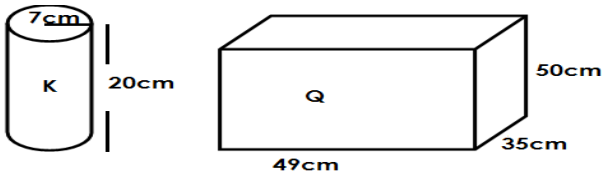
b) **Find the volume of the space left un occupied after packing all possible boxes in big box B,**

4. **Small boxes of match stick each measuring 4cm by 3cm by 2cm were packed in a certain box measuring 40cm by 28cm by 30cm.**



a) **Find the volume of the space left un occupied after packing all possible boxes of match sticks.**

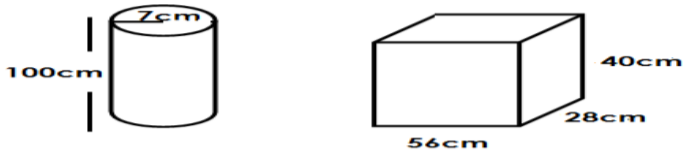
5. **Small tins K are to be packed in a big box Q.**



a) **Find the number of tins that can be packed altogether.**

b) **Find the volume of space left un occupied.**

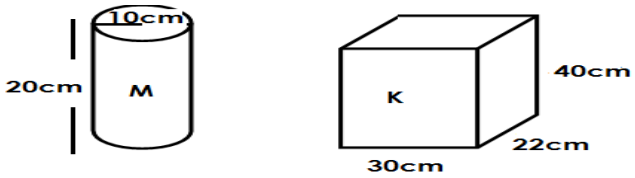
6. **Cylinder tins A were packed in box B.**



a) **How many cylindrical tins were packed altogether?**

b) **Find the volume of space left unoccupied.**

8. **Cylinder tins of blue band of radius 3.5cm and height 5cm were packed in a big measuring 70cm by 35cm by 30cm.**

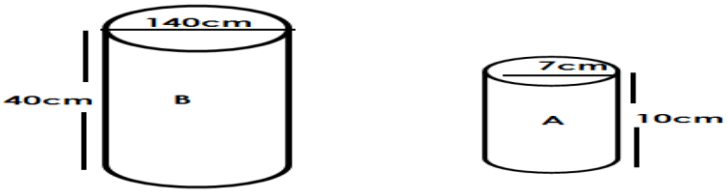


Find the volume of the space left unoccupied after packing all possible tins M in big box K.

PACKING CAPACITY

Example

1. **a trader bought cooking oil in tank B and sold it using small tins A as shown below.**



a) How many small tins A of cooking oil did he sell if all cooking oil was sold?

Soln.

No. of tins = $\frac{\text{volume of B}}{\text{Volume of A}}$

Volume of B = $\pi r^2 h$

= $\frac{22}{7}$ x $\frac{140\text{cm}}{2}$ x $\frac{140\text{cm}}{2}$ x 40cm

= 616000cm^3

Volume of A = πr^2

= $\frac{22}{7}$ x 7cm x 7cm x 10cm

= 1540cm3

No. of tins = $\frac{616000\text{cm}^3}{1540\text{cm}^3}$

= 400 tins

b) If each small tin of cooking oil was sold at sh. 2500, how much did he get after selling all the cooking oil?

Soln.

1 tin = sh. 2,500

400 tins = sh. 2500 x 400

= sh. 100,000

Activity:

1. Shell petrol station bought a rectangular container of petrol to its full capacity, measuring 440cm by 280cm by 100cm. later this petrol was sold in cylindrical drums of radius 70cm and height 40cm.



a) How many drums were sold if all petrol was sold?

b) If each litre was sold at sh.2500. how much was collected after selling all the petrol?

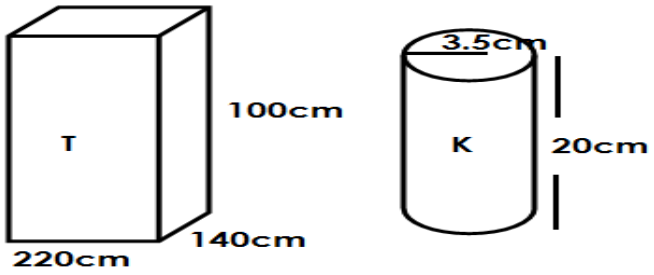
2. Malobo served visitors with water in a tank Q using cups B as shown below:



a) If all the water in tank Q was served to visitors and each visitor got a full cup of water.

b) Find the number of visitors served.

3. A farmer poured milk in jerrycan T and sold it using small tins B.



a). If all milk was sold using cups K each at sh. 1500. How much did the farmer get altogether?

4.Mulata served tea from big sauce pan A using small cups B as shown below.



a) If each pupil was given a full cup of tea. How many pupils were served altogether if the saucepan was emptied?

Scenarios

TOPIC 6 - NUMBER PATTERNS AND SEQUENCES

Subtopic: divisibility test.

Session 1

competences

- ✦ Test for divisibility of 3,6 and 9 2026
- ✦ Test for divisibility of 4 and 8
- ✦ Test for divisibility of 7 and 11

Methods

- Guided discovery
- Group discussion
- Think, pair and share.
- explanation

Lesson activities

- **adding**
- **multiplying**
- **dividing**
- **subtracting**

Content/facts

NUMBER PATTERNS AND SEQUENCE

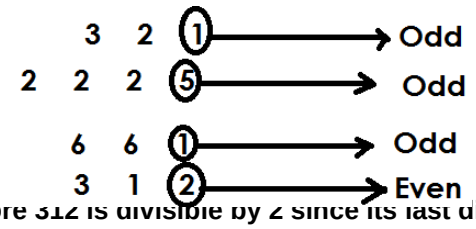
DIVISIBILITY TEST FOR TWO

A number is divisible by two if its last digit is even i.e. 0, 2, 4, 6 or 8

Example1

Given 321, 2225,661 and 312. Identify a number that is divisible by 2

Solution



Therefore 312 is divisible by 2 since its last digit is even.

DIVISIBILITY FOR 3

A number is divisible by 3 if the sum of its digits is a multiple of 3.

Example 1

Given the following numbers, identify a number that is divisible by 3 in

125, 241 and 426

125 sum
1+2+5
8
141 sum
2+4+1
7
426 sum
4+2+6
12

426 is divisible by 3 since the sum of its digit is a multiple of 3.

State whether 111 is divisible by 3

Solution

111 sum

= 1+1+1

= 3

Therefore 111 is divisible by 3 since the sum of its digits is a multiple of 3

State whether 241 is divisible by 3

Solution

241 sum

= 2+4+1

= 7

Therefore 241 is not divisible by 3 since the sum of its digits is not a multiple of 3.

Activity.

1. By showing the working, identify numbers divisible by 3

(a) 236 (b) 245 (c) 462 (d) 4659

2. Given 101, 100, 111 and 110. Without dividing, identify a number which is exactly divisible by 3

DIVISIBILITY TEST FOR 4

A number is divisible by 4 if it is even and the number formed by its last two digits is a multiple of 4 or last two digits are (00)

State whether 236 is divisible by 4

Solution



Therefo isible by 4 since the number formed by its last two digits is a multiple of 4.

State whether 111112 is divisible by 4

Solution



Therefore 111112 is divisible by 4 since the number formed by its last two digits is a multiple of 4

Example3

State whether 234 is a multiple by 4

Solution



Therefore ible by 4 since the number formed by tits last two digits is a not a multiple of 4.

Activity.

State whether the following are divisible by 4

(a) 144 (b) 512 (c) 9444 (d) 568

Without dividing, identify the numbers that are exactly divisible by 4 from:

62748, 10018, 232332 and 97658

DIVISIBILITY TEST FOR 5

A number is divisible by 5 if its last digit is 5 or 0

Given the following numbers, 556,104 and 205. Which number is divisible by 5?

Solution

5 5 6
1 0 4
2 4 5

Therefore **visible by 5 since its last digit is 5**

Which of the numbers below is exactly divisible by 5

503, 155,200 and 107

155 and 200 ends in either 0 or 5 and are divisible by 5.

Activity

Without dividing, identify numbers that are exactly divisible by 5 by the following.

(a) 15,27,30,38 and 45

(b) 1115, 5504 and 1053

DIVISIBILITY TEST FOR 6

A number is divisible by 6 if it is even and also divisible by 3

1. State whether 426 is divisible by 6.

Solution

4+2+6=12

Therefore 426 is divisible by 6 since it is divisible by 2 and 3

2. State whether 576 is divisible by 6

Solution

576----5+7+6

18

576 is divisible by 6 since it is divisible by 2 and 3.

3. State whether 646 is divisible by6

646

Solution

646 = 6+4+6

= 16

16 is not a multiple or 3.

Therefore 646 is not divisible by 6 since it's not divisible by 3.

Activity

State whether the following numbers are divisible by 6

(a) 936 (b) 463 (c) 12462 (d) 963 (e) 612 (f) 113

DIVISIBILITY TESTS FOR 7

A number is divisible by 7 when the last digit of a number is doubled and the result is subtracted from the number formed by the remaining digits and the outcome is divisible by 7.

1. State whether 861 is divisible by 7

Solution

861

1+1=2

```
 8 6
 - 2
 ---
 8 4
 ---
```
4+4=8

8-8=0

0 is divisible by 7 hence 861 is divisible by 7

2. Find whether 1792 is divisible by7

1792

2+2=4

```
 1 7 9
   - 4
 -------
 1 7 5
```
-5+5=10

```
     1 7
   - 1 0
   -----
       7
```
7 is divisible by 7 hence 1792 is exactly divisible by 7.

3. State whether 246 is divisible by 7

246

6+6=12

```
   2 4
 - 1 2
 -----
   1 2
 -----
```
12 is not divisible by 7 hence 246 is not divisible by 7.

4. Find whether 7962 is exactly divisible by 7

7962 **792**

2+2=4 **2+2=4**

```
 7 9 6
 -   4
 -----
 7 9 2
 -----
```

```
   7 9
 - 4
 ---
 7 5
```

75 is not divisible by 7 hence 7962 is not divisible by 7 exactly.

Activity

State whether the following numbers are exactly divisible by 7

(a)105 (b) 133 (c) 686 (d) 8792

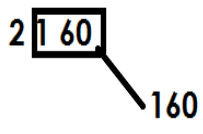
(e) 227955 (f) 18648 (g) 2864

(h) 18649 (i) 182

DIVISIBILITY TEST FOR 8

A number is divisible by 8 if the number formed by the last three digits is divisible by 8

1. State whether 2160 is divisible by8



160 is divisible by 8 hence 2160 is exactly divisible by8.

2. Find out whether 2080 is exactly divisible by 8

08 2 0 8 0 by 8 hence 2080 is divisible by 8 exactly.

3. Identify if 45094 is divisible by 8

4 5 0 9 4 isible by 8 hence 45094 is not divisible by 8.

Activity

1. State whether the following numbers are divisible by 8

(a) 7960(b) 18016 (c) 7320 (d) 65200 (e) 19098 (f) 8888

DIVISIBILITY TEST FOR 9

A number is divisible by 9 if the sum of its digits is divisible by 9.

1. State whether 198 is divisible by 9.

Sum=1+9+8

=18

18 is divisible by 9 hence 198 is divisible by 9 exactly.

2. Find whether 207 is divisible by 9.

Sum=2+0+7

=9

9 is divisible by 9 hence 207 is divisible by 9 exactly.

3. Find out if 21034 is exactly divisible by 9

Sum=2+1+0+3+4

=10

10 is not divisible by 9 hence 21034 is not divisible by 9 exactly.

Activity.

1. State whether the following numbers are exactly divisible by 9.

(a) 342 (b) 783 (c) 660

(d) 8757 (e) 4827 (f) 70308

(g) 54696 (h) 11034 (i) 10008

DIVISIBILITY TEST FOR 10

A number is divisible by 10 if it ends with 0.

1. State whether 120 is divisible by 10.

Solution

1 2 0

120 ends with 0 hence divisible by 10

2. State whether 255 is divisible by 10.

Solution

2 5 5

255 doesn't end with 0 hence 255 isn't divisible by 10 exactly.

Activity

State whether the following numbers are exactly divisible by 10.

(a)4050 (b)1980

(c)2020 (d)108

(e) 9110 (f) 1009

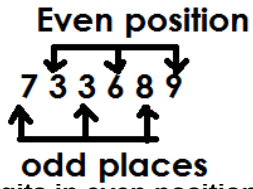
(h) 2345

DIVISIBILITY TEST FOR 11

A number is divisible by 11 if the difference between the sum of the digits in even places and the sum of the digits in the odd places is zero or divisible by 11.

Example

Identify whether 733689 is exactly divisible by 11.



Sum of the d ; =7+3+8=18

Sum of the digits in even positions =3+6+9=18

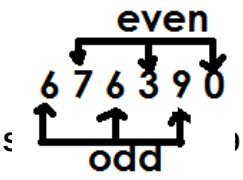
Difference=18-18

=0

Therefore 733689 id divisible by 11.

Example 2

Show whether 676390 exactly divisible by 11?



s)

=21

Sum of even=7+0+3

=10

Difference=21-10

=11

Since difference is divisible by 11 the number is divisible by 11.

Activity

Which of the following numbers is exactly divisible by 11?

(a) 2367 (b) 814 (c) 6425

(d) 7282 (e) 901938

Scenarios

Session 2

competences

- ✦ Types of numbers
- ✦ Finds consecutive counting and whole numbers
- ✦ Finds consecutive even numbers and odd numbers
- ✦ Solves word statements on consecutive numbers involving median, mean and range.

Methods

- Guided discovery
- Group discussion
- Think, pair and share.
- explanation

Lesson activities

- **adding**
- **multiplying**
- **dividing**
- **subtracting**

Content/facts

g TYPES OF NUMBERS

1. Whole numbers

It is where all other numbers are derived.

Whole numbers consist of 1 or more units and are not fractions

Examples

0, 1, 2, 3, 4, 5, 6, …

2. Counting/ natural numbers

These are numbers which show concrete quantity of something

For example

1, 2, 3, 4, 5, 6, …

N.B All whole numbers are counting numbers apart from zero (0)

3. Even numbers

These are numbers that are exactly divisible by 2

Even numbers have a pattern of +2 in ascending order and -2 in descending order.

Examples.

0, 2, 4, 6, 8, …

N.B

Even numbers end in an even digit.

Below is a set of the first 6 even numbers.

{0, 2, 4, 6, 8}

Find the sum of the first 4 even numbers

Solution

0,2,4,6

Sum = 0+2+4+6

= 12

ODD NUMBERS

These are numbers which when divided by 2, gives a remainder of 1. The first odd number is 1.

Below is a set the first 6 odd numbers.

{1, 3, 5, 7, 9}

All odd numbers end in an odd digit.

Odd numbers have a pattern of +2 in ascending order and -2 in descending order.

Find the product of the 4th and 6th odd numbers.

Solution

1, 3, 5, 7^{4th}, 9, 11^{6th}, 13

Product = 7x11

= 77

Find the next number in the sequence



PRIME NUMBERS

A prime number is a number with only 2 factors; 1 and 2 itself.

Below is a set of prime numbers

{2,3,5,7,11,13,17,19,23……}

Find the sum of prime numbers between 6 and 15

Solution

7, 11, 13

Sum = 7+11+13

= 31

Find the sum of prime numbers between 20 and 30

23, 29

Sum = 23+29

= 52

Find the next number in the sequence



COMPOSITE NUMBERS

These are numbers with more than 2 factors.

The first composite number is 4. The rest include;

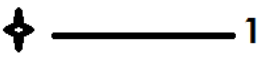
4, 6, 8, 9, 10, 12, 14, 15……

TRIANGULAR NUMBERS

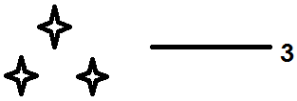
These are numbers got by adding consecutive counting numbers.

The first triangular number is 1

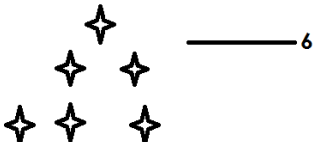
1.



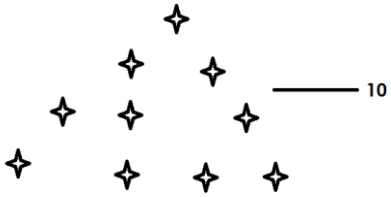
2.



3.



4.



Finding triangular numbers

Formula= $\frac{n(n+1)}{2}$

Examples

1. Find the 9th triangular number

$$\frac{n(n+1)}{2}$$
$$=\frac{9(9+1)}{2}$$
$$=\frac{9(\cancel{10}^{5})}{\cancel{2}_{1}}$$
$$=9\times5$$
$$=45$$

2. Find the 5th triangular number

$$\frac{n(n+1)}{2}$$
$$=\frac{5(5+1)}{2}$$
$$=\frac{5(\cancel{6}^{3})}{\cancel{2}_{1}}$$

3. Find the sum of the 7th and 3rd triangular number

$7^{th} = \frac{n(n+1)}{2}$

$= \frac{7(7+1)}{2}$

$= \frac{7(8)^4}{2_1}$

$=7x4$

$=28$

$\frac{n(n+1)}{2}$

$= \frac{3(3+1)}{2}$

$= \frac{3(4)^2}{2_1}$

$=3x2$

$=6$

Sum=28+6

=34

Activity

1. Find the 11th triangular number

2. Find the 17th triangular numbers

3. Find triangular numbers in these positions

(a) 8th (b) 21st

(c) 32nd (d) 4th

4. Find the sum of the 2oth and 15th triangular numbers.

5. What is the difference between the 10th and 6th triangular numbers?

6. Find the product of the 9th and 10th triangular numbers.

CONSECUTIVE NUMBERS

Finding consecutive counting numbers

N.B

Counting numbers have a pattern of +1 and -1 in ascending and descending orders respectively.

Examples

1. The sum of 3 consecutive counting numbers is 27. Find the numbers.

Let the first number be k

1st	2nd	3rd	sum
k	K+1	K+2	27

K + k + 1 + k + 2 = 27

K + k + k + 1 + 2 = 27

3k + 3 = 27

3k + 3 – 3 = 27 – 3

3k = 24

1st	2nd	3rd
k	K+1	K+2
=8	8+1	=8+2
	9	10

$$\frac{3k}{3} = \frac{24}{3}$$

K = 8

(b) Find their range

Range = H - L

= 10 - 8

= 2

2. Find the sum of 3 consecutive counting numbers is 39. If the 2nd number is P. find the numbers.

1st	2nd	3rd	sum
P-1	P	P+1	39

P-1+P+P+1 =39
P+P+P+1-1 =39
0
3P =39
13P =39 13
1 3 3 1
P =13

1st	2nd	3rd
P-1	P	P+1
13-1	=13	13+1
12		14

(b) Find the range of numbers

Range=H-L

=14-12

=2

(c) Find the mean of the numbers

Mean = sum of items

No.of items

$$= \frac{13}{\cancel{39}\ \ \cancel{3}\ \ 1}$$

=13

3. The sum of 3 consecutive counting numbers is 45 if the third number is k.

(a) Find the numbers

1st	2nd	3rd	sum
k-2	k-1	k	45

k+K-1+k-2 =45
k+k+k-1-2 =45
3k-3 =45
3k-3+3 =45+3
0
3k =48
16
1 3k =48
1 3 3 1
k =16

1st	2nd	3rd
k-2	k-1	k
16-2	16-1	=16
14	15	

(b) Find their range

Range=H-L

=16-14

=2

4. The average of 3 consecutive counting numbers is 8. Find the numbers

Let the first number be k

1st	2nd	3rd	average
k	K+1	K +2	8

Sum of items = Average

No. of items

$$\frac{k+k+1+k+2}{3} = 8$$

$$\frac{k+k+k+1+2}{3} = 8$$

$$\frac{3k+3}{3} = 8$$

$$\frac{3k+3\times 3}{3} = 8\times 3$$

$$3k+3 = 24$$

$$3k+3-3 = 24-3$$

$$3k = 21$$

$$\frac{3k}{3} = \frac{21}{3}$$

$$k = 7$$

1st	2nd	3rd
k	K+1	K+2
7	7+1	7+2
14	8	9

ACTIVITY

1. The sum of 3 consecutive counting numbers is 42.

(a) Find the numbers

(b) Find the range of numbers

2. The sum of 3 consecutive counting numbers is 66. If the middle number is k,

(a) Find the numbers

(b) Find the range of the numbers

3. Given that the sum of consecutive counting numbers is 54. If the 4th number is P.

(a) Find the numbers

(b) Calculate the range of the numbers.

4. If the average of 3 consecutive counting numbers is 12.

(a) Find the numbers

(b) Find the range of numbers

5. The mean of 3 consecutive counting numbers is 15.

(a) Find the numbers

(b) Find the product of the numbers.

FINDING CONSECUTIVE EVEN NUMBERS

N.B

Even numbers have a pattern of +2 and -2 in ascending and descending orders respectively.

Examples

1. The sum of 3 consecutive even numbers is 42.

(a) Find the numbers

Let the 1st number be k

$k+k+2+k+4 = 42$

$k+k+k+2+4 = 42$

$3k+6 = 42$

$3k+6-6 = 42-6$

$3k = 36$

$\frac{3k}{3} = \frac{36}{3}$

$k = 12$

1st	2nd	3rd
k	K+2	K+4
12	12+2	12+4
	14	16

(b) Find their range

Range =H-L

=16-12

=4

2. The sum of 4 consecutive even numbers is 44. If the last number is p.

(a) Find the numbers

1st	2nd	3rd	4th	Sum
p-6	p-4	p-2	p	44

P+p-2+p-4+p-6 = 44

P+p+p+p-2-4-6 = 44

4p – 12 = 44

4p – 12 + 12 = 44 – 12

4p = 32

$\frac{4p}{4} = \frac{32}{4}$

P = 8

1st	2nd	3rd	4th
P-6	P-4	P-2	P
14-6	14-4	14-2	=14
8	10	12	

(b) Find their range

Range=H-L

=14-8

=6

3. The average of 3 consecutive even numbers is 8

(a) Find the numbers

Let the first number be k

1st	2nd	3rd	4th
k	K+2	K+4	8

$$\frac{k+k+2+k+4}{3} = 8$$

$$\frac{k+k+k+2+4}{3} = 8$$

$$\frac{3k+6}{3} = 8$$

$$\frac{3k+6}{3} = 8$$

$$\frac{3k+6 \times 3}{3} = 8$$

$$3k+6 = 8 \times 3$$

$$3k+6 = 24$$

$$3k+6-6 = 24-6$$

$$3k = 18$$

$$\frac{3k}{3} = \frac{18}{3}$$

$$k = 6$$

Sum of items= average

No. of items

1st	2nd	3rd
k	K+2	K+4
=6	6+2	6+4
	8	10

Activity

1. The sum of 3 consecutive even numbers is 66.

(a) Find the numbers

(b) Find their range

2. The sum of 4 consecutive even numbers is 52.

(a) Find the numbers

(b) Find their range

3. The sum of 3 consecutive even numbers is 96. If the middle number is y.

(a) Find the numbers

(b) Find their range

4. Four consecutive even numbers add up to 28. If the 4th number is k

(a) Find the numbers

(b) Find their range

5. The average of 3 consecutive even numbers is 22

(a) Find the numbers

(b) Find the product of the numbers

Scenarios

TOPIC 7 - OPERATIONS ON WHOLE NUMBERS (HOLIDAY FOR TERM 1 2026)

Session 1

- ✦ Adds, subtracts, multiplies and divides whole numbers.
- ✦ Solves word statements involving operations on whole numbers.
- ✦ Works out numbers using mixed operations
- ✦ Works out numbers using number properties
- ✦ Writes numbers in standard form and vice versa
- ✦ Simplifies indices involving addition and subtraction

Methods

- Guided discovery
- Group discussion
- Think, pair and share.
- explanation

Lesson activities

- **adding**
- **multiplying**
- **dividing**
- **subtracting**

Content/facts

OPERATIONS ON WHOLE NUMBERS

Properties of operations

1. Commutative property.

It states that the order in which numbers are arranged does not change the result.

Example

5 + 6 = 11 ⟶ 6 + 5 = 11

4 x 3 = 12 ⟶ 3 x 4 = 12

Commutative property in under addition and multiplication only.

2. ASSOCIATIVE PROPERTY

- **It states that when carrying out addition or multiplication of more than two numbers the way in which numbers are grouped does not affect the sum or product.**

- **The alternation of brackets does not change answers in addition or multiplication of more than two numbers.**

Example 1.

(6 + 4) + 5 = 6 + (4 + 5)
10 + 5 = 6 + 9
15 = 15

Example II

(6 x 4) x 5 = 6 x (4 x 5)
24 x 5 = 6 x 20
120 = 120

DISTRIBUTIVE PROPERTY

- **The product of a factor and a sum or difference is equal to the sum of difference of the products respectively.**

Examples

Work out: 4 x 3 + 6 x 4
Identify the common factors.
(4 x 3) + (6 x 4)
4 (6 + 3)
4 (9)
4 x 9
=36
Work out: (5 – 4 x 54) + (46 x 5.4)
Soln.
5.4 (54 + 46)
5.4 x 100
$\frac{54}{10}$ **x 100**
540

Work out: 3.4 x 127 - 27 x 3.4
Soln.
(3.4 x 127) – (3.4 x 27)
3.4 (127 – 27)
$\frac{34}{10}$ **x 100**
349

Work out: 146 ÷ 5 – 46 ÷5
Soln.
(146÷5) – (46 ÷5)
(146 – 46)÷ 5
100 ÷5
= 20

Activity

Work out the following using distributive property only.

1. **2 x 17 + 2 x 13**
2. **75 x 29 + 25 x 19**
3. **17 x 19 + 19 x 13**
4. **(178 x 10) + (10 x 22)**
5. **(75 x 25) – (75 x 19)**
6. **4.6 x 18 – 8 x 4.6**
7. **12.4 x 146 – 46 x 12.4**
8. **0.6 x 119 – 19 x 0.6**
9. **(199 ÷ 8) – 7 ÷ 5**
10. **(179 ÷13) – (10 ÷13)**

11. 79 ÷8) – (7 ÷8)
12. (625 ÷20) + (75 ÷20)
13. (606 ÷12) + (18 ÷12)

WRITING NUMBERS IN STANDARD FORM/SCIENTIFIC NOTATION

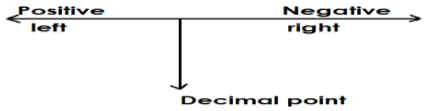
Standard form **is a way of expressing numbers as a product of powers of ten leaving one counting digit in the places of whole.**

The number of steps moved by a decimal point will determine the size of powers.

If a decimal point moves taking a left direction, a positive power is obtained.

If a decimal point moves taking the right direction, a negative power is obtained.

Illustration



Example 1 **Write 843 in scientific notation** **Solution** **843=** $8.43 \times 10^{?}$ $=8.43 \times 10^{2}$	**Example2** **Write 96.3 in standard form** **Solution** **96.3=** $96.3 \times 10^{?}$ 9.63×10^{1}	**Example 3** **Write 4676.16 in scientific form** **Solution** **4676.16=** $4676.16 \times 10^{?}$ 4.67616×10^{2}
Example 4 **Express 0.246 in standard form.** **Solution** **0.246=** $0.246 \times 10^{?}$ 2.46×10^{-2}	**5.Express 0.00478 in scientific notation** **Solution** **0.00478=** $0.00478 \times 10^{?}$ 4.78×10^{-3}	**6.Express 420000 in scientific notation** **Solution** **420000=** $420000 \times 10^{?}$ 4.2×10^{5}
7.Express 4.9 in standard form **Solution** **4.9** $= 4.9 \times 10^{?}$ $= 4.9 \times 10^{0}$		

Activity

1. **Express following in standard form**

(a) 423 (b) 3967(c) 497.3

(d) 97.04 (e) 9.6 (f) 560000

2. **Write the following in standard form**

(a) 0.96 (b) 0.09 (c) 0.00789

(d) 0.0499(e) 0.008637

FINDING ORIGINAL NUMBERS THAT WERE WRITTEN IN STANDARD FORM

Examples

1.Find the number that was written in standard form to give $8.73x10^2$? **Solution** $8.73x10^2$ = $\frac{873 \times 10 \times 10}{100}$ 873	**2.Find the number that was written in standard form to give $9.6x10^{-2}$** **Solution** $9.6 \times 10 = \frac{96 \times 1}{10\ 100}$ $= \frac{96}{1000}$ $= 0.096$	**3.Find the original number that was written inn scientific form to give $6x10^6$?** **Solution** $6x10^6 = 6 \times 10 \times 10 \times 10 \times 10 \times 10$ $= 6x1000000$ $= 6000000$
4.Which number was written in standard form to give $5.24x10^{2?}$ $5.24x10^2 = \frac{524 \times 10 \times 10}{100}$ $=524$	**5.Which number was written in standard form to give $6.234x10^{-5}$** **Solution** $6.234 \times 10^5 = \frac{6234 \times 1}{1000\ \ 100000}$ $= \frac{0.06234}{100000000}$ $=0.00006234$	

Activity

1.Find the number that was written in standard for to give

(a) $4.7x10^2$ (b) $5.4 x10^3$ (c) $5.678 x10^5$

(d) $6.04x10^3$ (e) $7.1x10^4$

2.Which numbers were written in scientific form to given the following?

(a) $7x\ 10^{-1}$ **(b)** 7.2×10^{-2} **(c)** 7.03×10^5

(d) 2.14×10^{-1} **(e)** 3.678×10^{-3} **(f)** $6.24x10^{-4}$

INDICES

Given;

To simplify in indices; leave the answer in power form.

To workout is to evaluate and give the value of the expression.

ADDITION AND SUBTRACTION BY EXPANSION

Note:

Expand the base by multiplying it as many times as the exponent.

Any number raised to exponent 0, gives one (1) and any number raised to exponent one (1) gives that same number.

1.Workout: 2^3+4^2 **= (2x2x2) + 4x4** **=8 +16** **=24**	**2.Workout: 10^3+5^2** **Solution** **= (10x10x10) + (5x5)** **=1000+25** **=1025**	**3.Workout: 4^3-3^2** **= (4x4x4)- (3x3)** **= 64-9** **= 55**	**4.Workout: 3^2-2^3** **= (3x3)- (2x2x2)** **= 9-8** **= 1**	**5.Workout: $2^3+9^0+2^1$** **(2x2x2) + 1+2** **8+3** **11**	**6.Workout: 14^2-12^0** **= (14x14) -1** **= 196-1** **= 195**

Activity

Workout the following

(a) 4^3+2^3 (b) $2^3+3^2+5^0$

(c) 4^2+3^2 (d) $3^4+4^2+2^1$

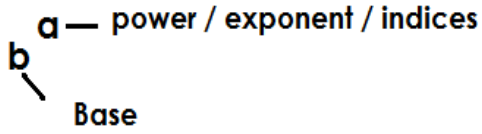
(e) 3^4+4^0 (f) 4^2-3^2

(g) 6^2-2^3 (h) 4^3-5^1

(i) 10^0-3^3

LAW OF INDICES IN MULTIPLICATION

- **Note: .**



When multiplying powers of the same base, maintain one common base and add the exponents.

Examples

1.Simplify: $2^2 \times 2^3$ **Solution** $2^2 \times 2^3$ 2^{2+3} 2^5 $(2\times2\times2)\times(2\times2)$ 8×4 32	**2. Workout: $10^1 \times 10^2$** $10^1 \times 10^2 = 10^{1+2}$ $= 10^3$ $10\times10\times10$ 1000	**3.Simplify: $3^2 \times 3^3$** $3^2 \times 3^3 = 3^5$ $=(3\times3\times3)\times(3\times3)$ $=27\times9$ $=243$	4.Simplify: 43×40 $4^3 \times 4^0 = 4^{3+0}$ $=4^3$ $=4\times4\times4$ $=64$

Activity

(a) $2^4 \times 2^2$ (b) $5^3 \times 5^1$

(c) $3^1 \times 3^3$ (d) $2^3 \times 2 \times 2^1$

(e) $10^4 \times 1^0$ (f) $5^3 \times 5^2$

(g) $3^2 \times 3^1 \times 3^0$ (h) $9^2 \times 9^0$

Laws of indices in division

Note:

When dividing powers of the same base, maintain one common base and then subtract the exponents.

Examples

1. Simplify: $4^3 \div 4^2$ $4^3 \div 4^2 = 4^{3-1}$ $= 4^1$	**2.Workout: $6^5 \div 6^4$** $6^5 \div 6^4$ $=6^{5-4}$ $=6^2$ $=6\times6$ $=36$	**3.Workout: $3^4 \div 3$** $3^4 \div 3^1 =$ $= 3^{4-1}$ $=3^3$ $=3\times3\times3$ $=27$	**4.Workout: $5^3 \div 5^0$** $5^3 \div 5^0 = 5^{3-0}$ $=5^3$ $=5\times5\times5$ $=125$ $\frac{2^3 \times 2^4}{2^5}$
5.Workout: $\frac{2^3 \times 2^4}{2^5}$ **Solution** $=\frac{2\times2\times2\times2\times2\times2\times2}{2\times2\times2\times2\times2}$ $=4$	**6.Workout:** $=\frac{5^2\times5^3}{5^7}$ $=\frac{5\times5\times5\times5\times5}{5\times5\times5\times5\times5\times5\times5}$ $=\frac{1}{25}$	**7. Workout:** $3^2 \div 3^5$ $=3^{2-5}$ $=3^{-3}$ $=\frac{1}{3^3}$ $=\frac{1}{3\times3\times3}$ $=\frac{1}{27}$	**8.Work out the following** (a) $n^7 \div n^4$ $= n^{7-4}$ $= n^3$ (b) $p^{3a} \div p^a$ $P^{3a}-p^a$ $=p^{2a}$

Activity

1.Workout the following

(a)$7^5 \div 7^4$

(b) $10^3 \div 10^2$

(c) $5^{12} \div 5^9$

(d) $3^7 \div 3^4$

(e) $4^3 \div 4^1$

(f)$5^2 \div 5^0$

2. Simplify the following.

(g) $n^{10} \div n^3$

(h) $132^k \div 13^k$

Scenarios

Session 2

competences

- ✦ Simplifies indices involving multiplication and division.
- ✦ Applies the laws of indices on multiplication and division involving unknowns
- ✦ Solves word statements involving LCM
- ✦ Solves word statements involving GCF
- ✦ Solves word statements involving LCM

Methods

- **Guided discovery**
- **Group discussion**

- **Think, pair and share.**
- **explanation**

Lesson activities

- **adding**
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- **dividing**
- **subtracting**

Content/facts

APPLICATION OF INDICES (A)

Examples.

Solve: $3^K = 81$

Soln.

- **Express 81 as power of 3.**

3^k =

3	81
3	27
3	9
3	3
	1

$3^k = 3 \times 3 \times 3 \times 3$

$3^k = 3^4$

$k = 4$

Solve: $2^a \times 2^2 = 16$

Soln.

$2^a \times 2^2$ =

2	16
2	8
2	4
2	2
	1

$2^{a+2} = 2 \times 2 \times 2 \times 2$

$2^{a+2} = 24$

$a + 2 = 4$

$a + 2 - 2 = 4 - 2$

$a = 2$

Activity:

Solve for the unknowns in the following: -

1. $2^{k} = 32$

2. $3^{n} = 27$

3. $4^{a-1} = 64$

4. $2^{21} = 25$

5. $2^{m} \times 2^{2} = 32$

6. $3^{2a} \times 3^{2} = 729$

7. $5^{n} \times 5^{n} = 25$

8. $4^{2} \times 4^{y} = 256$

9. $6^{a} \times 6^{2} = 216$

APPLICATION OF INDICES (B)

Examples

Simplify: $3^{m} \div 3^{2} = 9$

Soln.

$3^{m-2} =$

3	9
3	3
	1

$3^{m-2} = 3 \times 3$

$3^{m-2} = 3^{2}$

$m - 2 = 2$

$= 2 + 2$

Solve: $2^{p} \div 2^{1} = 8$

Soln.

$2^{p-1} = 8$

$2^{p-1} =$

2	8
2	4
2	2
	1

$2^{p-1} = 2 \times 2 \times 2$

$2^{p-1} = 2^{3}$

$p - 1 = 3$

$p - 1 + 1 = 3 + 1$

$= = 4$

Solve $2^{p} \div 8 = 32$

Soln.

$2^{p} \div$

2	8
2	4
2	2
	1

$=$

2	32
2	16
2	8
3	4
2	2
	1

$2^{p} \quad 2^{3}$

$2^{p-3} = 2^{8}$

$p - 3 + 3 = 8 + 3$

$p = 11$

Activity:

Solve for the unknown in the following: -

1. $2^x \div 2^2 = 4$
2. $3^p \div 3^1 = 9$
3. $2^b \div 2^2 = 8$
4. $3^n \div 3^2 = 27$
5. $6^k \div 6^1 = 36$
6. $2^{3y} \div 2^y = 16$
7. $3^{2m} \div 3^m = 27$
8. $5^{3p} \div 5^p = 625$
9. $3^p \div 3 = 9$
10. $3^m \div 9 = 31$
11. $5^y \div 25 = 625$

APPLICATION OF INDICES (C)

Example I

Solve: $2k \times 32 = 36$

Soln.

- **Express 36 as power of 2 and 3 i.e., factorize 36 using 2 and 3.**

$2^k \times 3^2$

= 2	36
= 2	18
3	9
3	3
	1

$2^k \times 3^2 = 2 \times 2 \times 3 \times 3$

$2^k \times 3^2 = 2^k \times 3^2$

$\frac{2^k \times \cancel{3^2}\ 1}{\cancel{3^2}\ 1} = \frac{2^2 \times \cancel{3^2}\ 1}{\cancel{3^2}\ 1}$

$2^k = 2^2$

$k = 2$

Activity:

Solve for the unknown in the following

1. $2^n \div 3^3 = 108$
2. $3^p \div 5^2 = 75$
3. $3^2 \times 7^y = 441$
4. $2^3 \times 5^{2k} = 200$
5. $2^m \times 5^2 = 100$
6. $2^3 \times 5^{k-1} = 200$

PRIME FACTORIZATION

Prime factorization is a way of dividing numbers using its prime factors

Prime factors of numbers are written in two ways;

- **Subscript form/ Set notation form**
- **Power form/ Multiplication form**

Example 1

Prime factorize 36 giving your answer in subscript form

$F_{36} = \{2_1, 2_2, 3_1, 3_2\}$

2	36
2	18
3	9
3	3
	1

Example 2

Prime factorize 50 and give your answer in power form

Solution

2	50
5	25
5	5
	1

$F_{50} = 2X5X5$
$= 2^1X5^2$

Example 3

Express 100 as a product of its prime factors.

2	100
2	50
2	25
5	5
	1

$F_{100} = 2X2X5X5$

Prime factorize 360 in power form

2	360
2	180
2	90
3	45
3	15
5	5
	1

$F_{360} = 2X2X2X3X3X5$
$= 2^3X3^2X5^1$

Activity

1. Prime factories the following and give your answer in subscript form

(a) 18 (b) 72 (c) 144

2. Prime factorize the following and give your answer in power form.

(a) 120 (b) 150 (c) 240

(d) Express each number as a product of its prime factors.

(a) 64 (b) 210 (c) 60

Finding umbers that were prime factorized

1. Given that FQ = $\{2_1, 2_2, 3_1, 5_1\}$ Find Q

$Q = 2 X 2 X 3 X 5$
$= 4X15$
$= 60$

2. If $k = 2^3x5^1$. Find k

K= (2x2) x (2x5)

= 4x10

= 40

3. Find the prime factorized number in the following

Fy = {2_1, 3_1, 3_2, 5_1}

Y = 21x31x32x51

= (2x3) x (3x5)

= 6x15

= 90

4. FK = 2^4x3^1

= (2x2) x (2x2) x3

= 4x4x3

= 16x3

=48

Activity

1. What number was prime factorized to give?

$\{2_1, 3_1, 5_1, 5_2\}$

2. Given k = 2^4 x 5^1. Find the value of k.
3. Find the value of Z if, FZ = 2^1 x 3^2 x5^1
4. Find the prime factorized numbers in the following

FM = {2_1, 2_2, 2_3, 3_2, 5_1}

K = 2x2x3x3x5

M = 3^2x5^2

Finding missing factors

1. If $F_{30}=\{2_1\ y_1\ 15_1\}$ Find y.

Solution

$yx2x5 = 30$

$10y = 30$

$\frac{10y}{10} = \frac{30}{10}$

$y=3$

$y=3_1$

2. Given that 2_1, 3_1, k and 5_1 are prime factors of 90. Find the value of k.

$kx2x3x5 = 90$

$30k = 90$

$\frac{30k}{30} = \frac{90^3}{30_1}$

$k = 3$

$k = 3_2$

3. If 12= (2_1, m, 3_1). Find m

$mx2x3 = 12$

$6m = 12$

$\frac{6m}{6} = \frac{12^2}{6_1}$

$m = 2$

Activity

1. If 2_1, 3_1, 5_1 and y are prime factors of 150. Find y
2. Given that {2_1 ,2_2, 2_3, n, 5_1} are prime factors of 120. Find the value of n
3. If {2_1, 2_2, k, 3_2, 5_1} are prime factors of 180. Find the value of k.
4. Given that 36= {2_1, 2_2, 3_1, x}. Find the value of x.
5. If 100 = {2_1, p, 5_1, 5_2}. Find the value of p.
6. The prime factors of 360 are 2_1, 2_2, 2_3, m, 3_2 and 5_1. Find the value of m.

Representing prime factors on a Venn diagram

1. Prime factorize 12 and 18 and represent the prime factors on a Venn diagram.

$P.F_{12}$ =

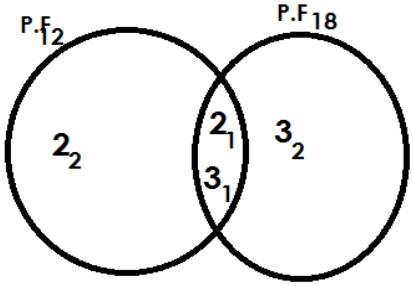
PF.1	NO
2	12
2	6
3	3
	1

$P.F_{18}$ =

P.F	NO
2	18
3	9
3	3
	1

$P.F_{12} = \{2_1 2_2 3_1\}$

$P.F_{18} = \{2_1 3_1 3_2\}$



(b) Find the G.C.F of 12 and 18
GCF = intersection product
= 2x3
= 6

(c) Find LCM of 12 and 18
LCM = union product
= (2X2) X (3X3)
= 4X9
= 36

2. Prime factorize 18 and 24 and represent the prime factors on a Venn diagram.

$P.F_{18}$ =

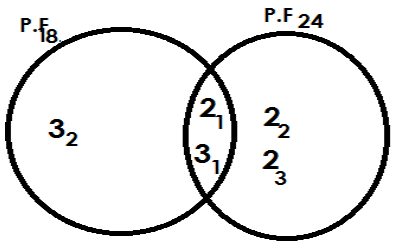
P.F	NO
2	18
3	9
3	3
	1

$P.F_{24}$ =

P.F	NO
2	24
2	12
2	6
3	3
	1

$P.F_{18} = \{2_1 3_1 3_2\}$

$P.F = 24\ \{2_1 2_2 2_3 3_1\}$



(b) Use the Venn diagram to find;

(i) GCF of 18 and 24

GCF=2X3

=6

(ii) LCM of 18 and 24

LCM=union product

= (3x2) x (3x2) x2

= (6x6) x2

=36x2

=72

Activity

1. Prime factorize 24 and 36 and represent the prime factors on a Venn diagram.

(b) **Use the Venn diagram to find**

(i) **GCF of 24 and 36.**

2. **Show the prime factors of 36 and 48 on a Venn diagram.**

(b) **Use the Venn diagram to find.**

(i) **G.C.F of 36 and 48**

(ii) **L.C.M of 36 and 48**

3. **Prime factorize 45 and 60 and represent the prime factors on a Venn diagram.**

(b) **Use the Venn diagram to find;**

(i) **GCF of 45 and 60**

(ii) **LCM of 45 and 60**

4. **Show the prime factors of 30 and 36 on a Venn diagram**

(b) **Using the Venn diagram; find**

(i) **HCF of 30 and 36**

(ii) **LCM of 30 and 36**

5. **Show the prime factors of 120 and 180 on a Venn diagram**

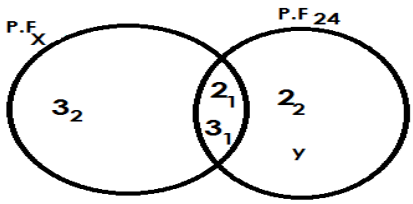
(b) **Use the Venn diagram to find,**

(i) **GCF of 120 and 180**

(ii) **LCM of 120 and 180**

Finding unknown values on a Venn diagram

1. **Use the Venn diagram below to answer questions that follow.**



(a) **Find the value of**

(i) **y**

$yx2x2x3 = 24$

$12y = 24$

$\frac{12y}{12} = \frac{24}{12}$ (2, 1)

$y = 2_3$

(ii) **x**

X = (3x2) x3

X = 6x3

X = 18

(b) **Find GCF of x and 24**

GCF=2x3

=6

(c) **Find LCM of x and 24**

LCM= (3x2) x (3x2) x y

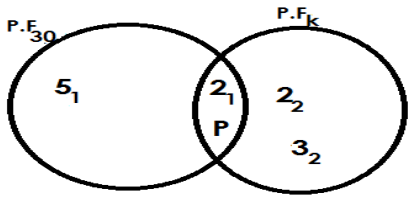
= (3x2) x (3x2) x2

= (6x6) x2

=36x2

72

2. Use the Venn diagram below to answer questions that follow.



(a) Find the value of

(i) P

$Px2x5 = 30$

$10P = 30$

$\frac{10P}{10} = \frac{30}{10}$

$P = 3$

(ii) K

K = 2XPX2X3

K = (2X3) X (2X3)

K = 6X6

K = 36

(b) Find HCF of 30 and k

HCF = 2xp

= 2x3

= 6

(c) Find the LCM of 30 and k

LCM = 5x2xpx2x3

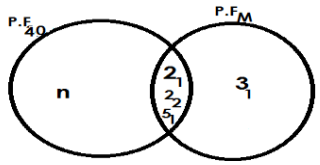
= (5x2) x (3x2x3)

= 10x18

= 180

Activity

1. **Use the Venn diagram below to answer questions that follow.**



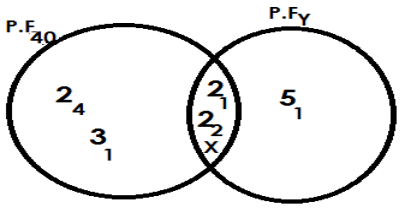
a) Find the value of;

(i) n **(ii) m**

(b) Find the GCF of 40 and m

(c) Find the LCM of 40 and m

2. **Below is a Venn diagram. Use it to answer questions that follow.**



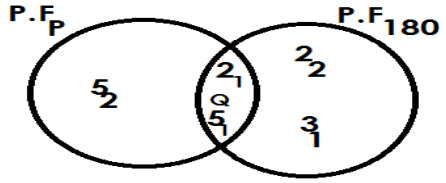
(a) Find the value of

(i) X **(ii) Y**

(b) Find the GCF of 48 and y.

(c) Find the LCM of 48 and y.

3. **Use the Venn diagram below to answer questions that follow.**



(b) Find the value of

(i) Q

(ii) P

(c) Find the GCF of p and 180

(d) Find the LCM of p and 180.

Finding lowest common multiple

1. **Find the LCM of 12 and 18**

2	12	18
2	6	9
3	3	9
3	1	3
	1	1

L x3)

=4x9

=36

2. **Find the LCM of 20 and 30**

2	20	30
2	10	15

LCM= (2x2) x (3x5)

=4x15

=60

3. Find the LCM of 40 and 50

2	40	50
2	20	25
2	10	25
5	5	25
5	1	5
	1	1

LCM= (2x2x2) x (5x5)

=8x25

=200

Activity

1. Find the LCM of the following numbers

(a) 18 and 21 (b) 18 and 24

(c) 16 and 20 (d) 45 and 50

(e) 20, 30 and 40 (f) 18,20 and 30

(g) 9, 12 and 15 (h) 32, 24 and 24

APPLICATION OF LCM

1. Find the smallest number when divided by 30 or 24 there is no remainder.

Solution

Smallest no. =LCM

2	24	30
2	12	15
2	6	15
3	3	15
5	1	5
	1	1

= (2x2x2) x3(3x5)

=8x15

=120

2. Find the least number of books which when divided by either 20 or 36,3 books remain.

Least No.=LCM + remainder

2	20	36
2	10	18
3	5	9
3	5	3
5	5	1
	1	1

= (2x2x3) x (3x5)

=12x15

=180

=180+3

=183 books

3. Two bells ring at interval of 20 minutes and 50 minutes for lower and upper section respectively.

(a) After how long will the bells ring together?

Common minutes=LCM

2	20	50
2	10	25
5	5	25
5	1	5
	1	1

= (2x2) x(5x5) minutes

=4x 25 minutes

=100 minutes

The two bells will ring together after 100 minutes.

(b) If they last rung at 8:00am, when will they ring together again?

60 minutes=1hr

100minutes $= \frac{5}{\cancel{60}\ 3}$

$= \frac{5}{3}$ 1 r2

$= 1\ \frac{2}{3}$ hrs

$\frac{2}{3} \times \frac{20}{60}$

= 40 min

= 1 hour 40 minutes

8:00
+1:40
9:40

They will ring together again at 9:40 am

4. Two soldiers were shooting at a target at an interval of 1 hour and 1 $\frac{1}{2}$ hours respectively.

(a) After how long will they shoot together?

Solution

1 hour=60 minutes 1hr=60 mins

1	2	60	90
	2	30	45
	3	15	45
	3	5	15
	5	5	5
		1	1

Common minutes = LCM

= (2x2) x (3x3) x5

= 4x9x5

= 180 minutes

(b) If they last shoot together at 10:00am, when did they shoot together again?

60minutes=1hr

180 min= $= \frac{\{180\}}{60}$ hrs

10:00
+3:00
13:00

13 00
- 12 00
1:00

They shoot together again at 1p

Activity

1. **Find the smallest number that when divided by 12 or 9 there is no remainder.**
2. **What is the least number of sweet that when shared by either 8 or 20 pupils, no sweet remains?**
3. **Find the smallest number of books that when divided by either 18 or 24, 7 books remain.**
4. **What is the lowest number that when divided by either 11 or 13, 5 becomes the remainder?**
5. **At ABC P/S two bells ring at an interval of 30 and 40 minutes for nursery and primary respectively.**

(a) After how long will the two bells will they ring together again

(b) If they first rung at 8:30 am, when will they ring together again?

6. Three bells for lower, middle and upper section at Kibuku P/S ring at an interval of 40 minutes, 50 minutes and 1 hour respectively.

(a) After how long will they be sounded together again?

(b) If they are first sounded at 8:00am, when are they sounded together again for the second time?

7. At a bus station three buses leave at an interval of 50 minutes, 1 hour and 1 $\frac{1}{2}$ hours respectively.

(a) After how long will the three buses leave at the same time?

(b) If they last left together at 10:00am, at what time will they leave together again?

FINDING GCF/HCF

1. Find the GCF of 8 and 12

CF	NO	NO
2	8	12
2	4	6
	2	3

= 2X2

= 4

2. Find the HCF of 12 and 15

CF	NO	NO
3	12	15
	4	5

HCF=3

3. Find the largest number that divides 6 and 8 without any remainder.

Largest no= HCF

CF	NO	NO
2	6	8
	3	4

HCF=2

4. Find the largest number that divides 18 and 24

Largest no. =HCF

CF	NO	NO
2	18	24
3	9	12
	3	4

= 2x3

= 6

Activity

1. Find the GCF of the following numbers

(a) 15 and 20

(b) 24 and 36

(c) 18 and 21

(d) 9 and 21

(e) 11 and 22

2. **Find the largest number that divides 9 and 24 without any remainder.**

3. **What is the highest number that divides 24 and 30 without a remainder?**

4. **Find the highest number of boys that you can give 21 and 24 pens equally?**

APPLICATION OF LCM AND GCF.

1. **The LCM of two numbers is 18. Find the second number.**

Solution

2nd no. = $\frac{\text{LCM} \times \text{GCF}}{\text{Given number}}$

$= \frac{{}^{2}36 \times 6}{{}_{1}18}$

$= 12$

2. **The LCM of two numbers is 72 and their GCF is 6. If one of the numbers is 24, find the second number**

2nd no. = $\frac{\text{LCM} \times \text{GCF}}{\text{Given number}}$

$= \frac{{}^{18}72 \times {}^{1}6}{{}_{1}24}$

2nd no. = 18

3. **The LCM of two numbers is 24 and their GCF is 2. If one of the numbers is 8. Find the second number**

2nd no.= $\frac{\text{LCM} \times \text{GCF}}{\text{Given number}}$

$= \frac{{}^{3}24 \times 2}{{}_{1}8}$

$= 6$

Activity

1. **The LCM of two numbers is 72 and their GCF is 6. If the first number is 18, find the second number.**
2. **The LCM of two numbers is 120 and their HCF is 10. If the first numbers is 40. Find the second numbers.**
3. **The LCM of two numbers is 225 and their GCF is 5. Find the second number if the first number is 25.**
4. **The LCM of two numbers is 300 and their GCF is 10. Given that the first number is 50, find the second number.**
5. **The LCM of two numbers is 144 and their GCF is 12. Find the second number if the first number is 36.**
6. **The product of two numbers is 216 and their cm is 36. Find their GCF.**
7. **The product of two numbers is 432 and their LCM id 72. Find their GCF.**

Scenarios

Term 2

Primary seven

TOPIC 8 - FRACTIONS;

Session 1

- ✦ Revision on the introduction of fractions
- ✦ Application on fractions
- ✦ Application of fractions involving taps

Methods

- **Guided discovery**
- **Group discussion**
- **Think, pair and share.**
- **explanation**

Lesson activities

- **adding**
- **multiplying**
- **dividing**
- **subtracting**

Content/facts

FRACTIONS

A fraction is a part of a whole

MIXED OPERATIONS ON COMMON FRACTIONS

1. Given addition and subtraction

Note:

Given addition and subtraction, we re-arrange starting with addition

Example I

Simplify: $\frac{1}{5} - 1\frac{1}{2} + 2$ **LCM = 10**

Soln.

$= \frac{1}{5} + \frac{2}{1} - 1\frac{1}{2}$

$= \frac{1}{5} + \frac{2}{1} - \frac{8}{2}$

$= (\frac{1}{5} \times 10) + (\frac{2}{1} \times 10) - (\frac{3}{2} \times 10)$

10

$= (2 + 20) - 15$

10

$= \frac{7}{10}$

Example II

Simplify: $\frac{1}{4} - \frac{2}{3} + \frac{1}{2}$

Soln. **LCM = 12**

$= \frac{1}{4} + \frac{2}{3} - \frac{2}{3}$

$= \frac{1}{4} + -\frac{1}{2} - \frac{2}{3}$

$= \frac{(\frac{1}{4} \times 12) + (\frac{1}{2} \times 12) - (\frac{2}{3} \times 12)}{12}$

$= \frac{(3+6) - 8}{12}$

$= \frac{9-8}{12}$

$= \frac{1}{12}$

Activity:

Simplify the following: -

1. $\frac{1}{2} - \frac{3}{4} + 1\frac{2}{3}$
2. $^2/_3 - 2\ ½ + 3\ ¼$
3. $^1/_5 - ½ + ¾$
4. $^1/_6 - ^2/_3 + ^4/_5$
5. $\frac{1}{9} - \frac{1}{3} + \frac{1}{2}$
6. $\frac{1}{5} - 2\frac{1}{4} + 2\frac{3}{4}$
7. $¼ - 1½ + 2$
8. $^1/_3 - ½ + 2¼$

2. GIVEN OTHER OPERATIONS

Note:

Given other operations, BODMAS is applied.

Examples

Simplify: $1\ ½ \times {}^1/_3 \div {}^4/_9$

Soln.

BODMAS

$1\ ½ \times {}^1/_3 \div {}^4/_9$

$\frac{3}{2}$ x ($\frac{1}{3} \div \frac{4}{9}$)

$\frac{3}{2}$ x $\frac{1}{3}$ x $\frac{9}{4}$

= $\frac{3 \text{ X } 3}{2 \text{ X } 4}$

= $\frac{9}{8}$ 1 rem 1

= $1\frac{1}{8}$

Simplify: $\frac{1}{9} \div \frac{2}{3}$ x ¼ $\div \frac{5}{6}$

Soln.

BODMAS

$(\frac{1}{9} \div \frac{2}{3})$ x $(\frac{1}{4} \div \frac{5}{6})$

$(\frac{1}{9}$ x $\frac{2}{3})$ x ($\frac{1}{4} \div \frac{5}{6}$)

$^{1}/_{9}$ x $^{3}/_{8}$ x ¼ x $^{6}/_{5}$

= $\frac{1}{20}$

Activity:

1. simplify: $\frac{1}{6}$ - $\frac{1}{3}$ x $\frac{1}{4} \div \frac{1}{2} + \frac{2}{3}$
2. $\frac{1}{8} - \frac{1}{4}$ *of* $\frac{2}{3} \div \frac{1}{3} + \frac{3}{4}$
3. $\frac{1}{2}$ - $\frac{1}{5}$ of $\frac{5}{6} + \frac{1}{4}$
4. $\frac{2}{3}$ of $\frac{3}{4} \div \frac{5}{6} + \frac{1}{5}$
5. $\frac{2}{3}$ x $\frac{1}{6} \div \frac{3}{4}$ x $\frac{1}{2}$
6. $\frac{1}{2} - \frac{2}{4} \div \frac{1}{5} + 1\frac{1}{3}$
7. $\frac{1}{3}$ x $\frac{1}{8} + \frac{1}{4} \div$
8. $\frac{3}{4} \div (\frac{1}{2} + \frac{1}{4})$
9. 2 ½ X (½ X $\frac{1}{3}$)
10. $\frac{3}{5} + \frac{1}{3} \div \frac{2}{3}$

CHANGING COMMON FRACTIONS TO DECIMALS

EXAMPLE I

Change $\frac{2}{3}$ to a decimal fraction

Soln.

$$
\begin{array}{r}
0.4 \\
5 \overline{)\,2.0} \\
0 \times 5 \quad -\,0 \\
\hline
20 \\
4 \times 5 \quad -\,20
\end{array}
$$

Example II

Change $\frac{5}{6}$ to a decimal fraction.

Soln.

$\frac{5}{6}$ =

$$
\begin{array}{r}
0.833\ldots \\
6\overline{)\,5.0} \\
0 \times 6 \quad \underline{-\ 0} \\
3.0 \\
8 \times 6 \quad \underline{-\ 48} \\
210 \\
3 \times 6 \quad \underline{-\ 18} \\
20 \\
3 \times 6 \quad \underline{-\ 18} \\
2
\end{array}
$$

ACTIVITY

Change the following common fractions to decimal fractions.

1. ½
2. $\frac{4}{5}$
3. $\frac{3}{4}$
4. $\frac{3}{7}$
5. $\frac{5}{9}$
6. $\frac{14}{15}$
7. $\frac{2}{3}$
8. $\frac{3}{11}$
9. $\frac{8}{15}$
10. $\frac{5}{11}$

RECURRING DECIMALS TO COMMON FRACTIONS

- **Recurring decimals are also called non terminating fractions / decimals.**
- **Vulgar fractions are common fractions i.e., fractions with numerators and denominators.**

Examples

1. Change 0.666 … to common fractions.

Soln.

Let the common fractions be n.

$n = 0.66\ldots$

$n \times 10 = 0.666\ldots \times 10$

$10n - n = 6.666\ldots$

$10n - n = 6.666\ldots$

$-0.666\ldots$

$9n = 6$

$\frac{9n}{9} = \frac{6}{9}$

$n = \frac{2}{3}$

2. Change 0.2424… to a common fraction.

Soln.

Let the common fraction be m.

$M = 0.2424\ldots$

$M \times 100 = 0.2424\ldots \times 100$

$100m = 24.2424\ldots$

$100m - m = 24.2424\ldots$

$= 0.2424$

$99m = 24$

$\frac{99m}{99} = \frac{24}{99} = \frac{8}{33}$

$m = \frac{8}{33}$

3. change 0.5333… to a common fraction.

Soln.

Let the common fraction be K.

$K = 0.5333\ldots$

$K \times 10 = 0.5333\ldots \times 10$

$10K = 5.333\ldots$

$10K \times 10 = 5.333\ldots \times 10$

$100K = 53.333\ldots$

$= 5.333\ldots$

$90K = 48$

$\frac{90K}{90} = \frac{48}{90} = \frac{8}{15}$

$K = \frac{8}{15}$

ACTIVITY:

Change the following recurring decimals to common fractions.

a) 0.3636…
b) 0.6363…
c) 0.4545…
d) 0.2121…
e) 0.5656…
f) 0. 233…
g) 0.333…
h) 0.8333…
i) 0.799…
j) 0.2525…

MIXED OPERATIONS IN DECIMALS

1. Given addition and subtraction

Example

1. work out: 0.6 – 3.6 + 4.4

Soln.

(0.6 + 4.4)-3.6

45.0
-3.6
1.4

0.6
+ 4.4
5.0

2. Work out: 4.6 – 12.1 + 9.44

Soln.

4.60
+ 9.44
14.04 **- 12.1**

14.04
- 12.10
1.94

ACTIVITY:

WORK OUT:

1. 8.2 – 12.6 + 5.4
2. 0.4 – 1.12 + 6.14
3. 6.8 – 8.4 + 7.43
4. 3.4 – 2.6 + 3.2
5. 4.2 – 4.6 + 3.8
6. 1.22 – 2.8 + 3.8
7. 7.2 – 11.4 + 5.4
8. 3.28 – 1.48 + 1.82
9. 14.8 – 2.2 + 1.2
10. 0.58 – 2.8 + 4.42

GIVEN OTHER OPERATIONS

Examples

1. Work out: 1.6 x 0.3 – 0.2 x 0.7

Soln.

BODMAS

(1.6 x 0.3) – (0.2 x 0.7)

$(\frac{16}{10}$ **x** $\frac{3}{10})$ **–** $(\frac{2}{10}$ **x** $\frac{7}{10})$

$(\frac{16}{10}$ **x** $\frac{3}{10})$ **–** $(\frac{2}{10}$ **x** $\frac{7}{10})$

$\frac{4.8}{100}$ **- 0.14**

0.48

- 0.14

0.34

2. Work out: 0.28 – 0.14 ÷ 0.2

Soln.

= 0.28 – (0.14 ÷ 0.2)

= 0.28 – $(\frac{14}{100} \div \frac{2}{10})$

= 0.28 – $(\frac{14}{100} \; x \; \frac{10}{2})$

$= \frac{28}{100}$ **-** $\frac{7}{10}$

$= \frac{(\frac{28}{100} \text{ x } 100) - (\frac{7}{10} \text{ x } 100)}{100}$

$= \frac{28 - 70}{100}$

$= \frac{-42}{100}$

=- 0.42

Activity:

Work out the following:

1. **0.12 x 0.34 + 0.36 ÷ 0.4**
2. **0.12 x 0.3 + (0.36 ÷ 0.4)**
4. **0.28 ÷ 0.14 ÷ 0.2 ÷ 0.4 + 0.5**
5. **0.15 ÷ 0.3 x 0.12**
6. **1.8 + (1.2 ÷ 0.3) + 8**
7. **3.2 x 0.24 + (3.2 ÷ 0.8)**
8. **1.4 x 0.2 – 0.2 x 0.3**

DECIMALISATION

Examples

1. Work out: $\frac{0.49 \text{ x } 0.28}{0.9 \text{ x } 0.7}$

Soln.

= (0.49 x 0.28) ÷ (0.9 x 0.7)

= ($\frac{49}{100}$ x $\frac{28}{100}$) ÷ ($\frac{9}{10}$ x $\frac{7}{10}$)

= $\frac{49}{100}$ x $\frac{28}{100}$ x $\frac{10}{7}$ x $\frac{10}{7}$

= $\frac{97 \text{ x } 4}{100}$

= $\frac{4928}{100}$

=0.28

2. Work out: $\frac{7.4-3.2}{0.2 \text{ x } 0.3}$

Soln.

(7.4 – 3.2) ÷ (0.2 x 0.3)

7.4 ÷ (0.2 x 0.3)

-3.2 ÷ (0.2 x 0.3)

4.2

$\frac{42}{10}$ ÷ ($\frac{2}{10}$ x $\frac{3}{10}$)

$\frac{42}{10}$ x $\frac{100}{6}$

70

2. Simplify: $\frac{8.4 \text{ x } 0.2}{0.14-0.02}$

Soln.

= (8.4 x 0.2) ÷ (0.14 – 0.02)

= (8.4 x 0.2) ÷ 0.14

- 0.02

0.12

= ($\frac{84}{100}$ x $\frac{2}{10}$) ÷ $\frac{12}{100}$

= $\frac{84 \text{ x } 2}{10 \text{ x } 10}$ x $\frac{100}{12}$

=14

Activity:

Work out the following:

1. $\frac{0.14 \text{ x } 0.8}{0.4}$

2. $\frac{0.63 \; x \; 0.49}{0.9 \; x \; 0.7}$

3. $\frac{0.42 \text{ x } 0.36}{0.7 \text{ x } 0.12}$

4. $\frac{0.49-0.04}{0.3+0.6}$

5. $\frac{0.48-0.12}{0.4+0.2}$

6. $\frac{8.4 \text{ X } 0.02}{0.14-0.02}$

7. $\frac{0.36 \text{ X } 0.2}{0.9-0.6}$

8. $\frac{4.5+3}{0.5 \text{ X } 2.5}$

9. $\frac{0.048 \text{ x } 0.2}{0.12 \text{ x } 0.1}$

9. $\frac{4.8 \text{ x } 0.2}{0.14-0.02}$

APPLICATION OF FRACTIONS

Examples

1. In a class of 40 pupils, ¼ are boys and the rest are girls. Find the fraction of girls.

Soln.

Fraction of boys = ¼

Fraction of girls = 1 – ¼

$= \frac{4}{4} - \frac{1}{4}$

$= \frac{4}{4} - \frac{1}{4}$

$= \frac{4-1}{4}$

$= \frac{3}{4}$

b) How many more girls than boys are in the class?

Soln.

No. of girls	No. of boys	More
$\frac{3}{4}$ x 40 3 x 10 30	$\frac{1}{4}$ x 40 1 x 10 10	30 - 10 20

2. After covering 40km, Alex's vehicle got a puncture this was $\frac{2}{5}$ of the whole journey. How long was the whole journey?

Soln.

Let the journey be K.

$\frac{2}{5}$ x k = 40km

$5 \text{ x } \frac{2}{5} \text{ k} =$ 40km x 5

$\frac{2k}{2} = \frac{200 \text{ km}}{2}$

= 100km

b) Find the remaining distance.

Soln.

Remaining fraction

1 - $\frac{2}{5}$

$\frac{5}{5} - \frac{2}{5} = \frac{3}{5}$

Remaining distance

= $\frac{3}{5}$ **x 100 km**

=3 x 20 km

=60 km

3. $\frac{2}{3}$ **of my salary is equal to** $\frac{3}{4}$ **of my uncle's salary. If my uncle's salary is sh. 100,000. How much is my salary?**

Soln.

Let my salary be n.

$\frac{2}{3}$ **of n =** $\frac{3}{4}$ **of sh. 100,000.**

$\frac{2n}{3}$ **= sh. 75000 x 3**

$\frac{2n}{2}$ **= sh.** $\frac{75000\ x\ 3}{2}$

n = sh. 112500

Activity:

1. In a class, $\frac{2}{3}$ **are girls and the rest are boys. There are 6 girls in the class. How many pupils are in the class?**

2. In a class, $\frac{3}{8}$ **are boys and the rest are girls. If there are 15 girls, how many boys are in the class?**

3. Othieno spent $\frac{3}{5}$ **of his income on food and he saved sh. 20,000. How much did he spend?**

4. Scala covered $\frac{3}{7}$ **of the journey by bus and he walked 8km. find the distance covered by bus?**

5. $\frac{1}{3}$ **of Okello's salary is equal to** $\frac{2}{5}$ **of John's salary. If John's salary is sh. 50,000, How much is Okello's salary?**

6. After covering $\frac{3}{4}$ **of the journey, Peter still had 40km to cover. How long was the whole journey?**

7. A woman sold $\frac{7}{10}$ **of her apples and she remained with 30 apples, how many apples did she have at first?**

8. $\frac{4}{11}$ **of the school team like red shorts and the rest like blue shorts. If 14 players like blue shorts, how many players are in a school?**

10. In a school, $\frac{5}{12}$ **of the workers can speak Kiswahili and the rest speak English. If 28 workers speak English, how many workers are there in the school altogether?**

FININD REMAINING FRACTIONS GIVEN TWO OR MORE FRACTIONS

Examples

1. A man spent ¼ of his income on food, $\frac{1}{3}$ **on fees and he saved the rest.**

a) Find the fraction saved.

Soln.

Fraction saved = $1 - (\frac{1}{4} + \frac{1}{3})$

$= 1 - (\frac{3+4}{12})$

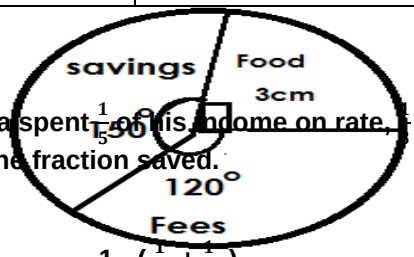
$= 1 - \frac{7}{12}$

$= \frac{12}{12} - \frac{7}{12}$

$= \frac{5}{12}$

b) Show the above information on a pie chart using a radius of 3cm.

Food	Fees	Savings
$\frac{1}{4}$ x 360° 90°	$\frac{1}{3}$ x 360° 120°	$\frac{5}{12}$ x 360° 150°



2. Mapala spent $\frac{1}{5}$ of his income on rate, $\frac{1}{3}$ on fees and saved the rest.

a) Find the fraction saved.

Soln.

$1 - (\frac{1}{5} + \frac{1}{3})$

$1 - (\frac{3+5}{15})$

$1 - \frac{8}{15}$

$\frac{15}{15} - \frac{8}{15}$

$\frac{7}{15}$

b) If he saved sh. 28,000, how much did he spend on rent?

Soln.

Let the income be m.

$\frac{7}{15}$ x m = sh. 28,000

5 x $\frac{7}{5}$ =sh. 28,000 x 15

$\frac{7}{7}$ m = sh. $\frac{28{,}000\ x\ 15}{7}$

m = $\frac{1}{5}$ x sh. 60,000

=sh. 12,000

3. $\frac{2}{3}$ of the farmers grow maize, $\frac{1}{6}$ grow rice and the rest grow beans. Find the number of farmers on the village if 48 farmers grow beans.

Soln.

Fraction of beans = $1 - (\frac{2}{3} + \frac{1}{6})$

$= 1 - (\frac{4+1}{6})$

$= 1 - \frac{5}{6}$

$= \frac{6}{6} - \frac{5}{6}$

$= \frac{1}{6}$

Let the no. of farmers be m

$\frac{1}{6}$ x m= 48

$\frac{m}{6} = \frac{48}{6}$

$6 \times \frac{m}{6} = 48 \times 6$

m = 2888 farmers

4. Konde spent 25% of his income on food $\frac{1}{5}$ on rent and the rest on transport. If he spent shs. 44,000 on transport, how much did he spend on food?

Soln.

Fraction on transport =1 – (25% + $\frac{1}{5}$)

$=1 - (\frac{25}{100} + \frac{1}{5})$

$=1 - (\frac{1}{4} + \frac{1}{5})$

$=1 - (\frac{1+4}{25})$

$=1 - \frac{9}{20}$

$=\frac{20}{20} - \frac{9}{20}$

$=\frac{11}{20}$

Let the income be K.

$\frac{11}{20} \times k$ = sh. 44000

$\frac{11k}{20}$ = sh. 44000

$20 \times \frac{11k}{20} = 44{,}000 \times 20$

$\frac{11k}{20} = sh. \frac{44000 \times 20}{11}$

K= sh. 80,000

Food= 25% of sh. 80,000

$= \frac{25}{100} \times$ sh. 80,000

= 25 x sh. 800

= sh. 20,000

Activity:

1. Mandevo covered $\frac{1}{6}$ of the journey by his bus, $\frac{1}{4}$ by taxi and he walked 35km. how long was the whole journey?

b) Find the distance covered by bus.

c) Show the above information on a pie chart of radius 3cm.

2. Kato spent ¼ of his salary on medical $\frac{1}{3}$ on rent, $\frac{1}{6}$ on fees and the rest on food. If he spent sh. 36,000, how much did he spend on medical?

b) Show the above information on a pie chart of radius 4cm.

3. SSekato spent $\frac{1}{5}$ of his salary on food $\frac{2}{3}$ on rent and saved the rest. If he saved 40,000, calculate his income.

4. Peter read $\frac{2}{5}$ of a book on Monday, $\frac{1}{4}$ on Tuesday and the rest on Wednesday. If he read 21 papers on Wednesday, how many papers make up a book?

5. $\frac{2}{3}$ of the farmers grow maize, $\frac{1}{6}$ grow rice and the rest grow beans. Find the number of farmers on the village if 48 farmers grow beans.

6. Okello spent 40% of his income on food, ¼ on fees and the rest on medical. If he spent sh.21,000 on medical. How much was his income?

7. Kapsico spent 20% of his income on food, ¼ on fees, $\frac{1}{10}$ on rent and he saved sh. 28,000. How much was his income?

APPLICATION ON FRACTIONS INVOLVING A REMAINDER

Examples

1. Wandera spent $\frac{1}{6}$ of his income on food, $\frac{2}{5}$ of the remainder on fees and saved the rest. Find the fraction he saved.

Soln.

Food	Remainder	Fees	saved
$\frac{1}{6}$	$1 - \frac{1}{6}$ $\frac{6}{6} - \frac{1}{6}$ $\frac{5}{6}$	$\frac{2}{5}$ of $\frac{5}{6}$ $\frac{2}{8} \times \frac{5}{6}$ $\frac{1}{3}$	$\frac{5}{6} - \frac{1}{3}$ $\frac{5-2}{6}$ $\frac{3}{6}$ $\frac{1}{2}$

b) If he saved sh. 60,000, how much did he spend on food.

Soln.

Let his income be K.

$\frac{1}{2} \times k$ = Sh. 60,000

$2 \times \frac{k}{2}$ = sh. 60,000 x 2

K = sh. 120,000

Food

$= \frac{1}{6}$ x sh. 120,000

=sh. 20,000

3. A man spent ¼ of his income on food, $\frac{1}{3}$ of the remainder on fees and he saves the rest. If he saved sh. 80,000 more than what he spent on food, find his income.

Soln.

Food	Rem	Fees	Saved
$\frac{1}{4}$	$1 - \frac{1}{4}$ $\frac{4}{4} - \frac{1}{4}$ $\frac{3}{4}$	$\frac{1}{3}$ of rem $\frac{1}{3} \times \frac{1}{4}$ $\frac{1}{4}$	$\frac{3}{4} - \frac{1}{4}$ $\frac{3-1}{4}$ $\frac{2}{4}$ $\frac{1}{2}$

Difference in fractions

$$= \frac{1}{2} - \frac{1}{4}$$

$$= \frac{2-1}{4}$$

$$= \frac{1}{4}$$

Let his income be K.

$\frac{1}{4}$ **x K = sh. 80,000**

$\frac{k}{4}$ **= sh. 80,000**

$\frac{k}{4}$ **x 4 = sh. 80,000 x 4**

K = sh. 320,000

b) How much did he save?

<u>Soln.</u>

Savings = ½ x sh. 320,000

=sh. 160,000

Matama spent $\frac{1}{5}$ of her income on Rent, 25% of the remainder on medical and she saved the rest.

a) Find the fraction saved.

Rent	Rem	Medical	Saved
$\frac{1}{5}$	$1 - \frac{1}{5}$ $\frac{5}{5} - \frac{1}{5}$ $\frac{4}{5}$	25% of rem $25\% \times \frac{4}{5}$ $\frac{25}{100} \times \frac{4}{5}$ $\frac{1}{5}$	$\frac{4}{5} - \frac{1}{5}$ $\frac{3}{5}$

If she spent sh. 20,000 on medical, how much did she save?

Soln.

Let her income be K.

$\frac{1}{5}$ **x k** **=** **sh. 20,000**

$\frac{k}{5}$ **=** **sh. 20,000**

$\frac{k}{5}$ **=** **sh. 20,000 x 5**

K **=** **sh. 100,000**

Medical

= $\frac{3}{5}$ **x sh. 100,000**

= **sh. 60,000**

Example IV

Owino spent $\frac{1}{3}$ **of his income on food, ¼ on fees. ¾ of the remainder on transport and he saved the rest. Find the faction saved.**

Soln

Feed	Fees	Rem	Transport	Saved
$\frac{1}{3}$	$\frac{1}{4}$	$1 - (\frac{1}{3} + \frac{1}{4})$ $1 - (\frac{4+3}{12})$ $1 - \frac{7}{12}$ $\frac{12}{12} - \frac{7}{12}$ $\frac{5}{12}$	$\frac{3}{4}$ of Rem $\frac{1}{4} \times \frac{5}{12}$ $\frac{5}{16}$	$\frac{5}{12} - \frac{5}{16}$ $\frac{20-15}{48}$ $\frac{5}{48}$

b) If he saved sh. 20,000, find his income.
Soln.

Let his income be p.

$\frac{5}{48}$ **x p** **=** **sh. 20,000**

$\frac{5p}{48}$ **=** **sh. 20,000**

$\frac{5p}{48}$ **= sh. 20,000 x 48**

5p **=** **sh. 20,000 x 48**

$\frac{5p}{5}$ **=** $\frac{sh.20,000 \times 48}{5}$

P **=** **Sh. 192,00**

Activity:

1. Akello covered $\frac{1}{8}$ of the journey by bus, $\frac{5}{7}$ of the remainder by a taxi and the rest by train. Find the fraction covered by train.

b) If the covered 20km by train, how long was the whole journey?

2. Opio spent $\frac{1}{3}$ of his salary on medical, $\frac{1}{4}$ of the remainder on rent and the rest on food. Find the fraction spent on fees.

b) If he spent sh. 40,000 on fees than on medical, how much is his salary?

4. Martha spent 20% of her income on food, $\frac{1}{3}$ of the remainder on fees and she saved sh. 20,000. Find her income.

5. Ssemutala spent 40% of his income on transport, $\frac{1}{4}$ of the remainder on medical and the rest on clothes. If he spent sh. 180,000 on clothes, how much did he spend on transport?

5. Pison spent $\frac{1}{3}$ of his income on food, $\frac{1}{5}$ on rent, $\frac{2}{3}$ of the remainder on fees and the rest on medical, find the fraction spent on medical.

b) If he spent sh. 35,000 on medical, how much did he spend on rent?

6. In a school, $\frac{1}{2}$ of the pupils eat meat, $\frac{1}{3}$ eat fish and $\frac{1}{4}$ of the remainder eat beans while the rest eat greens, if 30 pupils eat greens. How many pupils are in the class?

MORE ABOUT APPLICATION OF FRACTIONS

EXAMPLE I

A tank of water was ½ full, when 60 litres were added it became ¾ full. How many litres of water can the tank hold when completely full?

Soln.

Fraction added

= New - old

= $\frac{3}{4} - \frac{1}{2}$

= $\frac{3-2}{4}$

= $\frac{1}{4}$

Let the capacity of the tank be m

$\frac{1}{4}$ x m = 60 L

$\frac{m}{4}$ = 60 L

$\frac{m}{4}$ ×4 = 60L ×4

M = 240 L

b) How many litres of water were in the tank?

Soln.

$= \frac{1}{2}$ **x 240L**

=120 L

Example II

A tank was 2/3 full of water, when 200L of water were used, it became 5/12 full. Find its capacity.

Soln.

Fraction used

= **old – New**

= $\frac{2}{3} - \frac{2}{12}$

= $\frac{8-5}{12}$

= $\frac{3}{12}$

= $\frac{1}{4}$

Let its capacity be P.

$\frac{1}{4}$ **x p = 200L**

$\frac{p}{4}$ **= 200L**

$\frac{p}{4}$ **x 4 = 200L x 4**

P = 800L

Example III

A tank was $\frac{2}{3}$ **full of water, when** $\frac{1}{4}$ **of water in the tank was used, 2500L remained. Find the capacity of the tank.**

Soln.

Fraction used

= $\frac{1}{4}$ **x** $\frac{2}{3}$

= $\frac{1}{6}$

Remaining fraction.

= $\frac{2}{3} - \frac{1}{6}$

= $\frac{4-1}{6}$

= $\frac{3}{6}$

= $\frac{1}{2}$

Let the capacity be m.

½ x m = 2500L

= $\frac{m}{2}$**=2500L**

$\frac{m}{2}$ **x 2 = 2500L x 2**

m = 5000 L

Activity:

1. A tank was $\frac{1}{5}$ **full of water, when 8L of water were added it became** $\frac{1}{3}$ **full.**

a) How many litres of water can the tank hold when completely full of water?

b) How many litres of water were in the tank?

2. A jerry can was ¼ full of water, when 15L were added, it became 2/3 full. Find its capacity.

b) How many litres of water are in the tank?

3. A tank was $\frac{1}{3}$ **full of water, when 60L were drawn, the tank became** $\frac{2}{15}$ **full, find its capacity.**

4. A Jerry can was ½ full of water, when 10 L were used for cooking, the jerry can become $\frac{5}{14}$ **full, find its capacity.**

5. A petrol tank was $\frac{2}{3}$ **full, when ¼ of it was used to cover a certain distance, 120 L remained. Find the capacity of the tank.**

b) If a car consumes 5 litres of petrol to cover a distance of 19km, find the distance covered.

6. At a wedding party, a water tank was $\frac{3}{8}$ **full, when** $\frac{2}{3}$ **of it was used for cooking, 40L remained, if 10L were used to prepare 15kg of rice, how many kgs of rice were cooked?**

APPLICATION OF FRACTIONS INVOLVING OTHER PARTS OF FRACTIONS

In a class of 120 pupils, ¼ are girls and the rest are boys.

a) Find number of boys in the class.

Soln.

Fraction of boys= $1 - \frac{1}{4}$

$= \frac{4}{4} - \frac{1}{4}$

$= \frac{3}{4}$

No of boys = $\frac{3}{4}$ **x 120**

= 90 boys

b) 2/3 of the boys and 1/5 of the girls are in yellow color. How many pupils are in yellow color

No. of boys	Boys in yellow	No. of girls	Girls in yellow	Total
¾ x 120	2/3 x 90	¼ x 120	1/5 x 30	60
				+ 6
90	60	30	6	66

Activity:

1. In a school of 2000 pupils, $\frac{2}{5}$ **are girls and the rest are boys, ½ of the girls and** $\frac{1}{3}$ **of the boys are in upper. Find the number of pupils in lower.**

2. In a village of 2400 people, the ratio of female to male is 7:3, find the number of females in the village.

b) 40% of the females and 30% of the males are children, how many adults are in the village?

c) In each adult pays tax of sh. 20,000/=, how much will be collected by local council?

3. A certain country in Uganda has a population of 300,000 people, of whom. ½ are females, $\frac{3}{5}$**of the females are girls and**$\frac{2}{3}$ **of the males are boys. Find the number of children in the country.**

4. Out of 120 students who sat for the test, 1/3 were girls and the rest were boys.

a) How many girls took the test?

b) If 40% of the boys failed and 55% of the girls passed, how many students passed the test?

APPLICATION OF FRACTIONS INVOLVING TAPS THAT REQUIRE TO ADD.

Examples

3. Tap A takes 5min to fill a tank and tap B takes 15 min to fill the same tank. How long will it take both tapes to fill the tank it opened at the same time?

Tap A	Tap B	Both
5min = 1tank 1min = $\frac{1}{5}$tank	15min = 1tank 1min = $\frac{1}{15}$ tank	$\frac{1}{5}+\frac{1}{15}$ $\frac{3+1}{15}$ $\frac{4}{15}$ tank

$\frac{4}{15}$ tank = 1min

1 tank = (1 ÷ $\frac{4}{15}$)min

= 1 x $\frac{15}{4}$

= $\frac{15}{4}$ 3 rem 3

= 3 $\frac{3}{4}$ minutes

Tap X takes 9 min to fill the tank, tap Y take 12 min to fill and tap Z takes 18 min to fill the same tank. If all tapes are opened at the same time, how long will it take the tank to get filled?

Tap X	Tap Y	Tap Z	All taps
9 min = 1tank 1 min = $\frac{1}{9}$tank	12 min = 1tank 1min = $\frac{1}{12}$tank	18 min = 1tank 1min = $\frac{1}{18}$ tank	$\frac{1}{9}+\frac{1}{12}+\frac{1}{18}$ $\frac{4+3+2}{36}$ = $\frac{9}{36}$ = $\frac{1}{4}$ tank

¼ tank = 1 min

1 tank = (1 ÷ ¼) min

= (1 x $\frac{4}{1}$) min

= 4 min

b) 15 litres of water are filled in the tank in every 20 seconds, how many litres of water can the tank hold when completely full?

1 min = 60 sec

4 min = (4 x 60) sec

= 240 sec

20 sec = 15L

1 sec = $\frac{15}{20}$ L

240 sec = $\frac{15}{20}$ x 240L

=180 litres

c) If the tank was ¾ full of water, how long did the taps take to fill the tank.

Fraction to be filled. =1 - $\frac{3}{4}$

=$\frac{4}{4}$ - $\frac{3}{4}$

=$\frac{1}{4}$

No. of min

= ($\frac{1}{4}$ x 4) minutes

= 1 minute

Activity:

1. tap A takes 24 minutes to fill the tank and tap B takes 8 min to fill the same tank. How long will it take the same tank to get filled if both taps are left open at the same time?

b) One day the tank was $\frac{1}{3}$full, how long did the tank takes to be filled?

2. Martha takes 10 min to sweep the class, while Peter takes 15 min to sweep the same class. How long will both of them take to sweep the classroom if they started at the same time?

3. Tap A takes 8 min to fill the tank, tap B takes 6 minutes and tap C takes 24 minutes to fill the same tank. How long will it take the tank to get filled if they are opened at the same time?

b) If 16 litres of water are poured in the tank in 5 seconds, find the capacity of the tank.

4. Jane takes 12 minutes to clean the compound, Musa takes 8 minutes and Wasswa takes 24 minutes to clean the same compound, how long will they take to clean the class if they started cleaning at the same time?

APPLICATION OF FRACTIONS ON TAPS INVOLVING SUBTRACTION.

Example I

Tap F and tap A are connected on a water tank, tap F call fill the tank in 2 hours while tap A can empty the tank in 3 hours. One day the tank was$\frac{1}{5}$ full of water. How long did it take the tank to get fill if both taps were opened at the same time?

TAP F	TAP A	ALL TAPS
2 hours = 1tank 1 hr = ½ tank	3 hours = 1tank 1 hr = $\frac{1}{3}$ tank	$\frac{1}{2}$ - $\frac{1}{3}$ $\frac{3-2}{6}$ $\frac{1}{6}$

$\frac{1}{6}$ tank = 1 hr

1 tank = (1 ÷ $\frac{1}{6}$) hours

= (1 x $\frac{6}{1}$)hr

=6 hours

Fraction to be filled = 1 - $\frac{1}{5}$

= $\frac{1}{6}$ - $\frac{1}{5}$

=$\frac{4}{5}$

Time take = ($\frac{4}{5}$ x 6) minutes

= $\frac{24}{5}$ 4 rem 4

= 4 $\frac{4}{5}$ hours.

Example II

Tap A takes 12 min to fill a tank and tap B takes 9 minutes to fill the same tank while tap C takes 18 minutes to empty the tank. How long will it take the taps to fill the tank if all were left open at the same time?

Soln.

Tap A	Tap B	Tap C	ALL TAPS

12 min = 1 1 min = $\frac{1}{12}$	9 min = 1 1 min = $\frac{1}{9}$	18 min = 1 1 min = $\frac{1}{18}$	$(\frac{1}{12} + \frac{1}{9}) - \frac{1}{18}$ $(\frac{3+3}{36})$ - 2 $\frac{7-2}{36}$ $\frac{5}{36}$

$\frac{5}{36}$ tank = 1 min

1 tank = (1 $\frac{5}{36}$) min.

= $\frac{1 \times 36}{5}$ min

= $\frac{36}{5}$ min

= 7 $\frac{1}{5}$ minutes

Activity

1. Tap A takes 9 min to fill the tank and tap B takes 12 min to empty the tank one day the tank was $\frac{1}{12}$ full, how long did it take the tank to get filled with water?
2. Tap M takes 8 minutes to fill the tank and tap B draws water from the same tank in 12 minutes. How long will it take the tank to get filled with water if both taps are left open at the same time?
3. A water source has two taps. Tap B fills a jerrycan in 7 minutes and tap K empties the same jerrycan in 14 minutes. How long will it take the jerrycan to get filled which was $\frac{1}{7}$ full of water?

b) 20 litres of water were poured in the tank in 5 seconds. Find the capacity of the tank?

4. Tap A takes 12 mins to fill the tank and tap B takes 8 minutes to fill the same tank while tap C takes 24 min to draw water from the tank. How long will it take the tank to be filled if all taps left opened?
5. Tap K takes 6 min to fill the tank, and tap C takes 8 minutes to fill the same tank while tap F takes 4 minutes to empty the same tank. How long will it take to fill the tank if all taps are left open?

APPLICATION OF FRACTIONS ON TAPS THAT INVOLVES UNKNOWNS

Tap A takes K minutes to fill the tank and tap B takes 12 minutes to fill the same tank if both taps are left open, they take 4 minutes to fill the tank. Find the value of K.

Soln.

Tap A	Tap B	Both taps
K min = 1tank 1 min = $\frac{1}{k}$ tank	12 min = 1tank 1 min = ½ tank	4 min = 1tank 1 min = ¼ tank

$\frac{1}{k} + \frac{1}{12} = \frac{1}{4}$

$\frac{1}{k} + \frac{1}{12} - \frac{1}{12} = \frac{1}{4} - \frac{1}{12}$

$\frac{1}{k} = \frac{3-1}{12}$

$\frac{1}{k} = \frac{2}{12}$

$\frac{1}{k} = \frac{1}{6}$

1 x k = 1 x 6
k = 6 min

Example II

Tap F takes y minutes to fill the tank and tap E takes 3 minutes to empty the same tank. If both taps are left open, they take 6 minutes to fill the tank. Find the value of F.

Soln.

Tap F	Tap K	Both
Y min = 1tank 1 min = $\frac{1}{y}$ tank	3 min = 1tank 1min = $\frac{1}{3}$ tank	6 min = 1tank 1 min = $\frac{1}{6}$ tank

$\frac{1}{y} - \frac{1}{3} = \frac{1}{6}$

$\frac{1}{y} - \frac{1}{3} + \frac{1}{3} = \frac{1}{6} + \frac{1}{3}$

$\frac{1}{y} = \frac{1+2}{6}$

$\frac{1}{y} = \frac{3}{6}$

$\frac{1}{y} = \frac{1}{2}$

1 x y = 1 x 2
Y = 2 minutes

Activity:

1. Tap A takes 10 minutes to fill a tank and tap B takes Y min to fill the same tank. If both taps are left open, it takes 6 minutes for the tank to get filled. Find the value of y.

4. **Tap A takes X minutes to fill a tank and tap B takes 6 minutes to fill the same tank. If both taps are left open at the same time, it takes 2 minutes for the tank to get filled. Find the value of x.**
5. **Tap Q fill a tank in 8 minutes and tap W takes k minutes to empty Tank. If both taps are left open, they take 24 minutes to fill the tank. Find the value of K.**
6. **It takes y minutes for the tank to be filled by tap A and tap B takes 8 minutes to empty the same tank. If both taps take 24 minutes to fill the tank find the value of y.**

Scenarios

Session 2

- ✦ Introduction to ratios and proportions
- ✦ Application of fractions on percentages
- ✦ Application on simple interest

Methods

- **Guided discovery**
- **Group discussion**
- **Think, pair and share.**
- **explanation**

Lesson activities

- **adding**
- **multiplying**
- **dividing**
- **subtracting**

Content/facts

PROPORTIONS

Note:

1. **Proportion is the way of comparing two different things in terms of quantity.**
2. **Proportions is divided into three; -**
   - **Direct proportion**
   - **Indirect proportion**
   - **Constant proportion**

DIRECT PROPORTIONS

Note:

- **In direct proportions, formation of a scale is important.**
- **What we are looking for must be on right and what we have on left.**
- **The equivalence of one (1) is got by dividing what is on right by what is on left (1 - $\frac{R}{L}$)**

Examples:

1. **Three books costs sh. 1200. Find the cost of 5 such similar books.**

Soln.

3 books = sh. 1200

1 book = ($\frac{sh.1200}{3}$)

5 books = sh ($\frac{1200 \times 5}{3}$)

= Sh. 2,000.

2. **At a shop, 3 tomatoes cost sh. 600. How many tomatoes will one get from shs. 3000?**

Soln.

Sh. 600 = 3 tomatoes.

Sh. 1 = ($\frac{3}{600}$) tomatoes.

Sh. 3000 = ($\frac{3}{600}$ x 3000) tomatoes

= 15 tomatoes.

7. **A vehicle consumes 7 litres to cover a distance of 15 km. how many litres of petrol will it consume to cover a distance of 135km.**

Soln.

15km = 7 litres

1 km = ($\frac{7}{15}$) litres

135 km = ($\frac{7}{15}$ x 135) litres.

= (7 x 9) litres.

= 63 litres

Activity:

1. **Five books cost sh. 5000. Find the cost of 12 such similar books.**
2. **Mandela's car consumes 4 litres of petrol every 7km. how many litres of petrol will it consume to cover a distance of 35km?**
3. **6 dresses cost sh. 18,000. If Ouma had sh. 12,000, how many dresses did he get?**
4. **A taxi carries 14 passenger per trip. How many trips will it make to carry 392 passengers?**
5. **Seven shirts cost 49000. Find the cost of 3 such similar shirts.**
6. **3 pencils cost sh. 2,100, how many pencils will I get if I have sh. 84,000?**
7. **4 shirts cost sh. 24,000. Find the cost of 9 similar shirts.**
8. **James's car consumes 6 litres of petrol to cover a distance of 20km. how many litres of petrol will be needed to cover a distance of 240 km?**
9. **Opio bought 4 shirts at sh. 60,000, how many shirts will James get if he moved with sh. 180,000?**
10. **Two pens cost sh. 2,000. Find the cost of 8 similar pens.**

EXCHANGE RATES

An exchange rate is the value a country's currency in another country.

Example I

1. Given exchange rates at the forex bureau.

1 dollar ($) = Ug. Sh. 3000.

1 pound (£) = Ug. Sh. 4,000.

K sh. 1 = Ug. Sh. 25

a) A trader had 120 pounds, how much in Uganda shillings did he get at the bureau?

Soln.

1 pound = Ug. Sh. 4,000.

120 pounds = Ug. Sh. 4,000 x 120

= Ug. Sh. 480,000

b) Pande had Ug. Sh. 4,800,000, how many dollars did he get at the forex bureau?

Soln.

Ug. Sh. 3,000 = 1 dollar

Ug. Sh.1 = ($\frac{1}{3,000}$) dollar

Ug. Sh. 4,800,000 = ($\frac{1}{3,000}$ x 4,800,000) dollars.

= 1,600 dollars

c) Pomitar had 120 pounds. How many dollars did he get at the forex Bureau?

Soln.

1 pound = Ug. Sh. 4,000

120 pounds = Ug. Sh. 4,000 x 120

= Ug. Sh. 480,000

Ug. Sh. 3,000 = 1 dollar

Ug. Sh. 1 = ($\frac{1}{3,000}$) dollars

Ug. Sh. 480,000 = $\frac{1}{3,000}$ x 480,000

= 160 dollars

Example 2

The table below shows, the buying and selling rates at quality forex bureau.

Note:

Given the foreign currency, use buying rate and local currency use selling rates.

Currency	Buying	Selling
K Sh. 1	Ug. Sh. 20	Ug. Sh. 30
1 pound	Ug. Sh. 3000	Ug. Sh. 4,000
1 dollar	Ug. Sh. 2,000	Ug. Sh. 2500

a) A trader has Ug. Sh. 120,000. How much in Kenya shillings did he get at the Bureau?

Soln.

Ug. Sh. 30 = K sh. 1

Ug. Sh. 1 = ($\frac{1}{30}$)

Ug. Sh.120,000 = ($\frac{1}{30}$ x120,000) K.sh.

=K. sh. 4,000

b) A trader had 600 pounds. How much in Uganda shillings did he get?

Soln.

1 pound = Ug. Sh. 3,000

600 pounds = Ug. Sh. (3,000 x 600)

= Ug. Sh. 1,800,000

c) Wandudu had 180 dollars. How much in Kenya shillings did he get at the forex bureau?

Soln.

1 dollar= Ug. Sh. 2,000

180 dollars = Ug. Sh. 2000 x 180

= Ug. Sh. 360,000

Ug. Sh. 30 = K sh. 1.

Ug. Sh. 1 = K sh. ($\frac{1}{30}$)

Ug. Sh. 360,000 = $\frac{1}{30}$ x 360,000

= K sh. 12,000

Activity:

1. Below is a table showing exchange rate of different currencies at the forex Bureau.

1K sh. = Ug. Sh. 30

1 pound = Ug. Sh. 4500

1 dollar= Ug. Sh. 2500

a) A trader entered Uganda with 480 dollars. How much in K sh. Did he get?

b) A tourist had 46 pounds. How much in Uganda shillings did he get at the forex Bureau.

c) Kiberu had Ug. Sh. 1,350,000. How many pounds did he get at the Bureau?

2. Given the exchange rates below:

1 dollar ($) = Ug. Sh. 3000

K sh. 1 = Ug. Sh. 25

a) Pokopoko entered Uganda with 600 dollars. How much in Kenya shillings did he have?

3. At Kamukamu forex Bureau, the exchange rates are as follow: -

K sh. 1 = Ug. Sh. 25

1 dollar= Ug. Sh. 3,000

1 pound = Ug. Sh. 4,000

a) Musa entered Uganda with k sh. 2500. How much money did he get at the Forex Bureau?

b) Kasambo had Ug. Sh. 1200,000. How many dollars did he get at the forex Bureau?

c) Kintu entered Uganda with 360 pounds. How much in Kenya shillings did he get at the forex Bureau?

4. Given the exchange rates at the Forex Bureau.

1 US dollar = U. sh. 2500

1K sh. = Ug sh. 24

a) How much money in Uganda shillings will I get from the Bureau if I have 120 dollars?

b) Okello had 72,000 U.sh. how much in k sh. did he have?

c) If a trader had K sh. 12,000, how many dollars did he get from the Forex Bureau?

7. The table below shows the buying and selling rates of different currencies at kamu forex Bureau.

Currency	Buying	Selling
K sh.	Ug. Sh. 20	U sh. 30
US dollar ($)	Ug. Sh. 2000	Ug. Sh. 2500
R. Franc	Ug. Sh. 10	Ug. Sh. 15
Pound (£)	U sh. 400	Ug. Sh. 4200
T. Z sh.	U sh. 40	Ug. Sh. 42

a) Wandela had 120 dollars. How much in Ug. Shillings did he have?

b) A trader entered Uganda with 25 pounds. How much in Uganda shillings did he get?

c) Mr. Titto had Ug. Sh. 200,000, how much in dollars did he have?

d) Soweto had Tz.sh. 180,000. How much in Kenya shillings did he get?

INDIRECT PROPORTION (INVERSE PROPORTION)

Note:

In indirect proportion, we get the equivalence of one (1) by multiplying.

Examples

1. 3 men can do a piece of work in 24 minutes. How long will 6 men take to do the same piece of work?

Soln.

3 men = 24 min.

1 man = (24 x 3) min

6 men = ($\frac{24 \times 3}{6}$) min

= 4 x 3 min

= 12 minutes

2. Six boys can slash a compound in 5 hours. How many more boys are needed to do the same piece of work in 2 hours?

Soln.

Min girls

5 hours = 6 boys

1 hour = (5 x 6) boys

2 hours = ($\frac{5 \times 6}{2}$)boys

= 15 boys

More boys = (15 – 6) boys

= 9 boys

3. 12 boys take 6 hrs to slash the compound. How many more boys are needed to slash the same compound in 2 hours?

Soln.

6 hours = 12 boys

1 hours = (12 x 6) boys

2 hours = ($\frac{12\ x\ 6}{2}$) boys

= 36 boys

More boys = (36 – 12) boys

= 24 boys

Activity:

1. 6 girls take 8 minis to mop the classroom. How long will 12 girls take to do the same task?

2. Two children can dig the school garden in 8 days. How many children can dig the same garden in 4 days?

3. 4 boys take 9 minutes to off load a lorry. How many more boys are needed to do the same Job in 6 minutes?

4. 7 girls take 28 minutes to wash a basket of clothes. How many more girls are needed to wash the same basket in 4 minutes?

5. 25 boys constructed a house in 8 days. How many boys are needed to do the same work in 10 days?

6. In a home, 6 potters can clean and mop the house in 4 minutes. How many more potters will be needed to finish the same work in only 2 minutes?

7. 9 men can build a wall in 14 days. How long will 7 men take to build the same wall?

8. 5 men take 6 hours to dig a trench. How many more hours would it require 3 men to complete the same piece of work?

9. There is enough food for 20 pupils which can last for 5 days. If 10 more pupils join how long will the food last?

10. Nine men can construct a house in 18 days. How many less days will 27 men take?

PERCENTAGES

Note:

- **A percentage is a number after every a hundred.**
- **A percentage is a fraction with a denominator as 100.**
- **Percentage means out of hundred.**
- **A symbol for percent is %**

CHANGING PERCENTAGE TO COMMON FRACTION

Examples

1. Change 20% to a common fraction.

Soln.

20% = $\frac{20}{100}$

= $\frac{1}{5}$

2. Change 12 ½% to a common fraction.

Soln.

12 ½% = 12 ½ ÷ 100

= $\frac{25}{2} \div \frac{100}{1}$

= $\frac{25}{2}$ X $\frac{100}{1}$

= $\frac{1}{8}$

Activity:

1. change the following percentages to fractions.

a) 40%

b) 25%

c) 60%

d) 75%

e) 62 ½%

f) **2 ½%**

g) **$16\frac{2}{3}$%**

h) **$83\frac{1}{3}$ %**

i) **$33\frac{1}{3}$ %**

j) **50%**

CHANGING PERCENTAGES TO DECIMALS

Examples

1. **Change 66% to a decimal.**

Soln.

66% **=** $\frac{66}{100}$

= **0.66**

2. **change 12 ½ % to a decimal.**

Soln.

12 ½ % = **12 ½** **÷ 100**

= $\frac{25}{2} \div \frac{100}{1}$

= $\frac{25}{2}$ **x** $\frac{1}{100}$

= $\frac{1}{8}$

3. **Change 0.6% to decimal.**

Soln.

0.6% **=** **(** $\frac{6}{10}$ **)%**

= $\frac{6}{10} \div \frac{100}{1}$

= $\frac{6}{10}$ **x** $\frac{1}{100}$

= $\frac{6}{100}$

= **0.006**

Activity:

1. **Change the following percentages to decimals.**

a) **45%**

b) **22%**

c) **46%**

d) **1.2%**

e) **20%**

f) **2 ½%**

g) **1.6%**

h) **15 ¼ %**

CHANGING DECIMALS TO PERCENTAGES

Examples

1. **Change 0.5 to percentage form.**

Soln.

0.5 **=** $\frac{5}{10}$

= $\frac{5}{10}$ **x 100%**

= **5 x 10%**

= **50%**

2. **Write 0.036 as a percentage.**

Soln.

0.036 = 0.036 x 100%

= $\frac{36}{1000}$ x 100%

= $\frac{36}{10}$%

= 3.6%

Activity:

1. Change the following decimals to percentages.

a) 0.18

b) 0.49

c) 0.0063

d) 2.63

e) 0.036

f) 0.012

g) 0.4

h) 0.002

i) 6.2

CHANGING FRACTIONS TO PERCENTAGES

Examples

1. Change $\frac{3}{5}$ to a percentage.

Soln.

%age = $\frac{3}{5}$ x 100%

= 3 x 20%

= 60%

2. Change $\frac{1}{3}$ to percentage.

Soln.

%age = $\frac{1}{3}$ x 100%

= $\frac{100}{3}$

= 33 $\frac{1}{3}$ %

3. Change $\frac{2}{3}$ to a percentage.

Soln.

% = $\frac{2}{3}$ x 100%

= $\frac{200}{3}$ %

= 66 $\frac{2}{3}$%

Activity:

Change the following fractions to percentage.

1. $\frac{2}{5}$

2. $\frac{3}{10}$

3. $\frac{3}{7}$

4. $\frac{4}{9}$

5. $\frac{1}{8}$

6. $\frac{4}{5}$

7. $\frac{3}{4}$

8. $\frac{1}{2}$

9. $\frac{1}{4}$

10. $\frac{9}{10}$

EXPRESSING RATIOS AS PERCENTAGES

Examples

1. Express 4:5 as a percentage.

Soln.

Fraction = $\frac{4}{5}$

%age = $\frac{4}{5}$ x 100%

= 4 x 20%

= 80%

2. Express $\frac{1}{10} : \frac{1}{7}$

Soln.

Fraction = $\frac{1}{10} \div \frac{1}{7}$

= $\frac{1}{10}$ x $\frac{7}{1}$

= $\frac{7}{10}$

Percentage = $\frac{7}{10}$ x 100%

= 7 x 10%

= 70%

Activity:

Express the following ratios to percentage.

1. 2 : 5

2. 1 : 4

3. 7 : 20

4. 9 : 10

5. $\frac{1}{5} : \frac{1}{3}$

6. $\frac{1}{20} : \frac{1}{9}$

7. $\frac{1}{5} : \frac{1}{4}$

8. 5 : 7

9. 3 : 4

11. $\frac{1}{5} : \frac{1}{2}$

12. $\frac{1}{4} : \frac{1}{3}$

13. $\frac{2}{5} : \frac{1}{2}$

14. $\frac{4}{25} : \frac{1}{3}$

FINDING PARTS OF PERCENTAGES

Example I

In a village 40% of the farmers grow beans and the rest grow maize. Find the percentage of farmers who grow maize.

Soln.

Let the %age of farmers who grow maize be K

K + 40% = 100%

K + 40% - 40% = 100% - 40%

K= 60%

Example II

In a class, there 20% more girls than boys. Find the percentage of boys.

Let % of boys be P.

Boys	Girls	Total
P	P + 20%	100%

P + P + 20% = 100%

2p + 20% = 100%

2p + 20% - 20%= 100% - 20%

2p = 80%

$\frac{2p}{2p}$ **=** $\frac{80}{2}$ **%**

P = 40%

Activity:

1. In a basket, there are 3% more oranges than mangoes. Find the percentage of mangoes in the basket.

b) Find the percentage of oranges in the basket.

2. There are 10% more men than women in a meeting.

i) What is the percentage of women?

ii) Find the percentage of men in the meeting.

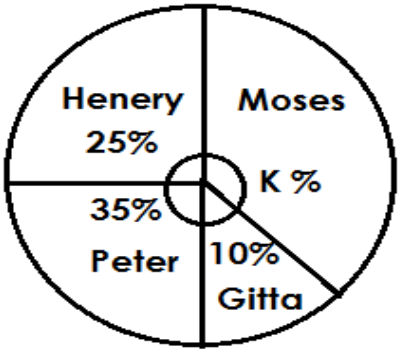
3. A family spent 40% more on school fees than on clothing. Find the percentage spent on clothing.

4. There are 10% more boys than girls in the class, find the percentage of boys in the class.

5. A child read 25% of a book on Monday, 38% on Tuesday and 16% on Wednesday. What percentage is left to be read.

6. 30% of the vehicles in town are white, 35% are red cars and the rest are black. What percentage of the cars are black?

7. The pie chart below shows how a man divided his land among his 4 children.



a) **What percentage of the land was given to Moses.**

b) **Find the fraction of the land that was given to Henry.**

EXPRESSING QUANTITIES AS PERCENTAGES

Example I

In a class of 40% pupils, 30 are girls and the rest are boys.

a) **Find the percentage of boys in the class.**

Soln.

No. of boys = 40 – 30

= 10 boys

%age of boys = $\frac{10}{40}$ x 100%

= 25%

b) **Find the percentage of girls.**

Soln.

%age of girls = $\frac{30}{40}$ x 100%

= 3 x 25%

= 75%

Example II

In a class, there are 7 desks, 8 pens and 10 books. Find the percentage of each item in the class.

Soln.

Total = 10 + 8 + 7 = 25

Desks = $\frac{7}{25}$ x 100%

= 28%

Pens = $\frac{8}{25}$ x 100%

= 32%

Books = $\frac{10}{25}$ x 100

= 40%

Activity:

1. **A man gave out sh. 5,000 to the three families A, B and C as follows: -**

A = sh. 1500

B = sh. 2000 and the rest to C

a) **Find the percentage of money that was given to C.**

b) **What percentage was given to B.**

2. **A bag contains 15 pens which are blue, 10 red pens and 25 black pens. Find the percentage of each type of pens in the bag.**

3. **In a class of 25 pupils, 20 are girls. Find the percentage of boys.**

b) **Find the percentage of girls.**

4. In a class of 80 pupils, 60 passed the exams. What percentage of the class failed exams?
b) Find the percentage of pupils who passed exams.
5. Juma scored the following marks in 4 tests, $\frac{9}{10}$ in the first test, $\frac{25}{50}$ in the second test, $\frac{64}{80}$ in the third test and $\frac{18}{25}$ in the fourth test. What percentage did he get in each test?

Expressing quantities as percentages of other quantities.

Note:
The quantity that comes after the word "of" is the denominator.
Example I

Express 20 as a percentage of 80.
Soln.
As a %age = $\frac{20}{80}$ x 100%

= 25%
Example II
What percentage of 400cm is 200cm?
Soln.
%age = $\frac{200}{400}$ cm x 100%

= 50%

Example III
Express 30cm as a percentage of 2m.
Soln.
1 m = 100cm
2m = (2 x 100) cm
= 200 cm
As a %age = $\frac{30cm}{200cm}$ x 100%

= 15%
Activity:
1. Express 3 as a percentage of 4.
2. What percentage of 20 is 4?
3. Express 36 books as a percentage of a gross.
4. What percentage of 2m is 40 cm?
5. Express 200g as a percentage of 2kg.
6. Express 250g as a percentage of 1kg.
7. Express 30 cm as a percentage of 2m.
8. What percentage of a score is 5.
9. What percentage of 10 weeks is 14 days?
10. Express 10 dm as a percentage of 25dm

SHARING QUANTITIES USING PERCENTAGES
Example I

Find 40% of 150.
Soln.
= 40% x 150

= $\frac{40}{100}$ x 150
= 60

Example II

What is 50% of 3 dozen of books?

Soln.

1 dozen = 12 books
3 dozen = (12 x 3) books
= 36 books
As %age = 50% x 36
= $\frac{50}{100}$ x 36 books
= 18 books

Example III

In a village of 800 people 40% are females and the rest are males.
a) Find the number of males in the village.

Soln.

Let the %age of males be K
K + 40% = 100%

K + 40% - 40% = 100 – 40%

K = 60%

No of males = 60% x 800

= $\frac{60}{100}$ x 800

= 480

b) How many more males than females are in the village?

Soln.

Male	Female	More
60%	40%	60% - 40% 20%

More males = 20% x 800

= $\frac{20}{100}$ X 800

= 160 males

Example IV

In a class there are 10% more girls than boys. If there are 600 pupils, find the number of girls in a class.

Soln.

Let %age of boys be M.

Boys	Girls	Total
M	M + 10%	100%

M + m + 10% = 100%

2m + 10% = 100%

2m + 10% - 10% = 100% - 10%

2m = 90%

$\frac{2m}{2} = \frac{90\%}{2}$

M = 45%

%age of girls = m + 10%

= 45% + 10%

= 55%

No. of girls = 55% x 600

= $\frac{55}{100}$ x 600

= 330 girls

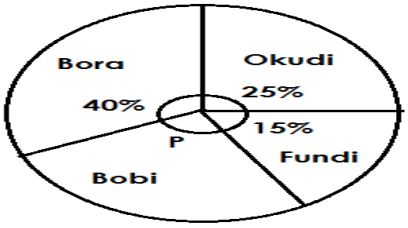
Activity:

1. What is 25% of 200 books?

2. In a class of 80 pupils, 20% are girls and the rest are boys. Find the number of girls in a class.

b) Find the number of boys in a class.

3. The pie – chart below shows how villages shared sh. 2,000,000 donated by president.



a) How much money was given to bobi?

b) How much more money was given to Bora than Okudi?

c) Express the amount of money given to fundi as a fraction.

4. In a class of 120 pupils, there are 30% more girls than boys.

a) How many boys are in the class.

b) Find the number of girls in the class.

5. In a club of 600 customers, there are 20% more men than women, how many women are in the club?

b) Find the number of men in the club.

8. Find 15% of 4dm.

9. What is 80% of 3 dozen of books?

10. Find 25% of 2 hours.

9. What is 50% of 1kg.

FINDING QUANTITIES SHARED USING PERCENTAGES

Example I

30% of a number is 12. Find the number.

Soln.

Let the number be K.

30% x K = 12

$\frac{30}{100}$ **x K = 12**

$\frac{30}{100}$ **K = 12**

$\frac{30}{100}$ **x 100 = 12 x 100**

$\frac{30}{30}$ **K = $\frac{12 \text{ x } 100}{30}$**

K= 40

Example II

In a class, 40% are girls and the rest are boys. If there are 12 boys, find the number of pupils in the class.

Soln.

Let the %age of boys be K.

K + 40% = 100%

K + 40% - 40% = 100% - 40%

K = 60%

Let the number of pupils be m.

60% x m = 12

$\frac{60}{100}$ **x m = 12**

$\frac{60\text{ m}}{100}$ **= 12**

$\frac{60\text{ m}}{100}$ **x 100 = 12 x 100**

60m = $\frac{12\ x\ 100}{60}$

M = 20 pupils

Example III

In a bag of 10% of the pens are blue and the rest are green. If there are 8 more green pens than blue pens. How many pens are in the bag?

Soln.

Let the %age of green pens be y.

Y + 10% = 100%

Y + 10% - 10% = 100% - 10%

Y = 90%

Green	Blue	More
90%	10%	90% - 10% = 80%

Let no of pens in the bag be m.

80% x m = 8

$\frac{80}{100}$ **x m = 8**

$\frac{80\ m}{100}$ **= 8**

$\frac{80\ m}{100}$ **x 100 = 8 x 100**

80m = 8 x 100

$\frac{80\ m}{80} = \frac{8 \times 100}{80}$

M = 10 pens.

Activity:

1. 30% of a number is 12, find the number.

2. In a village, 20% of the farmers grow maize and the rest grow beans. If 16 farmers grow bean, how many farmers are in the village?

b) How many farmers grow maize?

2. In a bus 40% are children and the rest are adults. If there are 24 adults, how many children are in the bus?

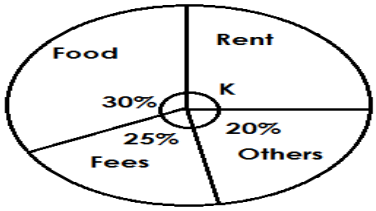
4. In a class, 30% are boys and the rest are girls. If there are 20 more girls than boys, how many boys are in the class?

b) Find the number of girls in the class.

6. In a class, there are 20% more boys than girls. If there are 24 boys, how many girls are in the class?

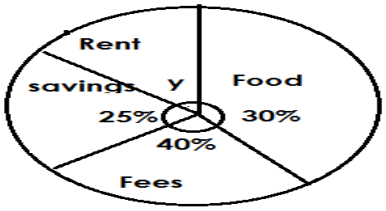
6. In a bus there are 40% more adults than children. If there are 12 children, how many adults are in the bus?

7. The pie chart below shows Mukasa's monthly expenditure.



If he spends shs. 20,000 on food more than rent. How much does he spend on others?

8. The pie chart below shows Mr. Kalungi's monthly expenditure.



a) If he spends shs. 180,000 on fees and rent, how much does he spend on food?

11. After spending 30% on food, Wamboka saved sh. 28,000. How much did he spend on food?

10. After covering 40% of the journey, a vehicle got a puncture and he remained with 30km. What distance had he covered?

INCREASING AND DECREASING QUANTITIES USING PERCENTAGES

Example I

Increase 60 oranges by 20%
Soln.
(100% + 20%) x 60 oranges

120% x 60

$\frac{120}{100}$ **x 60**

72 oranges

Example II
Increase 40 by 12 ½%
Soln.
(100% + 12½) x 40

$\frac{200 + 25\%}{2}$ **x 40**

$(\frac{100}{1} + \frac{25}{2})\%$ **x 40**

$\frac{225\%}{2}$ **x 40**

$(\frac{225}{2} \div \frac{100}{1})$ **x 40**
= $\frac{225}{2}$ **x** $\frac{1}{100}$ **x 40**

= **45**

Example III

Decrease 84 by 10%

Soln.

= (100% - 10%) x 84

= 90% x 84

$= \frac{90}{100}$ **x 84**

= $\frac{75}{10}$**%**

= 75.6

Example II

Decrease 300 by 33 $\frac{1}{3}$ **%**

Soln.

(100% - 33 $\frac{1}{3}$**%) x 300**

($\frac{100}{1}$ **-** $\frac{100}{3}$ **) % x 300**

$\frac{300-100\,\%}{3}$ **x 300**

($\frac{200}{3}\frac{100}{1}$ **) x 300**

$\frac{200}{3}$ **x** $\frac{1}{100}$ **x 300**

= 200

Activity:

1. **Increase 96 by 10%**
2. **Increase sh. 4000 by 60%**
3. **A school had 1200 pupils last year. Due to the good performance, the number increase by 15% find the new number of pupils in the school.**
4. **Increase 1200 by 22 ½%**
5. **Decrease 240 by 20%**
6. **A father gave his son sh. 4000 as pocket money. After one month at school, the pocket money had reduced by 12 ½. How much money did the boy remain with after that period?**
7. **Decrease sh. 14,000 by 5 ½%**
8. **The number of animals on a farm increased from 700 by 30%. Find the new number of animals on the farm.**
9. **Decrease sh. 24,000 by 66** $\frac{2}{3}$ **%**

10. Otieno bought a plot of land at sh. 1,500,000. After one year he increased the cost price by $2\frac{2}{3}$ % when he was selling it to Ojame. At how much did Ojame pay for the plot of land?

MULTIPLE INCREASE AND DECREASE OF QUANTITIES USING PERCENTAGES

Examples

1. increase 600 by 10% and then by 20%

Soln.

First %ageincrease = 100% + 10%

= 110%

Second %age increase = (100% + 20%)

= 120%

Final no. = 110% x 120% x 600

= $\frac{110}{100}$ x $\frac{126}{100}$ x 600

= 11 x 12 x 6

= 792

2. Decrease sh. 1200 by 10% then by 30%

Soln.

First % decrease = (100% - 10%)

= 90%

Second % decrease = 100% - 30%

= 70%

Final no. = sh. 1200 x $\frac{90}{100}$ x $\frac{70}{100}$

= sh. 12 x 9 x7

= sh. 756

3. Increase 600 by 40% and then decrease it by 10%

Soln.

Increase = (100% + 40%) x 800

= 140% x 800

= $\frac{140}{100}$ x 800

= 1120

Decrease = (100% - 10%) x 1120

= 90% x 1120

= $\frac{90}{100}$ x 1120

= 9 x 112

= 1008

8. **The number of pupils in a school decreased by 20% from 1200, this year they had a good performance and the number increased by 10%. What is the new number of pupils?**

Soln.

Decrease = (100% - 20%) x 1200

= 80% x 1200

= $\frac{80}{100}$ x 1200

= 80 x 12

= 960

Increase = (100% + 10%) x 960

= 110% x 960

= $\frac{110}{100}$ x 960

= 11 x 96

= 1056 pupils

Activity:

1.Increase 12000 by 20% then by 30%.
2.Increase 480 girls by 20% and then by 25%.
3.Decrease sh. 9000 by 40% and then by 10%
4.The number of pupils decreased by 20% for the first year and then by 15% for the second year. How many pupils are in the school if they were 1600?
5.Decrease sh. 70,000 by 70% and then by 30%
6.Decrease 8400 by 10% then by 10%
7.Peter's salary of sh. 400,000 was increased by 20% and later decreased by 10%. Find his new salary.
8.Askari's salary was increased from sh. 120,000 by 20% due to the poor performance during the duty, his salary was later reduced by 30%. Find his new salary.
9.Increase sh. 20,000 by 20% then decrease it by 80%
10.Jane's salary was increased by 40% from sh. 40,000 and later decreased by 60%. Find her new salary.

FINDING THE ORIGINAL NUMBER THAT WAS INCREASED OR DECREASED USING PERCENTAGES.

Examples

1.Which number was increased by 20% to become 36

Soln.

Let the number be K.

(100% + 20%) x k = 36

120% x K = 36

$\frac{120}{100}$ x k = 36

$\frac{120}{100}$ k = 36

100 x $\frac{120}{100}$ k = 36 x 100

$\frac{120}{120}$ k = $\frac{36\ x\ 100}{120}$

K = 30

2. After increasing a number by 5 ½% it became 422. Find the number.

Soln.

Let the number be n

100% + 5½ % x n = 422

(100% + $\frac{11}{2}$)% x n = 422

($\frac{200+11}{2}$)% x n = 422

($\frac{211}{2}$ ÷ $\frac{100}{1}$) x n = 422

$\frac{211\ n}{200}$ = 422

$\frac{211\ n}{200}$ x 200 = 422 x 200

$\frac{211}{2}$ n = $\frac{422\ x\ 200}{2}$

n = 400

3. Find the number which was decreased by 20% to become 48.

Soln.

Let the no be k.

(100% - 20%) of n = 48

80% x n = 48

100 x $\frac{80}{100}$ n = 48 x 100

$\frac{80}{80}$ = $\frac{48\ x\ 100}{80}$

n = 60

4. After decreasing Othieno's salary by 20% he now earns sh. 240,000. How much was his original salary?

Soln.

Let his original salary be m.

(100% - 20%) x m = sh. 240,000

80% x m = sh. 240,000

100 x $\frac{80}{100}$ m = sh. 240,000 x 100

$\frac{80}{80}$ m = sh. $\frac{240{,}000\ x\ 100}{80}$

m = 300,000

b) How much would be his salary if it was increased by 30%
soln.

=(100% + 30%) x sh. 300,000
=130% x sh. 300,000
=$\frac{130}{100}$ x sh. 300,000
= sh. 390,000

Activity:

1. **Which number was increased by 40% to become 42?**
2. **Find the number that was increased by 12 ½% to become 45.**
3. **Find the number that was increased by 20% and then by 10% to become 528.**
4. **Which number was decreased by 30% to become 63.**
5. **After decreasing a number by 12 ½%, it became 35. Find the number.**
6. **After increasing a number by 12 ½% it becomes sh. 81,000. Find the number.**
7. **What number when increased by 66 $\frac{2}{3}$% becomes 400?**
8. **If a man's salary is decreased by 35% it becomes sh. 15,000. What could be his salary if it increased by 40%.**
9. **Matama bought a radio at a certain amount of money and later sold it to Ogwal after increasing the cost price by 20%. After one year, Ogwal decreased the price by 30% after selling it to Owino who paid sh. 67200. At how much did Matama buy the radio.**

FINDING PERCENTAGE INCREASE AND DECREASE

Note:

Percentage increase = $\frac{\text{increase x } 100\%}{\text{original}}$

Increase = new – old

Decrease = old – new

Percentage decrease = $\frac{\text{decrease x } 100\%}{\text{oroginal}}$

Examples

1. **The number of animals on a farm increased from 50 to 80. By what percentage was the increase.**

Soln.

% increase= $\frac{\text{increase x } 100\%}{\text{original}}$

Increase = New – old
= 80 – 50
= 30

%age increase = $\frac{30}{50}$ x 100%

= 3 x 20%

= 60%

2. The number of pupils in class decreased from 800 to 600. Find the percentage decrease.

Soln.

Decrease = original – New

= 800 – 600

= 200

Percentage decrease = $\frac{\text{decrease}}{\text{original}}$ x 100%

= $\frac{200}{860}$ x 100%

= 25%

3. The number of animals on a farm decreased from 200 by 60. Find the percentage decrease.

Soln.

Percentage decrease = $\frac{\text{decrease}}{\text{original}}$ x 100

= $\frac{60}{200}$ x 100%

= 30%

Activity:

1. **Kambe's salary was increased from sh. 100,000 by sh. 20,000. Calculate the percentage increase in his salary.**
2. **The number of books in the library increased to 500 books by 100 books. Find the percentage increase in the number of books.**
3. **A father raised a boy's pocket money from sh. 8000 to sh. 10,000. At what percentage was the boy's pocket money raised.**
4. **When 200 birds are decreased, they become 150 birds. Calculate the percentage decrease.**
5. **The number of Malaria patients decreased from 50 by 10. Find the percentage in the patients.**
6. **When 200 birds are decreased, they become 150 birds. Calculate the percentage decrease.**
7. **Mandela's salary was increased from sh. 20,000 to sh. 29,000. What was his percentage increase?**
8. **A school collected 7.5 million last year. This year the collection dropped by 2.5 million. Find the percentage decrease.**
9. **On a farm of 400 animals, there was an outbreak and the number decreased to 350. By what percentage did the number decrease?**

PROFITS AND PERCENTAGE PROFIT

Definitions

1. **Profit is the extra amount of money added to the cost price of an item by a trader.**
2. **Profit is the extra money got after selling an item as compared to the buying price.**
3. **Profit is the same as increase, gain and raise.**
4. **Market price is the original /usual price of an item.**

Note:

Profit = selling price – buying price (cost price)

Profit = SP – BP

WHERE;

SP = selling price

BP = buying price / cost price.

Percentage profit = $\frac{\text{profit}}{\text{buying price}}$ x 100%

Examples

1. A trader bought a radio at sh. 80,000 and later sold it at sh. 100,000. Calculate his percentage profit.

Soln.

Buying price (BP) = sh. 80,000

Selling price (SP) = sh. 100,000

Profit = SP – BP

= sh. 100,000

= sh.80,000

= sh. 20,000

Percentage profit = $\frac{\text{profit}}{\text{buying price}}$ x 100%

= sh. $\frac{20{,}000}{80{,}000}$ x 100%

= 25%

3. After selling a shirt at sh. 12,000, a trader realized a profit of sh. 2,000. Calculate his percentage profit.

Soln.

Profit = sh. 2000

Buying = selling price – profit

= sh. 12000 – sh. 2,000

= sh. 10,000

Percentage profit = $\frac{profit}{buying\ price}$ x 100%

= $\frac{sh.2000}{sh.10{,}000}$ x 100%

= 20%

4. **A man bought a watch at sh. 20,000 and later sold it at a profit of sh. 5,000. Calculate his percentage profit.**

Soln.

Profit = sh. 5,000

Buying price = sh. 20,000

% profit $= \frac{\text{profit}}{\text{Buying price}}$ **x 100%**

$= \frac{\text{sh.5,000 x 100\%}}{\text{sh.20,000}}$

= 25%

Activity:

1. **a trader bought a dress at sh. 4,000 and later sold it at a profit of sh. 1,000. Calculate his percentage profit.**

2. **Wambi bought a plot of land at sh. 8,000,000 and later sold it as sh. 11,000,000. Find his percentage profit.**

3. **A boy bought a shirt at sh. 40,000 and sold it at sh. 50,000. Find his profit percent.**

4. **Okello bought a radio at sh. 100,000 and later sold it to Opio at sh. 120,000. Find his percentage profit.**

5. **A trader bought an item of sh. 7,500 and sold it at sh. 15,000. Find his percentage profit.**

6. **A woman bought 2 trays of eggs at sh. 15,000. She sold each egg at sh. 300. Calculate her percentage profit.**

7. **A trader bought a dress of sh. 1,600 and sold it at sh. 2,000. Find her profit percent.**

8. **The cost of a book is sh. 200. If a shopkeeper sold it at sh. 250, what was the percentage gain?**

9. **Matia sold his cow for sh. 47,700. If the cost of the cow was sh. 45,000. Find his percentage profit.**

10. **A trader bought a gross of books for sh. 36,000. He sold each book for sh. 500. Find the percentage profit made.**

LOSS AND PERCENTAGE LOSS

- **Loss is the less amount of money got compared to the cost price of an item.**
- **It is the same as decrease, deduction and sometimes discount in terms of calculation.**

Note:

Loss = buying price – selling price
Where

BP = buying price
SP= selling price

Percentage loss = $\frac{\text{loss}}{\text{buying price}}$ x 100%

Examples

3. After selling a dress at sh. 80,000. Wandera realized a loss of sh. 20,000. What was the percentage loss?

Soln.

SP = sh. 80,000
Loss = sh. 20,000
Loss% = $\frac{\text{loss}}{\text{B.P}}$ X 100%

=sh. 100,000

BP=SP + Loss
=sh. 80,000 + sh. 20,000

= $\frac{\text{sh.20,000}}{\text{sh.100,000}}$ x 100%

= 20%

2. A trader bought a watch at sh. 25,000 and later sold it at sh. 20,000. Calculate his percentage loss.

soln.
loss = B.P – S.P

= sh. 25,000
- sh. 20,000

= Sh. 5,000

%ageloss = $\frac{\text{loss}}{\text{B.P}}$ x 100%

= $\frac{\text{sh.5000}}{\text{sh. 25,000}}$ x 100%

= 20%

3. Kayima bought a pair of shoes at sh. 20,000 and sold it at a loss of sh. 5,000. Find his loss percent.

Soln.
Loss = sh. 5,000

Buying price = sh. 20,000

Loss % = $\frac{\text{loss}}{\text{B.P}}$ x 100%

= $\frac{\text{sh.5,000}}{\text{sh.20,000}}$ x 100%

= 25%

Activity:

1. **A school girl bought a pen at sh. 1000 and sold it at a loss of sh. 500. Find her percentage loss.**
2. **Kampyo bought a radio at sh. 15,000 and later sold it at sh. 12,000. Calculate his percentage loss.**
3. **Nagitta bought a pair of shoes at sh. 20,000 and sold it at a loss of sh. 4,000. Calculate her percentage loss.**
4. **After selling a watch at sh. 4,000, a trader realized a loss of sh. 1,000. Find his percentage loss.**
5. **A man bought 6 pots at sh. 9,000. Some 2 pots got broken and sold the remaining pots at sh. 2,000 each pot. Find his percentage loss.**
6. **The cost of 1 litre of milk is sh. 500. Aisha sold it at sh. 450. Calculate the percentage loss.**
7. **Kapere bought a motorcycle at sh. 600,000. After one year he sold it at sh. 480,000. Find his percentage loss.**
8. **After selling a set of chairs for sh. 150,000, a trader made a loss of sh. 30,000. Calculate the percentage loss.**
9. **A loss on a shirt which was sold at sh. 50,000 was sh. 6,000. Calculate the percentage loss.**

FINDING SELLING PRICE GIVEN PERCENTAGE PROFIT OR PERCENTAGE LOSS

Example I

Opio bought a radio at sh. 20,000 and later sold it at a profit of 30%. How much was his selling price?

(100% + 30%) X Sh. 20,000

130% X sh. 20,000

$\frac{130\%}{100}$ x sh. 20,000

Sh. 26,000

Example II

A trader bought a shirt at sh. 40,000 after selling it, he realized a percentage loss of 10%. How much money did he sell it?

Soln.

(100% - 10%) x sh. 40,000

90% x sh. 40,000

$\frac{90}{100}$ x sh. 40,000

Sh. 36,000

Example III

Tom bought a TV at sh. 100,000, he later sold it to Patunda at a profit of 15%. After one year, Patunda sold it to Okum at a loss of 10%.

a) At how much did Patunda pay for the TV?

Soln.

(100% + 15%) x sh. 100,000

115% x sh. 100,000

$\frac{115}{100}$ x sh. 100,000

= sh. 115000

b) Find amount at which Okum bought the TV.

Soln.

(100% - 10%) x sh. 115,000

90% x sh. 115000

$\frac{90}{100}$ x sh. 115000

=Sh. 103,500

Activity:

1. A man bought a shirt at sh. 12,000, he later sold it at a profit of 20%. Find his selling price.
2. Wandera bought a plot of land at sh. 4,000,000. After he sold it to Wako at a profit of 10%. At how much did Wako pay for the plot of land?
3. A trader bought a car at sh. 800,000. He later sold it at a loss of 30%. At how much did he sell the car?
4. Martha bought a dress at sh. 20,000. She sold it to Jane at a profit of 10% after 1-month Jane also sold it to Fatuma at a loss of 20%.
   a) At how much did Jane pay for the dress?
   b) Find the amount at which Jane sold the dress.
5. A tomato seller bought 60 tomatoes at sh. 12,000. At what price must she sell each tomato in order to get a profit of 30%.

6. Junju bought a plot of land at sh. 500,000 and be sold it to Mugabe at a profit of 20%. If Mugabe later sold it to Agaba at a loss of 20%. Calculate Agaba's buying price.

FINDING COST PRICE GIVEN PERCENTAGE LOSS OR PERCENTAGE PROFIT

Example I

After selling a shirt at sh. 26,000, a trader realized a profit of 30%. At how much did he buy the shirt.

Soln.

(100% + 30%) x CP = sh. 26,000

130% x CP = sh. 26,000

$\frac{130}{100}$ **x CP = sh. 26,000**

$\frac{130CP}{100}$ **= sh. 26,000**

$\frac{130}{100}$ **CP = sh. 26,000 x 100**

$\frac{130\ CP}{100}$**X100 = sh.** $\frac{26,000}{130}$ **x 100**

CP = sh. 20,000

Example II

James sold a cow at sh. 630,000 making a loss of 30%. How much would he have sold it in order to gain a profit of 10%

Soln.

Let cost price be cp

(100% - 30%) X CP = sh. 630,000

70% x CP = sh. 630,000

$\frac{70}{100}$ **x CP = sh. 630,000**

$\frac{70}{100}$ **CP = sh. 630,000**

$\frac{70PC}{100}$ **x 100 = sh. 630,000 x 100**

70CP = sh. 630,000 x 100

$\frac{70CP}{70}$ **= sh.** $\frac{630,000}{70}$ **x 100**

CP = sh. 900,000

= (100% + 10%) x sh. 900,000

= 110% x sh. 900,000

= $\frac{110}{100}$ **x sh. 900,000**

SP = 990,000

Activity

1. **After selling a dress at sh. 48,000, Matama realized a profit of 20%. At how much did she buy the dress?**

2. **Mega sold his radio to Aguti at sh. 54,000 making a loss of 10%. Aguti later sold it to Othieno at a profit of 15%.**

a) Calculate the amount of money Mega paid for the radio.

b) At how much did Aguti sell the radio?

3. **After selling a chair at sh. 89,000, a carpenter realized a profit of 30%. What was his cost price?**

4. **Kadaga sold a bag at sh. 64,000 to Jane making a profit of 60%. Jane later sold it to Kawuma at a loss of 20%**

a) At how much did Kadaga buy the bag?

b) Find the amount at which Kawuma paid for the bag?

5. **Othieno sold a shirt at sh. 36,000 realizing a loss of 10%. At how much did he buy it.**

6. A man sold his car at sh. 3,000,000, realizing a loss of 40%. At how much would he have sold it in order to realize a profit of 20%.

Discount and percentage discount

- **Discount is theamount of money reduced from the usual price of an item.**
- **Discount is the amount of money that is taken off from the cost price of an item.**

Discount=cost price – cash paid

= Market price – cash paid

Percentage discount = $\frac{\text{Discount}}{\text{Cost price}}$ x 100%

Example 1

A man bought a shirt that was marked sh. 40,000 at a discount of 20%. How much was the discount?

Soln.

Discount = 20% x sh. 40,000

= $\frac{20}{100}$ x sh. 40,000

= sh. 8,000

b) How much did he pay?

Soln.

Amount paid = (100% - 20%) x sh. 40,000

= 80% x sh. 40,000

= $\frac{80}{100}$ x sh. 40,000

= sh. 32,000

Martine bought a radio that was marked sh. 50,000 at a discount of sh. 6,000. Calculate his percentage discount.

Soln.

%age discount = $\frac{discount}{cash\ price}$ x 100%

= sh. $\frac{6,000}{50,000}$ x 100%

= 12%

b) How much did he pay?

Soln.

Amount paid = sh. 80,000

= sh. 6,000

= sh. 44,000

Example III

Nakita went to the market and bought the following items.

- **2kg of sugar at sh. 2,500 per kg.**
- **1 ½ kg of rice at sh. 2000 a kg.**
- **3 bars of soap at sh. 6,000**
- **500g of meat at 10,000 a kg.**

a) If she was given a discount of 10%, how much did she pay?

Soln.

Sugar	Rice	Soap	Meat
Sh. 2500 X 2 Sh. 5,000	Sh. 2000 x 1 ½ Sh. 2,000 x $\frac{2}{3}$ Sh. 3,000	Sh. 6000	$\frac{500}{1000}$ x 10000 Sh. 5,000

Total

Sh. 5,000
Sh. 3,000
Sh. 6,000
Sh. 5,000
Sh. 19,000

Amount paid

(100% - 10%) x sh. 19,000

90% x sh. 19,000

$\frac{90}{100}$ x sh. 19000

90 x sh. 190

Sh.17100

Activity:

1. **Okot went to the market and bought a radio marked sh. 40,000. If he was offered a discount of 5%. How much did he pay?**
2. **Kanda went to the market and bought 2kg of sugar at sh. 4000 each kg. how much was the discount if it was 10%?**
3. **The marked price of a dress is sh. 25,000 after a discount a customer paid sh. 20,000. How much was the discount?**

4. **A man bought shoes which were marked sh. 80,000 at a discount of 40%.**

a) **How much was the discount?**

b) **How much did he pay?**

4. **Kalinago paid sh. 36,000 for a shirt. If he was given a discount of 20%, how much was the cost price?**

6. **Wasswa went to the market and bought the following items.**

- **3kg of beans at sh. 2,000 per kg.**
- **2 ½ kg of posho at sh. 1,800 a kg.**
- **2 litres of cooking oil at sh. 4,000**
- **250gm of salt at sh. 2,000 a kg.**

a) **If he went with sh. 20,000 and he was offered a discount of 20%, how much was his change?**

7. **Keith went to the market and bought the following items: -**

- **2kg of rice at sh. 4,000 a kg.**
- **1 ½ kg of beans at sh. 2,000 a kg.**
- **500 ml of cooking oil at sh. 4,000 a litre.**
- **12 tomatoes at sh. 500 per heap of 3 tomatoes.**

If he was given a discount of 10%, how much did he pay?

SIMPLE INTEREST

Definitions

Principles (P)

This is the amount of money borrowed or lent or deposited in a bank.

Rate (R)

Rate is the percentage used to calculate interest.

Time (T)

Time is the period in years, months or week in which the money is kept, invested or used.

Simple interest (I)

Simple interest is the extra /additional amount of money offered or paid back.

Amount (A)

Amount = principle + interest.

CALCULATING SIMPLE INTEREST AND AMOUNT

Note:

Simple interest= principle x rate x time

SI = P x T x R

Examples

1. Othieno deposited sh. 20,000 in a bank at 20% per year for 2 years. Find the simple interest.

Soln.

P = sh. 20,000

T = 2yrs

R = 20%

I = P x T x R

= sh. 20,000 x 2 x $\frac{20}{100}$

= sh. 2,000 x 2 x 2

= sh. 8,000

2. Calculate the simple interest on sh. 24,000 at a rate of 15% per year for 8 months.

Soln.

P = sh. 24,000

R = 15% per year

T = 8 months

I = P x T x R

= sh. 24,000 x $\frac{8}{2}$ x $\frac{15}{100}$

= sh. 120 x 8 x 15

= sh. 14,400

Note:

- ***When rate is given as per year and time is in months, express the months as a fraction of the year i.e. Months divided by 12.***
- ***When rate is given as per month and time is also in months don't divide the month by 12.***
- ***If rate given is in months and period is in years, change period to months by multiplying by 12***

3. Matama borrowed sh. 400,000 from a bank at a rate of 12% per month for 4 months. Find the amount sh. Paid back after that period of 4 months.

Soln.

P = sh. 400,000

R = 12% per month

I = 4 months

I = P x T x R

= sh. 400,000 x 4 x $\frac{12}{100}$

= sh. 4,000 x 4 x 12

= sh. 192,000

Amount (A) = principal (P) + Simple interest I.

= sh. 400,000 + sh. 192,000

= sh. 400,000

= + sh. 192,000

= sh. 592,000

Activity:

1. Find the simple interest on sh. 100,000 deposited in a bank for 5 years at 5% per annum.

2. A man borrowed sh. 30,000 at a simple interest rate of 20% per month for 2 years. What will be his simple interest?

3. Kagame deposited sh. 60,000 in a bank that offers a simple interest rate of $5\frac{2}{3}$ % per year for 2 years. Calculate his simple interest.

4. Kaire borrowed sh. 360,000 from a bank at a simple interest rate of $33\frac{1}{3}$% per year for 4 months. Calculate his amount of money paid back.

5. Calculate the simple interest on sh. 200,000 kept in a bank for 2 months at 5% per month.

6. Calculate the simple interest on sh. 40,000 at 10% per month for 1½ years.

7. A woman finance company gives 15% interest on every money borrowed. Calculate the amount of money one would pay back after borrowing sh. 2,500,000 for $3\frac{1}{2}$ years.

8. calculate the simple interest on sh. 60,000 deposited in a bank for 2 ½ years at 5 ½% per year.

FINDING PRINCIPAL

Examples:

1. Oguti deposited a certain amount of money in a bank at a rate of 20% per year for 3 years. If he earned a simple interest of sh 12,000, how much money did he deposit?

Soln.

P x R x T = ST.

P x $\frac{20}{100}$ x 3 = sh. 12,000

10 x $\frac{6P}{10}$ = sh. 12,000 x 10

$\frac{6P}{6}$ = sh. $\frac{12000\ x\ 10}{6}$

P = sh. 20,000

2. After depositing a sum of money in the bank that offers a simple interest rate of 4% per year. Omara earned a simple interest of sh. 12,000 in 9 months. How much was deposited in the bank?

Soln.

P x R x T = ST

P x 4% x$\frac{9}{12}$ = sh. 12,000

P x $\frac{4}{100}$ x $\frac{9}{12}$ = sh. 12,000

$\frac{36P}{1200}$ = sh.12,000

1200 x $\frac{36P}{1200}$ = sh.12,000 x1200

$\frac{36P}{36}$= sh. $\frac{12,000\ X\ 1200}{36}$

P = sh. 400,000

Activity:

1. **Find the amount of money that can yield sh. 8,000 at a simple interest rate of 10% per year in one year.**

2. **What sum of money will yield a simple interest of sh. 18,000 after 3 months at a rate of 20% per year?**

3. **A man deposited a certain amount of money for 3 years at a simple interest rate of 12 ½ per year. If he got an interest of sh. 25,000, how much did he withdraw?**

4. **Oluk borrowed a certain amount of money at a simple interest rate of 33 $\frac{1}{3}$ per year for 6 months. If he paid an interest of sh. 3,000, how much did he borrow?**

5. **Find the amount of money that can yield sh. 144,000 at a simple interest rate of 20% per year for 2 years.**

6. **Kamya deposited a certain amount of money in a bank at a rate of 10% per year for 1½ years. If he earned a simple interest of sh. 72,000, how much money did he deposit?**

7. **What amount of money will yield sh. 144,000 in 2 years at 20% per annum?**

8. **Anent deposited a certain amount of money for 2 years at a simple interest rate of 5% per year. If he got an interest of sh. 20,000, how much did he withdraw?**

FINDING RATE

Examples

1. Calculate the rate if sh. 60,000 can yield sh. 6,800 in 2 years.

Soln.

P x R x T = ST

Sh. 60,000 x R% x 2 = sh. 6,800

Sh. 60,000 x $\frac{R}{100}$ x 2 = sh. 6,800

Sh. 60 x R x 2 = sh. 6,800

$\frac{120\ R}{120}$ = sh. $\frac{6,800}{120}$

R = 5 $\frac{2}{3}$%

OR

R = $\frac{1\ X\ 100}{P\ x\ T}$

= sh. $\frac{6800\ x\ 100}{60,000\ x\ 2}$

= $\frac{34}{6}$

= $5\frac{2}{3}$ %

2. Example 2

Calculate the rate of interest if sh. 30,000 can yield sh. 3000 in 2 ½ years.

Soln.

P x R x T = SI

Sh. 30,000 x R% x 2½ = sh. 3,000

Sh. 30,000 x $\frac{R}{100}$ x $\frac{5}{2}$ = sh. 3,000

Sh. 150 x R x 5 = sh. 3,000

Sh. $\frac{750\,R}{750}$ = sh. $\frac{3000}{750}$

R = 4%

ACTIVITY:

1. Sarah deposited sh. 50,000 on her saving account for 3 years and earned sh.15,000. Calculate the rate of interest.

4. Calculate the rate of interest if sh. 20,000 is deposited in a bank for 3 years can yield a simple interest of sh. 12,000.

5. A man deposited sh. 40,000 in a bank for 5 years. If he earned an interest of sh. 25,000, calculate the rate of interest.

6. Calculate the annual rate of interest if sh. 30,000 can yield sh. 1125 in 9 months.

5. Wamala deposited sh. 40,000 at a certain annual interest rate in a bank. If he earned sh. 19,200 interest after 4 months, what was the rate of interest?

6. The simple interest earned by Simon after depositing sh. 30,000 in the bank for 4 years was sh. 1,800. Calculate the rate of interest offered by the bank.

7. Calculate the rate of interest if sh. 30,00 can yield a simple interest of sh. 1125 in 9 months.

8. Tony deposited sh. 50,000 on his savings account. At the end of 3 years the simple interest earned was sh. 15,000. Calculate the rate of interest.

9. If the interest on a loan of sh. 140,000 for 6 months is sh. 5,600. Find the interest rate.

10. Kato deposited sh. 1,500,000 for 2 ½ years and earned an interest of sh. 375,000. Calculate his percentage rate.

FINDING TIME

Examples

1. Wandela deposited sh. 50,000 in a bank at a rate of 10% per year. If he earned a simple interest of sh. 15,000 for how long was the money in the bank?

Soln.

P x R x T = S.I

Sh. 50,000 x 10% x T = sh. 15,000

Sh. 50,000 x $\frac{10}{100}$ x T = sh. 15,000

Sh. $\frac{5000\ T}{5000}$ = sh. $\frac{15000}{5000}$

T = 3 years.

OR

T = $\frac{T\ x\ 100}{P\ x\ R}$

= sh. $\frac{15000\ x\ 100}{50{,}000\ x\ 10}$

= 3 years

2. In how many months can sh. 50,000 yield a simple interest of sh. 15,000 at 3% per year?

Soln.

P x T x R = S.I

Sh. $\frac{50{,}000\ x\ T\ x\ 3}{100}$ = sh. 15,000

Sh.$\frac{5000T}{500}$ =sh. $\frac{15000\ x\ 4}{500}$

T = 120 months

Activity:

1. **After depositing sh. 120,000 in a bank at a simple interest rate of 10% per year. Magufu earned a simple interest of sh. 7200. For how long was the money in the bank.**

2. **Owino got a simple interest of sh. 5000 after lending shs.20,000 to Mark at a simple interest rate of 10%. For how long did Mark use the money.**

3. **In what time will sh. 12,000 yield an interest of sh. 1800 at 5% per year?**

4. **A girl borrowed sh. 50,000 at a rate of 3% and paid sh. 15,000 as interest. How long did he use the money?**

5. **After what time will sh. 12,000 yield an interest of sh. 1800 at 5%?**

6. **A lady borrowed sh. 50,000 at a rate of 3% and paid sh. 15,000 as interest. How long did she use the money?**

7. **In what time will sh. 60,000 yield an interest of sh. 3,000 at 5% per year?**

8. **A man borrowed sh. 120,000 at a rate of 20% per annum. How long will he take to pay an interest of sh. 90,000 on the loan?**

FINDING PRINCIPAL GIVEN AMOUNT

Examples

1. A man deposited a sum of money in a bank that offer a simple interest rate of 20% p.a. after 3 years he withdrew sh. 960,000. How much did he deposit?

Soln.

P + S. I = amount

P + (P x R x T) = sh. 960,000

P + (P x 20% x 3) = sh.960,000

$$P + \left(P \times \frac{20}{100} \times 3\right) = \text{sh. } 960,000$$

$$\frac{P}{1} + \frac{6P}{10} = \text{sh. } 960,000$$

$$10 \times \frac{P}{1} + \frac{6P}{10} = \text{sh. } 960,000 \times 10$$

10P + 6P = sh. 960,000 x 10

16P = sh.960,000 x 10

$$\frac{16P}{16} = \text{sh. } \frac{960,000 \times 10}{16}$$

P = sh. 600,000

4. **Example 2.**

Find the amount of money that can amount to sh. 26,400 in 8 months at a rate of 15% per year.
Soln.

P + S. I = amount

P + (P x R x T) = sh. 26,400

P + P x 15% x $\frac{8}{12}$ = sh. 26,400

P + P x $\frac{15}{12}$ x 8 = sh. 26,400

P + P x $\frac{30}{300}$ = sh. 28,400

P x 300 + $\frac{30P}{300}$ x 300 = sh. 26400 x 3,000

300P + 30P = sh. 26,400 x 300

330P = sh.26,400 x 300

$\frac{330P}{330} = \frac{26400 \times 300}{330}$

P = sh. 24,000

ACTIVITY:

1. Find the amount of money that when deposited in a bank for three months at a simple interest rate of 20% per year can amount to sh. 378,000.

2. Kalevu borrowed a certain amount of money at a rate of 10% per year. If he paid sh. 9,000 after two years altogether. How much did he borrow?

3. Kyaligonza deposited a sum of money in a bank that offers a simple interest rate of 20% per year. After 4 years she withdrew sh. 216,000. How much did he deposit in the bank?

4. Kyamute borrowed some money from the bank at an interest rate of $33\frac{1}{3}$% per annum. If he paid sh. 21,000 in six months, how much did he borrow?

5. Kayondo deposited a certain amount of money in the bank which offer a simple interest rate of 2½ per annum. After 9 months on his account there was an amount of sh. 163,000. How much money did he deposit in the bank?

6. Find the sum of money such that when kept in a bank for 2 months at 10% p.a can amount to sh. 200,000.

7. Kemigisha borrowed a certain amount of money at a rate of 20% per year. If she paid sh. 10,000 after 5 years. How much did he borrow?

Scenarios

TOPIC 9 – GEOMETRIC CONSTRUCTION;

Session 1

- ✦ Constructing squares and rectangles
- ✦ Constructs triangles
- ✦ Constructs rhombuses and kites
- ✦ Constructs parallelograms and trapezium

Methods

- **Guided discovery**
- **Group discussion**
- **Think, pair and share.**
- **explanation**

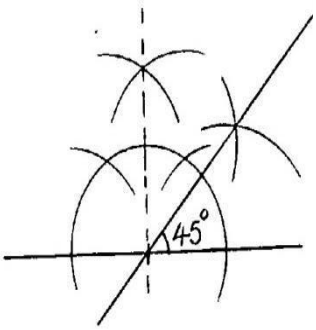
Lesson activities

- **adding**
- **multiplying**
- **dividing**
- **subtracting**

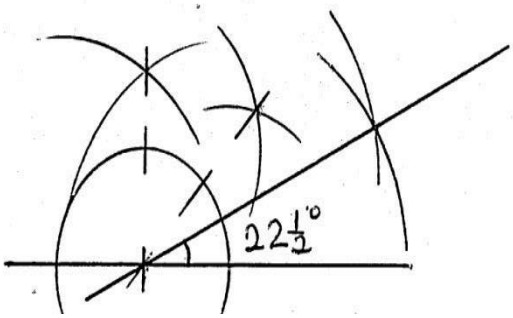
Content/facts

CONSTRUCTION OF

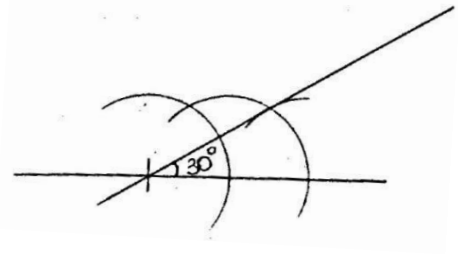
1. **Construct an angle of 45^0**



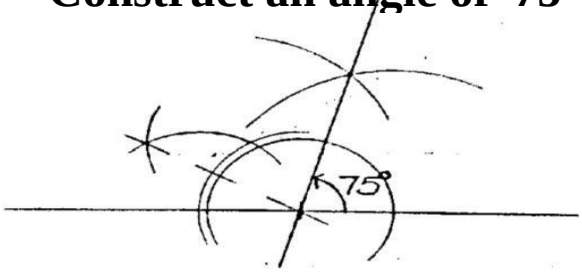
2. **Construct an angle of $22\frac{1}{2}^0$**

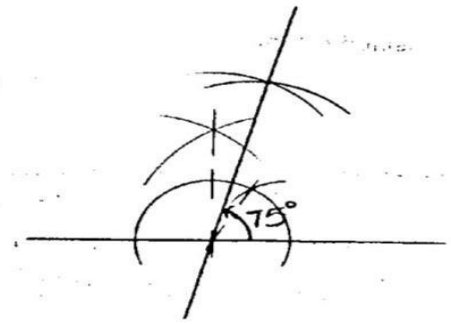


5. **Construct an angle of**



Construct an angle of 75^0





DROPPING A PERPENDICULAR FROM A GIVEN FIXED POINT

Guided Example I

1. **Using a ruler, a pencil and a pair of compasses, drop a perpendicular from M to meet line**

XY at p

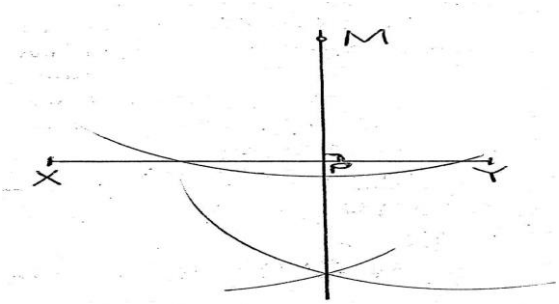
Procedure:

- **Place the needle at point M.**
- **Adjust the compass such that you make two acres on the given line.**
- **Transfer the needle to any of the arcs made and construct lower arc.**
- **Transfer the needle of the pair of compasses to the second arc and intersect the first arc to form a point of intersection.**
- **Join the point of intersection to point M using a pencil and a ruler.**

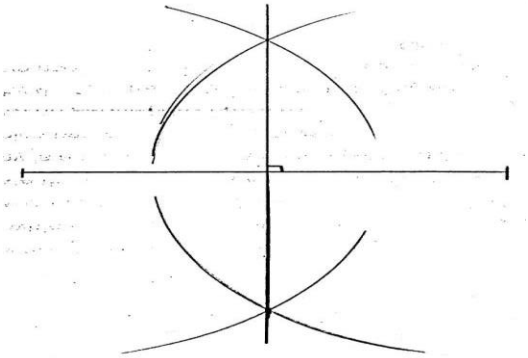
Guided example

2. **Using a ruler, a pencil and a pair of compasses, drop a perpendicular from M to meet line**

XY at p

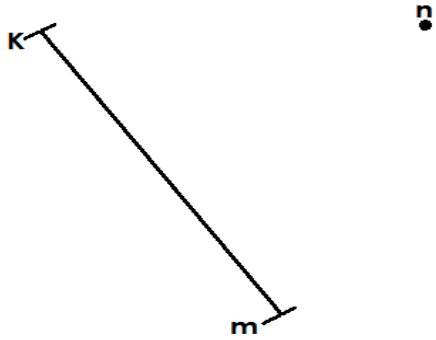


3. **Bisect the line below**



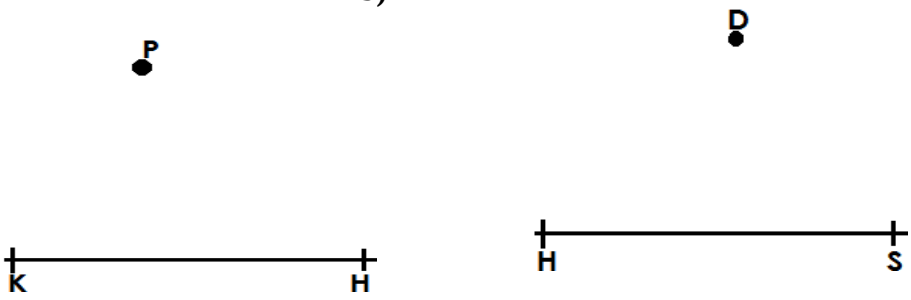
<u>Activity.</u>

1. **sing a pair of compasses, a ruler and a pencil only. Drop a perpendicular from n to meet line KM at point S.**

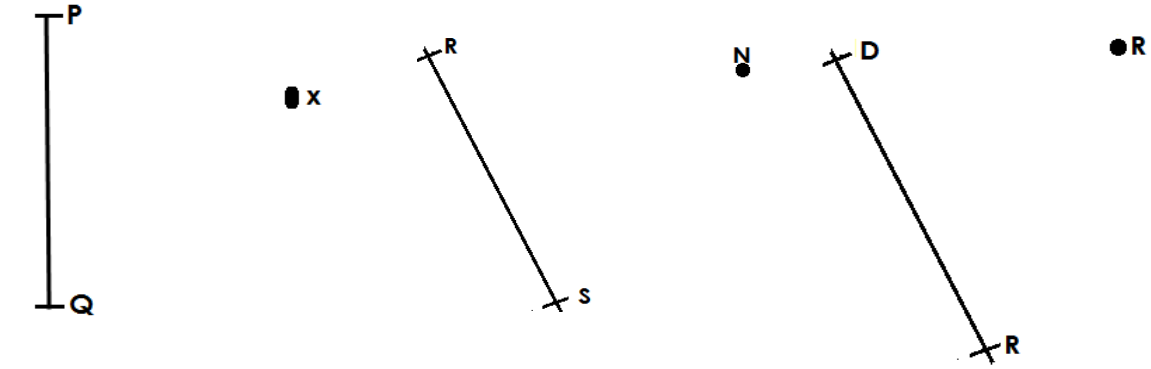


2. Using a pair of compasses, a ruler and a pencil only. Drop a perpendicular from a given point to meet a given line.

a) **b)**



c)



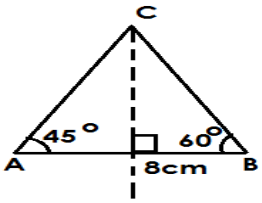
COMPETENCE TWO

CONSTRUCTION OF TRIANGLES, SQUARES, RECTANGLES

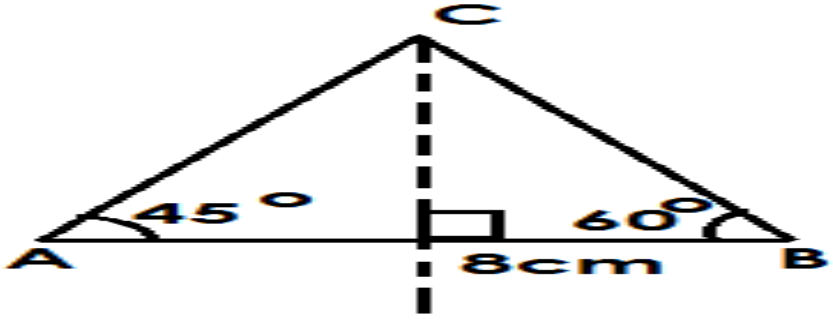
Guided Example

1. Using a pair of compasses, a ruler and a pencil only, construct a triangle ABC where *line* AB = 8cm, *L* ABC = 60^0 and *L* BAC = 45^0. Drop a perpendicular from C to meet AB at K.

sketch



Accurate diagram.



a) **Measure OK and calculate area of the triangle constructed.**

CK = 5.2cm

Area = ½ x b x h

= ½ x 8cm x 5.2cm

= ½ x 8cm x $\frac{52}{10}$cm

= $\frac{208}{10}$cm^2

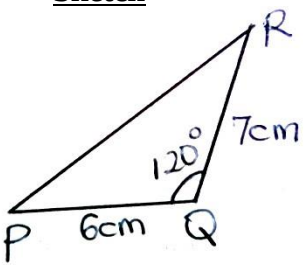
= 20.8cm^2

2. Using a pair of compasses, a ruler and a pencil only, Construct triangle PQR where line PQ = 6cm, angle PQR = 120

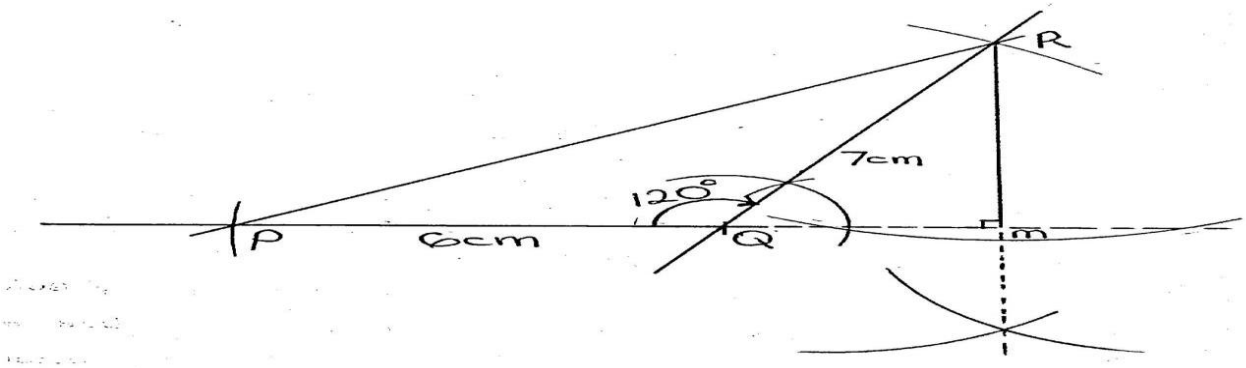
and QR=7cm.Drop a perpendicular from R to meet PQ at m

.

Sketch



Accurate diagram.



Measure QT

Soln.

QT = 4.4cm

Activity

Using a pair of compasses, a ruler and a pencil only, construct the following triangles:-

1. **Δ ABC , where $\overline{AC}$ = 7cm, *L* B AC = 45^0 and *L* ABC = 90°.**

ii) Measure $\overline{BC}$

2. **Δ XYZ where XY = 6.5cm angle YXZ = 60° and *L* XYZ = 45°. Drop a perpendicular from Z to meet XY at K.**

ii) Find the area of the triangle constructed.

3. **Δ MNO , where *L* M = 60^0, *L* N = 75^0, MN = 7cm. Drop a perpendicular from O to meet NM at Y.**

4. **Δ RST where ST = 5cm. *L*S = 120^0 and *L*T = 30^0. Drop a perpendicular from R to meet ST at V.**

ii) Measure VR.

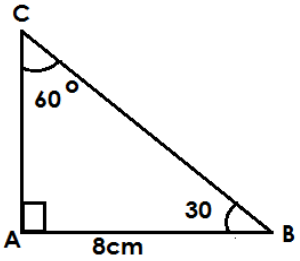
CONSTRUCTION OF TRIANGLES AFTER CALCULATIONS

GUIDED EXAMPLE

1. Using a ruler, a pencil and a pair of compasses only, construct a triangle ABC where AB = 8cm *L* A = 90^0, L C = 60°.

Soln

Sketch



Finding angle B.

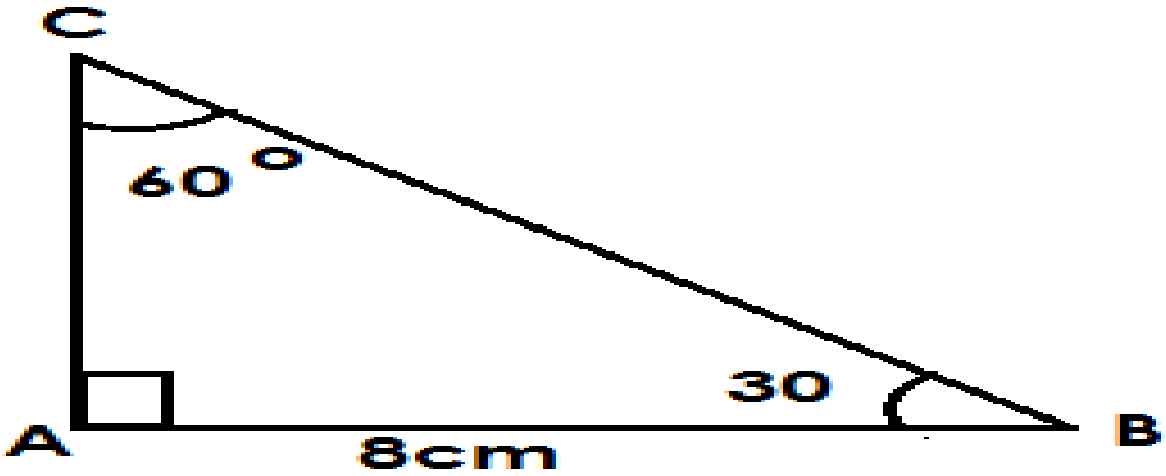
L B + 90° + 60° = 180°

L B + 150° = 180°

L B + 150° - 150° = 180° - 150°

L B = 30°

Accurate diagram



b) Measure:

i) AC = 4.5cm

ii) BC = 9.4cm

Activity

Using a ruler, a pencil and a pair of compasses only, construct the following triangles: -

1. **Triangle PQR, where PQ = 7cm, *L* P = 60^0, *L* R = 75^0. Measure lines PR and QR in cm.**
2. **Triangle EFG, where EF = 6cm, *L* E = 90^0, *L* G = 45^0. Measure GF in cm.**
3. **Triangle ABC, where AB = 55cm, *L* A = 45^0, *L* C = 90^0.**
4. **Triangle MNO, where MN = 4cm, *L* M = 120^0, *L*O = 30^0. Measure QN in cm.**
5. **Triangle XYZ, where XY= 5cm , *L*X = 60^0, *L* Z = 45^0. Measure lines XZ and YZ.**
6. **Triangle MNL , where MN = 6cm , *L*M = 60^0 and *L*L = 45^0. measure ML and LN in cm.**

CONSTRUCTING A SQUARE USING DIAGONALS

Note:

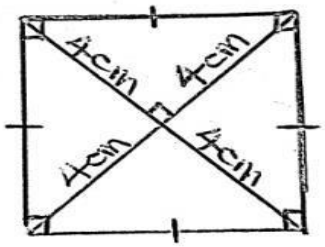
- ***Diagonals of a square are equal; they bisect each other at a right angle.***

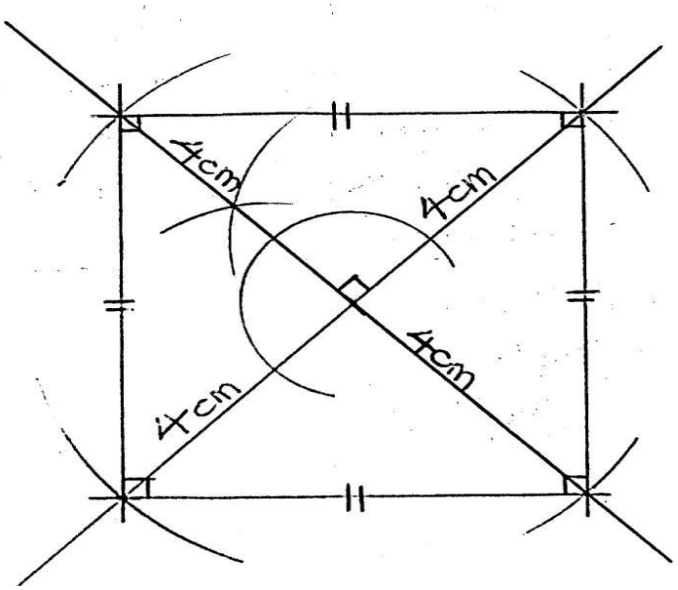
Guided example.

1.Using a ruler, a pencil and pair of compasses only, construct a square SRQP where diagonal PR = QS = 4cm.

Soln.

Sketch **ACCURATE DIAGRAM**



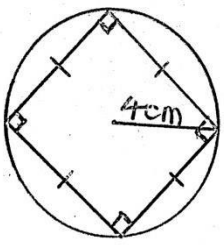


. Construct a square in a circle.

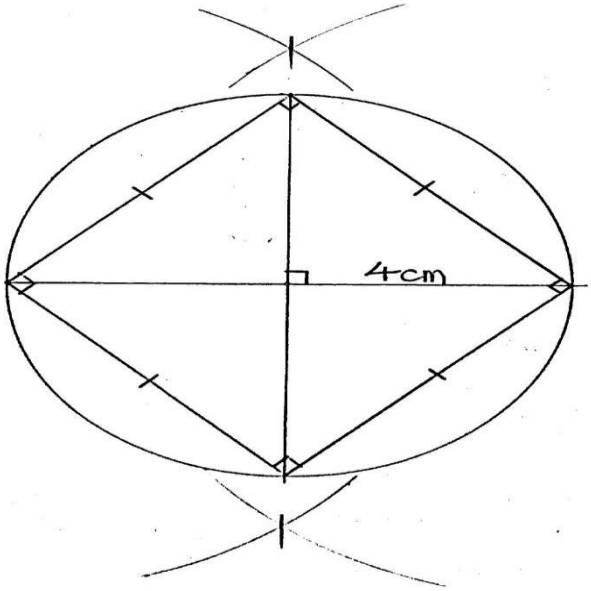
Step I

- ***Draw a circle of radius 4cm and on it draw diameter AB.***
- ***Draw a perpendicular bisector of AB and name the points of intersection with the circle CD.***
- ***Join the adjustment to form a square.***

Sketch



Accurate diagr



ACTIVITY

1. **Using a ruler and a pair of compasses only, construct the following squares: -**

ABCD of diagonals: -

a).6cm

(b) 8cm

2. Using a ruler, pencil and a pair of compasses only construct in a square in a circle of radius.

a) **3cm**

b) **4.5cm**

5cm

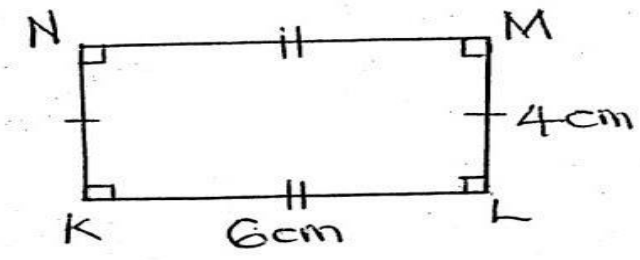
Construction of a rectangle

- **Opposite sides are equal and parallel.**
- **It has 4 right angles.**
- **Diagonals are equal and they bisect each other.**
- **Diagonals do not meet at right angles.**
- **It has 2 lines of folding symmetry.**
- **Opposite angles are equal.**

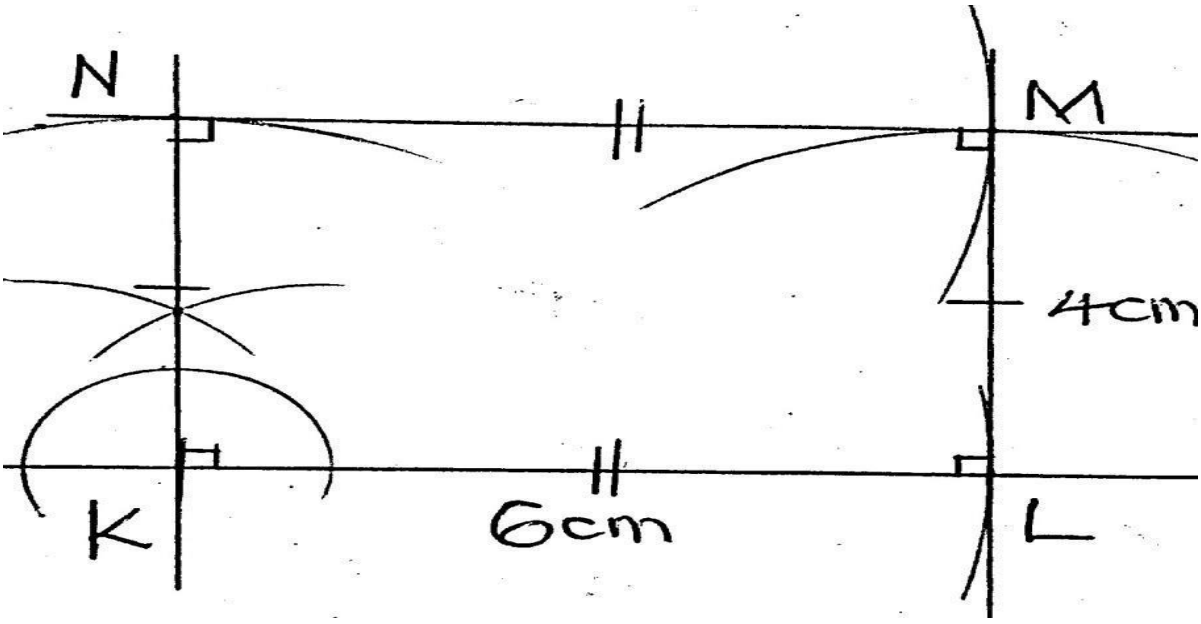
Guided example

1. **.Using a ruler, a pencil and pair of compasses only, Construct a rectangle KLMN where KL = 6CM, LM = 4CM.**

Sketch



Accurate diagram



Activity:

1. Using a ruler, a pencil and a pair of compasses only. Construct a rectangle MNOP where MN = 8cm and NO = 6cm.

a) Measure diagonal MO

b) Measure angle PMN

2. Using a ruler, a pencil and a pair of compasses only, construct a rectangle KLMN where KL = 7.5cm, LM = 5.5cm.

a) Measure diagonal NL

b) Measure the size of angle LKM

CONSTRUCTION OF A RHOMBUS GIVEN SIDE AND ANGLES

Note:

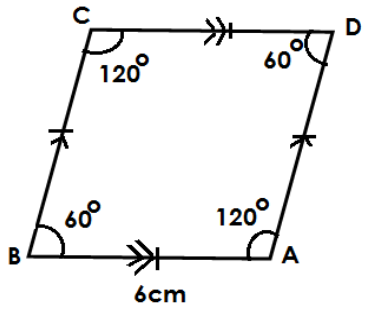
- **Opposite sides of a rhombus are parallel and equal.**
- **Opposite angles of a rhombus are equal.**

Example

Using a ruler, pencil and a pair of compasses only, construct a rhombus ABCD of sides 6cm and angle ABC = 60^0.

Soln.

Sketch



Activity:

1. **Construct a rhombus PQRS of side 5cm and angle PQR = 120º.**
2. **Construct a rhombus RPNS of side 6.8cm and angle RPN = 60º.**
3. **Construct a rhombus STNR of sides 5.8cm and angle TNS = 135º.**
4. **Construct a rhombus STUV of side 6.4cm and angle TSV = 120º.**
5. **Construct a rhombus ABC of side 6.5cm and angle DAB = 60º.**
6. **Construct a rhombus RMNO of side 7cm and angle RMN = 120º.**

CONSTRUCTING A RHOMBUS GIVEN DIAGONALS AND SIDE

Note:

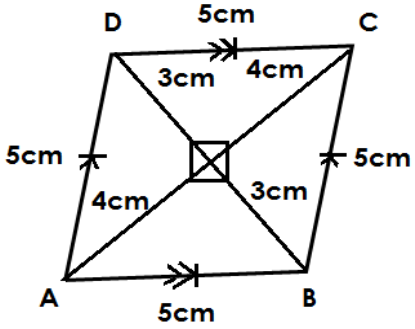
1. **Diagonals of a rhombus bisect each other at a right angle.**
2. **Diagonals of a rhombus are not equal.**

Example

Using a ruler and a pair of compasses only. Construct a rhombus ABCD of sides 5cm with diagonal AC = 8cm and BD = 6cm.

Soln.

Sketch



Activity:

Using a ruler, pencil and a pair of compasses only, construct:

1.A rhombus PQRS of side 6cm and diagonals PR = 9cm and QS = 8cm.

2.A rhombus ABCD of side 6.5cm and diagonals AC = 10cm, DB = 6cm.

3.A rhombus RSTU of side 6cm and diagonal RT = 7cm, SU = 6cm

4.A rhombus XYZP of side 5.8cm and diagonal XZ = 8cm, PY = 6cm.

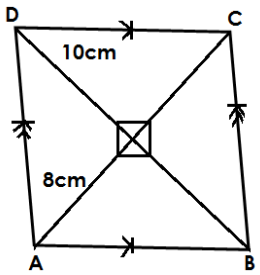
5.A rhombus AYZX of side 7cm and diagonals AZ = 10cm, XY = 8cm.

CONSTRUCTING A RHOMBUS GIVEN DIAGONAL ONLY

Example

Using a ruler, pencil and a pair of compasses only, construct a rhombus ABCD with diagonal BD = 10cm and AC = 8cm.

Sketch



b) Measure AB in cm.

AB = 6.3cm

Activity:

Construct the following rhombus below.

1. **A rhombus PQRS of diagonal PR = 8cm and SQ = 6cm.**

2. **A rhombus PNSR with diagonal PS = 12cm and RN = 10cm.**

3. **A rhombus STUV with diagonal SU = 9cm and VT = 7cm.**

4. **A rhombus RMNO with diagonal RN = 6cm and Mo = 4cm.**

5. **A rhombus AYZX with diagonal AZ = 8cm and XY = 6cm.**

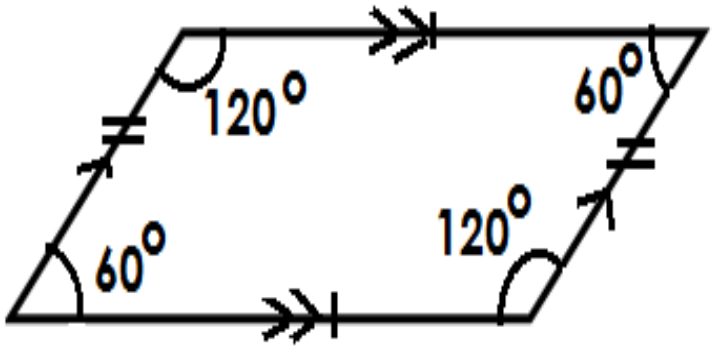
CONSTRUCTING A PARALLELOGRAM

A. Constructing a parallelogram given angles and side.

Note:

- **Opposite sides of a parallelogram are equal and parallel.**
- **Opposite angles of a parallelogram are equal.**

Illustration

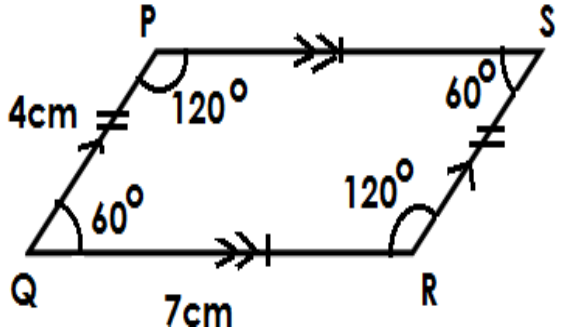


Example

Using a ruler, pencil and a pair of compasses only, construct a parallelogram PQRS where QR = 7cm, PQ = 4cm and angle PQR = 60.

Soln.

Sketch



Activity:

Using a ruler, pencil and a pair of compasses only, construct:

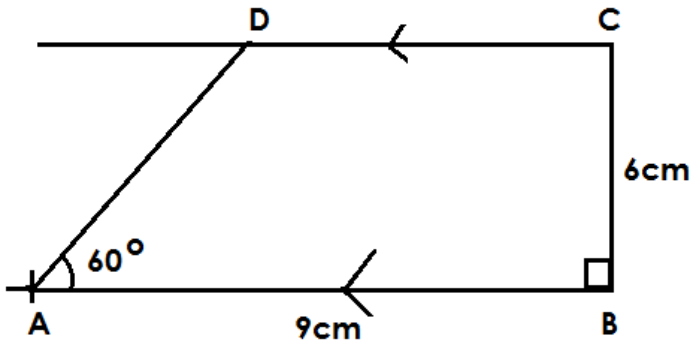
1. **A parallelogram JKLM where JK = 8cm, JM =4cm and angle MJK = 60º.**
2. **A parallelogram EFGH where EF=7.5cm, EH=6cm and angle HEF = 120º.**
3. **A parallelogram PQRS where QR = 7cm, PQ=3.5cm and angle PQR = 45º.**
4. **A parallelogram ABCD with AB = 6.8cm, BC = 5cm and angle ABC = 120º.**
5. **A parallelogram XYVM where XY=7.8cm YV = 6cm and angle XYV = 135º.**
6. **A parallelogram RMNO where RM=7cm, MN = 6cm and angle RMN = 75º.**

CONSTRUCTING TRAPEZIUM

Example I

1. **Using a pair of compasses, a ruler and a pencil only, construct a trapezium ABCD where AB = 9cm *L*ABC = 90º, *L*DAB = 60º and BC = 5cm.**

Sketch



b) Measure DC

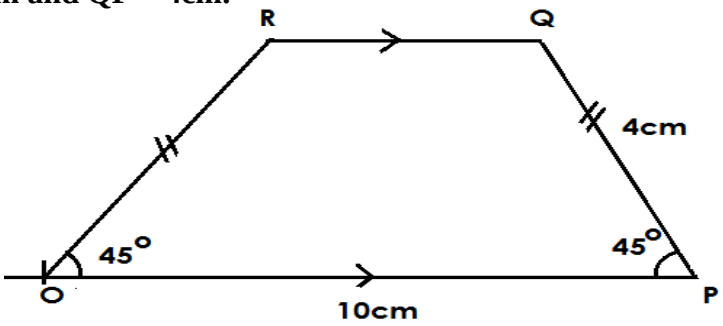
DC = 5.4cm

Example II

Using a pair of compasses, a ruler and a pencil only construct a quadrilateral OPQR where OP = 10 cm, L OPQ = L P

QOP = 45cm and QP = 4cm.

Sketch



Activity:

Using a ruler, pencil and a pair of compasses only, construct: -

1a) a trapezium PQRS where PQ = 8 cm angle PQR = 90º, angle = 60º and QR = 6.5cm.

b) Measure RS in cm.

2. A trapezium OPQR in which OP = 10cm, PQ = 6cm, angle ROP = 45º and OPQ = 90º.

b) Measure RQ in cm.

3. A trapezium WXYZ where WX = 11cm, and XY = ZW = 5.5cm, angle ZWX = 60^0 and angle YXW = 60^0.

b) Measure ZY.

c) Drop a perpendicular height from Z to meet WX at K.

d) Measure ZK in cm.

4. A quadrilateral MNOP in which MN = 9cm NO = PM = 5CM and angle MNO = 75^0.

b) Measure PQ in cm.

CONSTRUCTION OF A KITE

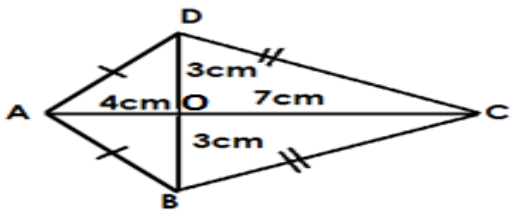
Note:

- ***Adjacent sides are equal.***
- ***Two of the opposite angles are equal.***
- ***Diagonals are not equal but they bisect at right angles.***

Example

1. using a pair of compasses, a ruler and a pencil only, construct a kite ABCD where AC meet BD at point O, AO = 4cm OC = 7cm and BD = 6cm.

Sketch



b) Measure AB

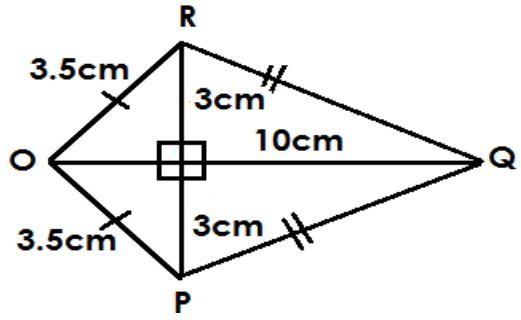
Soln.

AB =5cm

2. Using a pair of compasses, a ruler and a pencil only, construct a kite OPQR where OQ = 10cm, PR = 6cm and OP = OR = 35^0

Soln.

Sketch



a) Measure OR in cm.

OR $\overrightarrow{=}$ 4cm

Activity:

Using a ruler and a pair of compasses only, construct the following kites:

1. A kite ABCD where AK = 3cm, KC = 7cm and BK = KD = 4cm.

Measure AD and DC in cm.

2. A kite POQR where PX = 10cm and XQ = 5cm, OX = XR = 5cm.

Measure OP and RQ in cm.

3. A kite WXYZ where WY = 12cm and XZ = 8cm. measure XY in cm.

4. A kite OPMN where PN = 14cm and OM = 10cm.

Measure PM and MN in cm.

Scenarios

Session 2

- ✦ Introduction to angle properties (angles on parallel lines)
- ✦ Application of angles on parallel lines
- ✦ Introduction to polygons; interior angles, number of sides and no. of sides
- ✦ Triangulation

Methods

- **Guided discovery**
- **Group discussion**
- **Think, pair and share.**
- **explanation**

Lesson activities

- **drawing**
- **measuring**
- **subtracting**

Content/facts

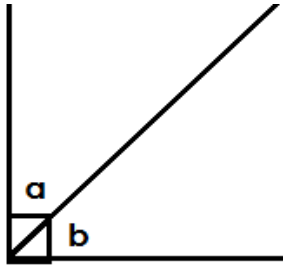
GEOMETRY

FINDING SUPPLIMENTS AND COMPLEMENTS OF COMPOUND ANGLES

Note:

- ***An angle is the amount of turning between two straight lines meeting at a fixed point.***
- ***Complementary angles are two or more angles that add up to 90º.***

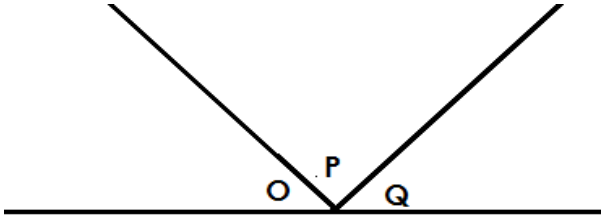
Illustration



***L* a and *L*b are complementary angles.**

- **Supplementary angles are two or more angles on a straight line that add up to 180º.**

Illustration



***LO,L* p and *L*Q are supplementary angles.**

Example I

Find the supplement of M + 40º

Soln

180º– (m + 40º

180 º – m – 40 º

180 º– 40 º – m

140 º– m

Example II

Find the complement of 2y + 80º.

Soln.

90º – (2y + 80º)

90º – 2y – 80º

90º – 80º – 2y

10º – 2y

Example III

Calculate the supplement of 3p – 60o.

Soln.

180º – (3p – 60º)

180º + 60 – 3p

240º – 3p

Example IV

Find the complement of $4y - 20^o$.

$90^o - (4y - 20^o)$

$90^o - 4y + 20^o$

$90^o + 20^o - 4y$

$110^o - 4y$

Activity:

A Find the supplements of the following angles:

1. $2k + 40^o$

2. $3m - 42^o$

3. $2y + 36$

4. $m + 70^o$

5. $3p + 110^o$

6. $2y - 120^o$

B Find the complementary angles of the following:

1. $3p - 20^o$

2. $60^o - p$

3. $4k + 50^o$

4. $m + 30^o$

5. $2w - 10^o$

6. $5y + 30^o$

MORE ABOUT COMPLEMENTS AND SUPPLEMENTS

Example I

Find the angle which is ½ of its supplement

Angles

Soln.

½ = 1:2 supplements

Total ratio = 1 + 2

= 3

Angle = $^1/_3$x 180°

= 60°

Example II

What angle is $\frac{1}{5}$ of its complement?

Soln.

$\frac{1}{5}$ = 1:5

Total ratio = 1+ 5

= 6

Angle $\frac{1}{6}$ = 1/6 x 90°

= 15°

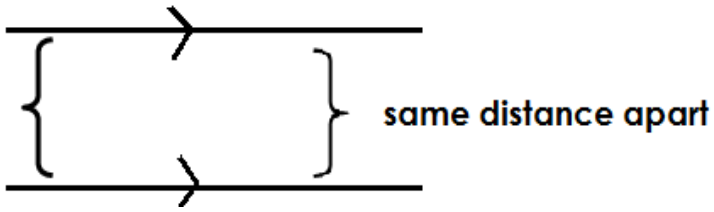
Activity:

1. **Find the angle which is$\frac{1}{3}$ of its supplement.**
2. **What angle is $\frac{3}{7}$ of its complement?**
3. **Which angle is $\frac{1}{4}$ of its supplement?**
4. **What angle is $\frac{2}{3}$ of its complement?**
5. **What angle is on $\frac{1}{8}$ of its supplements?**
6. **Find the angle which is $\frac{4}{5}$ of its supplement**
7. **Which angle is a $\frac{1}{9}$ of its complement?**

GEOMETRY (ANGLES AND LINES)

PARALLEL LINES

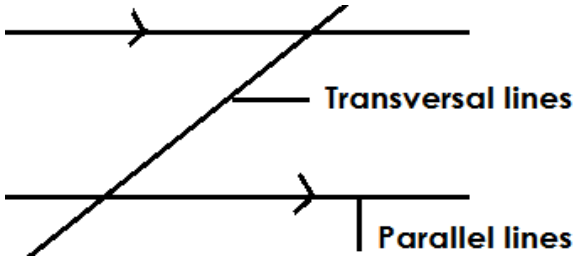
- **Parallel lines are lines that do not meet due to the same distance apart.**



When a transversal line intersects two parallel lines the different angles are formed and these include:

- **Co-interior angles**
- **Co-exterior angles**
- **Alternating angles**
- **Corresponding angles**
- **Vertically opposite angles**
- **Supplementary angles**
- **Complementary angles**

Illustration

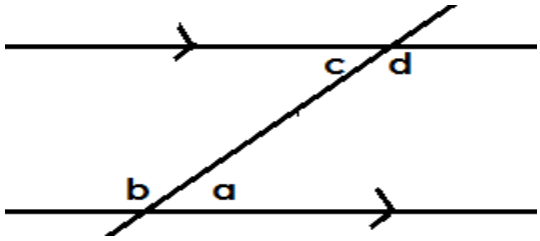


Co – interior angles

- **They add up to 180°**
- **They are found inside parallel lines.**
- **They are formed on the same side of the transversal line**

- **One angle is acute and another angle is obtuse.**

Illustration

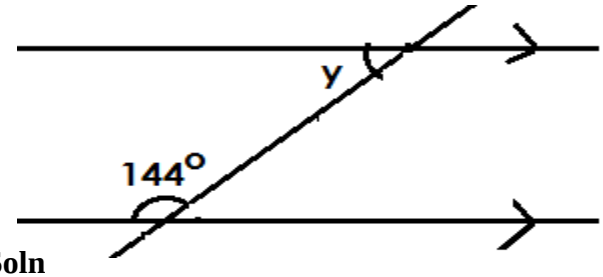


L **a + *L*d = 180º (co-int.*Ls*)**

L**c + *L*b = 180^0 (co-int.*Ls*)**

Example I

Find the value of y.



Soln

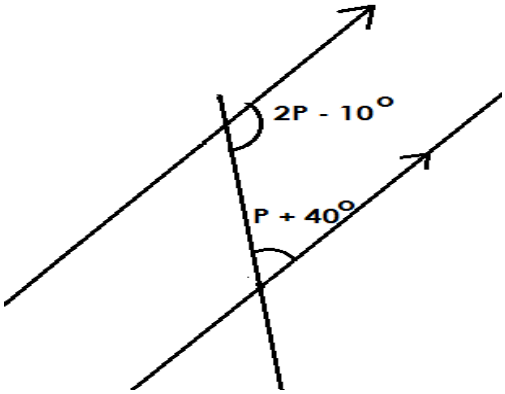
Y + 144º = 180º

Y + 144 º – 144º = 180 º – 144 º

Y = 36 º

Example I

Find the value of P.



Soln

$2P - 10^0 + P + 40^0 = 180^0$ (co-int Ls)

$2p + P - 10^o + 40^o = 180^o$

$3p + 30^o = 180o$

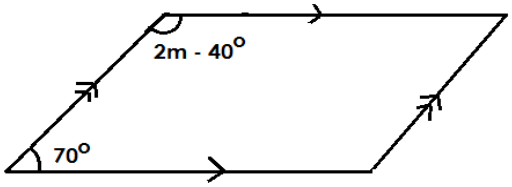
$3p + 30^o - 30^o = 180^o - 30^o$

$3p = 150^o$

$\frac{3P}{3} = \frac{150}{3}^o$

$P = 50^o$

In the figure below, find the value of M.



$2m - 40^o + 70^o = 180^o$ (co-int Ls)

$2m + 30^o = 180^o$

$2m + 30^o - 30^o = 180^o - 30^o$

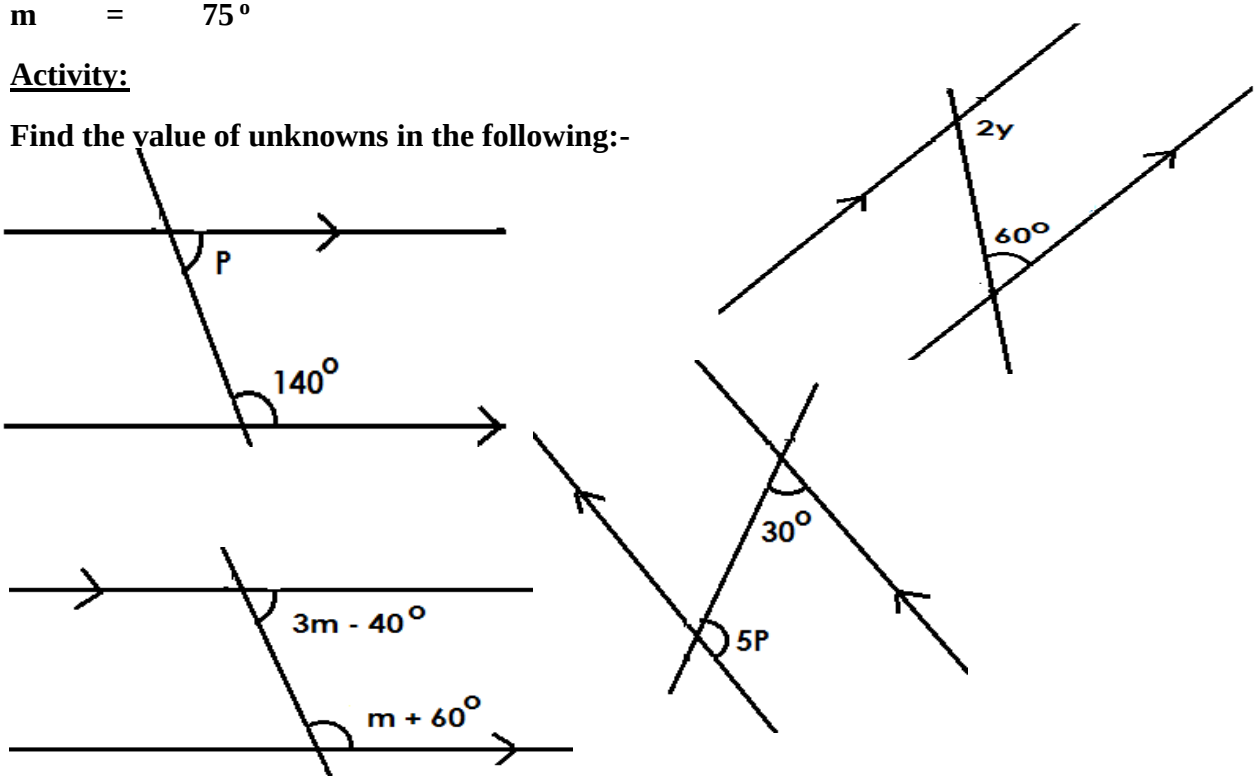
$2m = 150^o$

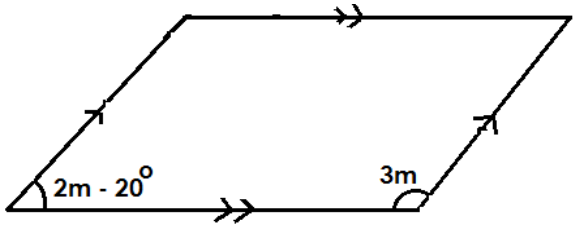
$\frac{2m}{2} = \frac{150}{2}$

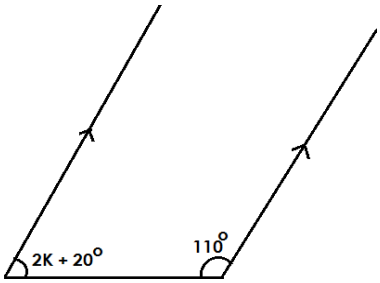
$m = 75^o$

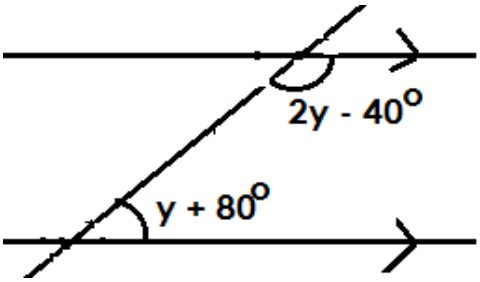
<u>Activity:</u>

Find the value of unknowns in the following:-





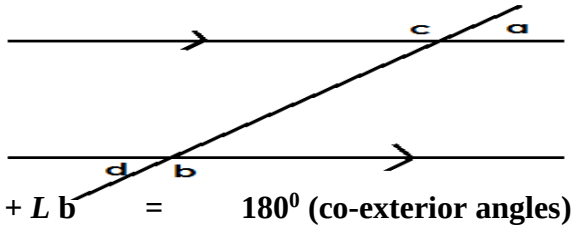




CO-EXTERIOR ANGLES

- ***Co-exterior angles add up to 180^0***
- ***They are found outside the parallel lines.***
- ***They are formed on the same side of the transversal line.***
- ***One angle is obtuse and another angle is acute.***

Illustration

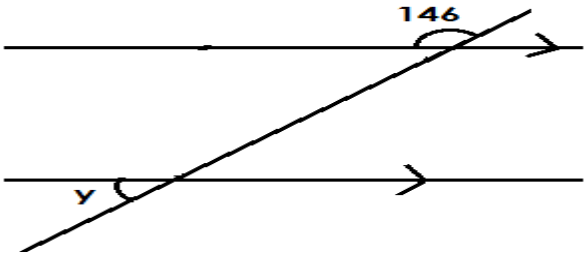


***L* a + *L* b = 180^0 (co-exterior angles)**

***L* d + *L* c = 180^0 (co-exterior angles)**

Example I

Find the size of angle y.



Soln

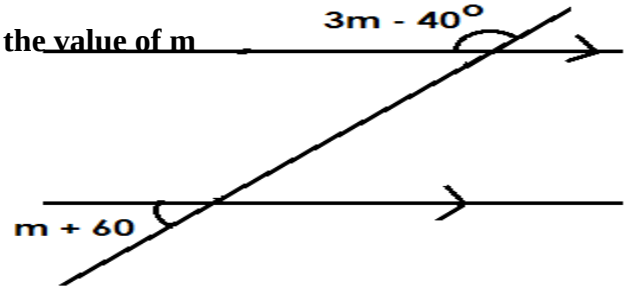
y + 146°= 180^0 (co-exterior *Ls*)

y + 146° – 146° = 180° – 146°

y = 34°

Example II

Find the value of m



Soln

$3m - 40^{o} + m + 60^{o} = 180^{o}$(co-ext Ls).

$3m + m - 40^{o} + 60^{o} = 180^{o}$

$4m + 20^{o} = 180^{o}$

$4m + 20^{o} - 20^{o} = 180^{0} - 20^{o}$

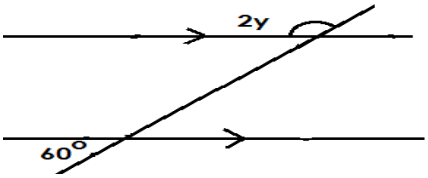
$4m = 160^{o}$

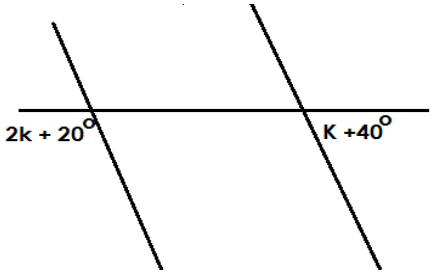
$\frac{4m}{4} = \frac{160}{4}$

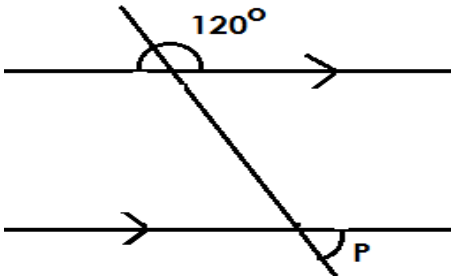
$m = 40^{o}$

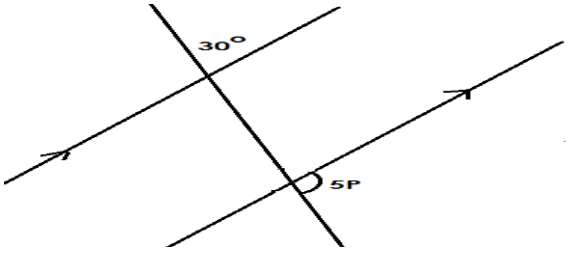
Activity:

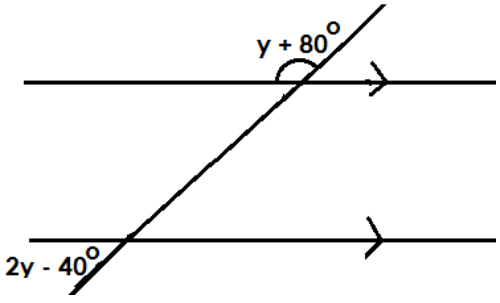
Find the value of unknowns in the following:

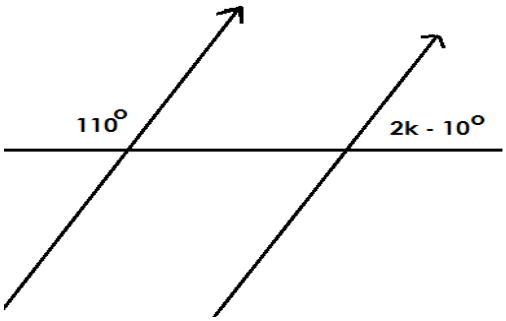










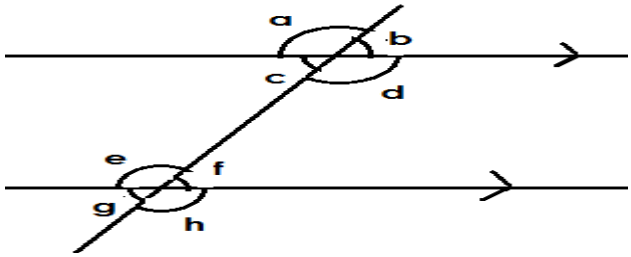


CORRESPONDING ANGLES

Note:

- ***Corresponding angles are equal in degrees.***
- ***Corresponding angles are formed on the same side of the transversal line.***
- ***One is found inside and another one is found outside parallel lines.***
- ***One angle is formed on the lower parallel line and another angle on the upper parallel line.***

Illustration



L **a = *L*** **e**

L **b = *L*** **f**

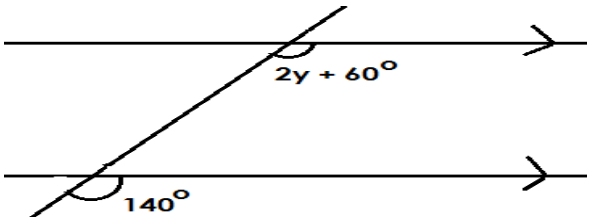
L **c = *L*** **g**

L **d = *L*** **h**

Pair of corresponding angles

Examples

1. **Find the value of Y**



Soln

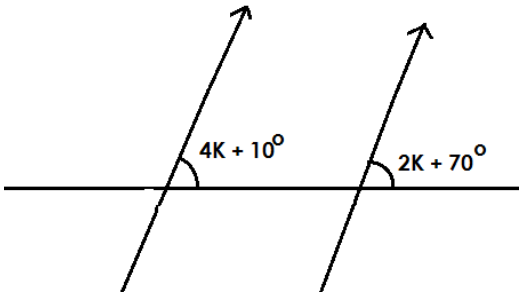
$2y + 60^o$ = 140° (corresponding angles)

$2y + 60^o - 60^o$ = $140^o - 60^o$

$\frac{2y}{2} = \frac{80}{2}$

y = 40°

2. Find the value of K in the diagram below.



Soln

$4K + 10^o$ = $2K + 70^o$ (corresponding angles)

$4k + 10^o - 10^o =$ $2K + 70^o - 10^o$

$4K = 2k + 60^o$

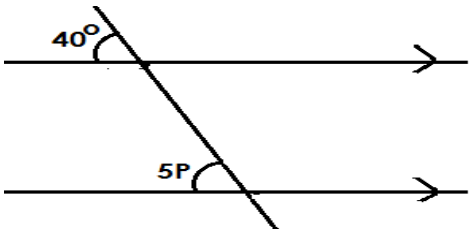
$4k - 2k = 2k - 2k + 60^o$

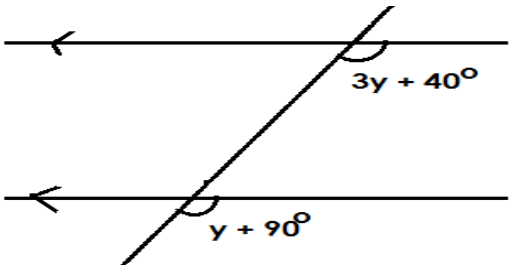
$\frac{2K}{2} = \frac{60}{2}$

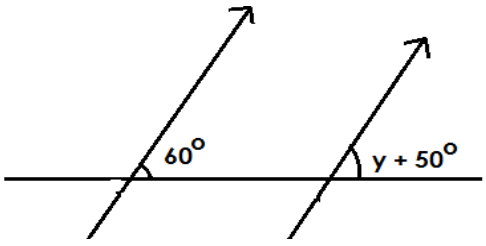
$K = 30^o$

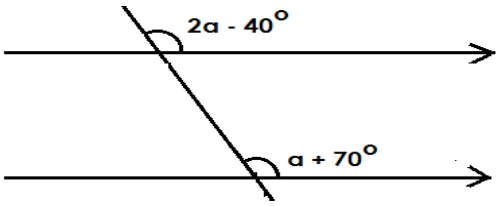
<u>Activity:</u>

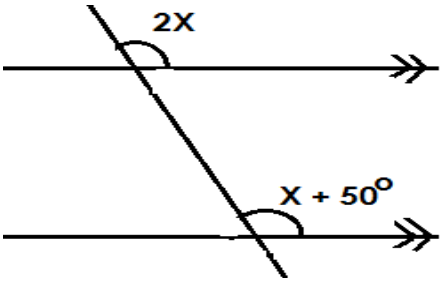
Find the unknown in each of the following diagrams: -

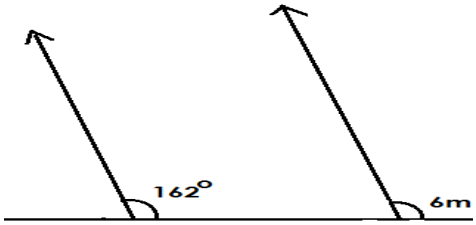


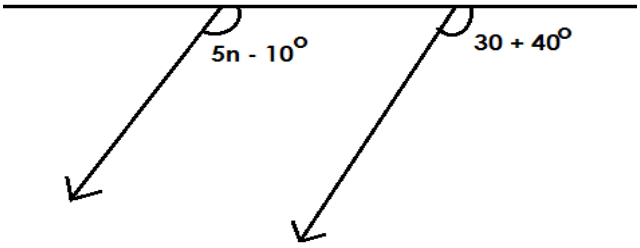


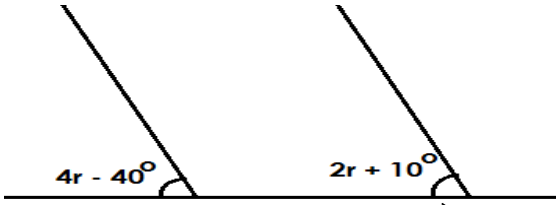


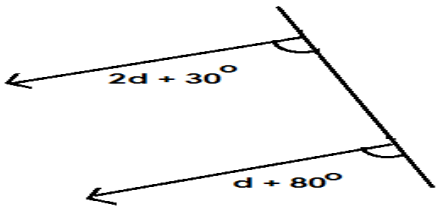


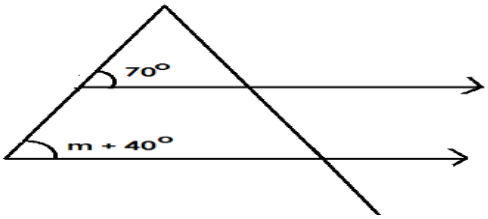












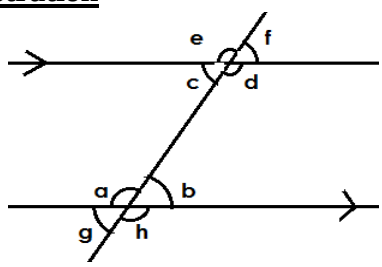
ALTERNATING ANGLES

(Alternate interior and alternate exterior angles)

Note

- ***Alternating angles are equal.***
- ***Alternating angles are formed on the opposite sides of the transversal line.***
- ***If they are inside, they are called alternate interior and if they are outside the called alternate exterior.***

Illustration



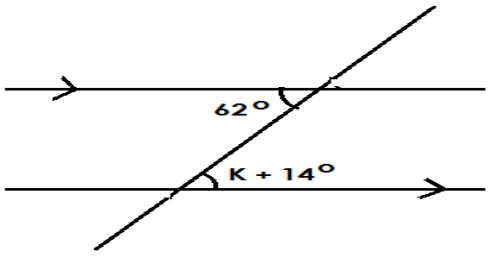
L a = L d
L c = L b } **Alternate interior angles.**

Then

L f = L g
L e = L h } **Alternate exterior angles.**

Examples

1. Find the value of K in the diagram below.



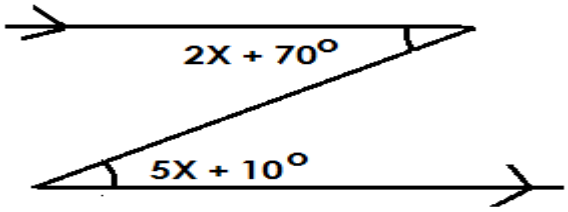
Soln

K + 14º = 62º (alternating interior angles)

K + 14º – 14º = 62º – 14º

K = 48º

2. Find the value of X



Soln

5X + 10º = 2X + 70º (alternating interior angles)

5X + 10º – 10º= 2X + 70º – 10º

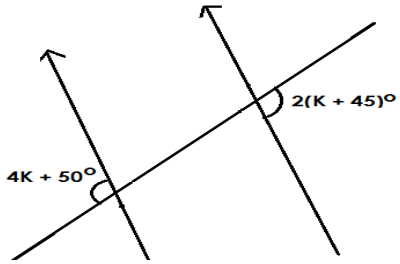
5X = 2X + 60º

5X – 2X = 2X – 2X + 60º

$\frac{3X}{3} = \frac{60}{3}$

K = 20º

5. Find the value of the unknown in the diagram below.



Soln

$4K + 50^o$ = 2(K + 45º) alternate exterior angles

$4k + 50^o$ = $2k + 90^o$

$4k + 50^o - 50^o$ = $2k + 90^o - 50^o$

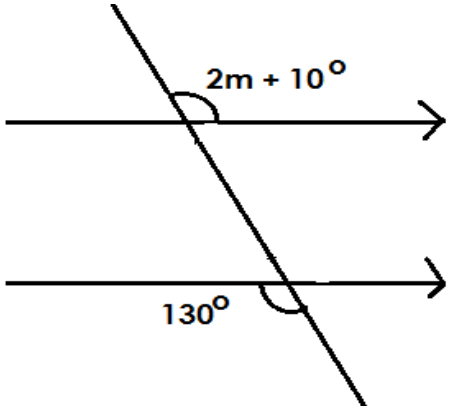
4k = $2k + 40^o$

4k – 2k = 2k – 2k + 40°

$\frac{2k}{2} = \frac{40}{2}$

K = 20o

4. Find the value of the unknown.



Soln

$2m + 10^o$ = 130^0 (alternate exterior angles)

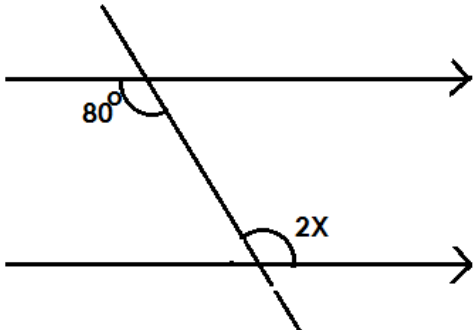
$2m + 10^o - 10^o$ = $130^o - 10^o$

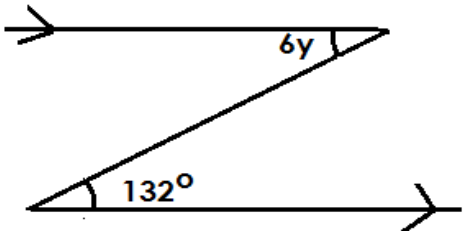
$\frac{2m}{2} = \frac{120}{2}$

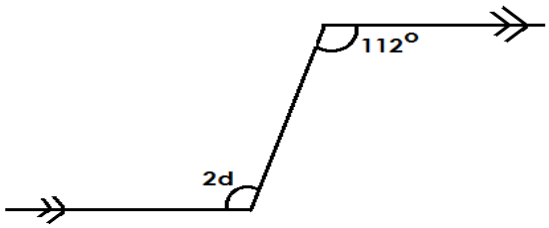
m = 60°

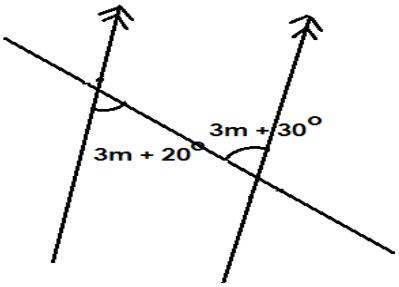
Activity:

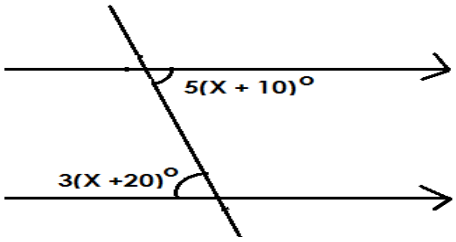
Find the value of the unknown in each of the diagrams below.

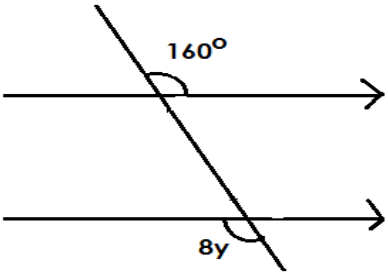


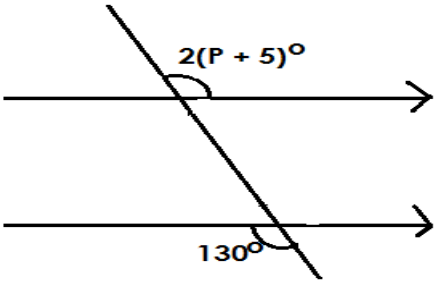


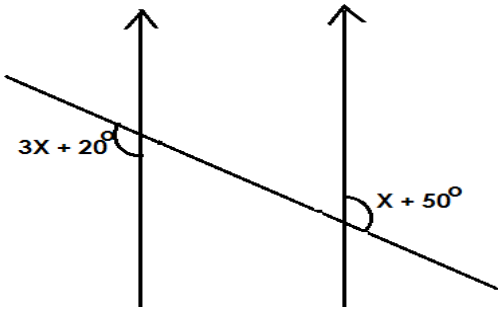








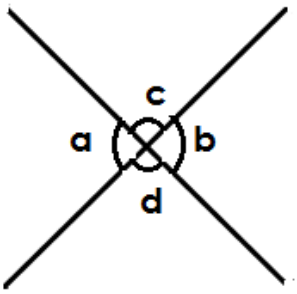




VERTICALLY OPPOSITE ANGLES

- **Vertically opposite angles are equal in degrees.**
- **They are formed by letter X (X – angles)**

Illustration

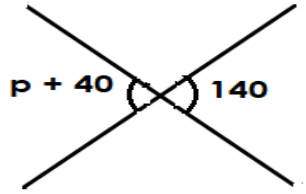


***L* a = *L*b Vertically opposite angles**

***L*c = *L* d**

Examples

Find the value of P in the figure below



Soln

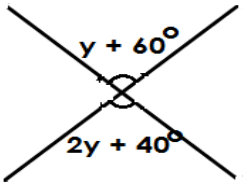
P + 40º = 140 º

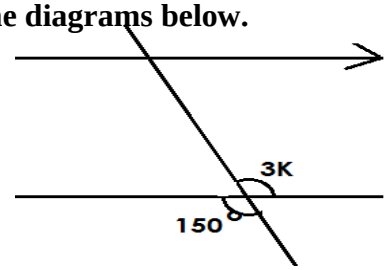
P + 40 º – 40 º

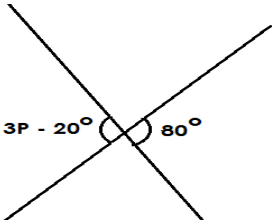
P = 100 º

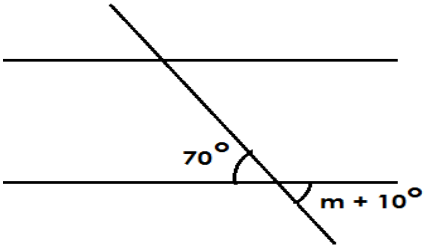
Activity

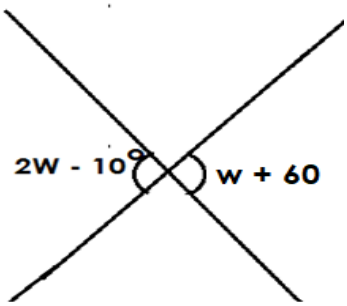
Find the unknown in each of the diagrams below.

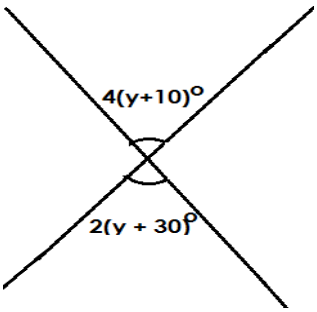


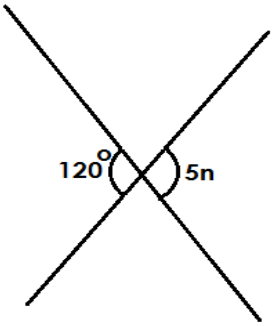






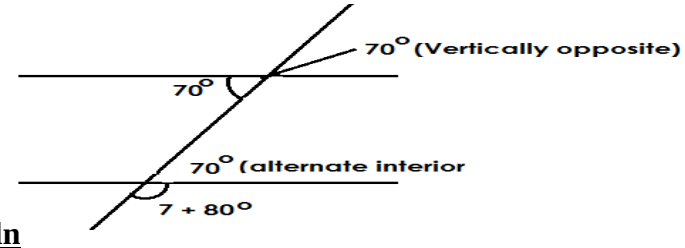






APPLICATION OF PARALLEL LINES

1. In the diagram below, find the value of y.



Soln

Y + 80º + 70 º = 180 º (Co-exterior angles)

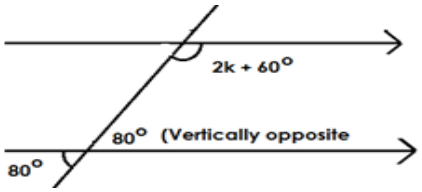
Y + 150 º = 180 º

Y + 150 º – 150 º = 180 º – 150 º

Y = 30 º

2. Example 2

Find the value of K



Soln

2k + 60o+80 o = 180o (Co-interior angles)

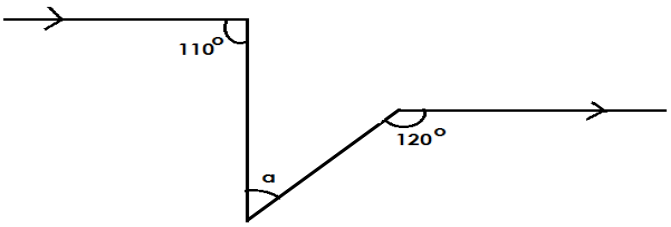
2k + 140 o = 180 o

2k + 140 o – 140 o = 180^0 – 140 o

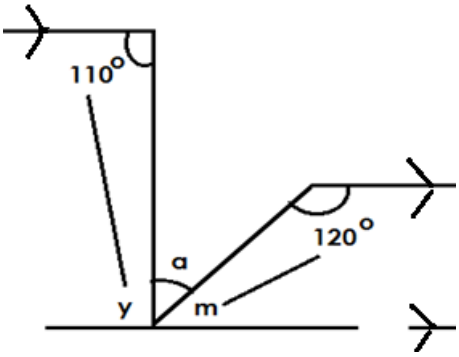
$$\frac{2k}{2} = \frac{40}{2}$$

K = 20 o

2. Find the value of a.



Soln



Y + 110º = 180 º (Co-interior angle)

Y + 110 º – 110 º = 110 º

Y = 70 º

m+ 120 º = 180 º (Co-interior angle)

m + 120 º – 120 º = 180 º – 120 º

m = 60 º

Therefore:

a + y + m = 180º (Supplementary angles)

a + 70 º + 60 º = 180 º

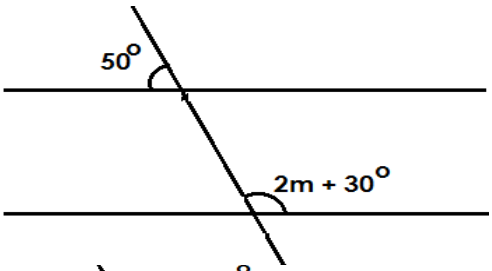
a + 130 º = 180 º

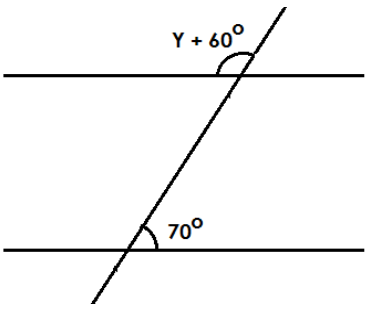
a + 130 º – 130 º = 180 º – 130 º

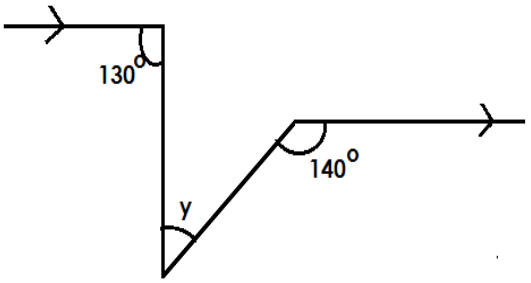
a = 50 º

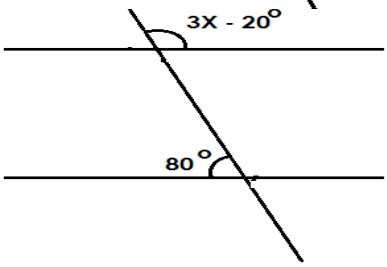
ACTIVITY:

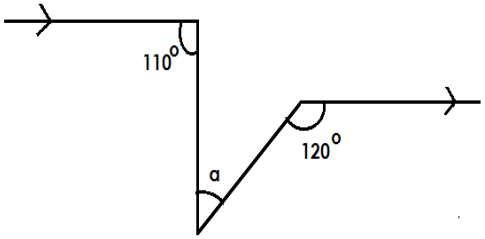
Find the value of the unknown in each of the diagrams below: -

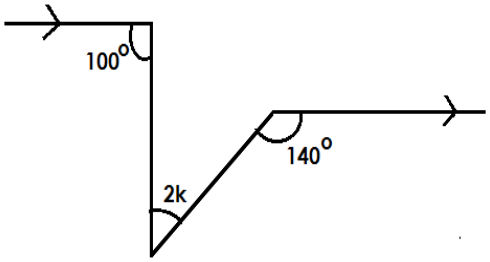


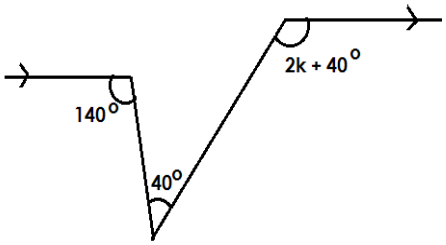


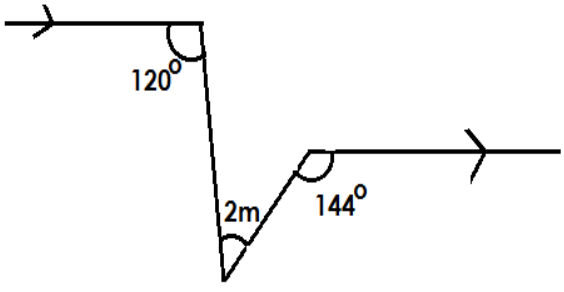








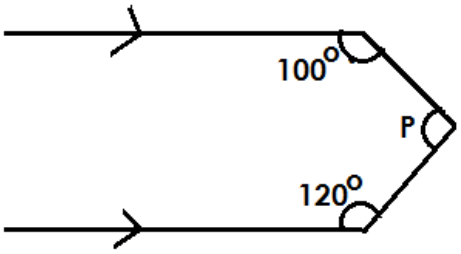




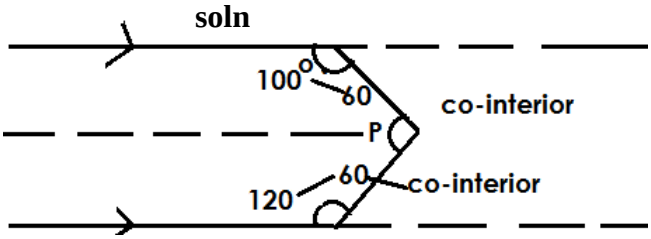
MORE ON APPLICATION OF PARALLEL LINES

Examples

Find the value of



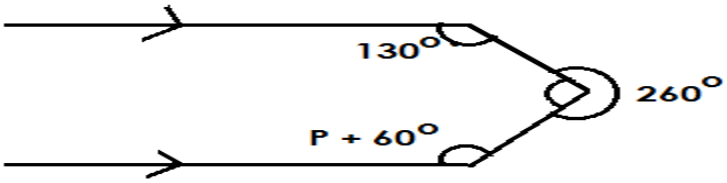
soln



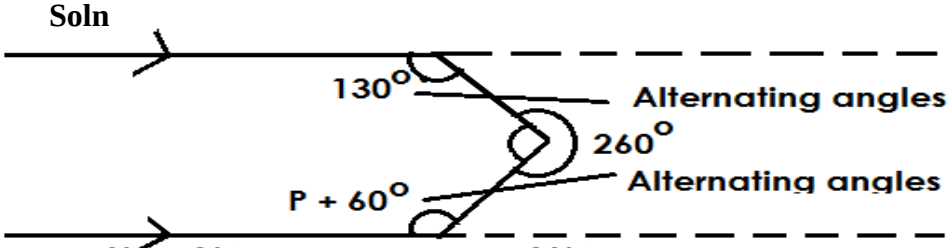
P = 60º + 80º

P = 140º

2. Find the value of P.



Soln



P + 60º + 130 º = 260 º

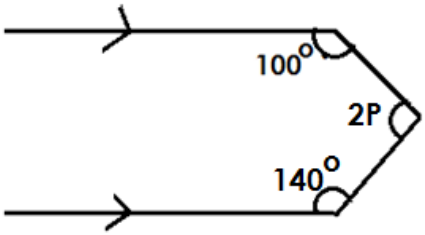
P + 190 º = 260 º

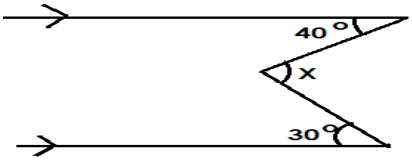
P + 190º – 190 º = 260 º – 190 º

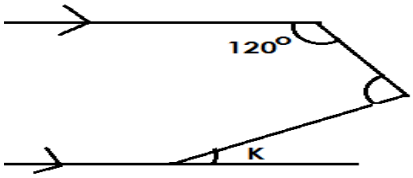
P = 70 º

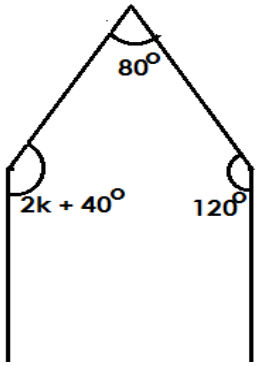
ACTIVITY

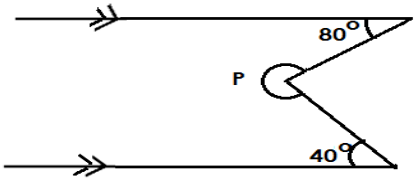
Find the unknown in each of the figures below.

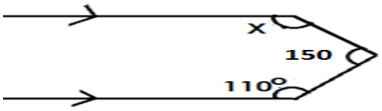


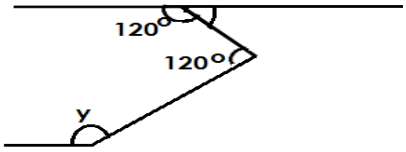


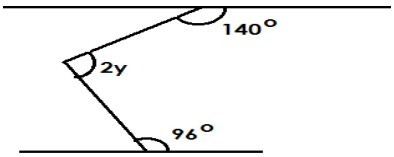


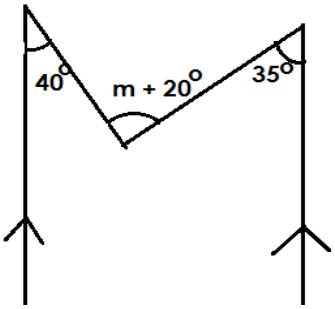








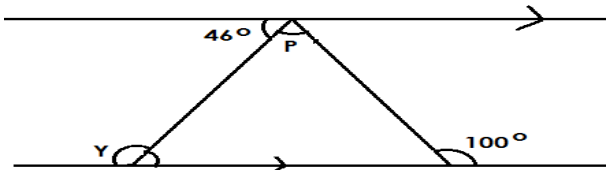




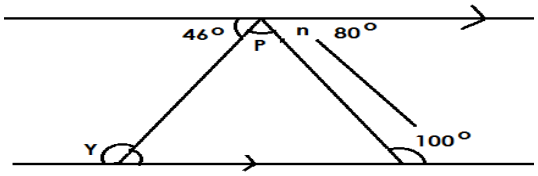
PARALLEL LINES AND TRIANGLES

Examples

1. Find the value of P.



Soln



$n + 100^{o} = 180^{o}$ (co-interior angles

$n + 100^{o} - 100^{o} = 180^{o} - 100^{o}$

$n = 80^{o}$

a) P + 80º + 46º = 180º (straight line angles)

P + 126º = 180º

P + 126º – 126º = 180º – 126º

P =54º

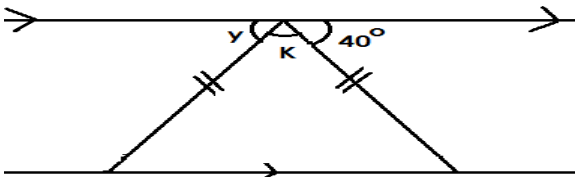
b) Find the value of Y

Y + 46º = 180º (Co-interior angles)

Y + 46º- 46º = 180º – 46º

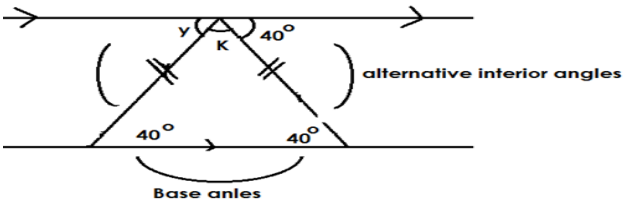
Y = 134º

2. Given the figure below:-



a) Find the value of K.

soln



K = 40º + 40 º = 180 º (interior angles sum of a triangle)

K + 80 º = 180 º

K + 80 º – 80 º = 180 º – 80 º

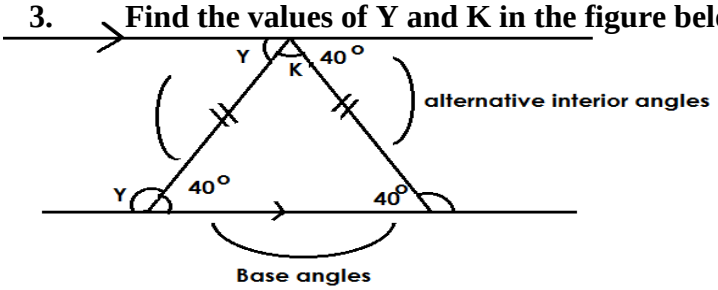
K = 100 º

b) Find the value of y

soln

Y = 40^0 (Alternate interior angles)

3. Find the values of Y and K in the figure below.



K + 40º + 40º = 180 º 9interior angle sum of a triangle)

K + 80 º = 180 º

K + 80 º – 80 º = 180o – 80 º

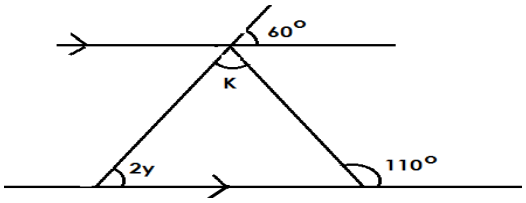
K = 100 º

b) Find the value of y

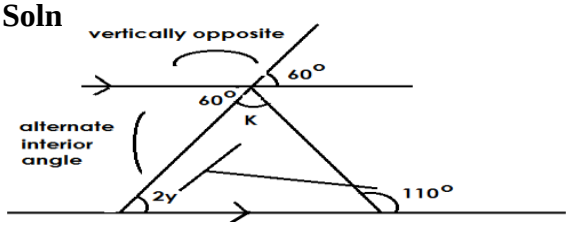
soln

Y = 40º (alternate interior angles)

3. Find the value of Y and K in the figure below;



Soln



$2y = 60^0$ (alternate interior angle)

$\frac{2y}{2} = \frac{60}{2}$

$Y = 30^o$

$K + 2y = 110^0$ (sum of 2 opposite interior angles)

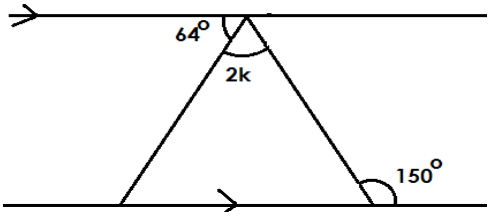
$K + 60^o = 110^o$

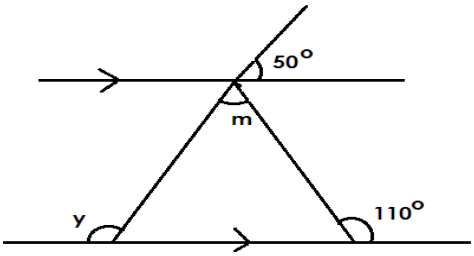
$K + 60^o - 60^o = 110^o - 60^o$

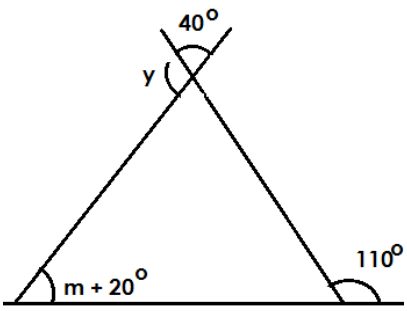
$K = 50^o$

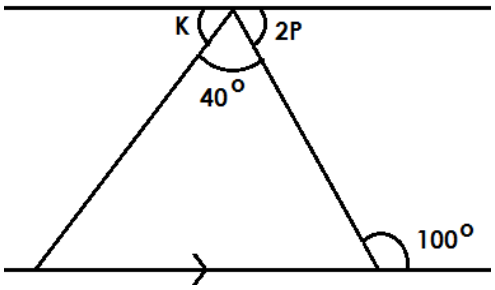
<u>Activity:</u>

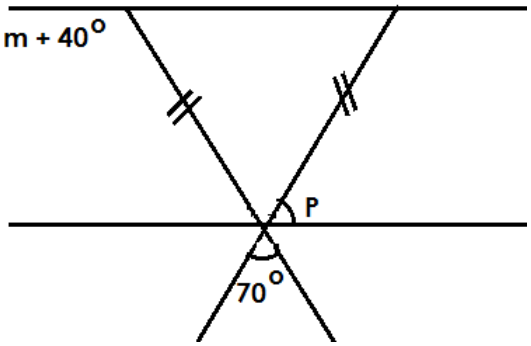
Find the unknowns in each of the diagrams below.

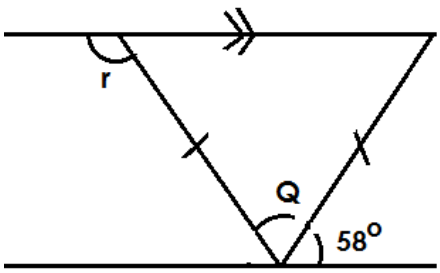


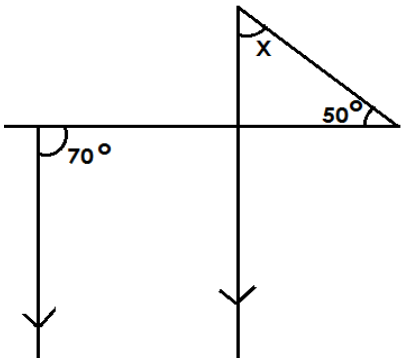


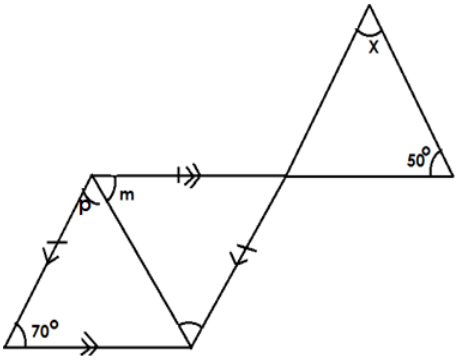


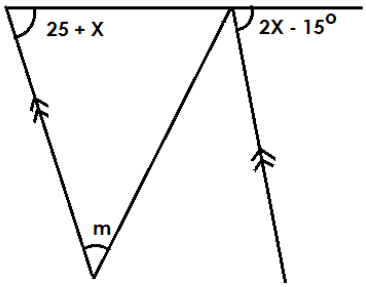


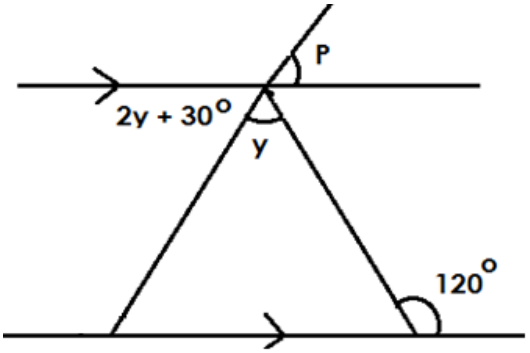










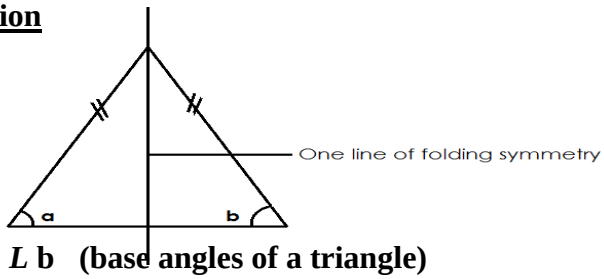


ANGLE PROPERTIES ON TRIANGLES

1. ISOSCELES TRIANGLES

- ***These are triangles with two sides equal.***
- ***Their base angles are equal.***
- ***An isosceles triangle has one line of folding symmetry.***

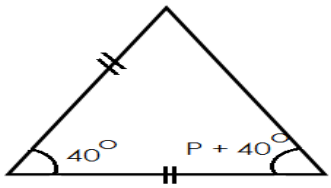
Illustration



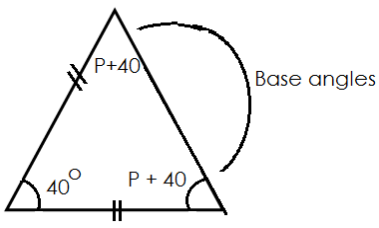
***L*a = *L* b (base angles of a triangle)**

Example I

Find the value of unknowns in the figure below.



Soln



P + 40º + P + 40 º +40 º = 180 º (interior angle sum of a triangle)

P + P + 40 º + 40 º + 40 º = 180 º

2P + 120 º = 180 º

2P + 120 º = 180 º

2P + 120 º – 120 º = 180 º – 120 º

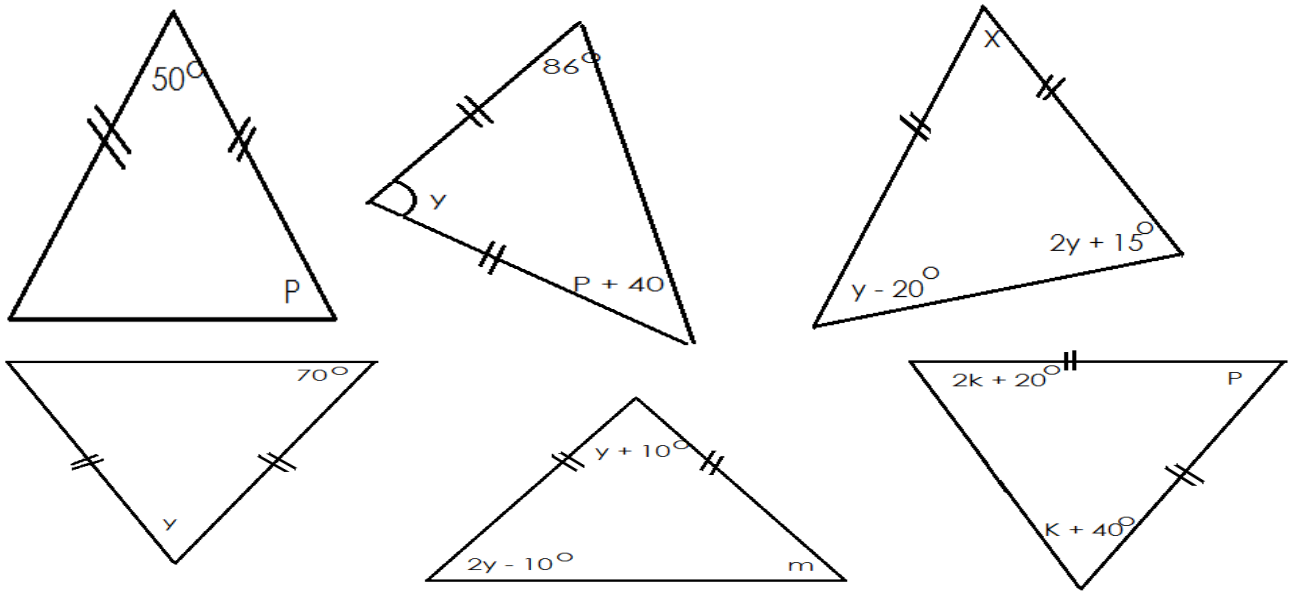
2P = 60 º

$\frac{2P}{2} = \frac{60}{2}$

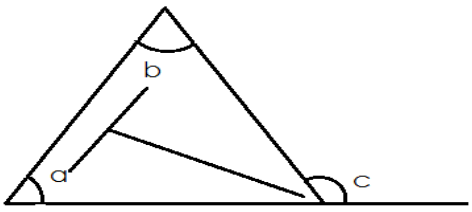
P = 30 º

Activity:

Find the value of unknowns in the following diagrams.



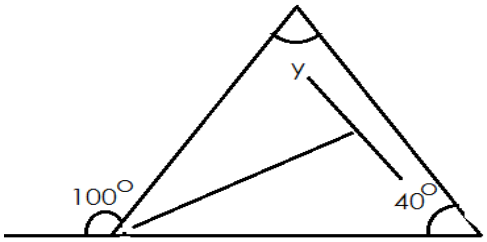
THE SUM OF TWO INTERIOR ANGLES BEING EQUAL TO ONE OPPOSITE EXTERIOR ANGLE



***L* a + *L* b = *Lc*(sum of 2 int.*Ls* of =1 opp. Ext.*L*)**

Example I

Find the value of y in the triangle below.



Soln

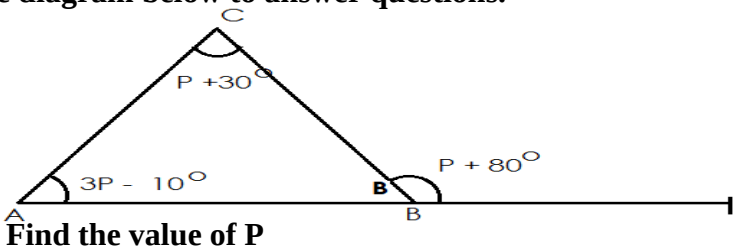
Y + 40º = 100 º (sum of 2 interior angles = 1 opposite exterior angle)

Y + 40 º – 40 º = 100 º – 40 º

Y = 60 º

Example II

Use the diagram below to answer questions.



a) Find the value of P

soln

3P – 10 º + P + 30 º = P + 80 º (sum of 2 interior angles = 10 opposite exterior angles)

3P + P – 10 º + 30 º = P + 80 º

4P + 20 º = P + 80 º

4P + 20 º = P + 80 º

4P + 20 º – 20º= P + 80 º – 20 º

4P = P + 60 º

4P – P = P – P + 60 º

3P = 60 º

$\frac{3P}{3} = \frac{60}{3}$

P = 20º

b) Find the size of angle ABC

soln

***L*B + P + 80º = 180 º (supp *L*s)**

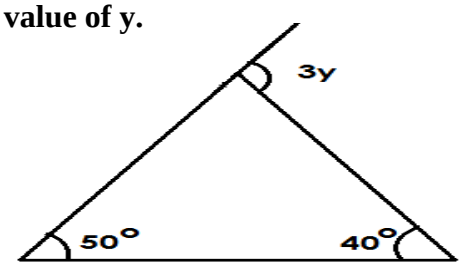
L b + 20º + 80º = 180º

B + 100º = 180º

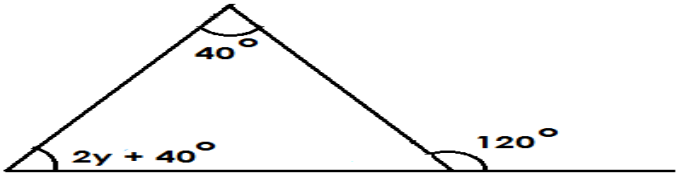
B = 80 º

<u>Activity:</u>

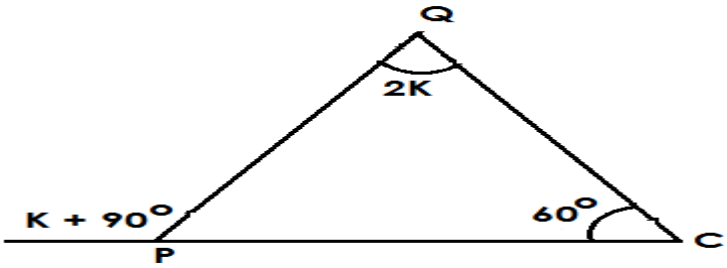
1. Find the value of y.



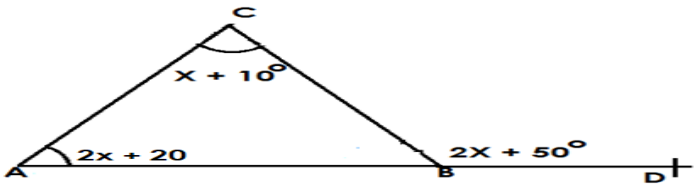
2. In the diagram below, find the value of K.



3. Use the diagram below to find the size of angle(C)P(Q).

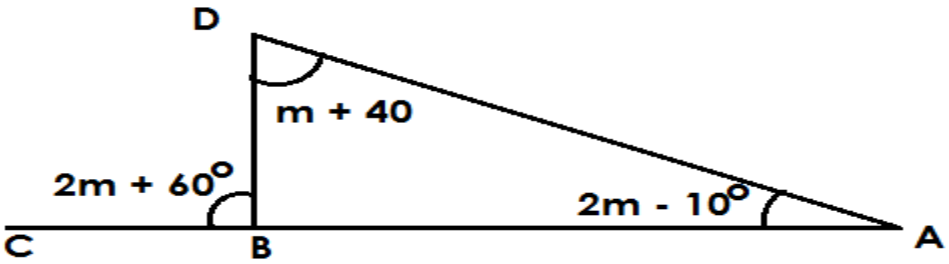


4. Find the value of X



b) **Find the size of angle ABC**

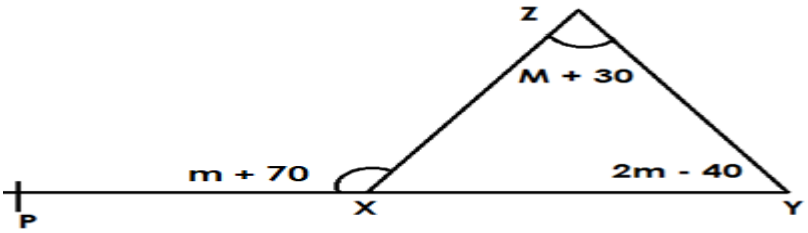
5. **Use the diagram below to answer questions.**



a) **Find the value of M.**

b) **Calculate the size of *L* BCA.**

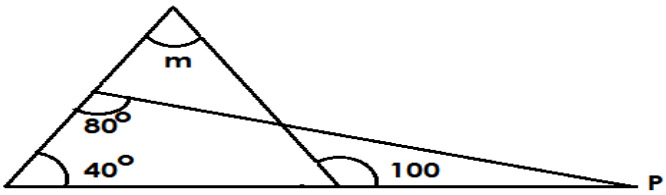
Use the diagram below to find the value of M.



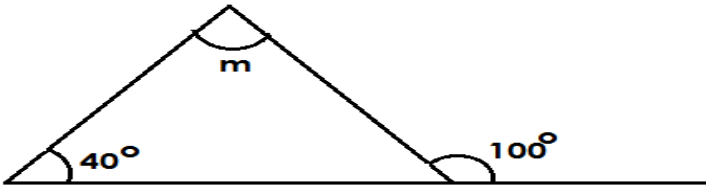
<u>MORE ABOUT SUM OF TWO INTERIOR ANGLES EQUAL TO ONE OPPOSITE EXTERIOR ANGLES</u>

<u>Example I</u>

Find the value of M.



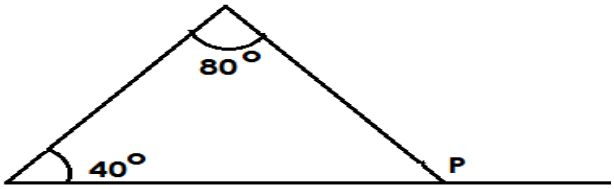
soln



m + 40º = 100º (Sum of 2 int. angles = 1 opp.Ext. angle)

m + 40 – 40 = 100 – 40º

m = 60º



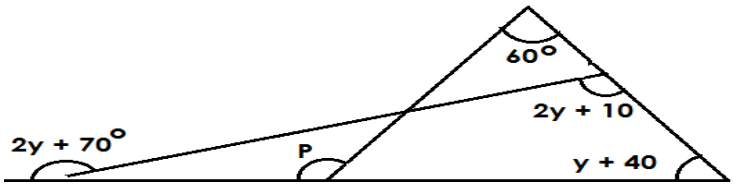
Soln

p = 80 + 40 (sum of 2 m + *L* = 1 oppext*L*)

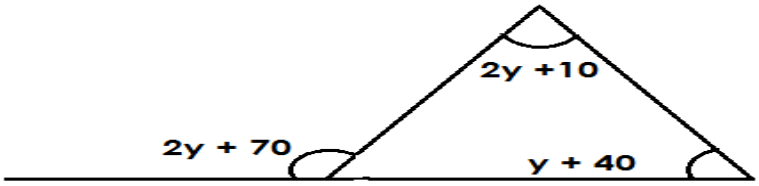
P = 120 º

<u>Example II</u>

Find the value of y.



Soln



2y + 10º + y + 40 º = 2y + 70 º (sum of 2 int. angles = 1opp. ext. angle)

2y + y + 10 º + 40 º = 2y + 70 º

3y + 50 º = 2y + 70 º

3y + 50 º – 50 º = 2y + 70 º – 50 º

3y = 2y + 20 º

3y – 2y = 2y – 2y + 20 º

Y = 20º

b) Find the value of P

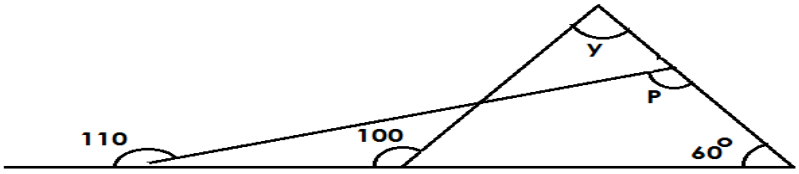
P = 60 + y + 40º (Sum of 2m + *L*s = 1opp ext*L*)

P = 60 + 20º +40º

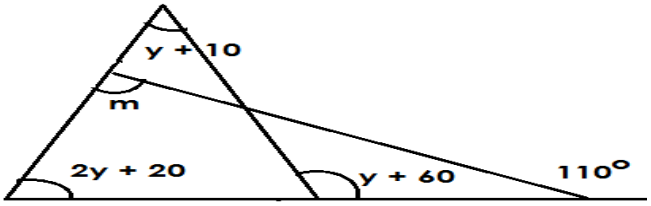
P = 120º

Activity:

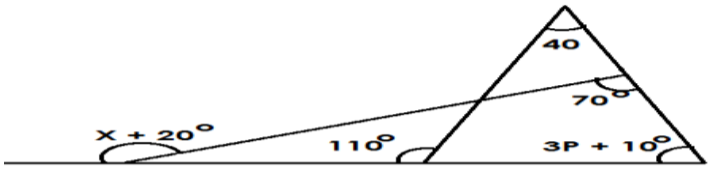
Find the value of unknowns in the following;



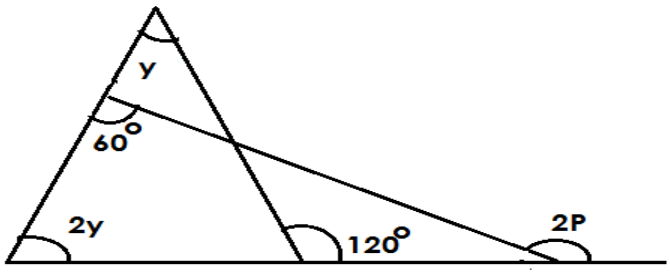
2.



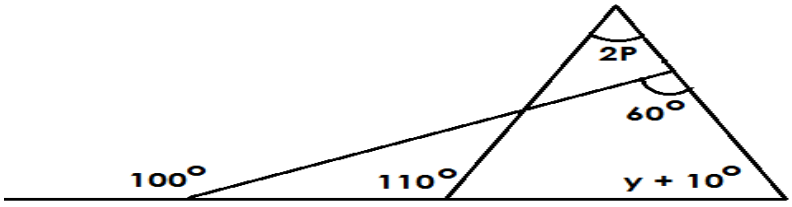
3.



4.



5.



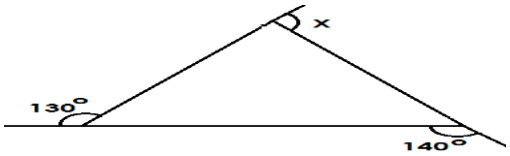
EXTERIOR ANGLES OF REGULAR POLYGON

Note:

- **The sum of all exterior angles of a regular polygon is 360º.**

Example I

Find the value of X.



X + 130º + 140 º = 360 º (sum of all exterior angles)

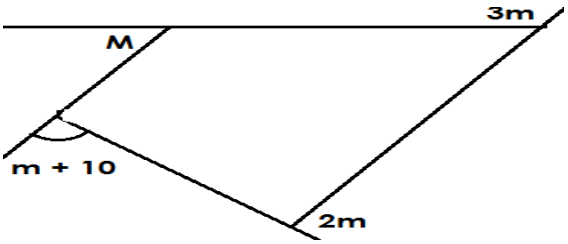
X + 270 º = 360 º

X + 270 º – 270 º = 360 º – 270 º

X = 90 º

Example II

Find the value of m in the diagram below.



2m + m + 10 º + m + 3m = 360 º (sum of all exterior angles)

2m + m +m + 3m + 10 º = 360 º

7m + 10 º = 360o

7m + 10 º – 10 º = 360o – 10 º

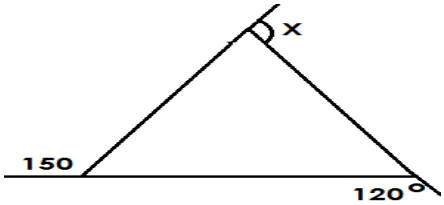
7m = 350 º

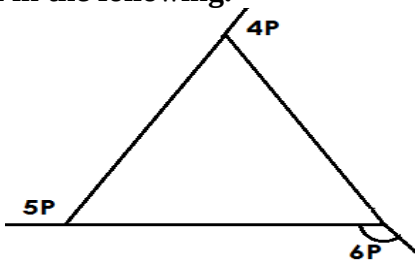
$\frac{7m}{7} = \frac{350}{7}$

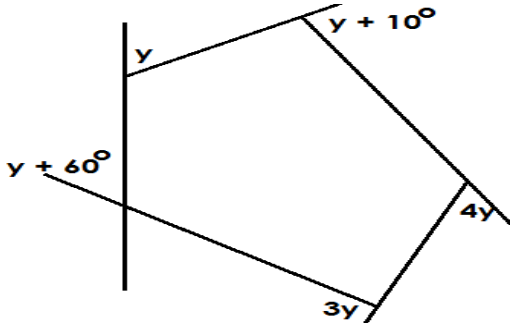
m = 50º

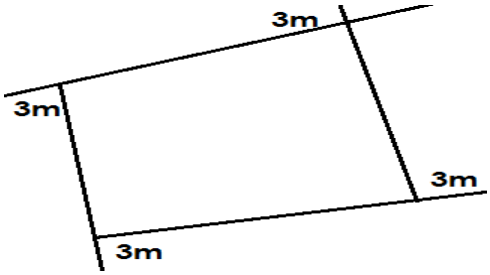
Activity:

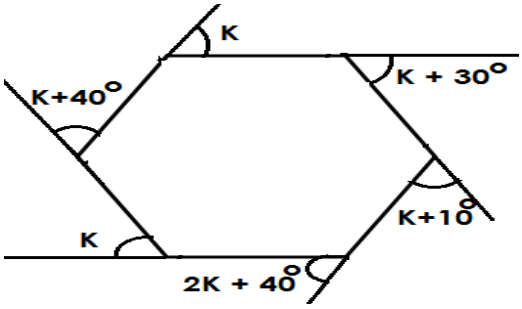
Find the value of unknown in the following.

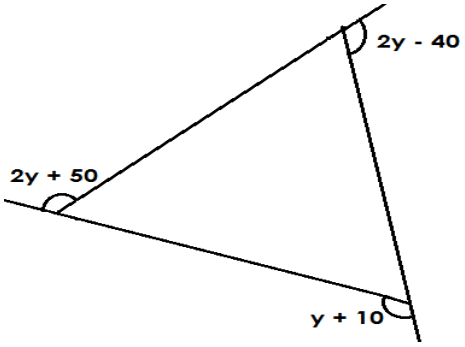


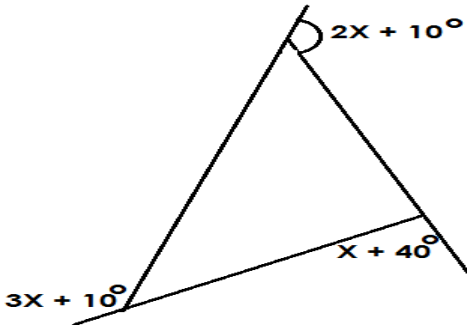












POLYGONS

- **A polygon is a closed sided figure.**
- **The word poly means many and gon means sides.**
- **All polygons are named according to their number of sides.**

Name of polygon	Number of sides
Triangle	3
Quadrilateral	4
Pentagon	5
Hexagon	6
Septagon / Heptagon	7
Octagon	8
Nonagon	9
Decagon	10
Nuo-decagon / Hendecagon	11
Duodecagon	12

FINDING EXTERIOR ANGLE OR CENTRE ANGLES OF POLYGONS

- **The sum of all exterior angles of any polygon is 360^0**

- **The sum of all centre angles of any polygon is 360^0**

One exterior angle $= \frac{\text{sum of ext Ls}}{\text{no.of sides}}$

One centre angle $= \frac{\text{Sum of all centres Ls}}{\text{No.of sides}}$

Example I

Find the size of each exterior angle of a regular pentagon.

Exterior L $= \frac{\text{sum of all ext.Ls}}{\text{No.of sides}}$

$= \frac{360}{5}^{o}$

$= 70^{o}$

Example II

Find the size of each exterior angle of a polygon with 30 sides.

Soln

Each ext.angle $= \frac{Sum\ of\ all\ ext.angles}{No.of\ sides}$

$= \frac{360}{30}$

$= 12^{o}$

Activity:

1. **Find the size of each exterior angle of a regular octagon.**

2. **Calculate the size of each exterior angle of a regular.**

a) **decagon**

b) **Equilateral triangle**

c) Nonagon

d) Duo decagon

2. A regular polygon has 10 sides. What is the measure of each of its exterior angle?

3. What is the size of each centre angle of a regular polygon with 18 sides?

4. Calculate the size of each centre angle of a regular hexagon.

6. If a regular polygon has 20 sides, what is the size of each exterior angle?

FINDING THE NUMBER OF SIDES AND NAMING THE POLYGON

Note:

No. of sides $= \frac{360}{Ext.L}$

Examples

1. Calculate the number of sides of a regular polygon whose exterior angle is 20°?

Soln

No. of sides $= \frac{360}{Ext.L}$

$= \frac{360}{20}$

$= 18^o$

5. What name is given to a regular polygon whose exterior angle is 60°.

Soln.

No. of sides= $\frac{360}{Ext.L}$

$= \frac{360}{60}$

$= 6.$

The polygon is hexagon.

Activity:

1. Name the regular polygon whose exterior angles are given below.

a) 72°

b) 36°

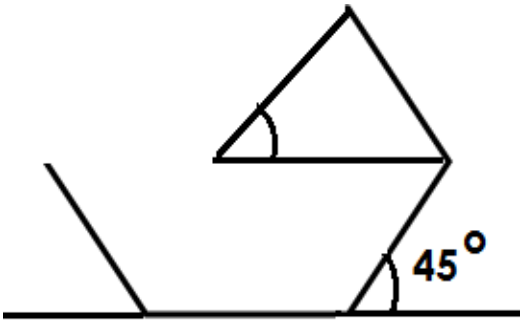
c) 90°

d) 45°

e) 30°

2. Each exterior angle of a regular polygon is 120. Calculate the number of sides of the polygon.

3. The figure below shows the exterior angle of a regular polygon. Calculate the number of sides.



4. One of the exterior angles of a regular polygon is18. How many sides has the polygon?

5. How many sides has a polygon whose exterior angle is 30°?

6. Calculate the number of sides of a regular polygon whose exterior angle is 40°.

FINDING EXTERIOR OR INTERIOR ANGLES OF REGULAR POLYGON

Note:

Exterior and interior angles of a regular polygon add up to 180° (They are supplementary)

Example I

The interior angle of a regular polygon is 144°.

Find the size of each exterior angle.

Let the exterior angle be K

Ext *L*	int *L*	sum
K	144^0	180^0

K + 144 º = 180º

K + 144 º – 144 º = 180 º – 144

K = 36º

b) Name the polygon

No. of sides = $\frac{\text{sum of all ext.Ls}}{\text{No.of sides}}$

= $\frac{360}{36}$

= 10 sides

∴ It is a decagon

Example II

The interior angle of a regular polygon is 120^0 more than its exterior angle. Name the polygon

Soln.

Let the ext. *L* be K

Ext. *L*	Int *L*	sum
K	K + 120º	180º

K + K + 120 º = 180 º

2K + 120 º = 180 º

2K + 120 º – 120º = 180 º – 120 º

2K = 60 º

$\frac{2K}{2}$ = $\frac{60}{2}$

K = 30º

No. of sides = $\frac{Sum\ of\ all\ ext.L}{Ext.L}$

= $\frac{360}{30}$

= **12 sides**

∴ It's a duo-decagon.

Example III

The interior angle of a regular polygon is twice its exterior.

b) Find the size of the exterior angle.

Let the exterior angle be M.

Ext. L	Int. L	Sum
m	2 x m 12m	180

m + 2m = 180 º

3m = 180 º

$\frac{3m}{3}$ = $\frac{180}{3}$

m = 60º

b) Name the polygon

Number of sides = $\frac{Sum\ of\ all\ ext.L}{No.of\ sides}$

= $\frac{360}{60}$

= **6 sides**

∴ It is a hexagon

Activity:

1. The interior angle of a regular polygon is 150º.

a) Find its exterior angle.

b) Nam the polygon.

2. If the interior angle of a regular polygon is 120° more than the exterior angle.

a) Find the exterior angle.

b) How many sides has the polygon.

3. The interior angle of a regular polygon is 90° more than its exterior.

a) Find the exterior angle.

b) Name the polygon

4. The interior angle of a regular polygon is 100° more than its exterior angle.

a) Calculate its exterior angle.

b) Name the polygon

5. The interior angle of a regular polygon is 4 times its exterior angle.

a) Find the size of each exterior angle.

b) Name the polygon

6. The exterior angle of a regular polygon is a ninth of its interior angle.

a) Find the exterior angle.

b) Name the polygon

7. If the interior angle of a regular polygon is 5 times its exterior angle. Name the polygon.

FINDING EXTERIOR ANGLE OR INTERIOR ANGLE GIVEN RATIOS

<u>Examples</u>

The interior and exterior angles of a regular polygon are in the ratio of 5: 1 respectively.

a) Find the size of its exterior angle.

Soln.

Total ratio = 5 + 1

= 6

Exterior angle = $\frac{1}{6}$ x 180°

$= 30^0$

c) Name the polygon

Soln

Number of sides $= \frac{\text{sum of all exterior angles}}{\text{1 exterior alnge}}$

$= \frac{360}{30}$

$=$ **12 sides**

∴ It's a duo-decagon

Activity:

1. The interior and exterior angles of a regular polygon are in the ratio 7:2 respectively.

a) Find the size of the exterior angle.

b) Name the polygon

2. The interior and exterior angles of a regular polygon are in a ratio of 4:1 respectively.

a) Find the exterior and interior angle.

b) Name the polygon.

3. The interior and exterior angles of a regular polygon are in a ratio of 3;2 respectively.

a) Find the size of the exterior angle.

b) Name the polygon

4. The interior and exterior angles of a regular polygon are in a ratio of 1:2 respectively. Name the polygon.

5. Name the polygon whose interior and exterior angles are in a ratio of 3:1 respectively.

6. The interior and exterior angles of a regular polygon are in a ratio of 2:1 respectively. Name the polygon.

FINDING INTERIOR ANGLE SUM OF POLYGONS

Note:

- **Interior angle sum = 180^0 (n – 2)**

Where;

- **180^0 is the interior angle sum of a triangle.**
- **n is number of sides.**
- **n – 2 is number of triangles in a regular polygon.**

Example

Find the interior angle sum of a hexagon.

Soln.

Interior angle sum = 180° (n – 2)

= 180° (6 - 2)

= 180° x 4

= 720°

Example II

Find the interior angle sum of a regular polygon whose exterior angle is 90^0.

Soln

No. of sides = $\frac{\text{sum of all exterior angles}}{1\ exterior\ angle}$

= $\frac{360}{8}$

= 6 sides

Interior angle sum = 180° (4 – 2)

= 180° (4 – 2)

= 180° x 2

= 360°

Activity:

1. Calculate the interior angle sum of the following polygons.

a) Quadrilateral

b) Octagon

c) Heptagon

d) Hendecagon

e) Decagon

f) Septagon

2. The interior angle of a regular polygon is 140°.

b) Name the polygon

c) Calculate its interior angle sum.

3. Given that the interior angle of a regular polygon is 144°.

a) Name the polygon

b) Calculate its interior angle sum.

4. The interior angle of a regular polygon is 135^{0}.

a) Name the polygon

b) Find its interior angle sum

5. The interior angle of a regular polygon is 90 more than its exterior angle. Calculate its interior angle sum.

NAMING POLYGON GIVEN INTERIOR ANGLE SUM

Examples

The interior angle sum of a regular polygon is 540^{0}. Name the polygon.

Soln.

180^{0} (n – 2) = interior angle sum

180^{0} (n – 2) = 540^{0}

$$\frac{180\,(n-2)}{180} = \frac{540}{180}$$

n – 2 = 3

n – 2 = 3

n – 2 + 2 = 3 + 2

n = 5 sides

∴ The polygon is a pentagon.

Activity:

1.Name the polygon whose interior angle sum is given below: -

1. **360º**
2. **900 º**
3. **180 º**
4. **720 º**
5. **1080 º**
6. **1620 º**
7. **1440 º**
8. **1260 º**

2.Find the number of sides of a regular polygon whose interior angle sum are;

a) 2880º

b) 2520º

c) 3240º

FINDING NUMBER OF TRIANGLES AND NUMBER OF RIGHT ANGLES.

Example I

Note:

- **Number of triangles = n – 2**
- **Number of right angles = 2(n – 2)**
- **Where n is number of sides of a regular polygon.**

Example I

How many triangles are in a regular octagon.

Soln

Number of triangles = n – 2

= 8 – 2

= 6 triangles

Example II

How many right angles are in a regular septagon.

Soln

Number of right angles = 2 (n – 2)

= 2(7 – 2)

= 2 x 5

= 10 right angles

Activity:

a) Find the number of triangles in the following polygons.

1. Pentagon

2. Triangle

3. Quadrilateral

4. Hexagon

5. Nonagon

6. Hendecagon

7. Duo decagon

8. Decagon

b) Calculate the number of right angles in the following polygons.

1. **Pentagon**
2. **Triangle**
3. **Quadrilateral**
4. **Hexagon**
5. **Nonagon**
6. **Decagon**
7. **Hendecagon**
8. **Duo decagon**

Scenarios

Session 3

- ✦ Introduction on bearing
- ✦ Diagrammatic application on bearing
- ✦ Bearing with scale drawing
  - ✦ Accurate scale drawing

Methods

- **Guided discovery**
- **Group discussion**
- **Think, pair and share.**
- **explanation**

Lesson activities

- **drawing**
- **measuring**
- **subtracting**

Content/facts

BEARINGS

Note:

- **Bearing is the direction / angle turned from a fixed point.**
- **In bearing, North is taken to be the fixed point.**
- **Bearing is measured from the last point mentioned.**

ROTATION / REVOLUTION.

- **A rotation is a complete turn from a fixed point and back to the very point.**
- **A complete revolution adds up to 360°.**

FINDING PARTS OF A REVOLUTION

Examples

1. What is $\frac{2}{3}$ of a revolution?

Soln.

$\frac{2}{3}$ x 360°

2 x 120°

240°

2. What is the supplement of $\frac{1}{12}$of a revolution?

Soln.

$\frac{1}{12}$ x 360^{0}

30°

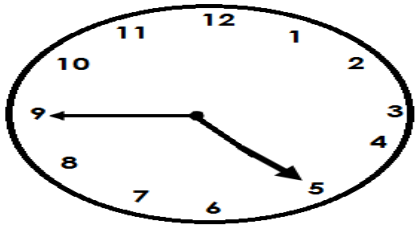
Let the supplement be K

K + 30° = 180°

K + 30° – 30° = 180° – 30°

K **=150°**

3. Below is a clock face.



Finding the angle turned by the minute hand.
Soln.

Angle turned $= \frac{45\ min}{60\ min}$ **x 360°**

$= \frac{45}{60}$ **x 360°**

= **45 x 6°**

= **270°**

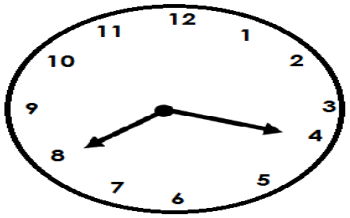
Activity:

1. Find $\frac{1}{4}$ of a revolution.

2. Find the complement of a $\frac{1}{10}$ revolution.

3. Find 20% of $\frac{1}{9}$ a revolution.

4. Below is a clock face



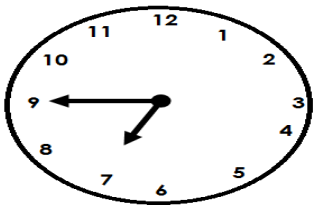
Find the angle turned by the minute hand.

5. What angle is turned by the hour hand on a clock face at 7 o'clock?

5. Find the supplement of the angle turned on the minute hand 25 minutes on the clock face.

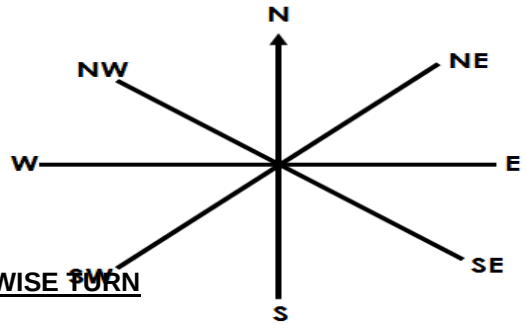
6. What is $\frac{1}{5}$ of the angle formed by the minute hand on a clock face at 30 minutes?

Below is a clock face.



7. **Find the complement of $\frac{1}{2}$ a revolution**
8. **Find the supplement of the angle turned by the minute hand of a clock in 30 minutes.**

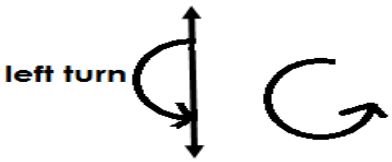
ANGLES ON A COMPASS DIRECTION



CLOCK WISE TURN



ANTI CLOCK WISE TURN

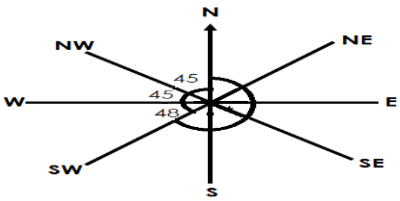


Example I

Musa turned from North to South West.

Find the smaller angle turned.

Soln.



$= 40^{O} + 45^{O} + 45^{O}$

$= 135^{O}$

Find the larger angle turned.

Soln.

1 Sector = 45°

5 Sectors = 45° x 5

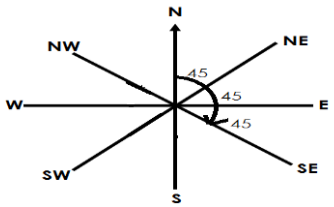
▪ = 225°

Example II

usa turned from North clockwise to face South East.

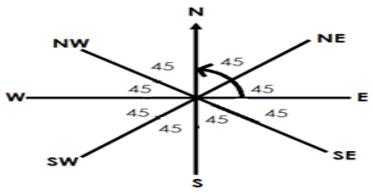
Through what angle did he go?

Soln



$$\text{Angle} = 45^{O} + 45^{O} + 45^{O}$$
$$= 135^{O}$$

Annet turned from East anticlockwise through an angle of 90^0.
In what direction is she facing now?



Direction = North

Soln.

<u>Activity:</u>

1.Kato turned from NE to SE. find the smaller angle he turned through.
Kumbe turned from North Clockwise to West. Through what angle did he turn?

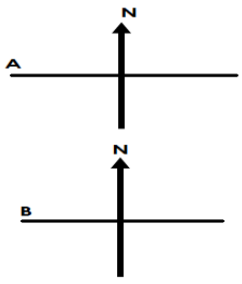
2.A boy turned from North West anticlock wise through an angle 0f 270°.
What is his new direction?
3.Find the smaller angle between North and South East.
Find the angle from East to West Clock Wise.
4.Mary was standing facing South Eastern direction. She turned clockwise through an angle of 90°.
In which direction is she facing now?
5.Ssembatya is facing North East. If he turned through an angle of 135o anti clock wise. Find his new direction.
6.Find the larger angle formed between North and East.
In which direction will I face if I turned anti clock wise from NE through 180°?

<u>ORDINARY BEARINGS</u>

<u>Note:</u>

- **Ordinary bearing is the description of direction starting from North or South but not East or West.**
- **If two directions take North, then they will have N 00°N.**

Illustration

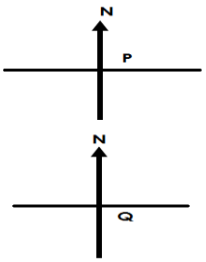


What is the direction of town A from B?

soln

N 00 ^{O}N

- **If two towns take South direction then they will have S 00 ^{o}S**



What is the direction of Q from P.?

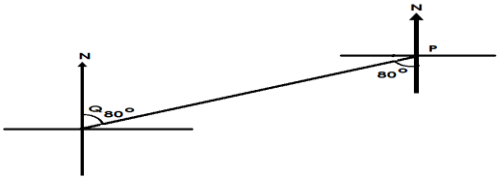
Soln.

S00^{O}S

- **If the two compasses are parallel to each other, angle properties of parallel lines will be applied.**

Examples

The diagram below shows the location of two towns.



What is the direction of P from Q?

Soln.

N80 ^{o}E

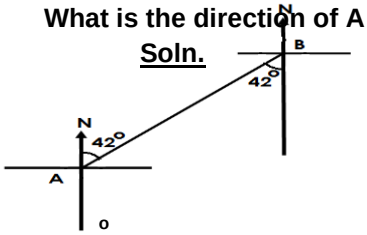
What is the direction of Q from P

Soln.

S 80 ^{o}W

What is the direction of A from B?

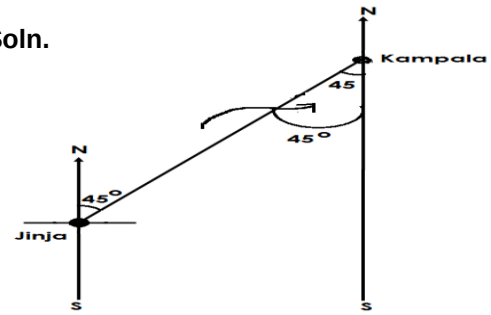
Soln.



S 42 ^{o}W

6. The direction of Kampala from Jinja is N 45^{0} E. What is the direction of Jinja from Kampala?

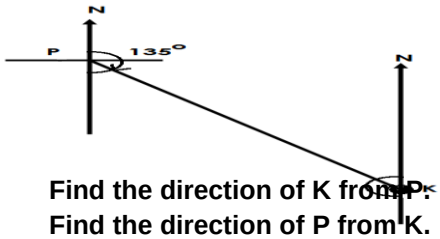
Soln.



Direction of Jinja from Kampala is S 45^0W.

Activity:

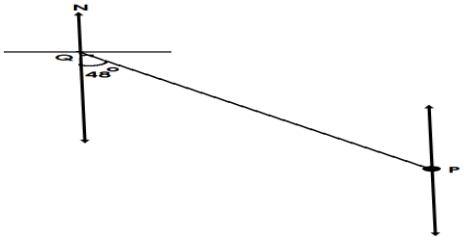
1. The diagram below shows location of two towns.



a) Find the direction of K from P.

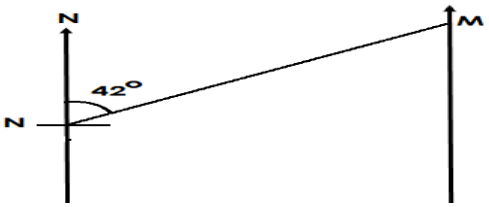
b) Find the direction of P from K.

c) What is the direction of P from Q in the diagram below.



d) What is the direction of Q from P.

7. Use the diagram below to answer questions that follow.



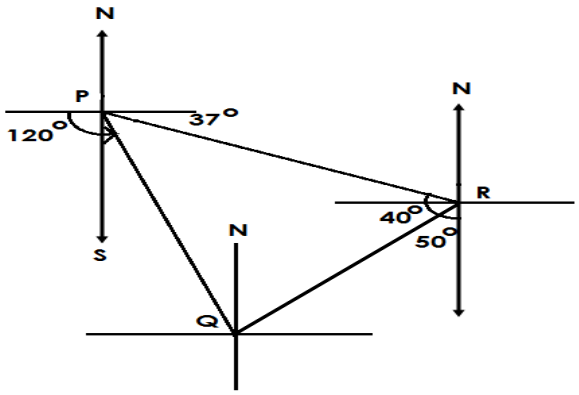
a) What is the direction of M from N?

b) What is the direction of N from M.

8. The direction of K from P is N 40^0 E. Find the direction of P from K.

9. The direction of Kampala from Jinja is S 52^0W. What is the direction of Jinja from Kampala?

10. The diagram below shows the location of three different towns.

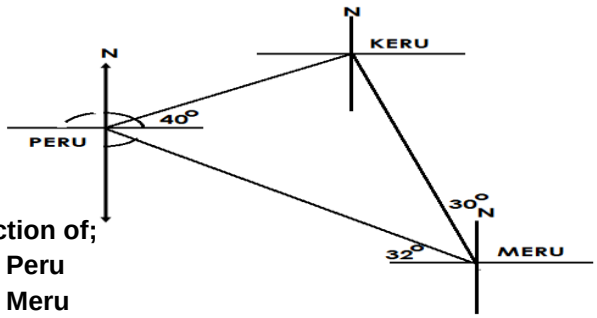


a) What is the direction of Q from R.

b) What is the direction of P from R.

c) Find the direction of Q from P.

11. The diagram below shows the location of three places.



Find the direction of;

a). Keru from Peru

b). Peru from Meru

c). Keru from Meru

TRUE BEARINGS

Note

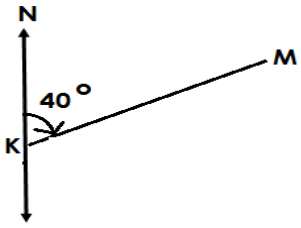
In true bearing, the point of origin is North only.

In true bearing all angles must be written in three digits.

In true bearing angles are obtained by measuring from North moving in a clockwise direction only.

Examples

1. Given the diagram below.

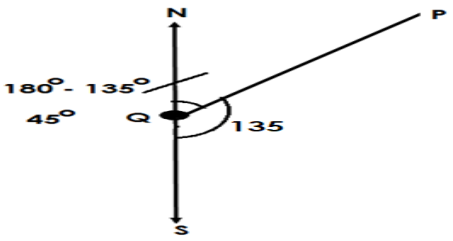


a). Find the bearing of M from K

Soln.

Bearing = 040°

2. Given the diagram below.



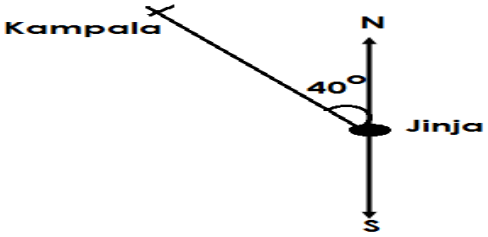
Finding the bearing of P from Q

Soln

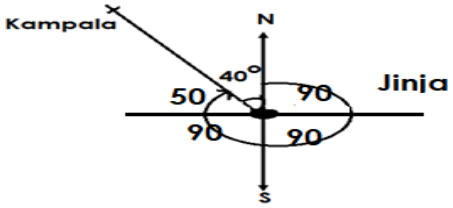
Bearing = $180^{o} - 135^{o}$

i. $= 045^{o}$

3. Use the diagram below to Find the bearing of Kampala from Jinja.



Soln

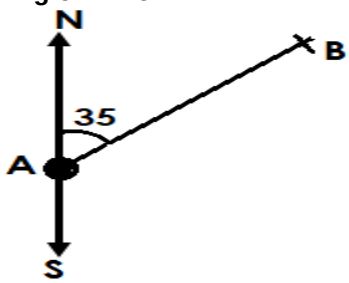


Bearing = $90^{0} + 90^{0} + 90^{0} + 50^{0}$

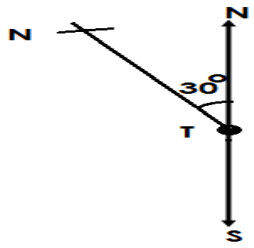
i. $=220^{o}$.

Activity:

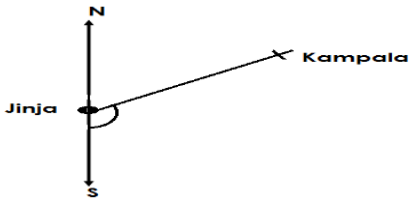
1.On the diagram below:
Find the bearing of B from A



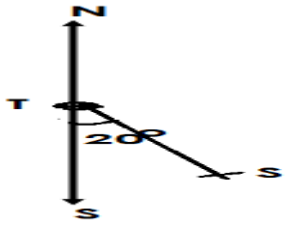
2. Find the bearing of N from T



3. Find the bearing of Kampala from Jinja.

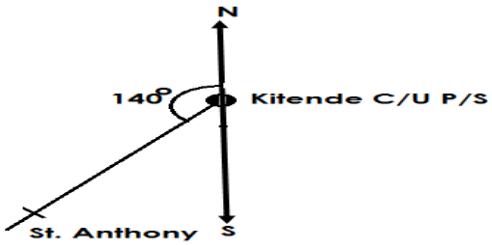


4. Find the bearing of S from T

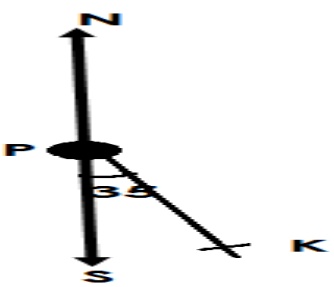


5. Draw a diagram to show that the bearing of Entebbe from Kajjansi is 135°.

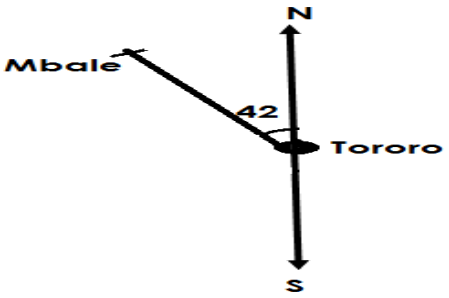
b) Use the diagram below to find the bearing of St. Anthony Primary School from Kitende C/U primary school.



6. What is the bearing of K from P in the diagram below.



7. What is the bearing of Mbale from Tororo in the diagram below?



FINDING OPPOSITE BEARINGS

Note:

- **In opposite bearing, if the given angle is more than 180°, its opposite bearing is less by 180°**

That is to say, subtract 180^0 from the given angle.

- **If the given angle is less than 180°, its opposite bearing is greater than 180 ^{o}**

That is to say, add 180 ^{o} to the given angle.

Examples

1. The bearing of town Q from B is 140^0. What is the bearing of B from Q?

Soln.

Bearing of B from Q = 140°

+ 180 ^{o}

320 ^{o}

2. The bearing of Mulu from Pulu is 272°. What is the bearing of Pulu from Mulu?

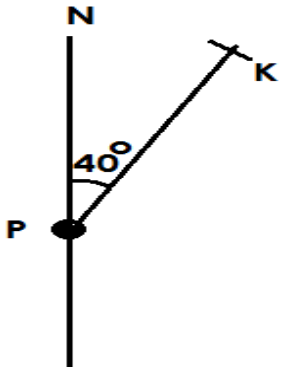
Soln.

Bearing of Mulu from Pulu = 272°

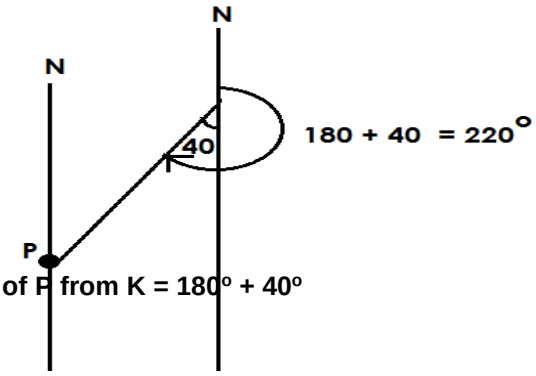
- 180°

092°

3. Use the diagram below to find the bearing of K from P.



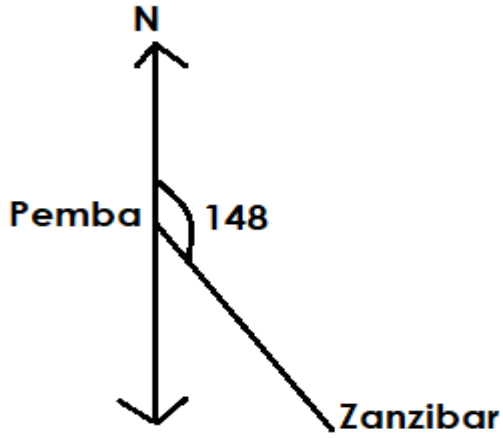
Soln



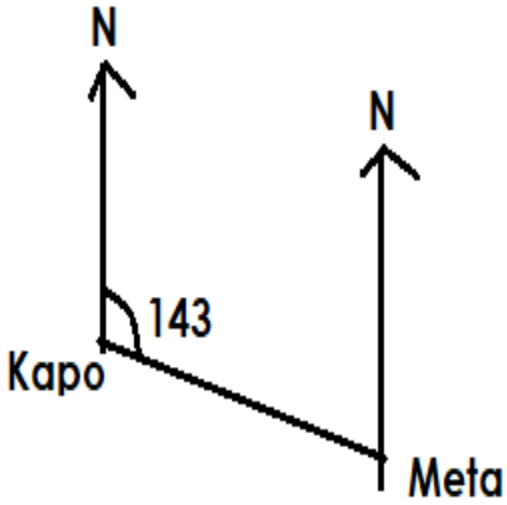
Bearing of P from K = 180° + 40°
= 220°.

Activity:

1. **The bearing of town A from Kampala is 043^0. Find the bearing of B from A.**
2. **The bearing of Karamoja from Kampala is 240^0. What is the bearing of Kampala from Karamoja?**
3. **The bearing of the Kitchen from class is 150^0. What is the bearing of class from the kitchen?**
4. **The bearing of Maya from Mukono is 110^0. What is the bearing of Mukono from Maya?**
5. **The bearing of A from K is 038^0. what is the bearing of K from A?**
6. **The diagram below shows the location of two different towns.**
   What is the bearing of Pemba from Zanzibar?

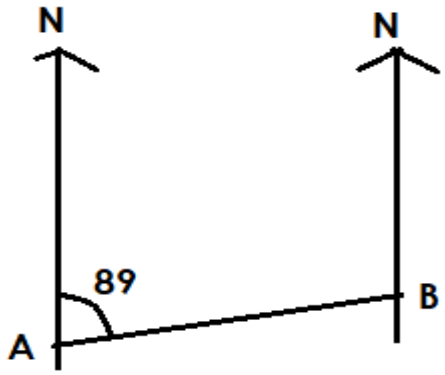


6. Given the diagram below:

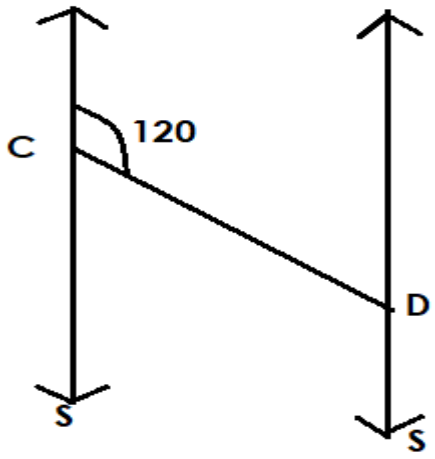


What is the bearing of Kapo from Meta?

8. In the diagram below, find the bearing of A from B?



9. Find the bearing of C from D in the diagram given below.



SCALE DRAWING

- **A sketch must be drawn.**
- **Kilometers or metres must be changed to centimetres using the given scale before drawing the accurate diagram.**
- **North has to be indicated on each compass**
- **Compasses drawn must be parallel to each other.**

Scale drawing is categorized into three: -

- **Given directions and distance.**
- **Given bearing and distance**
- **Given direction, distance and bearing**

A) GIVEN DIRECTION AND DISTANCE

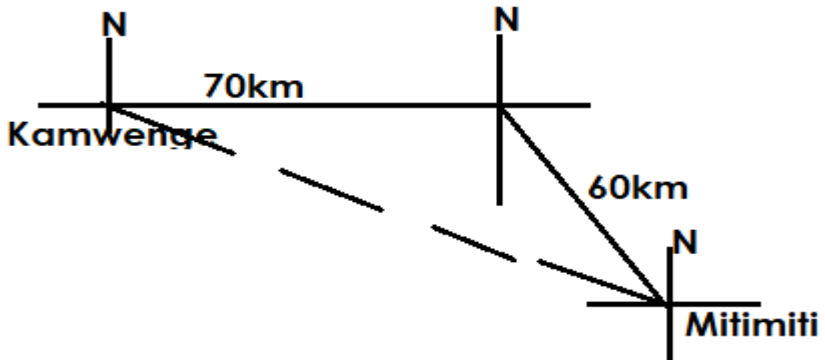
Note:

- **If semi/secondary cardinal points are given i.e., NE, NW, SE, SW, then they are got after bisecting.**

Example.

Kampala is 70km East of Kamwenge and Mitimiti is 60km South East of Kampala. Using a scale of 1cm to represent 10km, show the location of the three towns on an accurate diagram.

Sketch



Kampala – Kamwenge	Kampala – Mitimiti
10km = 1cm 1km = $\frac{1}{10}$ cm 70km = ($\frac{1}{10}$ x 70)cm = 7cm	10km = 1cm 1km = $\frac{1}{10}$ cm 60km = ($\frac{1}{10}$ x 60)cm = 6cm

Accurate.

Find the shortest distance from Kamwenge to Mitimiti.

Soln

Distance in cm = 12cm

1cm = 10km

12cm = (12 x 10) km

= 120km

Activity:

1. **Town A is in the West of town B a distance of 65km and C is in the South of town A a distance of 50km.**

Using a scale of 1cm to represent 10km, show the location of three towns on an accurate diagram.

2. **Town Q is in the East of P a distance of 100km and R is in the South West of Q a distance of 120km. Using a scale of 1cm to represent 10km, show the location of the three places on an accurate diagram.**

a) **Find the shortest distance from R to P.**

3. **Town Q is in the West of town P a distance of 70km and town R is in the North East of town P a distance of 45km. Using a scale of1:10 show the location of three towns on an accurate diagram.**

4. **Gulu town is in the South East of Kampala a distance of 140km and Mbale is in the North East of Gulu a distance of 100km. Using a scale of 1cm to represent 20km, show the location of the three towns on an accurate diagram.**

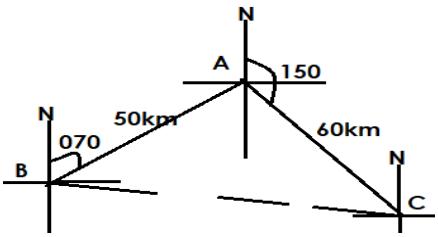
3. **St. Anthony Primary School is 620km West of Kawooto primary school and Disney primary school is 560km South of Kawooto primary school. Using a scale of 1cm to 100km, draw an accurate diagram to show the location of the three schools.**

B). <u>Given bearing and distance</u>

<u>Example I</u>

1. **Town A is on a bearing of 070^0 from B a distance of 50km and town C is on a bearing of 150^0 from A a distance of 60km. Using a scale of 1cm to 10km. show the location of the three towns on an accurate diagram.**

<u>Sketch</u>



<u>Accurate</u>

A to B	A to C
10km = 1cm 1km = ($\frac{1}{10}$) cm 5cm	10km = 1cm 1km = ($\frac{1}{10}$) cm 1km = ($\frac{1}{10}$)cm 60km = $\frac{1}{10}$ x 60 = 6cm

b) Find the shortest distance from C to B.

1cm = 10km

8.2cm = (8.2 x 10) km

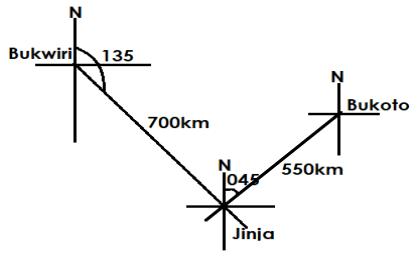
= ($\frac{82}{18}$ X 10) km

= 82km

<u>Example 2</u>

Jinja on a bearing of 135^0 from Bukwiri, a distance of $045^0$from Jinja a distance of 550km, using a scale of 1cm to represent 100km, show the location of the three places on an accurate diagram.

<u>Sketch</u>



Bukwiri – Jinja	Jinja - Bukoto

100km = 1cm 1km = ($\frac{1}{100}$) cm 1km ($\frac{1}{100}$)cm 7cm	100km = 1cm 1km = ($\frac{1}{100}$) cm 1km = ($\frac{1}{10}$)cm 550km = $\frac{1}{100}$ x 550) cm 5.5cm

Activity:

1. Town Q is on a bearing of 120 from R a distance of 120km. town P is on a bearing of 240° from Q a distance of 100km. using a scale of 1cm to represent 20km. Show the location of the three towns on an accurate diagram.

b) Find the shortest distance from town P to R.

c) Find the bearing of town R from P.

2. A plane flew from airport K to airport T on a bearing of 120°. the distance from K to T is 600km. it then left airport T for airport on a bearing of 210° a distance of 500km. Using a scale of 1cm to represent 100km. draw an accurate diagram to show the journey made by the place.

b) Find the bearing of airport R from K.

3. Town Q is on a bearing of 140° from P a distance of 140° from P a distance of 140km and town R is on a bearing of 250° from Q a distance of 100km. Using a scale of 1cm to 20km, show the location of the three towns on an accurate diagram.

b) Find the bearing of town R from P.

c) What is the shortest distance between P and R?

4. The bearing of town B from A is 240° and town B is 40km from A. the bearing of town C from B is 330° and town C is 50km from B. construct an accurate route showing the position of the three towns and measure the shortest distance.

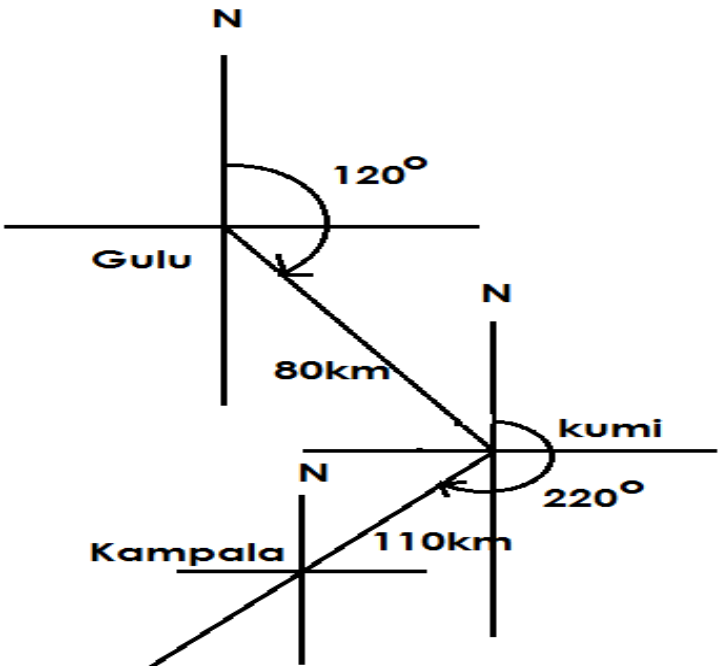
MORE ABOUT BEARINGS WHEN GIVEN TIME AND SPEED

A bus travelled from Gulu on a bearing of 120^0 to Kumi at an average speed of 40km/hr. in 2 hours. It then turned on a bearing of 220° travelling towards Kampala at an average speed of 55km/hr. for 2 hours.

a) Using a scale of 1cm to 20km, draw a sketch showing the three towns.

Time	Speed	Distance
2hrs	40km/hr.	D = S x T D = $\frac{40km}{hr.}$ x 2hrs = 80kms
2hrs	55km/hrs.	D = S X T D = $\frac{55km}{Hr.}$ x 2hrs D = 110km

Sketch.



b) Draw an accurate diagram to show the position of the three towns.

Distance	Length
80km	$= \frac{80km}{20km}$ = 4cm
110km	$= \frac{110km}{20km}$ = 5.5cm

Accurate diagram

c)Calculate the shortest distance between Kampala and Gulu.

Gulu to Kampala = 6cm

1cm rep 20km

6cm rep (20 x 6)cm

120km

Activity:

1. Iganga is a bearing of 130^0 from Lira. A motorist left Lira to an average speed of 30km / hr to Iganga. At Iganga he turned on a bearing of 045^0 travelling at an average speed of 45km/hr for 2 hours to Tororo town.

a) Draw a sketch showing the three towns.

b) Using a scale of 1cm to represent 15km, draw an accurate diagram showing the three town.

c) Calculate the shortest distance between Lira and Tororo.

2. **An army helicopter flew from airbase B on a bearing of 120^0 at 100km / hr for 2 hrs and then flew for 4 hrs at the same speed in the NE direction to airbase C. From C, it flew in West direction at a speed of 200km/hr for 1 ½ hrs to air base D. construct an accurate route of the helicopter showing the air bases.**

3. **Alax left town Z and drove East ward at a speed of 30 km/hr for 2 hrs to town Y. He then changed direction and travelled on a bearing of 120^0 to town X at 40km / hr for 2 hrs. Using a scale of 1cm to represent 10km, construct an accurate diagram showing Alex'sJourneys.**

4. **A motorist left town K on a bearing of 060^0 travelling at an average speed of 60km/hr for 1 ½ hrs to town M. At town M, he moved West wards at a speed of 50km/hr for 2hrs to town R.**

a) **Draw a sketch to represent the three towns.**
b) **Using a scale of 1cm to represent 10km, draw an accurate diagram showing the three towns.**
c) **Calculate the shortest distance between R and K**

Scenarios

TOPIC 10 – DATA HANDLING;

Session 1

- ✦ Introduction to data handling
- ✦ Application on average
- ✦ Interpreting pie-charts
- ✦ Drawing pie-charts

Methods

- **Guided discovery**
- **Group discussion**
- **Think, pair and share.**
- **explanation**

Lesson activities

- **drawing**
- **measuring**
- **subtracting**

Content/facts

DATA HANDLING

ARITHMETIC MEAN

Mean of whole numbers

Mean/average

Median

Mode

Modal frequency

Frequency

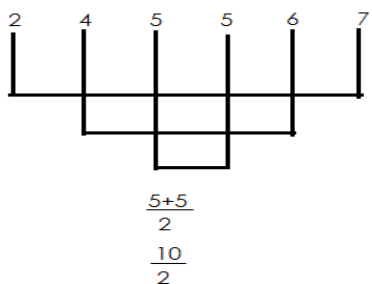
Range

MEDIAN

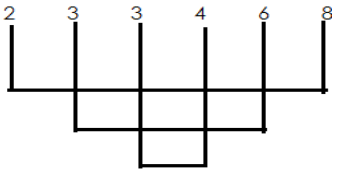
Medina is the middle number got after arranging in either ascending or descending order.

Find the median of 5, 4, 5, 6, 7 and 2

Solution



2. Find the median of 2, 3,6,3,8 and 4

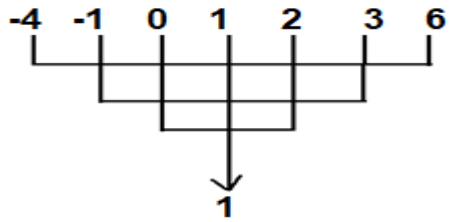


$\frac{3+4}{2}$

$\frac{7}{2}$ 3 r1

$3\frac{1}{2}$

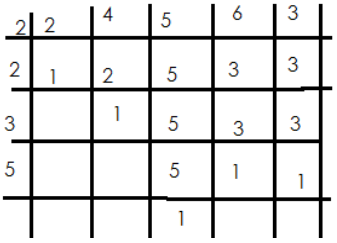
3. Find the median of -4, 2, 0, 3, 6, -1 and 1



Median = 1

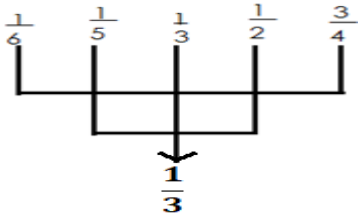
4. Find the median of $\frac{1}{2}, \frac{3}{4}, \frac{1}{5}, \frac{1}{6}, \frac{1}{3}$

$\frac{1}{2} \times 60$ = 30	$\frac{3}{4} \times 60$ = 45	$\frac{1}{5} \times 60$ = 12	$\frac{1}{6} \times 60$ = 10	$\frac{1}{3} \times 60$ = 20



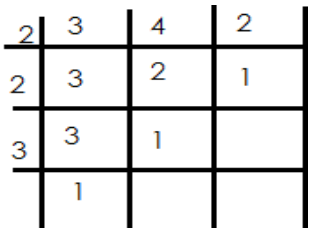
L.C.M=(2X2)X (3X5)
=4X15
=60

Ascending order

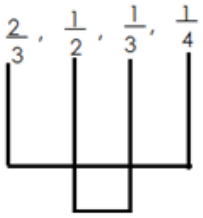


5. Find the median of $\frac{1}{3}, \frac{2}{3}, \frac{1}{4}$ and $\frac{1}{2}$

$\frac{1}{3} \times 12$ = 4	$\frac{2}{3} \times 12$ = 8	$\frac{1}{4} \times 12$ = 3	$\frac{1}{2} \times 12$ = 6



L.C.M = 2X2X3
=12



$$\frac{\frac{1}{2} + \frac{1}{3}}{2}$$

$$\left(\frac{1}{2} + \frac{1}{3}\right) \div 2$$

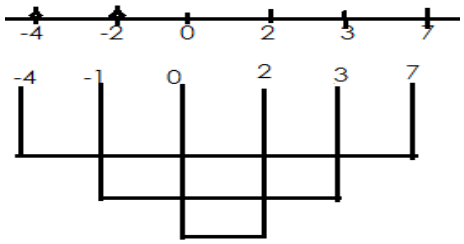
$$\frac{(3+2)}{6} \div 2$$

$$\frac{5}{6} \div 2$$

$$\frac{5}{6} \times \frac{1}{2}$$

$$\frac{5}{12}$$

6. Find the median of -4, 3, 0, -2, 7, 2



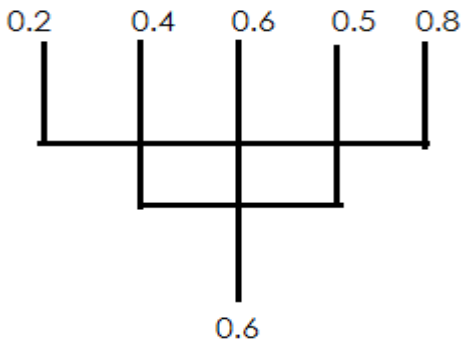
$$\frac{0+2}{2}$$

$$\frac{2}{2}$$

Median = 1

7. Find the median of 0.2, 0.6, 0.4, 0.5, and 0.8

Solution



Activity

1. Find the median of the following

(a) 20, 30, 60, 40, 50

(b) **-1, 0, 1, 2, 4, -3**

(c) **3, 3, 7, 8, 4, 5, 7, 8**

(d) $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \frac{1}{6}$

(e) $\frac{2}{5}, \frac{1}{2}, \frac{3}{4}, \frac{1}{6}$

(f) **-3, -4, 0,3,2,6**

(g) $\frac{1}{3}, \frac{3}{4}$, **1,** $\frac{1}{2}, \frac{2}{6}, \frac{1}{2}$

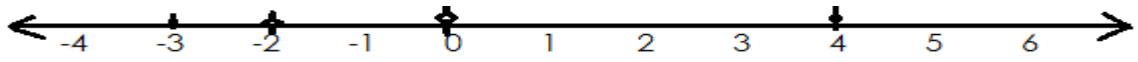
(h) **0.4, 0. 3, 0.7, 0.6, 0.2**

RANGE

Range is the difference between the highest and lowest numbers given.

Range = Highest - Lowest

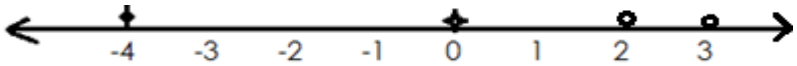
Find the range of -2, -3, 4, 0, -4 and 6



-4 -3 -2, 0, 4 ,6

range= H-L
=6- (-4)
=6+4
=10

2. **Find the range 2, -4, 0, and 3**

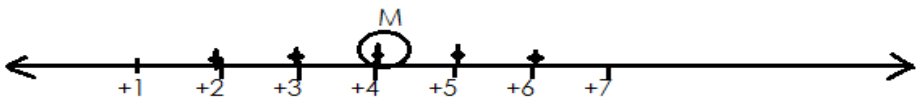


-4, 0, 2, 3

range=H-L
=3 - (-4)
=3+4
=7

3. **The median of 5 consecutive positive integers is +4**

(a) **What are the numbers?**



2, 3, 4, 5, 6

(b) **Find their range**

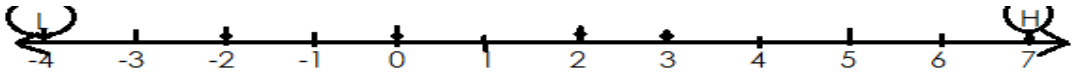
Range = H-L

= 6-2

=4

4. Find the range of

-4, 3, 0,-2, 7, and 2



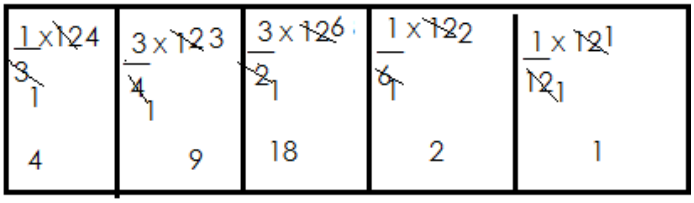
-4,-2, 0, 2, 3, 7

Range=H-L

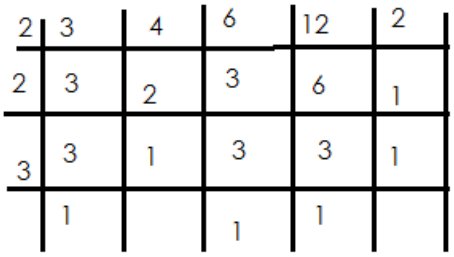
=7 – (-4)

=7+4

5. Find the range of $\frac{1}{3}, \frac{3}{4}\ 1\frac{1}{2}, \frac{1}{6}, \frac{1}{12}$



$\frac{1}{12}$ $\frac{1}{6}$ $\frac{1}{3}$ $\frac{3}{4}$ $1\frac{1}{2}$



L.C.M =2X2X3
=12

Range= H-L

$= 1\frac{1}{2} - \frac{1}{12}$

$= \frac{3}{2} - \frac{1}{12}$

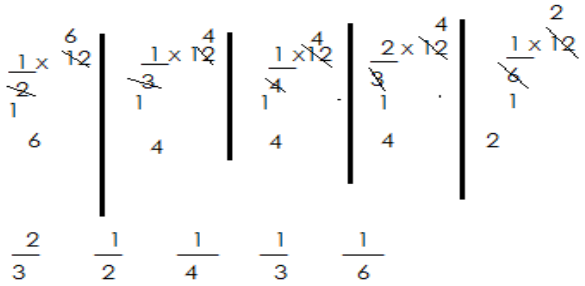
$= \frac{(3 \times 12) - (1 \times 12)}{12}$

$= \frac{18 - 1}{12}$

$= \frac{17}{12}$ 1 r 5

$= 1\frac{5}{12}$

6. Find the range $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{2}{3}, \frac{1}{6}$



$\frac{2}{3}$ $\frac{1}{2}$ $\frac{1}{4}$ $\frac{1}{3}$ $\frac{1}{6}$

Range= H-L

$$= \frac{\frac{2}{3} - \frac{1}{6}}{\frac{(\frac{2}{3} \times 6) - (\frac{1}{6} \times 6)}{6}}$$

$$\frac{4-1}{6}$$

$$\frac{3}{6} \quad 1$$

$$2$$

$$\frac{1}{2}$$

Activity

1. Find the range of the following

(a) 33, 38, 34, 36, 37, 35

(b) -5, 2, 3, 0, -2

(c) 5, 6, -2, -1, 3, 7

(d) $\frac{1}{2}, \frac{1}{4}, \frac{1}{6}, \frac{1}{5}, \frac{3}{4}$

(e) $\frac{1}{3}, \frac{1}{2}, \frac{1}{4}, \frac{1}{10}$

(f) -4, -2, 0, -1, 5

<u>MEAN/AVERAGE</u>

Mean is quotient of sum of all items and number of all items

Mean=<u>sum of all items</u>

No. of items

<u>Exmples</u>

Find the mean of 2, 4, 5, 6, 8, 7

Average=sum of all items

No. of items

=2+4+5+6+3+8+7

7

$$= \frac{35}{7} = 5$$

2. Find the average of 4, 0, -3, 8 and 6

Average=sum of all items

No. of items

= 4+0+8+6-3

3

= 18-3

5

$$= \frac{15}{5} = 3$$

3. Calculate the average of -4, 1, 0, 9 and -6

Average=sum of all items

No. of items

= 9+1+0-4-6

5

$$= \frac{10 - 10}{5} = \frac{0}{5} = 0$$

Activity

1. Find the average of the following

(a) 3, 6, 7, 4, 5

(b) 7, 8, 7, 8, 5, 2 and 5

(c) 40%, 20% 60% and 80%

(d) -2, 4, 6, 7 and 5

(e) -7, 0, 1, -5, 9 and 2

(f) 3, -4, -2, 7 and 6

(g) Find the average of 2y-2, 3y, 4 and 3y+6

(h) Find the average of k-3, k+5,2kand 2.

AVERAGE/MEAN OF GROUPED DATA

1. The table below shows marks scored by different pupils.

Marks	**40**	**60**	**30**	**80**
No of pupils	**3**	**2**	**4**	**1**

a) How many pupils are in class.

Soln.

No of pupils = (3+2) + (4+1)

=5+ 5

=10pupils

b) Find their range in marks

Soln.

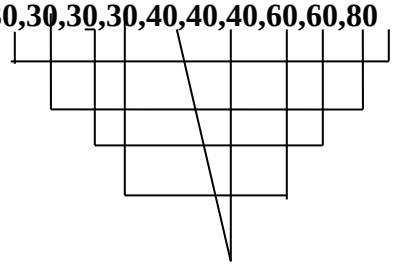
Range = H-L

= 80- 30

= 50

c) Find the median mark

30,30,30,30,40,40,40,60,60,80



$= \frac{40+40}{2}$

$= \frac{80}{2}$

= 40

d) Calculate the average mark.

Soln.

Average $= \frac{\text{sum of items.}}{\text{no of items}}$

$= \frac{40x\ 3) + (60\ x\ 2) + 30\ x\ 4) + (\ 80\ x\ 1)10}{10}$

$= \frac{120 + 120 + 1\ 20 + 80}{10}$

$= \frac{440}{10}$

= 44

2. Use table below to answer questions that follow.

No of pupils	**2**	**2**	**4**	**2**	**1**
marks of scored	**40**	**70**	**50**	**80**	**90**

a) How many pupils did the test.

= 2+ 2+ 4+ 2+1

= 11 pupils

b) How many pupils scored above the average mark

Av = sum of items

No of items

= (2x40) + 70 x 2) + (50 x 4) + 80x 2) + (90x 1)

11

= 80+ 140+ 200+ 160+ 90

11

= 670

11

= $60\frac{1}{11}$

No of pupils = 2+2+1

= 5 pupils

3. The table shows marks scored by different pupils in a test.

marks	**60**	**70**	**80**	**50**	**40**
No of pupils	**5**	**3**	**4**	**3**	**2**

a) How many pupils did the test?

5+ 3+ 4+3+ 2

= 17 pupils

b) Find the range of marks scored

Range = H - L

=80- 40

= 40

c) Find the mean mark of pupils who scored below 70

Mean = Sum of items

No of items

= (60x 5) +(50 x 3) + (40 x2)

5+ 3+ 2

= 300+ 150+ 80

10

= 530

10

= 53

Activity

1. The table below shows marks scored by some pupils in class.

marks	**60**	**70**	**80**	**50**	**40**
no of pupils	**2**	**2**	**1**	**2**	**3**

a) How many pupils are in class?

b) Find the median mark

c) Calculate the average mark

2. Use the table below to answer the given questions.

marks	**60**	**70**	**50**	**80**
no of pupils	**2**	**3**	**4**	**1**

a) Find the number of pupils in class

b) Find the median mark

c) Find the range of marks scored

d) How many pupils scored above the average mark?

3. The table shows scores of different pupils in a class.

No of pupils	**60**	**80**	**30**	**70**
Marks scored	**2**	**1**	**4**	**2**

a) Find the median score.

b) Find the range of the score.

c) Calculate the average mark of pupils who scored above the range.

4. Different weight of pupils were recorded in a certain class. Use it to answer questions that follow.

weight in kg	**25**	**30**	**40**	**50**	**55**
No of pupils	**4**	**4**	**2**	**1**	**1**

a) How many pupils are in class?

b) Find the median weight of pupils who are above 35kg

c) Calculate the average weight of pupils who are below 50kg

MORE ABOUT GROUPED DATA.

Complete the table below

Marks	No of pupils	Total
40	3	120
50	2	100
30	4	120
90	1	90

No of pupils= 120

40

= 3

Marks = 100

2

Total = 30 x 4

= 120

No of pupils= 90

90

= 1

b) Find the range

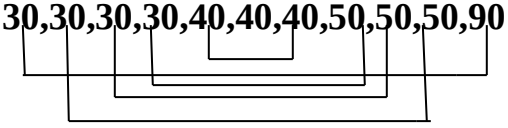
Range = H- L

= 90- 30

= 60

c) Find the median

30,30,30,30,40,40,40,50,50,50,90



= 40 + 40

2

= 80

2

= 40

d) Calculate the average mark.

Average= sum of all items

No of items

= 40 x 3) + 50 x 2) + (30 x 4) + 90 x 1)

10

= 430

10

= 43

e) Find the modal mark

30

f) Find the modal frequency.

4.

Activity.

Complete the table below and answer the following questions.

Marks	No of pupils	Total
40	_____	160
50	2	_____
_____	3	180
80	_____	80

a) Complete the table above

b) Find the modal mark

c) Find the modal frequency

d) Find the range

e) Find the mean mark

f) Find the median mark

AVERAGE OF GROUPS.

1. The average number of girls in 4 classes is 60. Find the total number of girls

Solution

Total = Average number

= 60 x 4

= 240 girls.

2. The average weight of 4 boys is 48kg.Find their total weight.

Solution

Total Weight = Average x number

= 48kg x 4

= 192kg

3. The average mass of 3 pickups is 4 tonnes. Find their total mass

Solution

Total = Average x 40

= 3 x 4

= 12 tonnes.

4. The average weight of 5 teachers is 60kg.Find their total weight.

Total weight = Average x number

= 60 x 5

= 300 kg

The average weight of 4 men is 50 kg. If 3 of them weighs 40kg,45kg and 45kg.Find the weight of the 4th man.

Soln.

Let weight of 4th be k

$$\frac{\text{sum of all items}}{\text{number of all items}} = \text{average}$$

$$\frac{k+40kg+45kg+45kg}{4} = 50kg$$

$\frac{k+130kg}{4}$ = 50kg.

$\frac{k+130kg}{4} \times 4$ = 50kg × 4.

k+130kg = 200kg

k + 130- 130 = 200 – 130

k = 70kg

The average weight of 6 boys is 60kg and the average weight of other 4 boys is 30 kg. Find the average of all boys

No	Average	sum
6	60kg	(60 x 6) kg 360 kg
4	30	30 x 4 120kg

Average = $\frac{\text{sum of all items}}{\text{No of items}}$

= $\frac{360kg + 120\ kg}{6+ 4}$

= $\frac{480kg}{10}$

= 48kg

2. The average weight of 6men is 45kg and average weight of other 4 men is 70 kg. Find the average weight of all men.

Solution.

NO	Average	sum
6	45kg	(45 x 6) kg 270kg
4	70kg	70 x 4 kg 280 kg

Average= sum of all items.

No of items

= 27 kg+ 280kg 0

6 + 4

= $\frac{550kg}{10}$

= 55kg

3. The average age of 3 girls is 12 years. When Wasswa joins them, their average becomes 13 years. How old is Wasswa?

No	Average	sum
3	12	3 x12) years 36 years
3+ 1 4	13	(4 x 13) years 52years.

Wasswa's weight= 52

- 36

16 years

Wasswa is 16 years.

4. The average age of 5 girls is 10 yrs. When Nakato and Babirye who are twins join them, their average becomes 12 yrs. How old is Babirye?

No	mean	sum
5	10	5 x 10 50 years
5 + 2 7	12	(7 x 12) years 84years.

Weight of twins= 84

- 50

34 years

Weight of Babirye = $\frac{34}{2}$

=17kg

5. The average weight of 5 boys is 20 kg. When John leaves the group, the average weight becomes 22kg.How old is John.

No **5**	**Average** **20**	**Total** **5 x 20) kg**
5- 1 **4**	**22**	**100kg** **(4 x 22) kg** **88kg**

John = 100 kg

- 88kg

12kg

Activity.

1. The average weight of 6 boys is 60 kg. Find their total weight.

2. The average age of 5 boys is 40 years. Find their total age.

3. The average of 6men is 45kg.Find their total weight

4. The average age of 5 girls is 12years and average age of 3 girls is 4 years. Find the average of all girls.

5. The average weight of 4 boys is 23kg and the average of 2 girls is 6kg.Find the average of 2 girls is 6kg.Find average age of all pupils.

6. The average weight of 3 woman is 60kg and the average weight of 3 women is 60kg.Find the average weight of all people.

7. The average weight of 4 boys is 42kg.When their teacher joins them, the average becomes 40kg.How heavy is the average.

8. The average age of 5 boys is 14 years. When their teacher joins them, the average becomes 40kg.How heavy is the teacher.

9. The average weight 5 boys is 50kg.When Nakato and Babirye joins the group their average become 40kg.Find the weight of Nakato.

10. The mean age of 3 boys is 14 years age of 2 of them is 13 years,15 years. Find the age of the 3rd boy.

11.The average weight of 5 pupils is 40kg.If 4 of them weigh 40kg,30kg,45kg and 35 kg. Find the weight of the 5th pupil.

FINDING UNKNOWNS GIVEN AVERAGE

Note

In this context, terms are separated using commas (,) and the word "and"

1. The average of n, 3,4 and 2 is 3. Find the value of x

Solution

$$\frac{\text{sum of all items}}{\text{No of items}} = \text{Average}$$

$$\frac{n+3+4+2}{4} = 3$$

$$\frac{n+9}{4} \times 4 = 3 \times 4$$

$$n + 9 = 12$$

$$n + 9 - 9 = 12$$

$$n + a + 9 = 12 - 9$$

$$n = 3$$

2. The mean of 6,2k, 7 and 3 is 6. Find the value of k.

solution

$$\frac{\text{sum of all items}}{\text{no of items}} = \text{mean}$$

$$\frac{2k+6+7+3}{4} = 6$$

$$\frac{2k+17}{4} \times 4 = 6 \times 4$$

$$2k + 16 = 24$$

2k+16-16 = 24-16

$\frac{2k}{2} = \frac{8}{2}$

K = 4

3. The average of 3p,6 and 2p-1 is 10. Find the value of p

Solution

sum of all items = Average

No of items

3p+ 2p-1+6 = 10

3

3p+ 2p+6-1 = 10

3

5p+ 5 x 3 = 10 x 3

3

5p + 5 x 3 = 30-5

3

5p = 25

P 5

P = 5

b) Find their range

soln.

3P **3 X P** **3 x 5** **15**	**2P – 6** **(2 x P) – 6** **(2 x 5) – 6** **(2 x 5) – 6** **10 – 6** **4**	**6**

Range = **H – L**

= **15 – 4**

= **9**

Activity:

1. **The average of 2,9,11, x,14 and 18 is 11. Find x**
2. **Find the value of k if the average of 7, x,3,9,8 and 10 is 8.**
3. **Given that the average of 5m, m+2,7-m and 3 is 8. Find the value of m.**
4. **If the average of m,6-2m,4m,3-2m and 4 is 4. Find the value of m.**
5. **The average of 5,3y,2-2y and 3 is 4. Find the value of y.**
6. **Find the value of p if the mean of 6,0,2p, p+4 and 2-p is 4.**
7. **The mean of n, 6, 7, 8 and 9 is 7. Find the value of n.**
8. **The mean of k, K + 4 and K + 2 is 6. Find K.**

b) Find their range.

9. **The average of 3, 4 and y is 5. Find the value of y.**
10. **The mean of 2 – X, 3X + 4, 0 and 6 is 5. Find the value of X.**
11. **The mean of K, K + 2 and 3 is 5. Find the value of K.**
12. **The average pf K – 6, 2K, K – 3, 6 and 15 – K is 10. Find the value of K.**

MORE ABOUT AVERAGE

The table below shows marks scored by different pupils.

Marks	**40**	**50**	**P**	**20**
No. of pupils	**3**	**2**	**4**	**1**

a) How many pupils did the test.

= 3 + 2 + 4 + 1

= 10 pupils

b) If the average mark was 36, find the value of P.

soln.

$$\frac{\text{Sum of all items}}{\text{No.of items}} = \text{Average}$$

$$\frac{(4 \times P)+(40 \times 3)+(50 \times 2)+(20 \times 1)}{10} = 36$$

$$\frac{4P+120+100+20}{10} = 36$$

$$\frac{4P+240}{10} = 36 \times 10$$

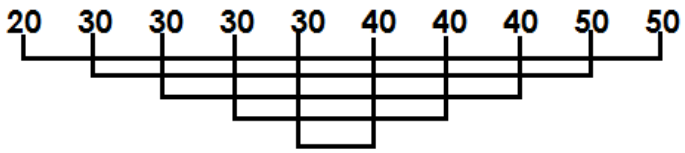
$$4P + 240 - 240 = 360 - 240$$

$$\frac{4P}{4} = \frac{120}{4}$$

$$P = 30$$

Find the median score

Soln.



$$\frac{30+40}{2}$$

$$= \frac{70}{2}$$

Median =35

2. The table below shows marks scored by different pupils

Marks	60	Y	50	40
No. of pupils	1	3	2	4

a) If the average score was 33. Find the value of y.

soln.

$$\frac{\text{Sum of all items}}{\text{No.of items}} = \text{Average}$$

$$\frac{(3 \times y)+(60 \times 1)+(50 \times 2)+(40 \times 4)}{10} = 38$$

$$\frac{3y+60+100+160}{10} = 38$$

$$\frac{3y+320}{10} = 38$$

$$\frac{3y+320 \times 10}{10} = 38 \times 10$$

$$3y + 320 = 380$$

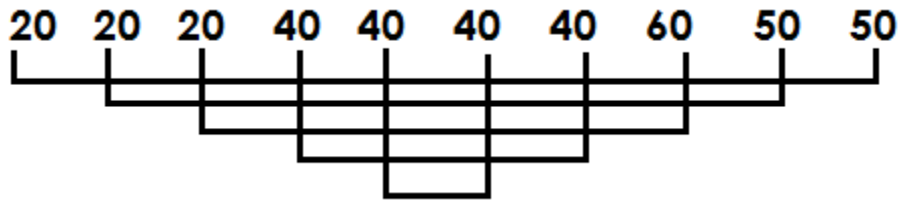
$$3y + 320 - 320 = 380 - 320$$

$$\frac{3y}{3} = \frac{60}{8}$$

$$y = 20$$

b) How many pupils scored above the median mark.

Median



$$= \frac{40+40}{2}$$

$$= \frac{80}{2}$$

$$= 40$$

Above 40 = 3 pupils

3. The table below shows marks scored by different pupils in a test where average score was 30.

Marks	40	20	30	50
No. of pupils	2	M	3	1

Find the value of m.

Soln.

$$\frac{\text{sum of all iems}}{\text{No.of all items}} = \text{Average}$$

$$\frac{(20 \times m) + (40 \times 2) + (30 \times 3) + (50 \times 1)}{(m+6)} = 30$$

$$\frac{20m+80+90+50}{m+6} = 30$$

$$\frac{(20m+220)\ x\ (m+6)}{m+6} = 30\ (m+6)$$

$$20m + 220 = 30\ x\ m + 6\ x\ 30$$

$$20m + 220 = 30m + 180$$

$$20m - 30m + 220 = 30m - 30m + 180$$

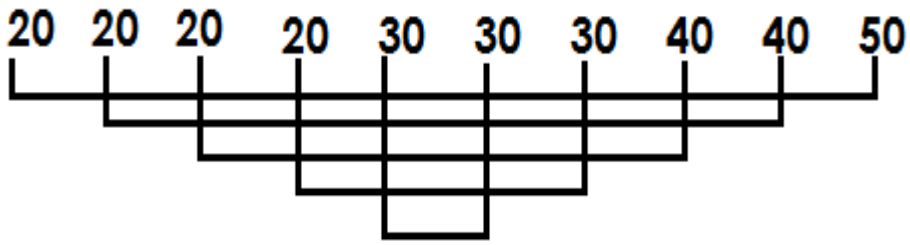
$$-10m + 220 = 180$$

$$-10m + 220 - 220 = 180 - 220$$

$$\frac{-10m}{-10} = \frac{-40}{-10}$$

$$M = 4$$

b) Find the median mark.



$$= \frac{30+30}{2}$$

$$= \frac{60}{2}$$

$$= 30$$

4. The table below shows marks scored by different pupils

No. of pupils	2	n	1	4
No. of marks	60	40	80	30

a) If the average mark was 44. Find the value of n.

Soln.

$$\frac{\text{Sum of all items}}{\text{No.of items}} = \text{Average}$$

$$\frac{(40\ x\ n) + (60\ x\ 2) + (80\ x\ 1) + (30\ x\ 4)}{n+7} = 44$$

$\frac{40n+120+80+120}{n+7} = 44$

$\frac{40n+320}{n+7} = 44$

$\frac{40n+320+(n+7)}{(n+7)} = 44\,(n+7)$

$40n + 320 = 44 \times n + 7 \times 44$

$40n + 320 = 44n + 308$

$40n + 320 - 320 = 44n + 308 - 320$

$40n = 44n - 12$

$40n - 44n = 44n - 44n - 12$

$\frac{-4n}{4} = \frac{-12}{-4}$

$n = 3$

Activity:

1. The table below represents marks scored by a group of boys in end of term examination.

Marks	30	25	n	35
No. of pupils	1	2	3	4

a) **How many boys did the examination?**

b) **If the average mark of the boys was 34. Find the value of n.**

c) **Find the median mark**

2. Study the table below.

Marks	70	80	m	90
No. of pupils	2	3	4	1

If the average mark of the pupils was 71. Find the value of m.

b) **Find the median mark.**

3. The table below shows marks scored by different pupils in a test.

Marks	**50**	**60**	**40**	**80**
No. of pupils	**K**	**3**	**4**	**1**

If the mean mark of the pupils was 52. Find the value of K

4. The table below shows marks scored by pupils in a mathematics test.

Marks	**80**	**70**	**90**	**60**
No. of pupils	**2**	**1**	**n**	**4**

a) If their average score is 74. Find the value of n.

b) Find their median score.

FINDING AVERAGE / MEAN OF COMPOUND TERMS

1. Find the mean of 2y, 6, y + 4, and y + 6

Soln.

$$\text{Mean} = \frac{\text{Sum of all items}}{\text{No.of items}}$$

$$= \frac{2y+6+y+4+y+6}{4}$$

$$= \frac{2y+y+y+6+4+6}{4}$$

$$= \frac{4y+16}{4}$$

$$= \frac{4y}{4} + \frac{16}{4}$$

$$= y + 4$$

2. Find the mean of 3P + 4, P + 6, 3, 2 and P + 5

Soln.

$$\text{Mean} = \frac{\text{Sum of all items}}{\text{No.of items}}$$

$$= \frac{3P+4+P+6+3+2\ P+5}{5}$$

$$= \frac{3P+P+P+4+6+2+P+5}{5}$$

$$= \frac{3P+P+P+4+6+2+5+3}{5}$$

$= \frac{5P+20}{5}$

$= \frac{5P}{5} + \frac{20}{5}$

$= P + 5$

3. Find the mean of k, 3k, 2 and K + 10

Mean $= \frac{\text{Sum of all items}}{\text{No.of items}}$

$= \frac{K+3K+2+K+10}{4}$

$= \frac{5K+12}{4}$

4. Find the mean of 2m, 7, 0, 3 and 3m.

Soln.

Mean $= \frac{\text{Sum of all items}}{\text{No.of items}}$

$= \frac{2m+3m+7+3+0}{5}$

$= \frac{5m+10}{5}$

$= \frac{5m}{5} + \frac{10}{5}$

$= m + 2$

Activity:

1. Find the mean of the following: -

a) m, 6 and 2m.

b) 2w, 8, 3w, 4, 2 and w +4

c) 6, 4, and 3m + 8

d) 3k + 2, 5k + 4, 2k + 4 and 2k + 2

e) 2k + 4, 0, 3k + 6 and 2

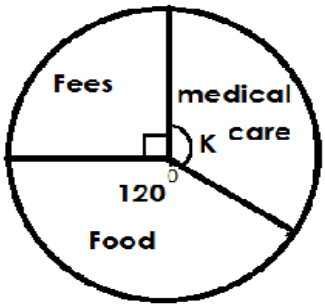
f) k – 2, k + 6, 4 and 2k

PIE CHARTS

- **Is a diagram in which quantities or proportions are illustrated as sectors of a circle**
- **It is also a circular graph representing data**
- **The sectors of a circle are mostly represented in degrees, fractions, percentages and quantities.**
- **Given degrees, all sectors must add up to 360º.**
- **Given percentages, all sectors must add up to 100%**
- **Given fractions, all sectors must add up to 1 (a whole).**

Interpretation of circle graph

The pie chart below shows how a man spent sh. 180,000. Use it to answer questions.



a) Find the values of K.

K + 90º + 120º = 360º

K + 210º = 360º

K + 210º – 210º = 360º – 210º

K = 150º

b) How much did he spend on food?

Soln.

$\frac{120}{360}$ **x sh. 180,000**

= sh. 60,000

c) How much more money was spent on medical care than fees?

Soln.

More = 150º – 90º

= 60º

More money $= \frac{60}{360}$ **x sh. 180,000**

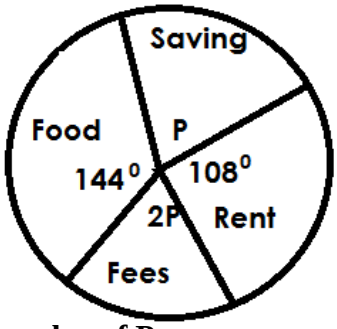
$=$ **sh. 30,000**

d) Express the amount spent on fees as a fraction

$= \frac{90}{360}$

$= \frac{1}{4}$

2. The pie chart below shows how a man spent sh. 120,000.



a) Find the value of P.

$P + 144^o + 2P + 108^o = 360^o$

$3P + 252^o = 360^o$

$3P + 252^o - 252^o = 360^o - 252^o$

$3P = 108^o$

$\frac{8P}{3} = \frac{108}{3}$

$P = 36^o$

b) How much did he spend on rent?

Rent $= \frac{108}{36P}$ **x sh. 120,000**

$=$ **sh. 36,000**

c) How much did she spend on food than fees?

Food	Fees	More in degrees
144°	2 X P 2 x 36° 72°	$144^o - 72^o$ 72°

More money $= \frac{72}{360}$ **x sh. 120,000**

$=$ **sh. 24,000**

d) What percentage represents saving?

As a percentage $= \frac{P}{360}$ **X 100%**

$= \frac{36}{360}$ **X 100%**

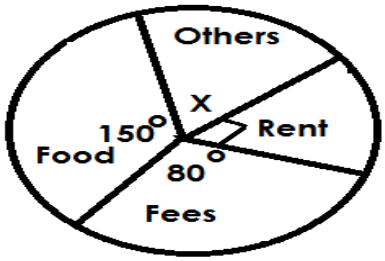
= 10%

e) Express his expenditure on Rent as a fraction.

As a fraction $= \frac{108}{360}$

$= \frac{3}{10}$

3. Use the pie chart below showing how Mugisha's spends his income to answer questions.



a) Find the value of X

X + 90° + 80° + 150° = 360°

X = 40°

b) How much does he spend on fees if he earns sh. 180,000?

Fees $= \frac{80}{360}$ **x sh. 180,000**

$=$ **sh. 40,000**

c) How much more money did he spend on food than Others.

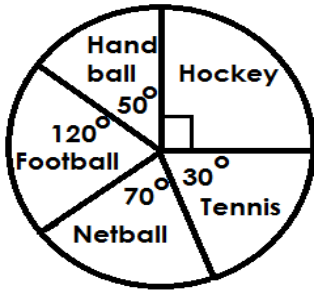
More = 150° – 40°

More money $= \frac{110}{360}$ **x sh. 180,000**

$=$ **sh. 55,000**

Activity:

1. The Pie chart below shows number of pupils who participated in games. There are 720 pupils altogether.



a) **What is the most played game?**

b) **How many pupils play?**

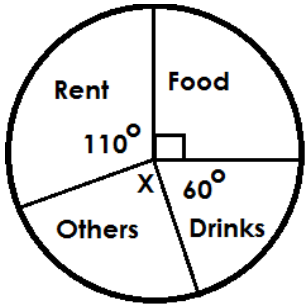
i) **Football**

ii) **Hand ball**

c) **If all girls play netball, how many girls are in the school?**

d) **How many more pupils play hockey than tennis?**

3. The pie chart below shows a man's expenditure of sh. 180,00 over the weekend.



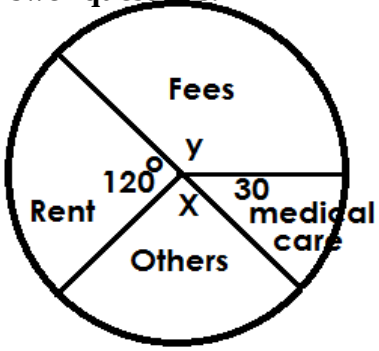
a) Find the value X.

b) How much money is spent on food.

c) How much more money is spent on rent than drinks.

d) What fraction represents drinks.

3. Below is a pie chart showing how Modoolo spent his monthly salary of sh. 360,000. Use it to answer questions.



a) **Find the value of;**

i) **X**

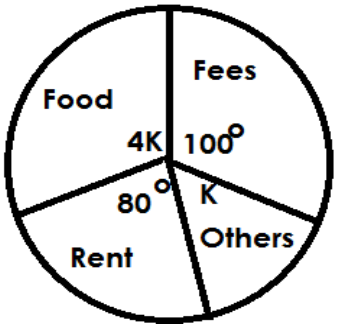
ii) **Y**

b) **How much money was spent on Rent?**

c) **How much more money was spent on Fees than others?**

d) **What percentage of the money was spent on medical care?**

4. **The pie chart below shows how a man spent his income of sh. 360,000.**



a) **Find the value of K**

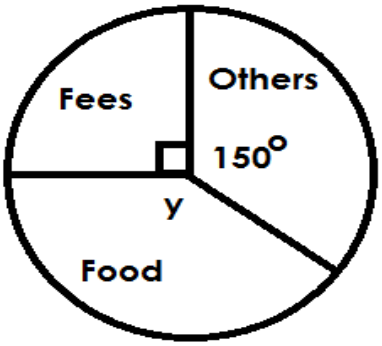
b) **How much does he spend on food?**

c) **How much more money did he spend on rent than others?**

d) **Express the amount spent on food as a percentage of his income.**

MORE ABOUT INTERPRETATION OF PIE-CHART

1a) **The Pie chart below shows how a man spent his monthly salary**



a) **Find the value of Y**

$y + 90^o + 150^o = 360^o$

y + 240 º = 360 º

y + 240 º – 240 º = 360 º – 240 º

y = 120 º

b) If he spent sh. 45,000 on others, how much was his income?

Soln.

Let his income be K.

$\frac{150}{360}$ **x K = sh. 45,000**

$\frac{150K}{360}$ **= sh. 45,000**

$\frac{150K}{360}$ **x 360º = sh. 45,000 x 360**

150K = sh. 45,000 x 360

$\frac{150K}{150} = \frac{sh.45,000 \times 360}{150}$

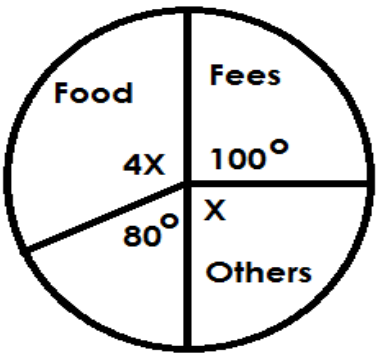
K = sh. 108,000

b) How much did he spend on fees?

Fees = $\frac{90}{360}$ **x sh. 108,000**

= Sh. 27,000

2. The pie chart below shows how Musa spent his salary.



a) Find the value of X.

4X + X + 100º + 80 º = 360 º

5X + 180 º = 360 º

5X + 180 º – 180 º = 360 º – 180 º

$5X = 180^{o}$

$\frac{5X}{5} = \frac{180}{5}$

$X = 36^{o}$

b) If he spent sh. 160,000 on rent. Find his salary.

Let his salary be K.

$\frac{80K}{360}$ **x K = sh. 160,000**

$\frac{80K}{360}$ **= sh. 160,000**

$\frac{80K}{360}$ **x 360 = sh. 160,000 x 360**

80K = sh. 160,000 x 360

$\frac{80K}{80} = \frac{sh.160,000 \times 360}{80}$

K = sh. 720,000

c) How much does he spend on food?

= 4 x K

= 4 x 36°

= 144°

Food = $\frac{144}{360}$ **x sh. 720,000**

= sh. 288,000

d) How much more money was spent on Fees than Rent?

More = 100° – 80°

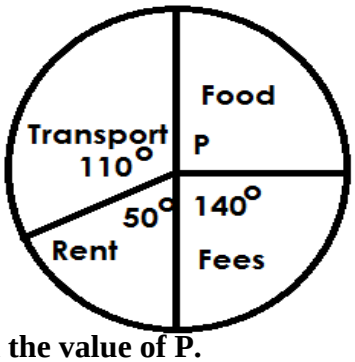
= 20°

More money = $\frac{20}{360}$ **x sh. 720,000**

= sh. 40,000

Activity:

1. The pie chart below shows how Alex spends his salary.

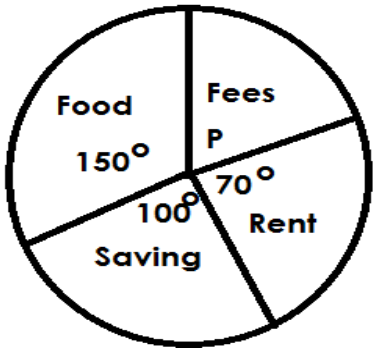


a) Find the value of P.

b) If he spends sh. 120,000 on food, how much does he earn?

c) How much money does he spend on fees?

2. The circle graph below shows how Okello spends his income.



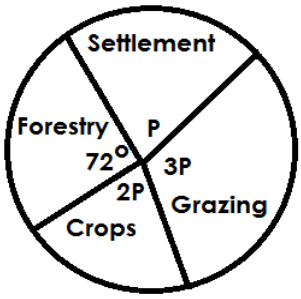
a) Find the value of P

b) If he spends sh. 21,000 on rent, find his income.

c) How much money does he spend on Rent?

d) How much more money is saved than spent on fees?

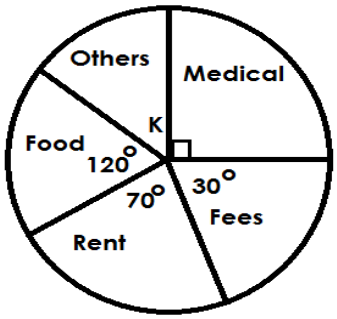
3. Below is a pie chart showing how Kato divided his piece of land.



a) If he uses 4 acres for grazing, find the size of his land in acres?

b) How much more land is used for grazing than crops.

4. The pie chart below shows how Nandudu spent her income.



a) Find the value of K.

b) If she spent sh. 40,000 on others more than fees. Find his income.

b) How much did he spend on food?

CONSTRUCTION OF PIE CHARTS

Note:

While constructing pie charts, first change quantities, percentages, fractions into degrees.

CONSTRUCTION OF PIE CHARTS GIVEN FRACTIONS

- **Given fractions, convert them to degrees by multiply the given fraction by 360°.**

Examples.

A man spent $\frac{1}{4}$ of his salary on food.

$\frac{1}{3}$ on fees and $\frac{5}{12}$ on rent. Using a radius of 3cm. Show the man's expenditure on a pie chart.

Food	Fees	Rent
$\frac{1}{4}$ x 300 90°	$\frac{1}{3}$ x 360° 120°	$\frac{5}{12}$ x 360° 150°

2. Osama spent $\frac{1}{6}$ of his income on rent, $\frac{1}{4}$ on fees, $\frac{1}{3}$ on food and $\frac{1}{4}$ on medical. Using a radius of 3cm. Show Osoma's expenditure on a pie chart.

Rent	Fees	Food	Medical
$\frac{1}{6}$ x 360 60°	$\frac{1}{4}$ x 360 90°	$\frac{1}{3}$ x 360 120°	$\frac{1}{4}$ x 360 90°

3. A man spent 0.2 of his salary on food, 0.4 on fees, 0.3 on rent and he saved 0.1. using a radius of 4cm. show the above information on a pie chart.

Food	Fees	Rent	Saved
0.2 x 360o $\frac{2}{10}$ x 360 72°	0.4 x 360o $\frac{4}{10}$ x 360o 144°	0.3 x 360o $\frac{3}{10}$ x 360o 108°	0.1x 360 $\frac{1}{10}$ x 360 36°

Activity:

1. A man spent 2/5 of his income on food, 3/10 on Rent. 1/5 on fees and 1/10 on saving. Using radius of 3.5cm, construct a pie chart showing how a man spent his income.

2. Mukasa spent $\frac{5}{12}$ of his salary on rent, $\frac{1}{3}$ on food and the rest on fees. Construct a pie chart and represent Mukasa's expenditure. Use radius of 4cm.

3. The school's expenditure per terms is as follows:

- **$\frac{1}{3}$ on scholastic materials.**
- **$\frac{1}{12}$ on renovations**
- **$\frac{1}{6}$ on others**

Using a radius of 3.5cm, construct a pie chart showing the above information.

4. A family spends 0.2 of its income on medical care, 0.1 on rent, 0.2 on saving and 0.5 on other things. Using radius of 4cm, construct a pie chart showing the above information.

5. Maleeka uses 0.3 of his land for agro forestry, 0.1 for settlement, 0.2 for crop growing and the rest of the land for grazing. Construct a pie chart of radius 3.5cm representing the above information

CONSTRUCTION OF PIE CHARTS GIVEN PERCENTAGES.

Examples.

A man divided his land as follows: -

20% for construction, 25% for crop growing, 10% for children and the remainder for grazing. Show the above information on a pie chart of radius 4cm.

Construction	Crop growing	Children	Grazing
20% x 360° $\frac{20}{100}$ x 360° 72°	$\frac{25}{100}$ x 360° 90°	$\frac{10}{100}$ x 360° 36°	100% - (25% + 20% + 10%) 100% - 55% $\frac{45}{100}$ x 360° 162°

2. In a village, 25% are men, 32% are children and 40% are women. Show the above information on a pie chart of radius 3.5cm.

Men	Children	Women
25% x 360° $\frac{25}{100}$ x 360^0 90°	20% x 360° $\frac{20}{100}$ x 360° 72°	40% x 360° $\frac{40}{100}$ x 360° 144^0

Activity:

1. 45% of the pupils in a school passed in grade I, 30% of the pupils passed in grade III and 10% pupils passed in grade IV. Represent this information on a pie chart/

2. In a basket, 30% of the fruits are manages, 25% are apples, 10% are oranges and 35% are pineapples. Using a radius of 3.5cm, show the above information on a pie chart.

3. Okello spends 30% of his salary on food, 15% on rent, 35% on fees and saves the rest. Use the information and draw a circle graph.

4. In a village meeting, 35% of the people are women, 40% are men and the rest are children. Represent this information on a pie chart.

CONSTRUCTION OF PIE CHARTS GIVEN QUANTITIES

- **Given quantities, get total number of items.**
- **Express each item as a fraction of total and multiply by 360° to get number of degrees.**

Examples.

In a bag, there are 2 pencils, 3 pens, 1 set and 3 books. Show the above information on a pie chart of radius 3cm.

Total = 2 + 3 + 1 + 3

= 9

Pencils	Pens	Sets	Books
$\frac{2}{9}$ x 360° 80°	$\frac{3}{9}$ x 360° 120°	$\frac{1}{9}$ x 360° 40°	$\frac{3}{9}$ x 360° 120°

2. On a farm, three are 2 cows, 3 bulls, 5 goats and 8 rabbits. Using a radius of 3.5cm. show the above information on a pie chart.

Total = 2 + 3 + 5 + 8

= 18

Pencils	Pens	Sets	Books
$\frac{2}{18}$ x 360° 40°	$\frac{3}{18}$ x 360° 60°	$\frac{5}{18}$ x 360° 100°	$\frac{8}{189}$ x 360° 160°

Activity:

1. In a village, 4 farmers grow beans, 3 grow maize, 7 grow simsim and 6 farmers grow cassava. Using radius of 4cm. show the above information of a pie chart.

2. Abigail gave sh. 4,000 to her husband sh. 2,000 to her son. Sh. 3,000 to her daughter and sh. 1,000 to her driver. Represent this information on a pie chart of radius 4.5cm.

3. There are 5 bulls, 9 calves, 10 cows and 6 heifers on a farm. Represent the above data on a pie chart of radius 3.5cm.

4. A farmer earned sh. 4,000 from the sales of beans sh. 7,000 from peas sh. 3000 from tomatoes and sh. 6,000 from others. Use the information to draw a pie chart of radius 4.4cm

Scenarios

Session 2

- ✦ Interpreting travel graphs
- ✦ Drawing a travel graph
- ✦ Plotting coordinates on a cartesian grid
- ✦ Completing cartesian tables.

Methods

- **Guided discovery**
- **Group discussion**
- **Think, pair and share.**
- **explanation**

Lesson activities

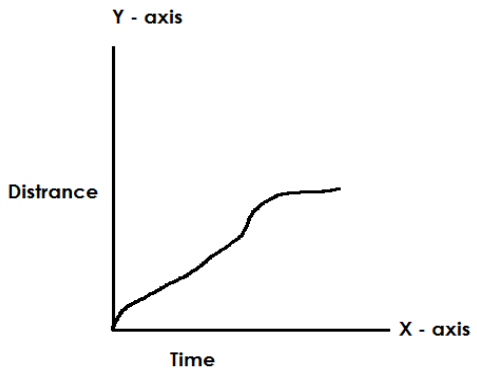
- **plotting**
- **drawing**
- **measuring**
- **subtracting**

Content/facts

TRAVEL GRAPHS

A travel graph is a distance, time graph where distance is plotted against time.

Illustration



- **Distance is plotted on the Y- axis (vertical axis)**
- **Time is plotted on the x-axis (horizontal axis)**

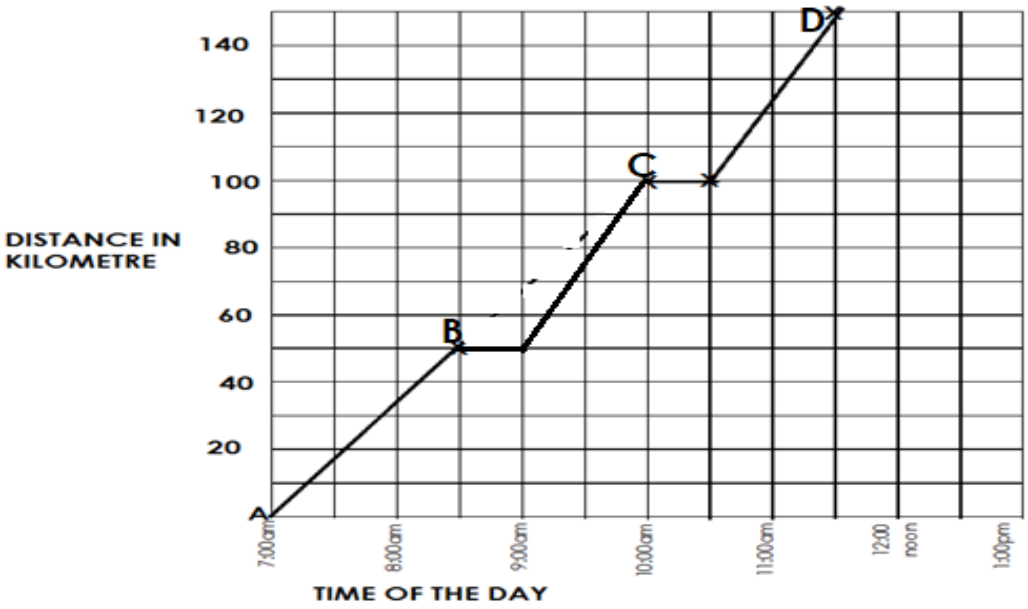
INTERPRETATION OF TRAVEL GRAPHS

Note:

Before answering any question about a travel graph, identify both the vertical and horizontal scales used.

Example I

The graph below shows Mandela's journey. Use it to answer questions below.



a). State the vertical scale

Soln

2 squares = 20km

1 square = $\frac{(20)km}{2}$

=10km

∴ 1 square rep. 10km

b) State the horizontal axis

soln

2 squares rep one hour.

1 square rep (½) hrs

∴ 1 square rep. ½ hrs

c) For how long was the journey from town A to town B?

soln

1 square = ½ hr.

3 squares = (½ x 3) hrs

$= \frac{3}{2}$ **hrs**

=1 ½ hrs

∴ It was for 1 ½ hrs.

d) How far is town B from town D?

square= 10km

squares = (10 x 9)km

= 90km

∴Town B is 90km far from Town D.

e) At what time did he leave town C?

soln

At 10:30a.m

f) Calculate his average speed for the whole journey.

Soln

Average speed $= \frac{\text{Total distance}}{\text{total time}}$

$= \frac{140\text{km}}{1\frac{1}{2}\text{hrs}+\frac{1}{2}\text{hrs}+1\text{ hr}+\frac{1}{2}\text{hrs}+1\text{ hr}}$

$= \frac{140km}{4\frac{1}{2}hrs}$

= **140km ÷ $\frac{9}{2}$ hrs**

= **$15\frac{5}{9}$ km / hr.**

f) What was his average speed for the whole journey while travelling?

Note:

- ***When finding average speed for the whole journey while travelling, exclude the resting time.***

Soln

Average speed = $\frac{\text{Total distance}}{\text{Total time}}$

= $\frac{140\text{km}}{4\frac{1}{2}\text{hrs} - 1\text{ hr}}$

= $\frac{140\text{km}}{3\frac{1}{2}\text{hrs}}$

= **140km ÷ 3 ½ hrs**

= **140km ÷ $\frac{2\text{hrs}}{7}$**

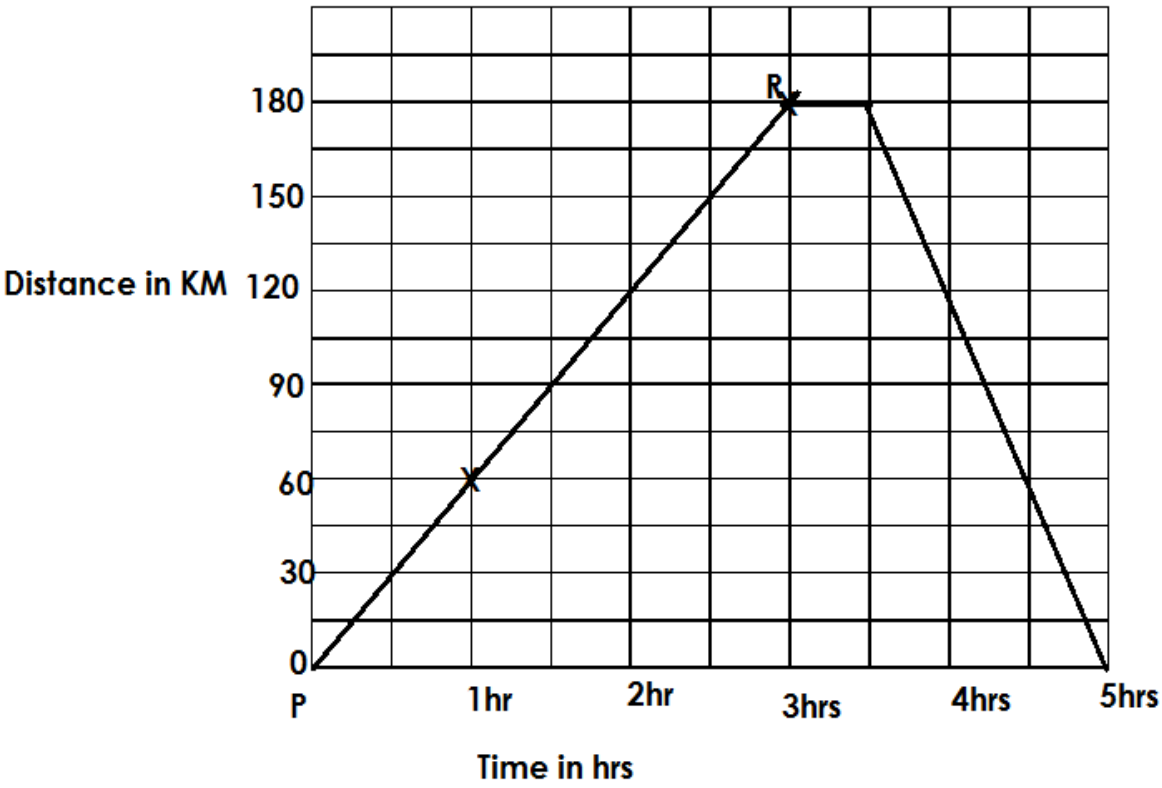
= **20km x $\frac{2\text{hrs}}{7}$**

= **20km x 2hr**

= **40km / hrs.**

Example 2

The graph below shows Mukulu's journey.



a) Calculate his speed from P to Q

soln

S $= \frac{D}{T}$

= 60km ÷ 1 hr

= 60km x $\frac{1}{1}$ hr

= 60km/hr.

b) For how long was the stop over for the whole journey?

Soln

1 squares = 1hr

1 square =(½) h0urs

∴1 square = ½ hours

1 sq = ½ hr

2 squares = (1/2 x ½) hours

= 1hr

c) Calculate his average speed for the whole journey?

Soln.

Average speed = $\frac{\text{total distance}}{\text{total time}}$

$= \frac{180\text{km}+180\text{km}}{5\text{hrs}}$

$= \frac{360\text{km}}{5\text{hrs}}$

= 72km/hrs

d) Calculate his average speed for the whole journey while moving.

Soln.

Average speed = $\frac{\text{total distance}}{\text{total time}}$

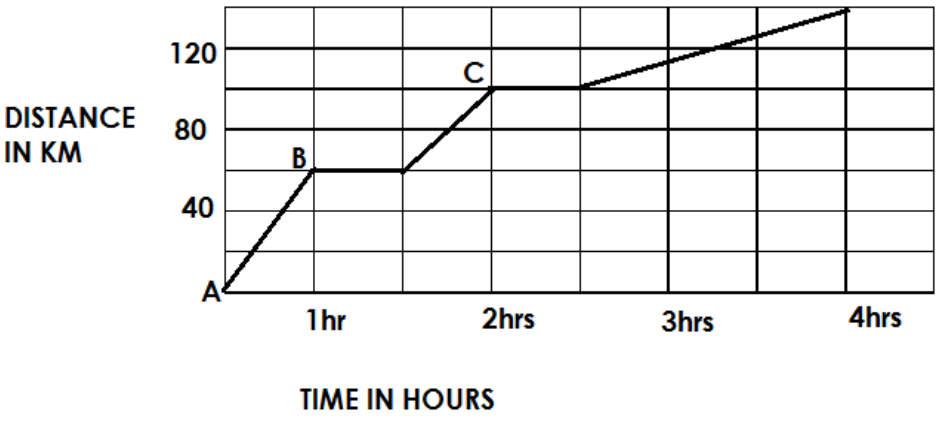
$= \frac{180\text{km}+180\text{km}}{5\text{ hrs}-1\text{ hr}}$

$= \frac{360\text{km}}{4\text{ hrs}}$

= 90 km/hr.

Activity:

1. The graph below shows Mandela's journey.



a) State the vertical scale.

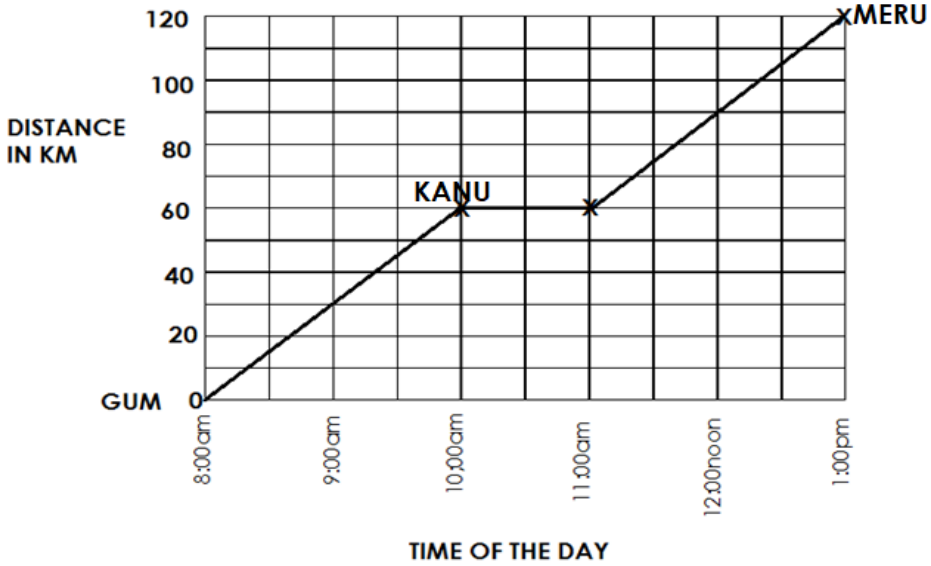
b) What is the horizontal scale?

c) How far is Town B from A?

d) Calculate his average speed from Town A to town B.

e) For how long was the stop overs for the whole journey.

2. The graph below shows a tax's journey. Use it to answer questions.



a) **State the vertical scale.**

b) **At what time did the taxi leave Gumu town?**

c) **Find the distance between Kanu and Meru.**

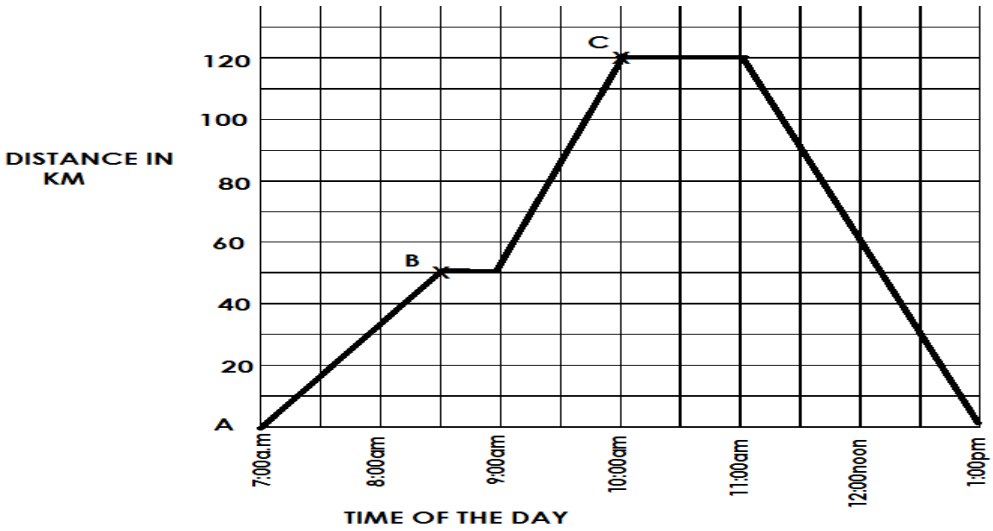
d) **For how long did the taxi stay at Kanu?**

e) **Find the time taken by a taxi to move from Gumu to Kanu.**

f) **Calculate the taxi's average speed for the whole journey.**

g) **Find the taxi's average speed for the whole journey while traveling.**

3. **The graph below shows a bus' journey. Use it to answer questions.**



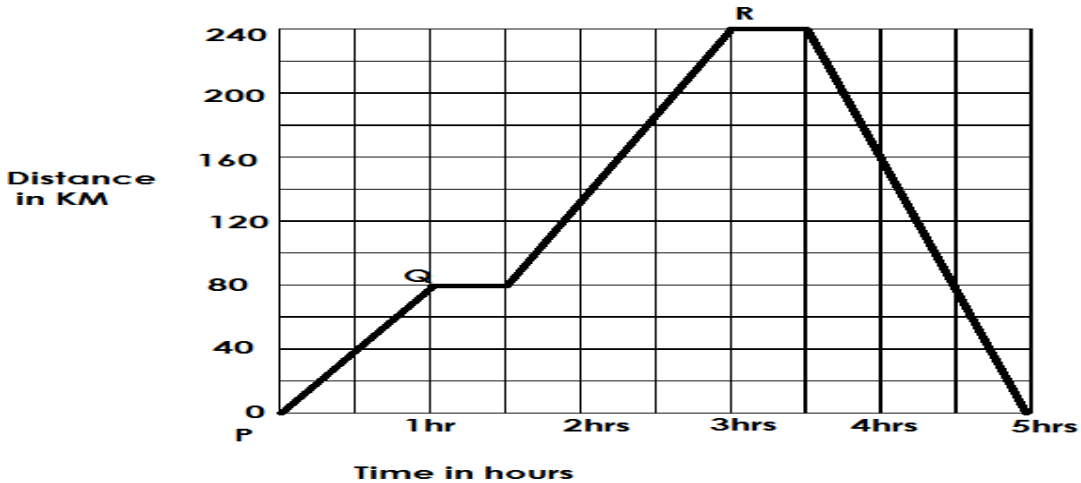
a) **How far is town C from B?**

b) **At what time did the bus set for B?**

c) **For how long was the stop over at C?**

d) **Calculate the bus' speed for the journey from A TO C.**

e) **For how long was the journey from C to A.**



a) **At what time did Mukulu leave town P?**

b) **Calculate Mukulu's speed from P to Q.**

c) **For how long was the stop over at R.**

d) **Calculate his average speed for the whole journey.**

e) **Calculate Mukulu's average speed for the whole journey while travelling.**

REPRESENTING INFORMATION ON A TRAVEL GRAPH.

Note:

- **Both vertical and horizontal scales have to be formed.**
- **A scale is a unit measure of something.**
- **Journeys moved have be labeled.**

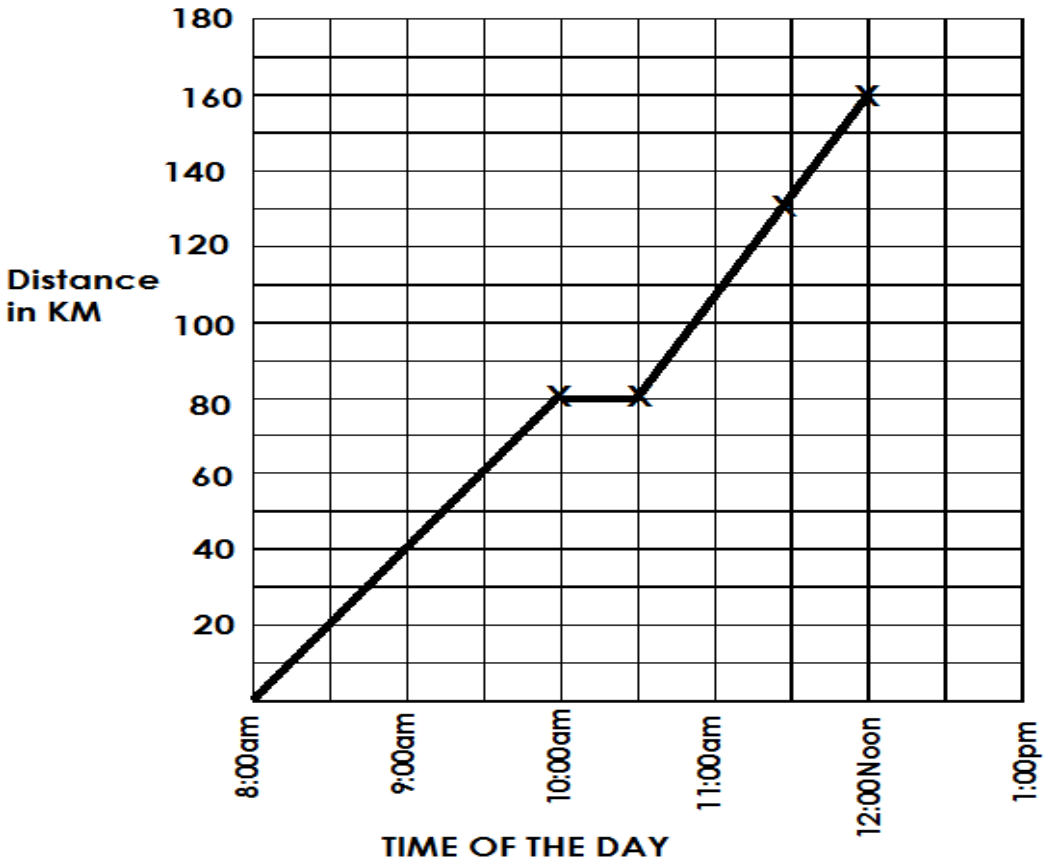
Example I

Matama left Town A at 8:00am travelling at an average speed of 40km/hr in 2 hours to town B. He rested for ½ hours and later continued to C at a speed of 60km/hr in 1 ½ hours.

a) **Represent Matama's journey on a graph.**

Journey	**Time**	**Distance**
A to B	**2hrs**	**D = S X T** **= $\frac{40km \times 2\ hrs}{1hr}$** **= 80km**
Rest	**½ hours**	
B to C	**1 ½ hrs**	**D = S x T** **= $\underline{60km} \times \frac{3}{2}hr$**

		hr **= 90km**



b) At what time did she reach Town B at 10:00a.m.

c) Calculate her average speed for the whole journey while travelling.

Soln.

A.S = $\frac{\text{Total distance}}{\text{Total time}}$

= $\frac{170\text{km}}{3\frac{1}{2}\text{hr.}}$

= 170km ÷ $\frac{7hr}{2}$

= 170km x $\frac{2}{7}$hr

= $\frac{340km}{7hr}$

= 48 $\frac{4}{7}$km / hr

Example II

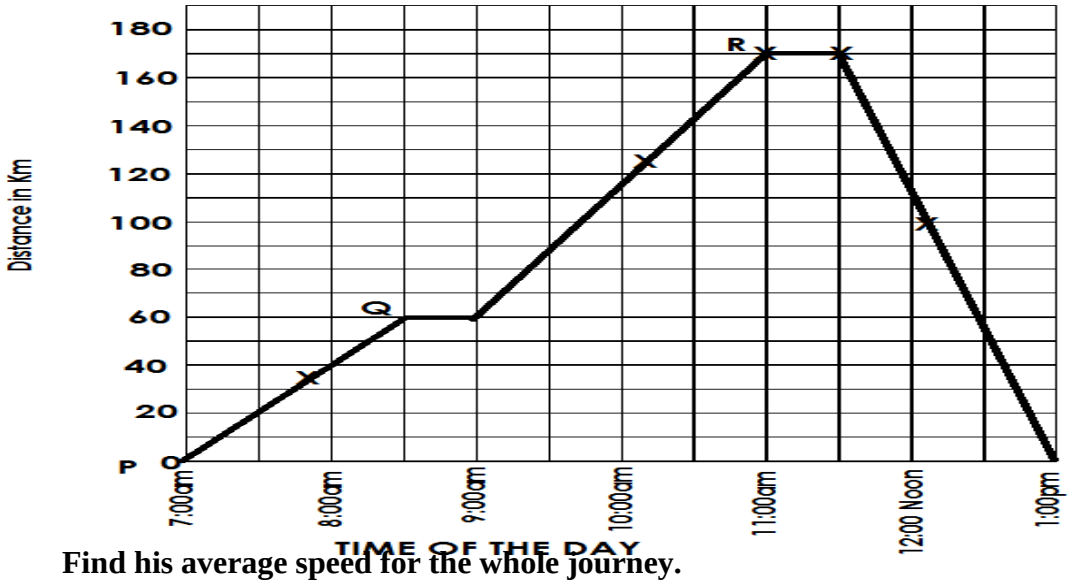
A cyclist left Town P at 7:00a.m travelling at a speed of 40km / hr for 1 ½ to Q. He rested for ½ hrs and later continued to Town K which is 120km away at a speed of 60km/hr. He rested for ½ hrs and later returned to P at a speed of 120km/hr. Represent his journey on a graph.

Soln

Journey	Time	Distance
P - Q	1 ½ hr	D = S X T = $\frac{40km}{1hr}$ x $\frac{3}{2}$hrs = 60km
Rest	½ hrs	-----
Q to R	T = D ÷ S = 120km ÷ $\frac{60km}{1hr}$ = 120km x $\frac{1hr}{60km}$ = 2hrs	120km
Rest	½ hrs	-------
R to P	T = D ÷ S	120km + 60km 180km

	= 180km ÷ $\frac{120km}{1hr}$ = 120 km = $\frac{180km}{120}$ 1 ½ hrs	

A cyclist's journey



b) Find his average speed for the whole journey.

Soln

Average speed = $\frac{\text{Total distance}}{\text{Total time}}$

= $\frac{180\ km+180km}{6hrs}$

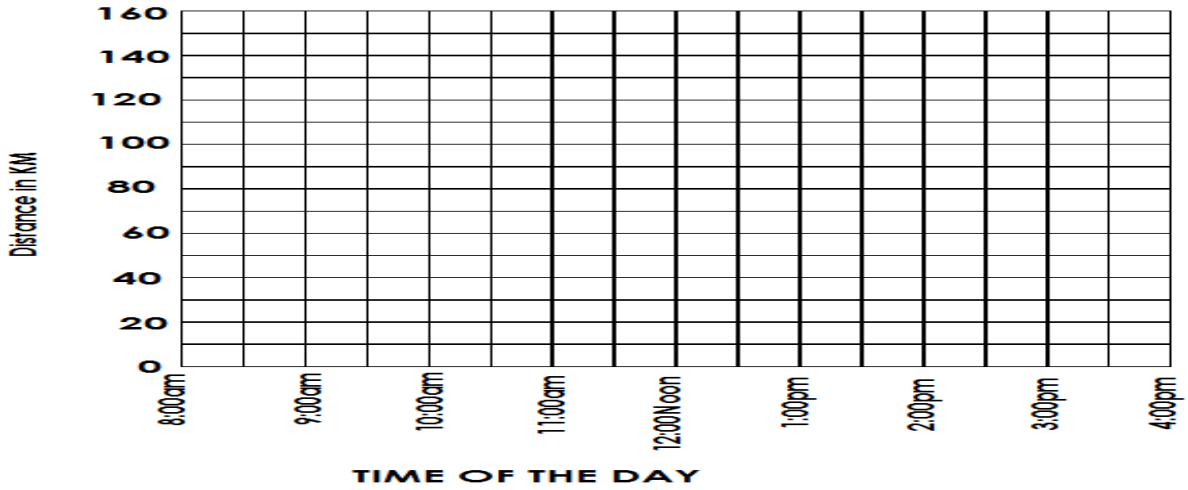
= $\frac{360km}{6hrs}$

= **60km/hr**

Activity:

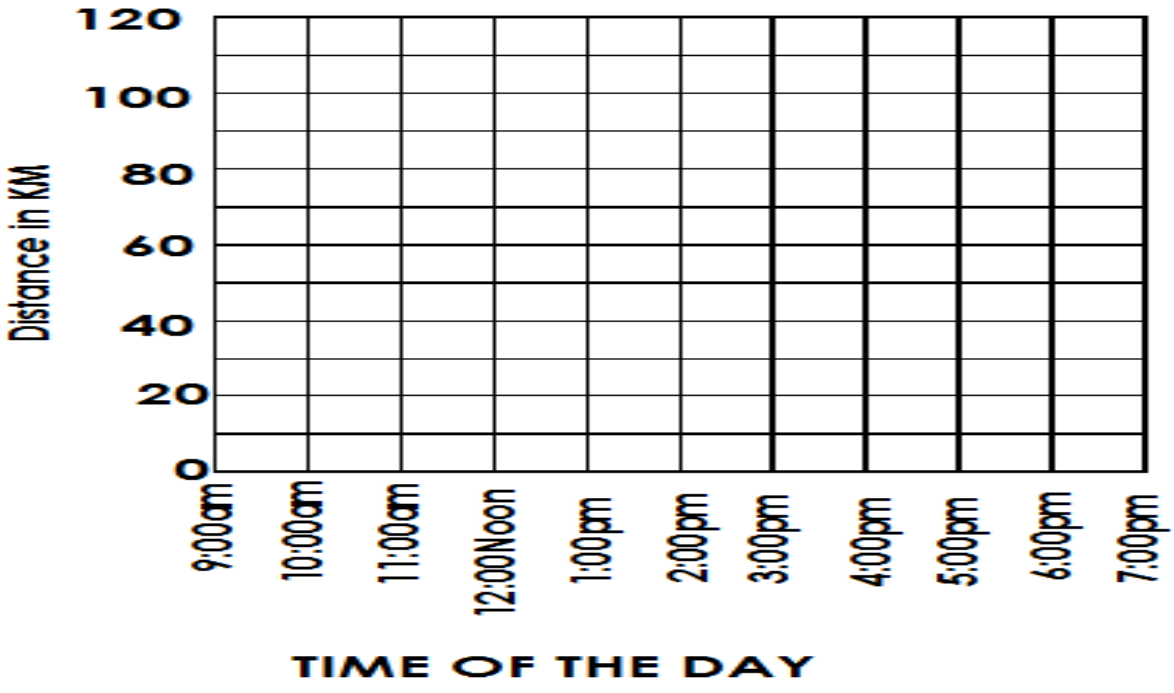
1. A motorist left town A at 8:00am travelling at a speed of 40km/hr for 1 ½ hr to town B. He rested at town B for ½ hr. and then travelled to Town C at an average speed of 45km/hr for 2 hrs. he rested for an hour at town C before returning to Town A at a speed of 50km/hr.Show the above journey in the grid below.

A motorist's journey



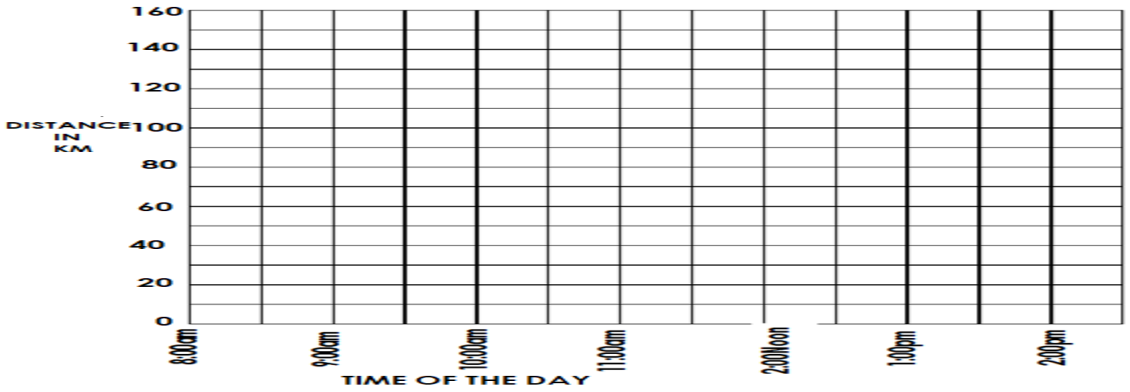
b) Calculate his average speed for the whole journey.

2. Kamu left Town A at 9:00am travelling at a speed of 20km/hrs in 2 hrs. he rested for one hour and later continued with his journey at a speed of 10km/hr for 5hrs. Aisha also left town A at noon and goes after Kamu. She catch up with him at 2:00pm. Show Aisha and Kamu's journey on a travel graph using the grid below.



3. A and B are two towns which are 160km apart. Sarah leaves town A at 8:00am travelling to town B. After 1 ½ hrs while travelling at a speed of 60km/hr, she rests for 30 min and then continues to B at 35 km/hr. On the same day Peter leaves town B at 9:00am travelling at a steady speed of 80km/hr travelling to Town A.

a) Draw a graph to show the two journeys in the grid below.



b) At what time did they meet?

c) Calculate Sarah's average speed for the whole journey.

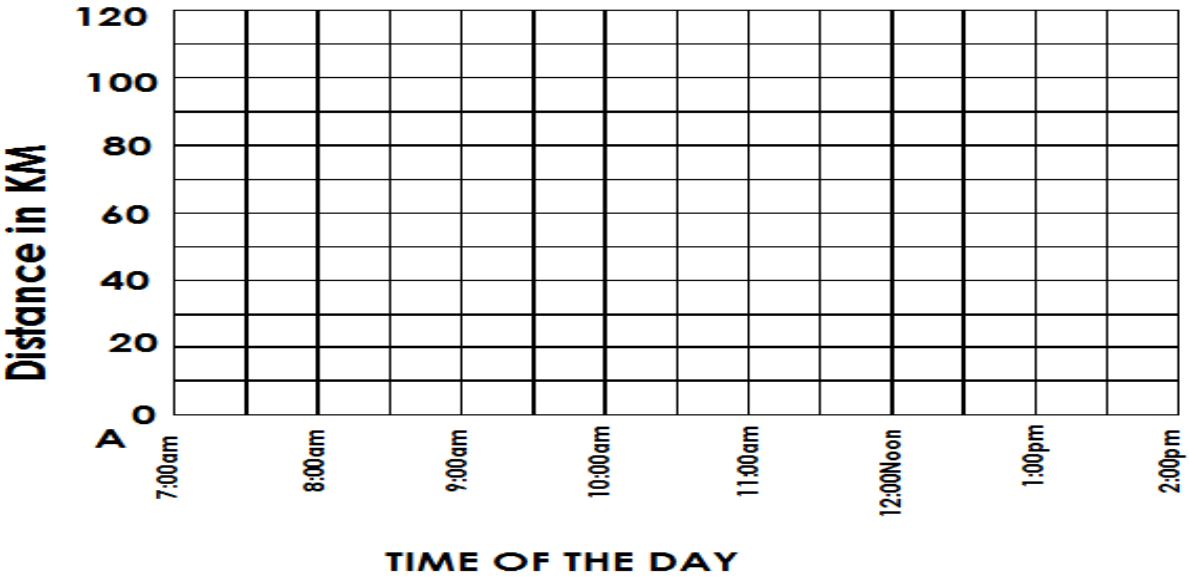
d) At what time did Peter reach town A?

e) What distance from A was Peter when Sarah started her second journey to B?

<u>MORE ON TRAVEL GRAPHS</u>

Kamu left town A at 7:00am travelling to town B which is 120km apart. After covering a distance of 40km. He rested for 1 ½ hrs and later continued for the rest of the journey at 40km/hr. Meanwhile, Betty also left town A at 8:00 am going after Kamu at a speed of 60km/hr.

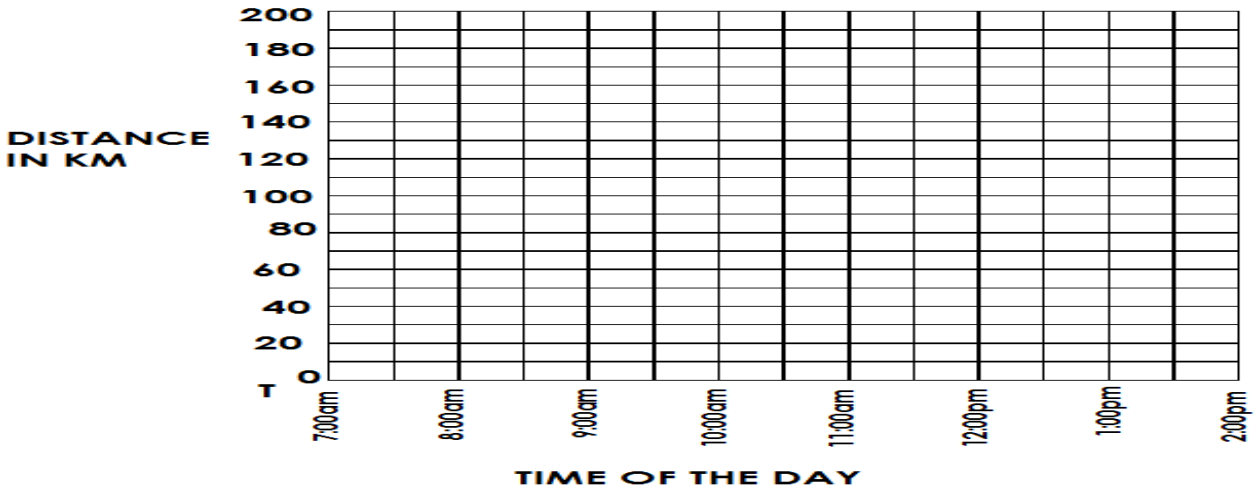
Show the above journeys on a travel graph.



b) At what time did Betty meet Kamu?

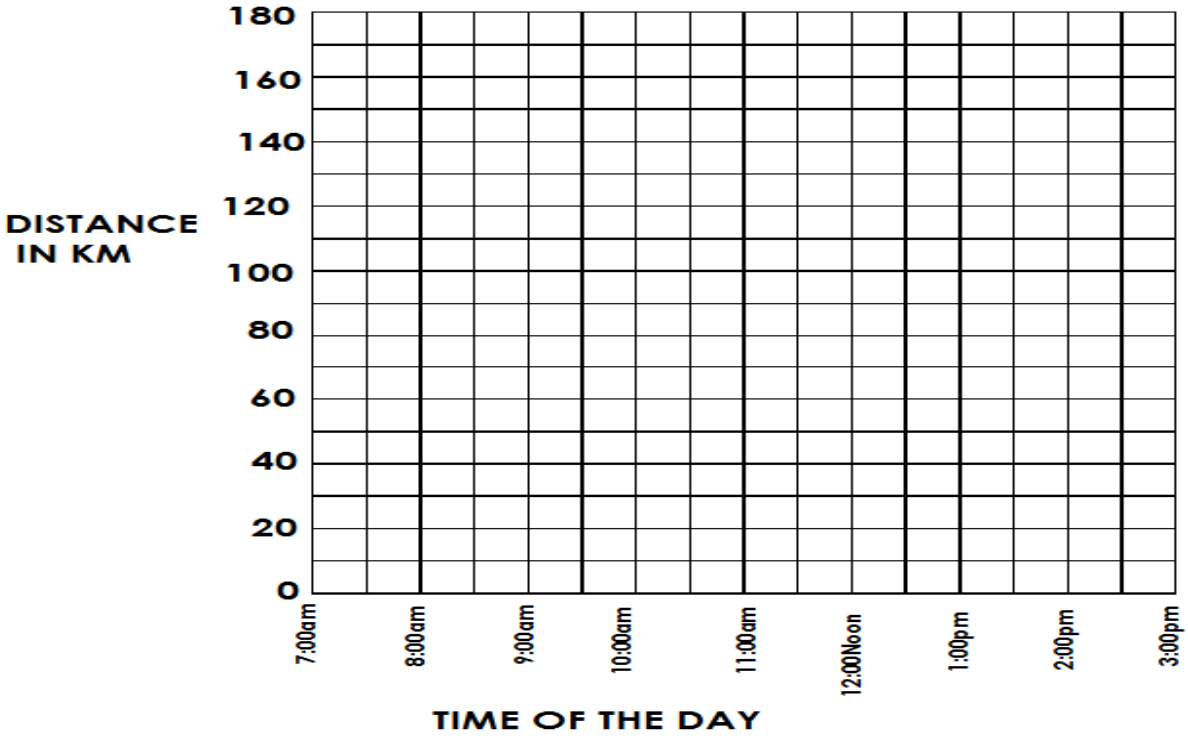
2. Town T and Town M are 200km apart. A car left town T at 7:00am travelling at a speed of 60km/hr. After some time, it got a puncture and the repair took 30 minutes. It continued at 100km/hr for the rest of the journey. On the same day a lorry left Town M at 7:30 am travelling at steady speed of 50km/hr.

a) Show the two journeys on a travel graph. Using the grid below.



2. Town K and M are 180km apart. A car left town K at 7:00am for M while travelling at an average speed of 40km/hr for 2 ½ hrs. It got a puncture and the repair took 2hrs and later it continued at 80km/hr for the rest of the journey. On the same day a bus left town M at 8:00am travelling at a speed of 40km/hr for 2hrs. it stopped for 1 hr and later it continued to town K in 2 hours.

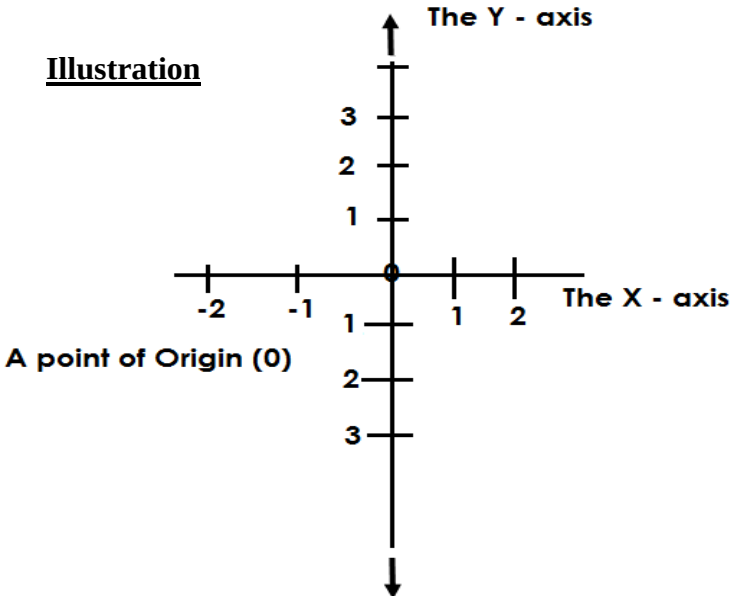
Show the two journeys on a travel graph. Using the grid below.

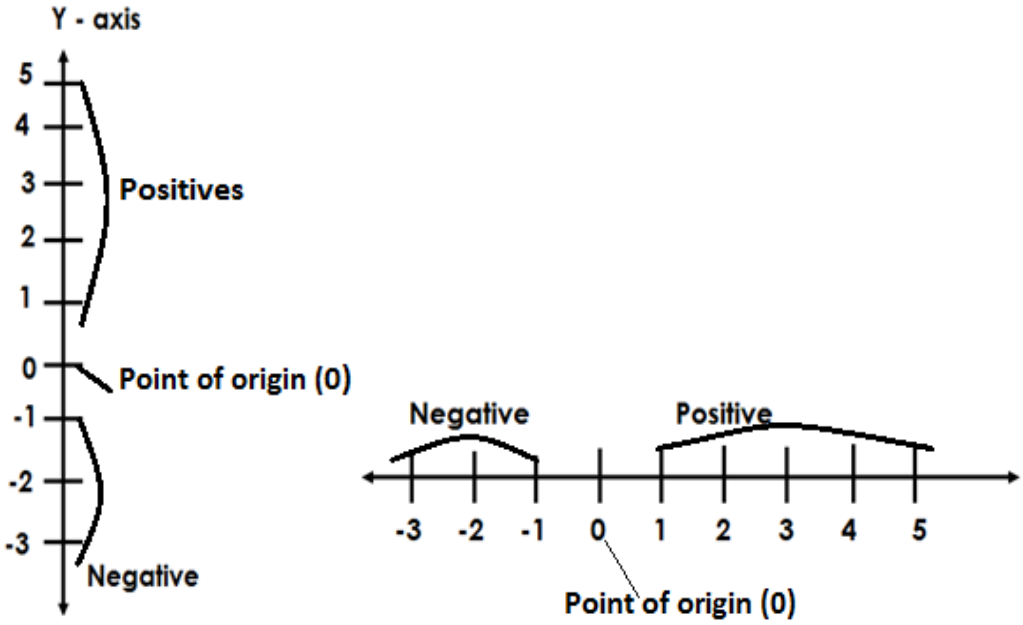


COORDINATE GRAPHS

- The coordinate graph has two axes; vertical axis (Y–axis) and Horizontal axis (X–axis)
- A co-ordinate is a pair of ordered integers that can be used to plat a point.
- Co-ordinates are in form of (X, Y) where X is an integer plotted on the x-axis and Y is an integer plotted on the y-axis.
- The two axes cross each other at a point of origin mark O.

Illustration



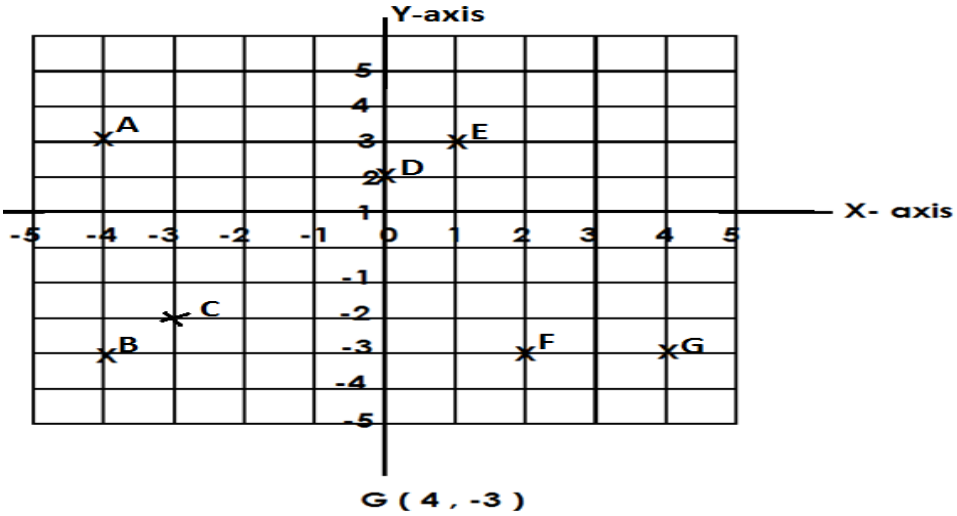


NAMING CO-ORDINATES PLOTTED

Note:

(X, Y)

Starting point of X-axis Steps to be moved on the y-axis



A (-4 , 3)

B (-4 , b)

C (-2 , -2)

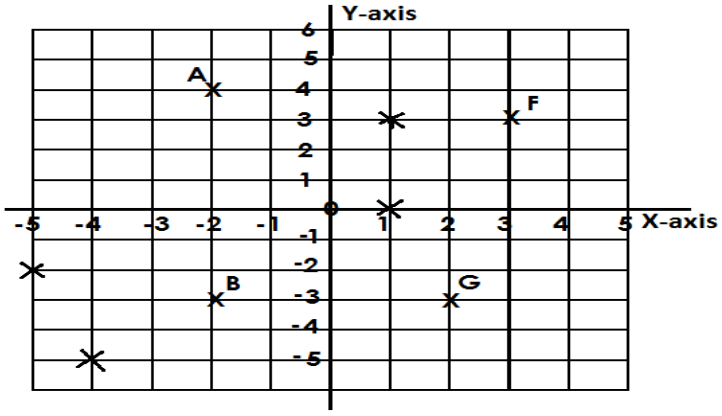
D (0 , 2)

E (1 , 3)

F (2 , -3)

Exercise:

Use the graph below to name the co-ordinates for the given points.

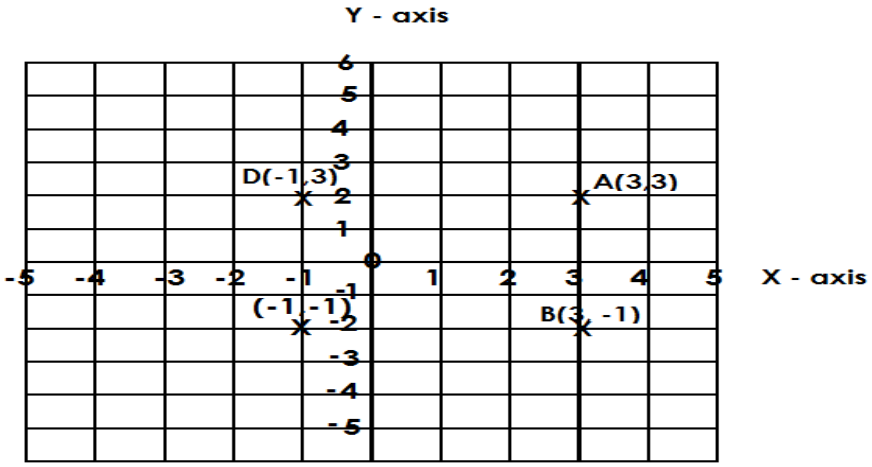


FORMATION OF FIGURE BY PLOTTING

Example

Using a 10 by 10 grid graphs plot the following points.

A(3 , 3) , B (3 , -1) , D (-1 , 3) C (-1 , -1)



b) Join A to B to C and C to D and D to A.

c) Name the figure formed.

It's a square

d) Find its area

soln.

Area = 5 x 5

= 4sq X 4sq

= 16 Sq units

Activity:

1. Draw a grid coordinate graph and plot the following points

a) A (4 , 4) , B (0 , 0) , C (4 , 0)

b) P (-5 , 2) , Q (-5 , -5) , R(3 , -5) , s (3, -2)

c) U (-6 , 0) , V (-3 , 0) , W (-3 , -2) X (2 , -2) Y (2 , -4) , Z(-6 , -4)

d) D(-7 , 1) E (-3 , 1) F(-4 , 5) G(0 ,5)

2. Draw co-ordinate graph and plot the given ordered pairs, name the figure formed and work out their area.

a) A(-1 , 2) , B (-4 , -3) , C(3 , -3)

b) P(-1 , 1) Q (-1 , 4) R(5 , 1) S(5,4)

c) W(-2 , -2) X(-2 , 3) ,Y (1 , 3) Z (4 , -2)

d) A(-3 , 2) B(0,0) , C(0,4) D(5, 2)

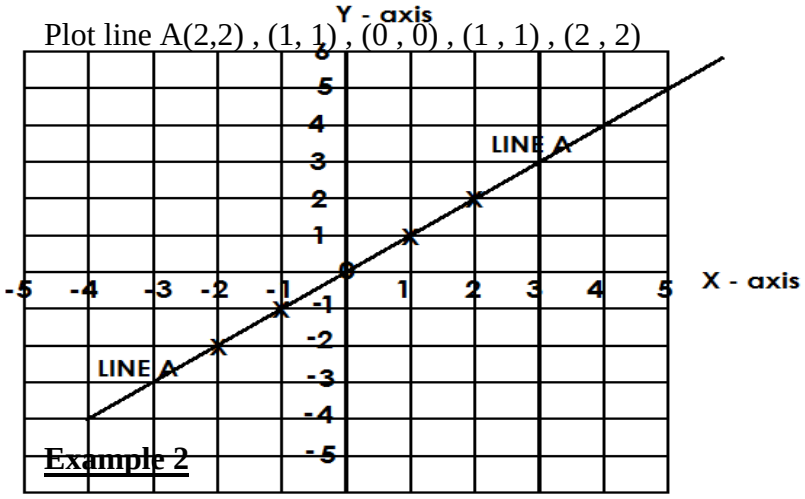
e) K(-4 , -2) I(-1 , 1) M(4 , -2) , N(7 , 1)

f) O(2 , -1) , P(5, -1) Q(2 , -5) , R(5 , -5) S(9 , -3

PLOTTING COORDINATES THAT FORM LINES

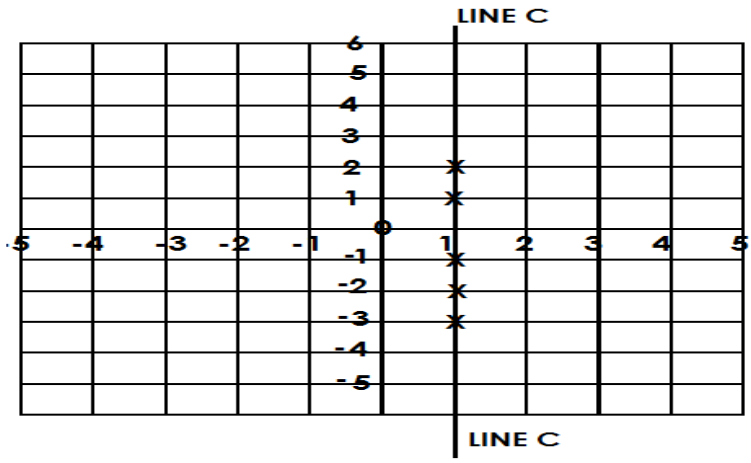
Example I

Plot line A(2,2) , (1, 1) , (0 , 0) , (1 , 1) , (2 , 2)



Example 2

Plot line C = (1 , 2), (1,1) , (1,-1) , (1, -2) , (1 , -3)



Activity:

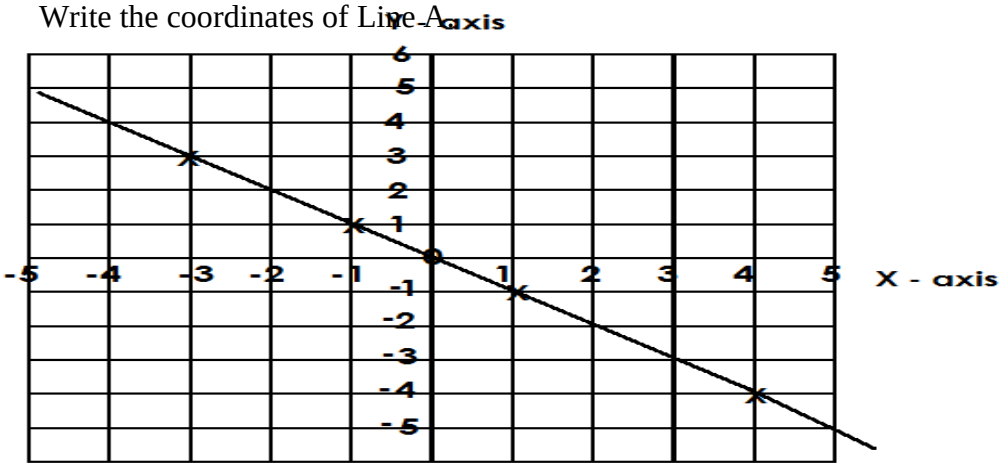
Draw coordinate graphs, plot the given coordinates and join to form the given lines below:-

1. Plot line D = (-3, 1), (-2, 0), (-1, -1), (0, -2), (1, -3)
2. Line B = (-3, 2), (-2,2), (-1, 2), (0,2), (1, 2), (2,2)
3. Line P = (-3 , 3) , (-2 , 2) , (-1 , 1) (0,0) , (1 , -1) , (2 , -2)
4. Line Q = (-2, -4), (-1, -3), (0,-2), (1, -1), (2, 0), (3, 1)
5. Line F = (3, -3), (-2,-3), (-1, -3), (0, -3), (1, -3)
6. Line M = (-3, -4), (-3,3), (-3, 2), (-3,1), (-3,0), (-3, -1)

INTERPRETING COORDINATES THAT FORM LINES

Example

Write the coordinates of Line A.

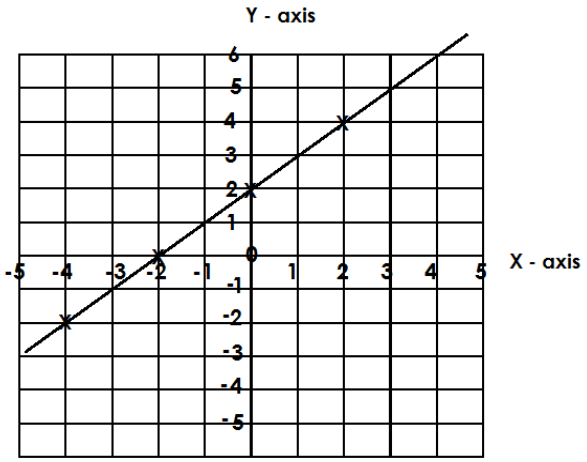


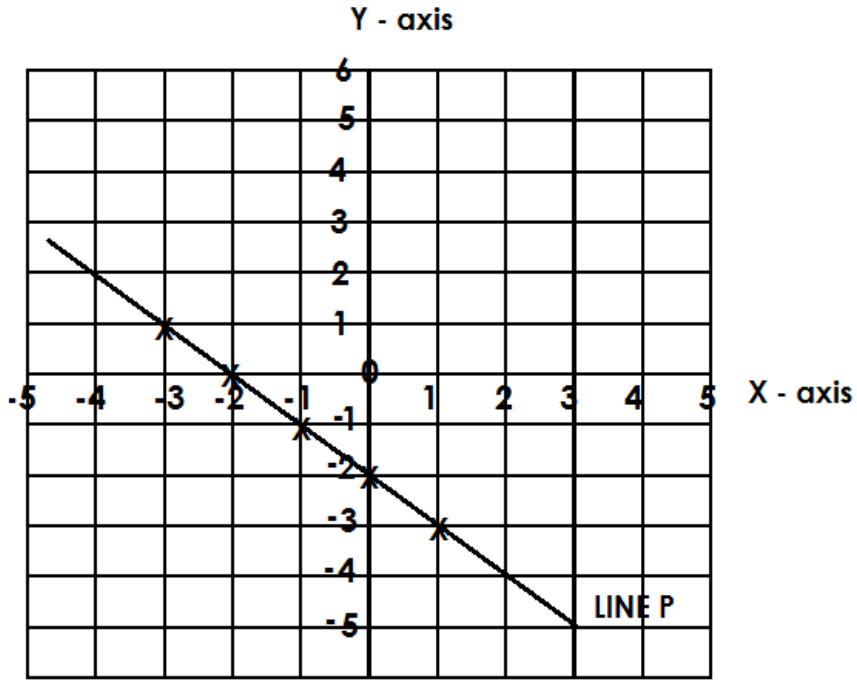
Soln.

Line A = (-3, 3), (-1 ,1), (1, -1), (4, -4)

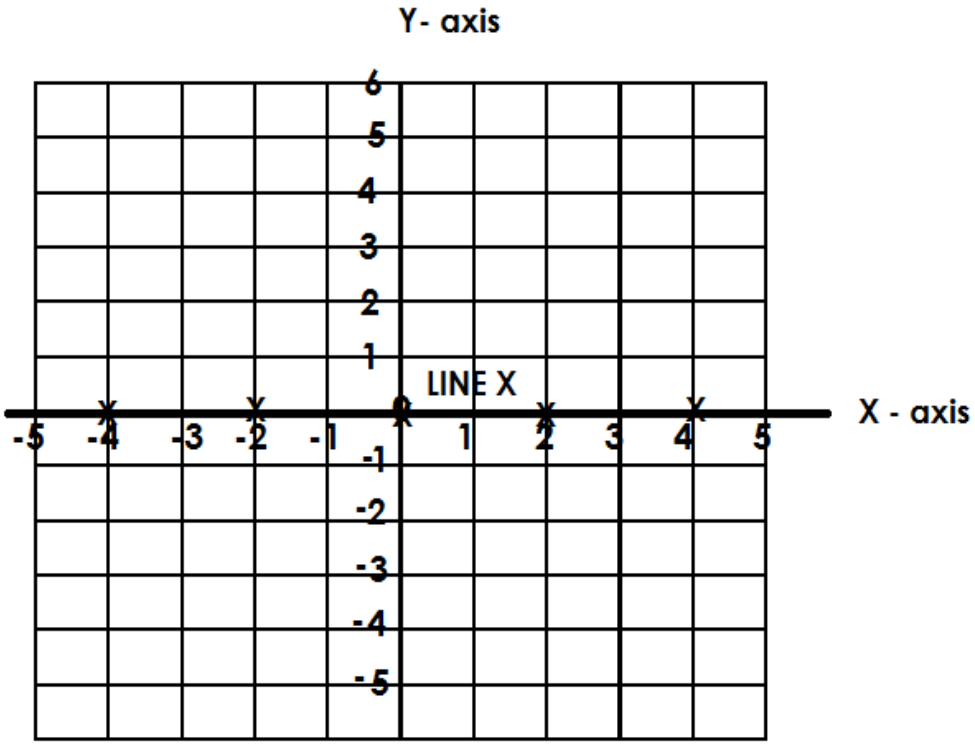
Activity:

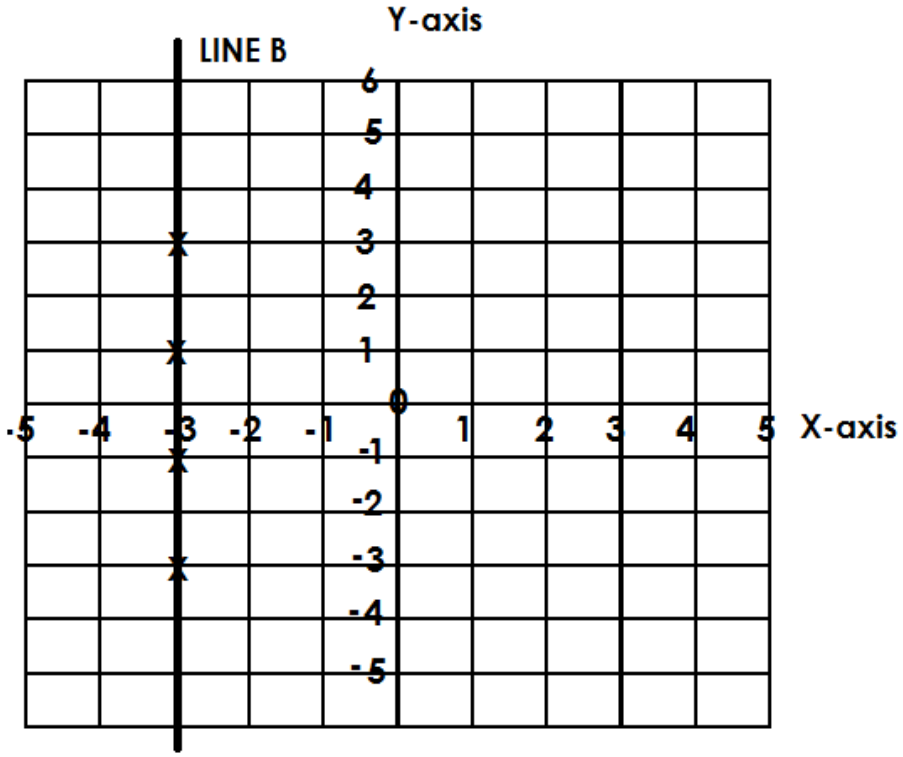
1. Write the coordinates of the lines drawn in each of the coordinate graphs below.





LINE X





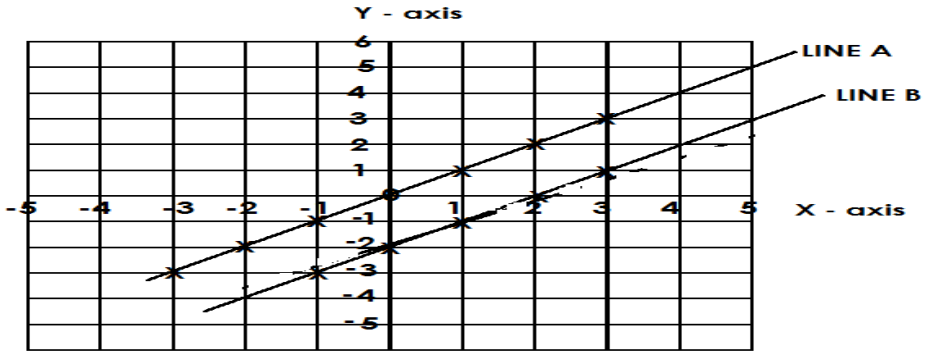
FORMING EQUATIONS OF GIVEN LINES ON A GRAPH

Example I

Given Lines A and B.

Line A passes through (-3 , -3), (-2 , -2), (-1 , -1), (0,0), (1, 1) among others.

Line B passes through (-2 , -4) , (-1 , -3) , (0 , -2), (1,-1), (2,0), (3,1).



X	-3	-2	-1	0	1	2	3
Y	-3	-2	-1	0	1	2	3

X	-2	-1	0	1	2	3
Y	-4	-3	-2	-1	0	1

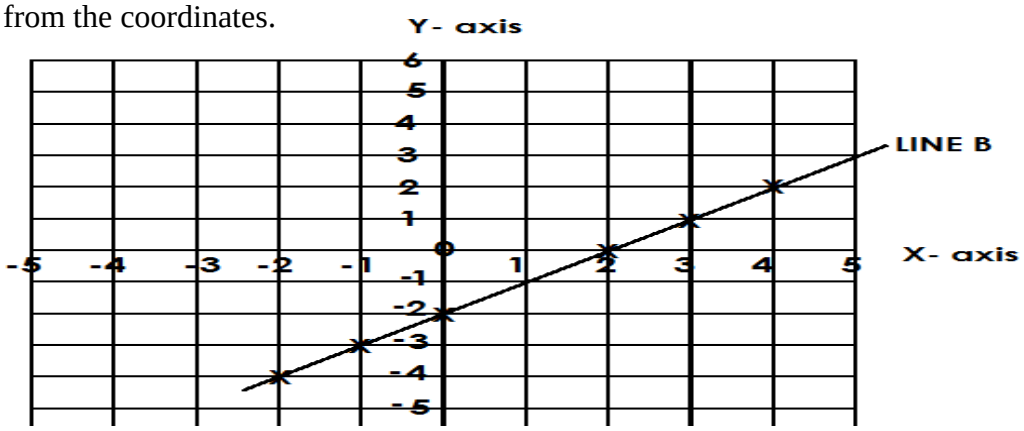
The equation of line A is the equation of Line B.

Line A is Y = X – 3

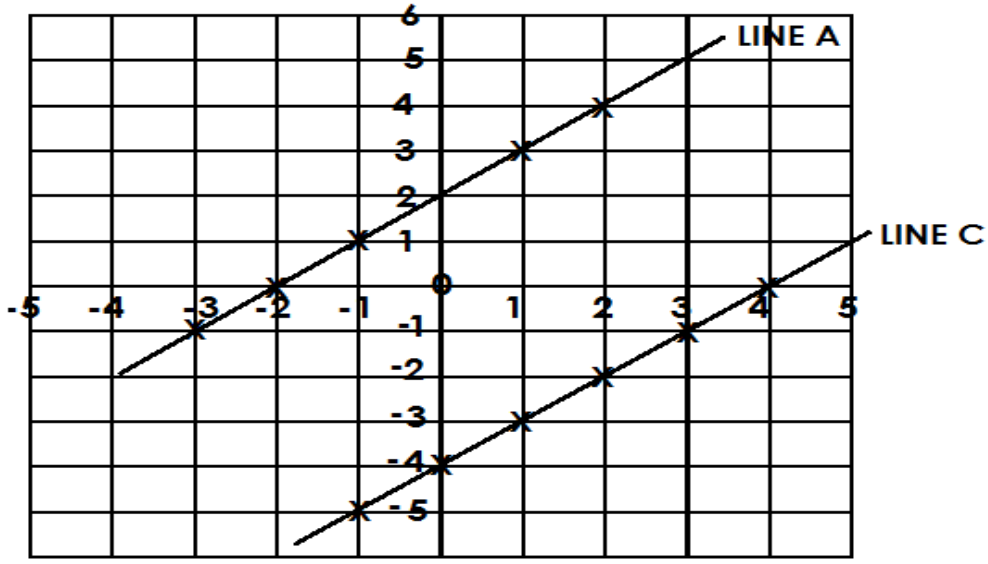
Activity:

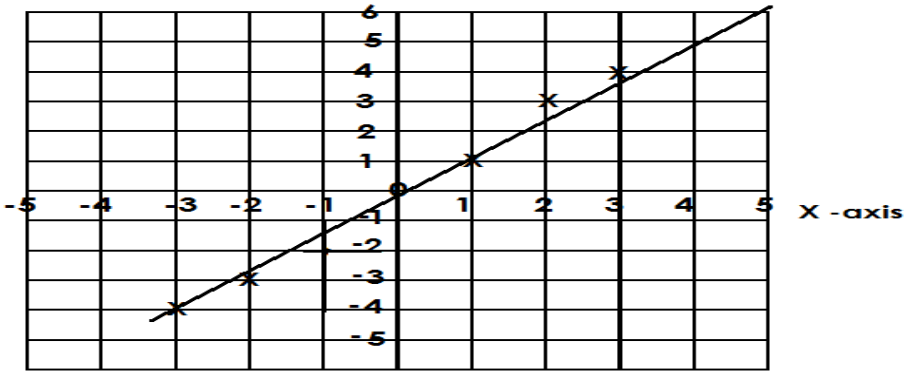
Study the graphs below and write the coordinates for each Line and form the equation of each line from the coordinates.

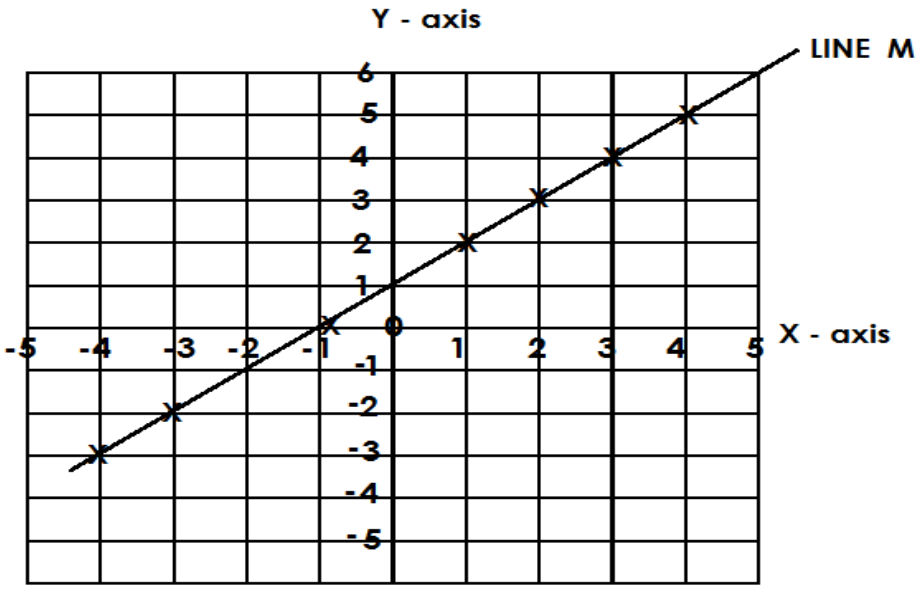
1.



2.







DRAWING LINES FROM THE GIVEN EQUATIONS.

Example I

Given the equation Y = X +1

a) Complete the table below

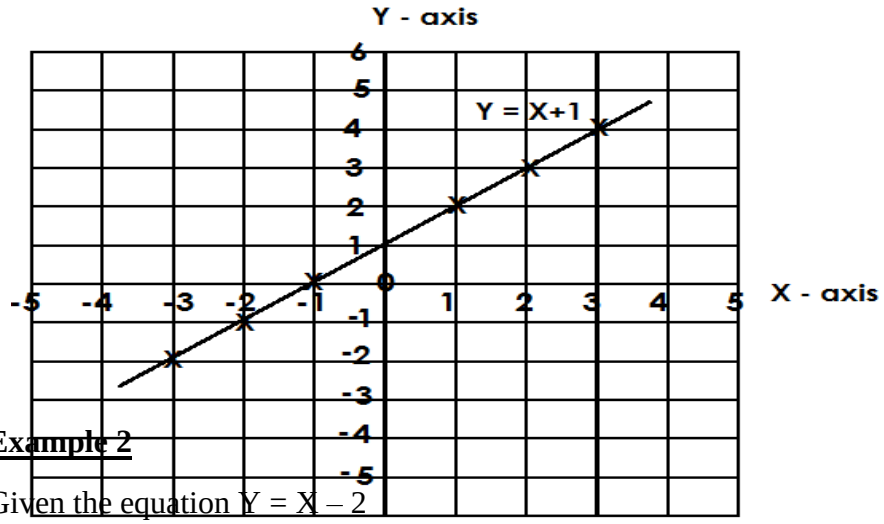
X	-1	0	1	2	3	-3	-2	-1
Y	0	1	2	3	4	-2	-1	0

Soln.

When x=-1	**When Y=1**	**When X=1**	**When Y=3**
Y=X+1 Y=-1+1 Y=0	1=X+1 1-1=X+1-1 0=X X=0	Y=X+1 Y=1+1 Y=2	Y=X+1 3=X+1 3-1=X+1-1 2=X X=2
When X=3	**When X=-3**	**When Y=-1**	**When X=-1**
Y=X+1 Y=3+1	Y=X+1 Y=-3+1	Y=X+1 -1=X+1	Y=X+1 Y=-1+1

Y=4	Y=-2	-1-1=X+1-1 -2=X X=-2	Y=0

Draws a graph for the above equation.



Example 2

Given the equation Y = X – 2

a) Complete the table below.

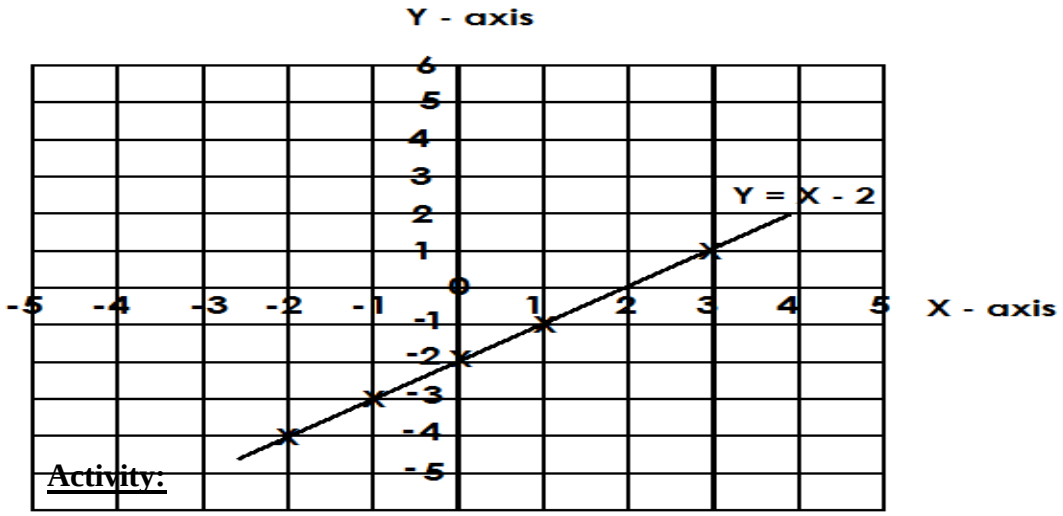
X	0	-1	2	3	-1	-2
Y	-2	-1	0	1	-3	-4

Soln.

When X=0	**When Y= -1**	**When X=2**
Y=X-2 Y=0-2 Y=-2	Y=X-2 -1=X-2 -1+2=X-2+2 -1=X X= -1	Y=X-2 Y=2-2 Y=0
When Y=1	**When X= -1**	**When Y= -4**
Y=X-2 1=X-2 1+2=X-2+2	Y=X-2 Y= -1-2 Y= -3	Y=X-2 -4=X-2 -4+2=X-2+2

3=X X=3		-2=X X= -2

Below is a graph for the above equation Y = X – 2



Activity:

1. Draw graphs for each of the following equations: -

a) Y = X + 2

b) Y = X – 3

c) Y = X + 4

d) Y = 2 x -3

e) Y = 2x – 1

2. Given the equation Y = 2x + 2

a) Complete the table below.

X	0	-1	-	-3	1	2
Y	2	-	-2	-	-	-

b) Draw a graph for the above equation

Scenarios

TOPIC. 11: TIME

Session 1

- ✦ Converts 12 to 24-hour clock system and vice versa.
- ✦ Finds duration
- ✦ Finds distance, time and speed.

Methods

- **Guided discovery**
- **Group discussion**
- **Think, pair and share.**
- **explanation**

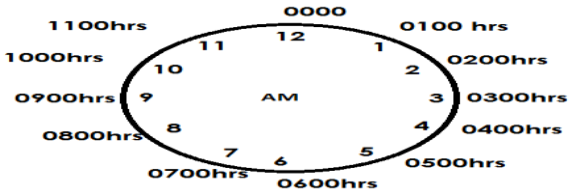
Lesson activities

- **adding**
- **multiplying**
- **subtracting**

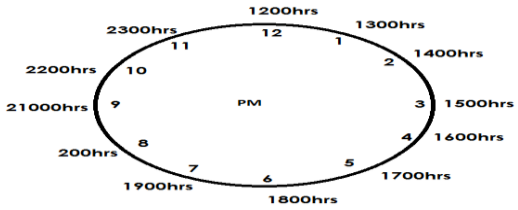
Content/facts

CHANGING TIME FROM 12 HOURS CLOCK SYSTEM TO 24 HOUR CLOCK SYSTEM

AM CLOCK FACE



PM CLOCK FACE

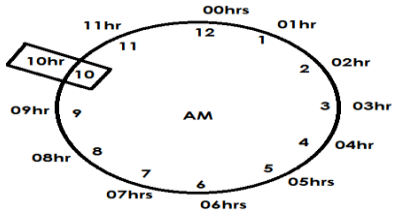


Note:

If the given time is in am, use the am part and when it is pm use pm clock face.

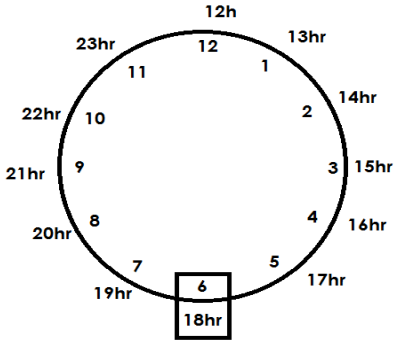
Examples.

1. Change 10:40am to 24 hours clock system



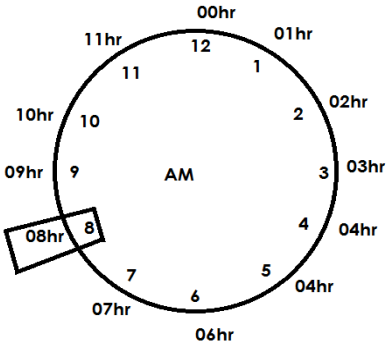
10:40am = 1040hrs

2. Change 6:20pm to 24 hour clock



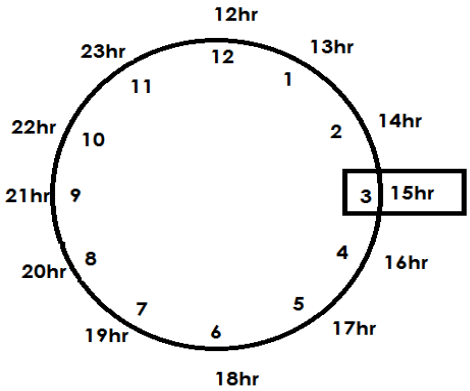
6:20pm = 1820hrs

3. Change 8:14am to 24 hours clock system



8: 14am = 0814hrs

4. Change 3:57 pm to 24 hours clock



3:57pm = 1557hrs

Activity:

1. Change the following to 24-hour clock system

a) **9:00am**

b) **1240pm**

c) **11:45pm**

d) **7:30am**

e) **9:42pm**

f) **8:50pm**

g) **10:59pm**

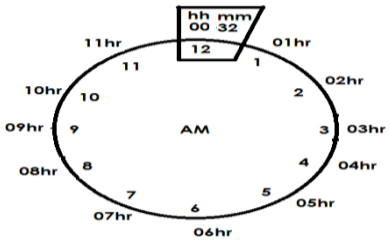
h) **4:00pm**

CHANGING TIME FROM 24 HOUR CLOCK SYSTEM TO 12 HOUR CLOCK SYSTEM

Note:

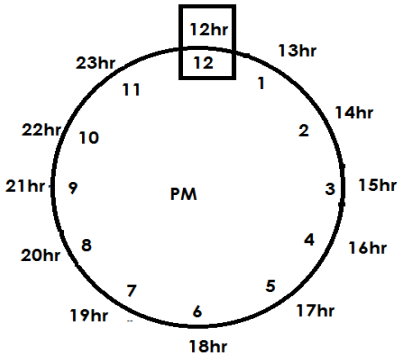
- **If the given time in 24-hour clock system is below 12 hours, use am clock face.**
- **If the given time in 24 hour clock system is 12 hours and above, use the pm clock face.**
- **While writing time in 12-hour clock system, the part of the day MUST be indicated (am, pm)**

1. **Change 0032hrs to 12hr o'clock system**



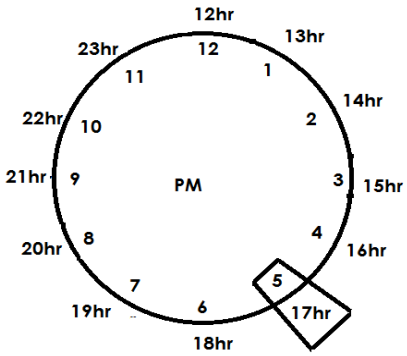
00 32hrs = 12:32am

2. **Change 1240hrs to 12 hour clock system**



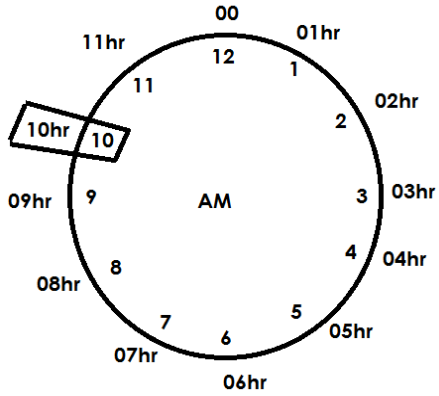
12 40hrs = 12:40pm

3. Change 17 50 hours to 12 hour clock system



17:50hrs = 5:50pm

4. Change 10:40hrs to 12hr clock system



1040hrs = 10:40am

<u>Activity:</u>

Change the following to 12-hour clock system.

a) 1026hrs

b) 0930hrs

c) 0845hrs

d) 1818hrs

e) 2340hrs

f) 1734hrs

g) 12:00hrs

TIME, DISTANCE AND SPEED

TIME:

FINDING DURATION WITH IN THE SAME PART OF THE DAY.

1. A lesson started at 8:00am and ended at 10:00am. For how long was the lesson.

Soln.

Duration = Ending time – starting time

= E.T - S. T

Hrs	min
10	00
-8	00
2	00

∴ It was for 2 hours

2. A meeting which stated at 9:20 am ended at 11:00am. For how long was the meeting?

Soln.

Duration = E.T – S.T

HRS	min
10	60
11	00
-9	20
1 :	40

It was for 1hr 40min

3. A baby went to bed at 2:40pm and it woke up at 4:10pm. For how long was it in bed?

Duration = E. T - S.T

HRS	Min	
3	(60	
4	10)	
-2	40	70
1	30	-40

It was for 1 hour 30 min

4. A math test started at 8:20am and ended at10:50 am. For how many hours was the test?

Soln.

Duration = E.T - S.T

HR	min
10	50
- 8	20
2 :	30

It was for 2hrs $\frac{30}{60}$

It was for 2 ½ hrs.

Activity:

1. A math lesson started at 12:30pm and ended at 12:40pm. How long was the lesson?

2. An interview started at 7:20am and ended at 10:00am. For how long was the interview.

3. A Journey started at 3:40pm and ended at 7:20pm. For how long was the lesson.

4. A workshop which started at 7:20am and ended at 11:50am. For how long was the workshop?

5. A football match started at 8:20pm and ended at 11:10pm. For how long was the match.

FINDING DURATION ACROSS DIFFERENT PARTS OF THE DAY.

Examples.

Askari reported on duty at 8:00am and ended at 1:00pm.

Soln

	Hrs. : min
Duration =	4 : 00
	+ 1 : 00
	5 : 00

HRS		MIN
12	:	00
-8	:	00
4	**:**	**00**

He was on duty for 5 hrs

OR

8:00am = 8 : 00

00 : 00hrs

08 00hrs

1:00pm = 1 : 00

+12 : 00hr

13 00hr

Duration = E.T - S. T

HR - Min

13 : 00

- 08 : 00

5 : 00

He was on duty for 5 hours

2. A meeting started at 10:40 a.m. and it ended at 2:30pm. For how long was the meeting.

From 10:40 ⟶ Mid-day

HR	Min
12	00
- 10	40
1	20

Duration = Hrs Min

1 : 20

+ 2 : 30

3 : 50

The meeting took 3 hours 50 minutes

3. How many hours are between 9:20am and 2:30pm?

Soln.

HR		MIN
11	:	60
12	:	00
- 9	:	20
2	**:**	**40**

Duration =

HR		MIN
2	:	40
+2	:	30
5	**:**	**10**

70 ÷ **60** = 1 r 10

$\frac{10}{60}$

Its $5\frac{1}{6}$ hrs

ACTIVITY

1. A workshop which started at 11:40 am and ended at 3:20pm. For how long was the workshop.

2. A science lesson started at 8:20 am and 5:10pm, For how long was the lesson?

3. A tax collection started at 11:00am and ended at 4:20pm. For how long did the collection take?

4. A church serves started at 10:40am. Fr how long was the church service?

5. Athletics competition started at 9:30am and ended at 2:10pm. For how long was the competition?

FINDING ENDING TIME GIVEN STARTING TIME AND DURATION

1. A test which started at 8:15 am lasted for 2hrs. at what time did it end.

Ending time = starting time + Duration

E. T = S.T TD

HR		MIN
8	:	15
+2	:	00
10	**:**	**15**

It ended at 10:15 am

2. A meeting which started at 10:15am lasted for 1 hr 45min. At what time did it end.

E.T = S.T + D

HR		min
10	:	15
+ 1	:	45
12	:	00

1hr

60) 60

60

It ended at 12:00 noon

3. Sarah leaves work for home at 6:30p.m. it takes her 1 hour 40 minutes to get home. What time does she get home?

E.T = S.T + D

HR		min
6	:	30
+ 1	:	40
8	**:**	**10**

1hr

60) 70

60

10min

She gets home at 8:10pm

4. A bus left Kampala at 11:40am travelling to Masaka for $2\frac{1}{5}$ hrs. At what time did it reach?

1hr = 60 min

$\frac{1}{5}$hr = $\frac{1}{5}$ x 60

= 12 min

$2\frac{1}{5}$ hr = 2 hr 12 min

E.T = S.T + D

HR		min
11	:	40
+ 2	:	12
13	**:**	**52**

HR		Min
13	:	52
- 12	:	00
1	:	52

It ended at 1:52 pm

5. Okello started performing at 3:30pm and took 90 minutes. At what time did it end?

E.T = S.T + D

HR		MIN
3		30
+ 1		30
5	:	00

60) 60

- 60

30 mm

90 mm = 1hr 30min

it ended at 5:00 pm

Activity:

1. A football match started at 1:30pm and lasted for 1hr 30 minutes. At what time did it end?

2. A forty minutes lesson started at 11:00am. At what time did it end?

3. A Kamba bus started at 4:30pm and moved for 2 hr 30 minutes. At what time did he reach?

4. A science test that lasted for 2 ¼ hours, started at 9:40 am. At what time did it end?

5. A meeting started at 11:42 am and it lasted for 2 ½ hours. At what time did it end?

6. A parent's meeting which lasted for 5 1/5 hours started at 10:40am. At what time did it end?

<u>FINDING STARTING TIME GIVEN ENDING TIME AND DURATON</u>

1. A fifty minutes lesson ended at 11:20a.m at what time did it start?

Soln.

S. T = E.T - D

HR	MIN
10	60
~~11~~ :	20
-	50
10 :	30

80
- 50
30

The lesson started at 10:30 a.m.

A birthday party which lasted for 2 ½ hours ended at 11:20am. At what time did it end?

Soln.

S. T = E.T - D

HR	MIN
10	60
11 :	20
-2	30
8:	50

80
- 30
50

The lesson started at 8:50 a.m.

3. A function which lasted for 4 $\frac{2}{5}$hrs ended at 3:05pm. At what time did it start?

Duration = 4hrs ($\frac{2}{5}$ x 60) min

= 4 hrs 24 min

S. T = ET - D

		HR		Min
30	:	05pm	3	: 05
			+ 12	: 00
			15	: 05hrs

HR		MIN	
14		60	65
~~15~~	:	05	-24
- 4	:	24	**41**
10	:	**41**	

It started at 10:41 a.m.

Activity:

1. A marathon which lasted for 2 hours ended at 10:00am. At what time did it start?

2. A lesson which lasted for 56 minutes ended at 11:20am. At what time did it start?

3. A drive which lasted for 2 $\frac{1}{4}$ hrs ended at 10:30pm. At what time did it start?

4. A function which lasted for 5 $\frac{2}{3}$ hours ended at 2:05pm. When did it start?

5. A man covered a certain journey from town A to town B in 4 $\frac{4}{5}$ hours. If she reached town B at 3:10pm, At what time did she leave town A?

6. A security guard who works for 7 $\frac{2}{3}$ hours left work at 2:00pm. At what time did he start working?

FINDING DISTANCE GIVEN SPEED AND TIME

Distance = speed x Time

1. A taxi covered a certain distance at a speed of 60km/hrs in 3 hours.

Distance = S x T

$= \frac{60\text{km}}{\text{hr}}$ **x 3 hrs**

= 180km

2. A car takes 2 ½ hours to cover a journey at a speed of 40km/hr. find the distance travelled.

Distance = S x T

$= \frac{40\text{km}}{\text{hr}}$ **x** $\frac{45}{60}$ **hrs**

= 90km

4. Calculate the distance travelled in 7 seconds at 8m/sec.

D = S x T

$= \frac{8\text{m}}{\text{sec}}$ **x 7 sec**

= 56m

Activity:

1. Find the distance travelled by a car in 4 hours at a speed of 30km/hr.

2. Asio's journey took 3 hours at an average speed of 45km/hr. how long was the journey?

3. Find the distance covered by a car at a speed of 80km/hr in ¼ hours.

4. A taxi covered a certain distance in 1 ½ hours at a speed of 80km/hr. find the distance covered.

5. John takes 40 minutes to cover a journey at a speed of 60km/hr. How long was the journey?

6. Amooti takes 48 minutes travelling at 80kph. How far does he go?

7. A car moves at a speed of 80km/hr. what distance does it cover in 15 minutes?

8. The speed of a car is 30m /sec. calculate the distance it covers in 40 minutes.

9. A horse moves at a speed of 12 m/sec. calculate the distance it covers in 20 minutes.

FINDING TIME, GIVEN DISTANCE AND SPEED

NOTE:

Time = distance ÷ speed

Example 1

Find time taken to cover a distance of 120km at a speed of 40km/hr.

Soln.

Time = distance ÷ speed

$$=120\text{km} \div \frac{40\text{km}}{\text{hr}}$$

$$=120\text{km} \times \frac{\text{hr}}{40\text{km}}$$

$$= 3 \text{ hrs.}$$

2. If a bus moves at 30km/hr. and covers a distance of 240km; How long does it takes to cover the journey.

Time = D ÷ S

$$= 240\text{km} \div \frac{30\text{km}}{\text{hr}}$$

$$= 240\text{km} \times \frac{\text{hr}}{30\text{km}}$$

= 8 hours

3. A taxi covered 100km at a speed of 40km/hr. for how long did it move?

Time = D ÷ S

$$= 100\text{km} \div \frac{40\text{km}}{\text{hr}}$$

$$= 100\text{km} \times \frac{\text{hr}}{40\text{km}}$$

$$= 2\frac{1}{2}\text{hrs.}$$

Activity:

1. How long will it take a cyclist to cover a distance of 80km at a speed of 20km/hr.

2. The speed of a cyclist is 70km/hr. how long will he take to cover a distance of 350km?

3. Find the time taken to cover a distance of 90km at a speed of 40km/hr.

4. A lex covered a distance of 90km at a speed of 20km/hr. find the time taken.

5. How long will a car take to cover a distance of 180km at a speed of 60km/hr?

6. Find the time taken by a taxi to cover a distance of 140km at a speed of 50km/hr.

MORE ABOUT TIME TAKEN

1. Car covered a distance of 120km at an average speed of 60km/hr. how much longer does it take if it moves a speed of 40km/hr.?

Time at 60km/hr

T = D ÷ S

=120km ÷ $\frac{60km}{hr}$

=120km x $\frac{hr}{60km}$

= 2 hours

More hours = 3hrs – 2hrs

= 1 hour

Time at 40km/hr

T = D ÷ S

=120km ÷ $\frac{40km}{hr}$

=120km x $\frac{hr}{40km}$

= 3 hours

2. At a speed of 30km/hr, a car covers a distance of 180km. how many less hours does it take to cover the same distance at a speed of 90km/hr?

Time at 30km/hr

T = D ÷ S

= 180km ÷ $\frac{30km}{hr}$

=180km x $\frac{hr}{30km}$

= 6 hours

Time at 90kmkm/hr.

T = D ÷ S

= 180km ÷ $\frac{90km}{hr}$

= 180km x $\frac{\text{hr}}{\text{90km}}$

= 2 hours

Less hours = 6hours – 2 hours

= 4 hours

3. Okello covered a distance of 240km at a speed of 80km/hr. How many more hours will he take to cover the same distance at a speed of 30km/hr?

Time at 80km/hr

T = D ÷ S

= 240km ÷ $\frac{\text{80km}}{\text{hr}}$

= 240km x $\frac{\text{hr}}{\text{90km}}$

= 3 hours

Time at 30km/hr.

T = D ÷ S

= 240km ÷ $\frac{\text{30km}}{\text{hr}}$

=240km x $\frac{\text{hr}}{\text{30km}}$

= 8 hours

More hours = 8 hours – 3 hours

=5 hours

ACTIVITY:

1. At a speed of 80km/hr, a car covers a distance of 240km. how many less hours does it take to cover the same distance at a speed of 120km/hr?

2. A cyclist covered a certain distance of 750km at a speed of 30km/hr. how many less hours can the same car take to cover the same distance at a speed of 50km/hr.?

3. How many more hours will a car travelling at a speed of 70km/hr. take to cover a journey of 350km if its speed is reduced to 50km/hr?

4. Find how many less hours will a taxi travelling at a speed of 50km/hr take to cover a journey of 300km. if its speed is increased?

MORE ABOUT DISTANCE AND TIME

1. A taxi and a bus left station A for station B a distance of 240km apart travelling at a speed of 40km/hr and 60km/hr respectively. How far from B will a taxi be when the bus arrives at station B?

Time of taxi	Time of bus	More time
$T = D \div S$ $240km \div \frac{40km}{h}$ $240km \times \frac{hr}{40km}$ 6 hours	$T = D \div S$ $240km \div \frac{60km}{h}$ $240km \times \frac{hr}{60km}$ 4 hours	6 hours – 4 hours = 2 hours

Distance of taxi in 2 hours.

$D = S \times T$

$= \frac{40km}{hr} \times 2\ hrs$

$= 80km$

2. Town P and town Q are 480km apart. A taxi and a cyclist left town P travelling at 60km/hr. and 40km/hr. respectively. How far from town Q will a cyclist be when a taxi arrives there?

Time of a taxi

$T = D \div S$

$= 480km \div \frac{60km}{hr}$

$= 480km \times \frac{hr}{60km}$

$= 8\ hours$

Time of cyclist

$T = D \div S$

$= 480km \div \frac{40km}{hr}$

$= 480km \times \frac{hr}{40km}$

$= 12\ hours$

More hours = 12 hours – 8 hrs

= 4 hours

Distance of cyclist in 4hrs

T = D x S

$= \frac{40km}{hr}$ **x 4hrs**

= 160km

3. A cyclist and a motorist travelled from Peru to Meru a distance of 120km at a speed of 20km/hr and 60km/hr respectively. Find the distance that will have been covered by the cyclist when the motorist arrives at Meru.

Time by motorist

T = D ÷ S

$= 120km \div \frac{60km}{hr}$

$= 120km \times \frac{hr}{60km}$

= 2 hours

Distance by cyclist in 2hrs

D = S x T

$= \frac{20km}{hr}$ **x 2hrs**

= 40km

b) How far from Meru will a cyclist be when a motorist arrives there?

120km
- 40km
80km

Activity:

1. Towns K and P are 360km apart. A bus and a taxi left town K travelling at a speed of 120km/hr. and 40km/hr. respectively. How far from P will a taxi be when the bus arrives there?

2. A motorist and a cyclist left town X for Y a distance of 240km apart. While moving at a speed of 30km/hr. and 20km/hr. respectively. How far from town Y will a cyclist be when the motorist arrives town Y.

3. Bus station A and Bus station B are 180km apart. Gaga bus and Akamba left station A for B while moving at a speed of 90km/hr. and 30km/hr. respectively.

a) Find the distance that would have been covered by the Akamba bus when the Gaga bus arrives station B.

b) How far from station B will Akamba bus be when Gaga bus arrives there?

4. Othieno and his Friends home are 72km away from town Othieno and his Friend rode bicycles to town at a speed of 36km/hr. and 12km/hr. respectively.

How far from town will Othieno friend be when Othieno arrives town?

5. Mutasa and Mutaali were riding bicycles from home to the nearby town which 40km away at a speed of 8km/hr. and 5km/hr. respectively.

a) Find the distance that Mutaali would have covered when Mutasa arrived town.

b) How far from Town will Mutaali be when Mutasa arrives town?

FINDING SPEED GIVEN TIME AND DISTANCE

Speed is the rate at which a certain distance is covered at a given time.

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}}$$

OR

Speed = Distance ÷ Time if given fraction.

1. Find the speed used to cover a distance of 120km. in 3 hrs.

$$S = \frac{D}{T}$$

$$= \frac{120\text{km}}{3\text{hrs}}$$

$$= 40\text{km/hr.}$$

2. A taxi left town K to town P a distance of 90km. What speed did it use to cover that distance in 2hrs?

$$\text{Speed} = \frac{D}{T}$$

$$= \frac{90\text{km}}{2\text{hrs}}$$

$$= 45\text{km/hr.}$$

3. Find the speed used to cover a distance of 100km in $2\frac{1}{2}$ hrs

Speed = D ÷ T

= 100km ÷ 2 ½ hrs.

= 100km ÷ $\frac{5}{2}$ hrs

= 100km x $\frac{2}{5hr}$

= 40km/hrs.

4. Wandela covered a distance of 120km in 45 mins. Find the speed used in km/hr.

S = D ÷ T

= 120km ÷ $\frac{45hrs}{60}$

= 120km x $\frac{60}{45hrs}$

= 160km/hr.

5. Find the speed used to cover a distance of 180km from 8:00am to 10:00am.

Duration = E.T – S.T

HR		MIN
10	:	00
- 8	:	00
2:	**:**	**00**

Speed = D ÷ T

= $\frac{180km}{2hrs}$

90km/hr

6. A taxi left Kampala at 8:00am t raveling to Iganga a distance of 120m. if it reached Iganga at 10:30am. Calculate its speed.

Soln.

Duration = E.T - S.T

HR		MIN
10	:	30
- 8	:	00
2	**:**	**30**

2Hr $\frac{30h}{60}$

D = $2\frac{1}{2}$ hrs

Speed = $\frac{D}{S} \div T$

= $120km \div 2\frac{1}{2} hr$

= $120km \div \frac{5}{2} hr$

= $120km \times \frac{2}{5hrs}$

= 88km/hr.

Activity:

1. Matama covered a distance of 120km in 2 hours. Calculate her speed in km/hr.

2. A car covered a distance of 180km in 4 ½ hrs. find the speed used.

3. Find the speed used to cover a distance of 100km in 1 ½ hrs.

4. Agaso bus covered a distance of 240km in 2 ½ hrs. calculate the average speed for the whole journey.

5. A taxi left Entebbe at 7:00a.m travelling to Kampala a distance of 120km. If it reached Kampala at 10:00am. Calculate its speed.

6. Musa covered a distance of 40km in 20 min. Find its speed in km/hr.

Scenarios

Session 2

- ✦ Solves word statements on distance, speed and time
- ✦ Solves word statements involving average speed.
- ✦ Converts in m/s to km/h and vice versa.
- ✦ Interpreting travel, class, work and personal time tables.

Methods

- **Guided discovery**
- **Group discussion**
- **Think, pair and share.**
- **explanation**

Lesson activities

- **adding**
- **multiplying**
- **subtracting**

Content/facts

CHANGING KM/HR TO M/SEC

1. Change 26km/hr to m/sec.

$$\frac{36km}{1hr} = \frac{m}{sec}$$

1km = 1000m

36km = (36 x 1000) m

= 36000m

1hr = 3600sec

$$\text{Speed} = \frac{D}{T}$$

$$= \frac{36000m}{3600S}$$

= 10m/sec

2. Convert 90km/hr. to m/s

$$\frac{90km}{hr} = \frac{m}{s}$$

1km = 1000m

90km = (90 x 1000) m

= 90,000m

1hr = 3600sec

Speed = $\frac{D}{T}$

$$= \frac{90{,}000m}{3600S}$$

= 25 m/s

3. Osako covered 360km in 2hours convert his speed in m/s

$$\frac{360km}{2hr} = \frac{m}{s}$$

1km = 1000m

360km = (360 x 1000) m

1hr = 3600 sec

2hrs = (3600 x 2) sec

Speed = $\frac{D}{T}$

$$= \left(\frac{360 \times 1000}{3600 \times 2\ sec}\right)m$$

= 50m/s

4.A distance from Mbale to Iganga is 216km.A car takes 2 hours to cover the distance. At what speed in m/sec does it take?

1km = 1000m

130km = (216 x 1000) m

1hr = 360sec

2hrs = (3600 x 2) sec

Speed = $\frac{D}{T}$

$= \frac{(216 \times 1000)m}{3600 \times 2\ sec}$

$= 30\ m/s$

Activity:

1. Convert the following km/hr. to m/s

a) 72km/hr.

b) 54km/hr.

c) 144km/hr.

d) 108km/hr.

e) 162km/hr.

f) 126km/hr.

2. Musono covered 72km in 2hrs. What speed did he use in m/s?

3. The distance from Mbarara to Masaka is 360km. if it takes a taxi 4hours to cover the journey. Calculate its speed in m/s.

CHANGING SPEED FROM M/S TO KM/HR

1. Changing 10m/sec to km/hr.

$\frac{10m}{s}$ **to** $\frac{km}{hr}$

1000m = 1km

1m = $(\frac{1}{1000})$m

10m = $(\frac{1 \times 10}{1000})$km

= $(\frac{10}{1000})$km

36000 sec = 1hr

1 sec = $(\frac{1}{3600})$ hr

Speed = D ÷ T

$= \frac{10}{1000}$ km ÷ $\frac{1}{3600}$ hr

$= \frac{10km}{1000} \times \frac{3600}{1hr}$

$= 36km/hr.$

2. Change 40m / sec to km/hr.

$\frac{40m}{sec}$ **to** $\frac{km}{hr}$

1000m = 1km

1m = ($\frac{1}{1000}$)km

40m = ($\frac{1}{1000}$ x 40) km

= $\frac{40}{1000}$ km

3600 sec = 1hr

1 sec = ($\frac{1}{3600}$)hr

Speed = D ÷ T

= $\frac{40}{1000}$km ÷ $\frac{1}{3600}$hr

= $\frac{40}{1000}$km x $\frac{3600}{1hr}$

= 144km/hr.

3. A plane covered 2000m in 80 seconds. Calculate its speed in km/hr

Soln.

1000m = 1km

1m = ($\frac{1}{1000}$)m

2000m = ($\frac{1}{1000}$ X 20000) km

= ($\frac{2000}{1000}$)km

3600 sec = 1hr

1 sec = ($\frac{1}{3600}$)hr.

80 sec = ($\frac{1}{3600}$)hr

= ($\frac{80}{3600}$)hr

Speed = D ÷ T

= ($\frac{2000}{1000}$)km ÷ $\frac{80}{3600}$ hr

$= \quad \left(\frac{2000}{1000}\right)\text{km} \div \frac{80}{3600}\text{hr}$

$= \quad \frac{2000}{1000}\text{km} \times \frac{3600}{30\text{hr}}$

$= \quad 90\text{km/hr.}$

Activity

1. Change the following m/sec to km/hr.

a) 15m / sec

b) 20m / sec

c) 30m / sec

d) 50m /sec

e) 100m / sec

2. A distance of 40m was covered in 2 seconds. Calculate the speed used in km/hr.

3. Okot covered 120m in 2 seconds. Calculate his speed in km/hr.

FINDING AVERAGE SPEED

PART A

- **Involving a return journey.**

Example 1

A motorist travelled from Kampala to Entebbe a distance of 240km in 4hrs. He later returned in 6 hours. Calculate his average speed for the whole journey.

Soln.

Journey	Time	Distance
K – E	4 hrs.	240km
E – K	6hrs	240km

Average speed $= \quad \frac{\text{Total distance covered}}{\text{total time taken}}$

$= \quad \frac{240\text{km}+240\text{km}}{4\text{hrs}+6\text{hrs}}$

$= \quad \frac{480\text{km}}{10\text{hrs}}$

= **48km /hr.**

2. Taire covered a distance of 150km in 2hrs and then returned in 4hrs. calculate his average for the whole journey.

Soln.

Journey	Time	Distance
1st	2hrs	150km
2nd	4hrs	150km

Average speed = $\frac{\text{total distance covered}}{\text{total time taken}}$

= $\frac{150\text{km}+150\text{km}}{2\text{hrs}+4\text{hrs}}$

= $\frac{300\text{km}}{6\text{hrs}}$

= **50km/hr.**

3. Esther covered a certain distance from home to town at a speed of 120km in 1½hrs then returned in one hour.

Calculate his average speed for the whole journey.

Journey	Time	Distance
Home – town	1 ½ hrs.	S x T 120km x 1 ½ hrs. 120km x $\frac{3}{2}$ 180km
Town – home	1hr	180km

Average speed = $\frac{\text{total distance covered}}{\text{total time taken}}$

= $\frac{180\text{km}+180\text{km}}{1\frac{1}{2}\text{hr}+1\text{ hr}}$

= **(360km) ÷ 2 ½ hr.**

$= \quad 360\text{km} \div \frac{2}{5\text{hr}}$

$= \quad 360\text{km} \times \frac{2}{5\text{hr}}$

$= \quad 144\text{km/hr.}$

Activity:

1. A taxi traveled from town A to town B a distance of 180km in two hours it later returned to town A in 3 hours. Calculate its average speed for the whole journey.

2. Amooti covered a distance of 200km from home to school in 2 hours. He later returned using the same route in 3 hours. Calculate his average speed for the whole journey.

3. A taxi travelled from town A to town B at a speed of 80km/hr. in 3 hours it later returned to town A in 2 hrs. calculate its average speed for the whole journey.

4. A bus travelled from Kampala to Bumba at a speed of 60km/hr. in 3 hours and then he returned in 2 hours. Calculate its average sped for the whole journey.

MORE ABOUT AVERAGE ON RETURN JOURNEY.

Examples.

1. A taxi covered a certain distance at a speed of 60km/hr. in 2hrs from town A to B then later returned at speed of 40km/hr. calculate its average speed for the whole journey.

Journey	Time	Distance
A – B	2hrs	D = S x T $\frac{60\text{km}}{1\text{hr}}$ x 2hr 120km
B – A	D ÷ S $\frac{120\text{km}}{40\text{km/hr}}$ 3hrs	120km

Average speed $= \quad \frac{\text{total distance covered}}{\text{total time taken}}$

$= \quad \frac{120\text{km}+120\text{km}}{2\text{hrs}+3\text{hrs}}$

$= \quad 48\text{km/hr.}$

2. Mandela left home travelling to town at a speed of 60km/hr. in 4hrs. he rested for an hour and later returned at a speed of 80km/hr. calculate his average speed for the whole journey.

Journey	Time	Distance
H – T	4hrs	S x T $\frac{60km}{1hr}$ x 4hrs 240km
Rest	1hr	-----
T - H	D ÷ S 240km ÷ $\frac{hr}{80km}$ 3hrs	

Average speed = $\frac{\text{Total distance covered}}{\text{Total time taken}}$

= $\frac{240km+240km}{4hrs+3hrs+1hr}$

= $\frac{480km}{8hrs}$

= 60km/hr.

3. A motorist covered a certain distance at a speed of 80km/hr. in 2½ hrs. from town P to town Q. He rested for ½hour he later returned at a speed of 100km/hr. calculate his average speed for the whole journey.

Journey	Time	Distance
P – Q	2 ½ hrs.	S x T $\frac{80km}{hr}$ x $\frac{5}{2}$ hrs 200km
Rested	$\frac{1}{2}$hrs.	
Q – P	D ÷ S $\frac{200km}{100km/hr}$ 2hrs	200km

Average speed $= \frac{\text{total distance}}{\text{total time}}$

$= \frac{200\text{km}+200\text{km}}{2\frac{1}{2}\text{hrs}+\frac{1}{2}\text{hrs}+2\text{hrs}}$

$= \frac{400\text{km}}{5\text{hrs}}$

$=$ **80km/hr.**

Activity:

1. A bus travelled from Kampala to Mukono at a speed of 60km/hr. in 3 hrs. and then later he returned at a speed of 90km/hr. calculate its average speed for the whole journey.

2, A cyclist travelled at a speed of 90km/hr. in 2 hrs. from Kimonolike to Mbale, it later returned using the same route at a speed of 60km/hr. calculate its average speed.

3. A motorist travelled at a speed of 120km/hr. in 3hrs from Kibuku to Mbale it stopped for 2 hours and later returned at a speed of 90km/hr. calculate its average speed for the whole journey.

4. A motorist covered a certain distance at a speed of 60km/hr. in 2hrs he rested for ½ hr. and later returned. Using the same route at a speed of 80km/hr. calculate his average speed for the whole journey.

FINDING AVERAGE SPEED OF CONTINUING JOURNEY

1. A taxi travelled at speed of 60km/hr. from Kampala to Jinja in 2hrs. It then continued to Iganga a distance of 80km in 3 hrs. Calculate its average speed for the whole journey.

Journey	Time	Distance
K – J	2hrs	S x T $\frac{60\text{km}}{1\text{hr}}$ X 2hrs 120km
Jinja – Iganga	3hrs	80km

Average speed $= \frac{\text{Total distance}}{\text{total time}}$

$= \frac{120\text{km}+80\text{km}}{2\text{hrs}+3\text{hrs}}$

$=$ **40km/hr.**

2. A man travelled at a speed of 40km/hr. from P – Q in 3 hrs. he rested for 1hour later continued to town X at a speed of 120km/hr. in 2hrs. Calculate his average speed.

Journey	Time	Distance
P – Q	**3hrs**	**S x T** $\frac{40km}{hrs}$ **x 3 hours** **120km**
Rested	**1hr**	**------**
Continued to town X	**2hrs**	**S x T** $\frac{120km}{hr}$ **x 2hrs** **240km**

Average speed = $\frac{\text{Total distance}}{\text{total time}}$

= $\frac{120km+240km}{3hrs+1hr+2hrs}$

= $\frac{360km}{6hrs}$

= **60km/hr.**

ACTIVITY

1. Crown bus travelled from town A to town B at a speed of 100km/hr. in 1 ½ hrs. he then continued to town C at a speed of 40km/hr. in 1 ½ hrs. workout its average speed for the whole journey.

2. Job covered a certain distance from Kampala to Iganga in 3 hours at a speed of 50km/hr. he rested for one hour and later continued to Mbale at a speed of 40km/hr. in 6hrs. calculate his average speed for the whole journey.

3. Kato left town A driving at 60km/hr. after 2hrs. his car got a puncture and delayed for 1 hour. He then continued at 80km/hr. for 3 hours to town B.

a) What distance had Kato covered before his car got a puncture.

b) How far is town B from A?

c) Calculate Kato's average speed for the whole journey.

Scenarios